WORLDWIDE INFLUENZA CENTRE
THE FRANCIS CRICK INSTITUTE
MILL HILL LABORATORY
LONDON

(formerly: the WHO CC for Reference and Research on Influenza, MRC National Institute for Medical Research, London)

Report prepared for the WHO annual consultation on the composition of influenza vaccine for the Southern Hemisphere 2016

21st – 23rd September 2015

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The epidemiological section of this report was prepared using data reported to the European Surveillance System (TESSy) and presented in Flu New Europe (http://www.flunewseurope.org) the joint ECDC-WHO Europe weekly influenza update, with the assistance of members of the Influenza and Other Respiratory Pathogens Group at the WHO European Office:
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Southern Hemisphere Vaccine Recommendation Meeting for 2016 influenza season.

The Crick Institute (WIC – Mill Hill Laboratory): Final Report for the WHO VCM

21-23 September 2015.

Influenza Activity in the WHO European Region

Weekly reporting on influenza activity started in week 40/2014. The course of the 2014-2015 season saw levels of transmission increasing slowly following week 50/2014 with a peak of activity in weeks 07-08/2015 (**Figure E1-A**). Maps showing the intensity of influenza activity and the levels of geographic spread in week 08/2015 for EU/EEA countries are shown in **Figure E2**.The season lasted for 21 weeks (weeks 51/2014 to 19/2015) with a distinct West to East pattern in the spread across the Region (**Figure E3**).

For the whole WHO European Region, up to week 20/2015, over 107,000 influenza detections had been reported to the European Surveillance System (TESSy) - with type A detections (69%) predominating over type B (31%) (**Table E1**). Within type A, the A(H3N2) subtype (77%) predominated over A(H1NI)pdm09 viruses (23%) **(Table E1)** and this pattern was evident for both sentinel and non-sentinel influenza detections **(Figure E1-B, C).** Within type B, 98% of the viruses for which lineage-determination was performed were of the Yamagata lineage **(Table E1).** The majority of countries in the Region reported influenza A virus detections to outnumber those of influenza B; however, the proportion of influenza B detections was considerably greater than that observed over the 2013-2014 season. Influenza A(H3N2) detections greatly outnumbered A(H1N1)pdm09 detections in most countries, with the exception of some countries in Southern Europe and West Asia **(Figures E3 & E4).**

Up to week 20/2015, eight countries (Finland, France, Ireland, Romania, Slovakia, Spain, Sweden and the UK) reported a total of 6023 laboratory-confirmed hospitalized influenza cases, with type A detections (79%) predominating over type B (21%). Within type A, the A(H3N2) subtype predominated over A(H1N1)pdm09 viruses, in proportions similar to that observed in ILI and ARI patients (**Table E1 & Figure E1-B, C**). For hospitalized patients where age was reported the group aged ≥65 years accounted for the greatest number of cases (**Figure E5**).

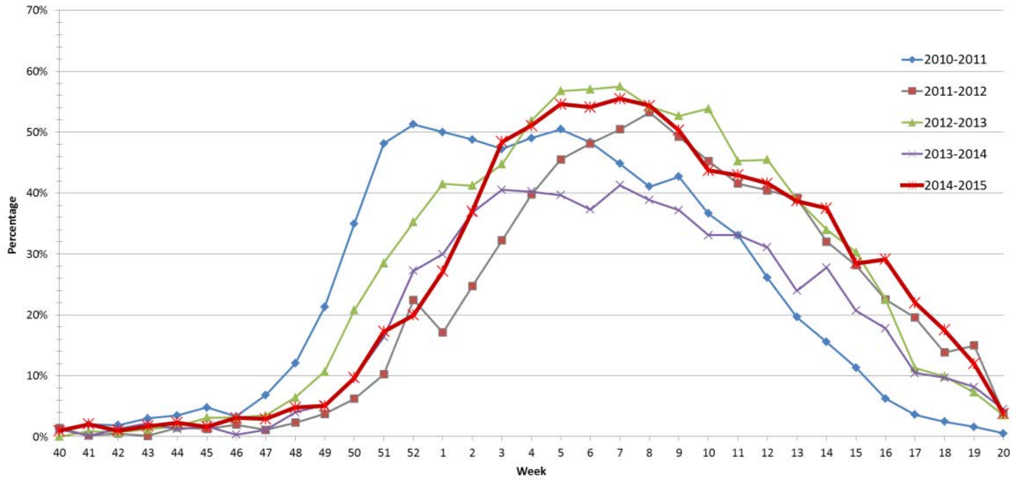
Based on monthly reports after week 20/2015 there have been relatively low numbers of influenza detections in the Region (**Figure E1-B, C**) and type B viruses have been detected in greater numbers than type A (**Table E2**).

Since week 40/2014, TESSy received reports on 2616 viruses having been screened for susceptibility to the neuraminidase inhibitors oseltamivir and zanamivir: 1535 A(H3N2), 566 A(H1N1)pdm09 and 515 influenza B viruses. Only six viruses showed evidence of reduced susceptibility to at least one neuraminidase inhibitor: three A(H3N2) viruses carried neuraminidase E119V amino acid substitution and showed reduced inhibition by oseltamivir; one A(H3N2) virus carried neuraminidase R292K amino acid substitution and showed reduced inhibition by oseltamivir and zanamivir; and two A(H1N1)pdm09 viruses carried neuraminidase H275Y amino acid substitution, causing resistance to oseltamivir, were identified. All viruses assessed for susceptibility to M2 ion channel blockers (adamantanes) were resistant due to S31N amino acid substitution in the M2 protein.

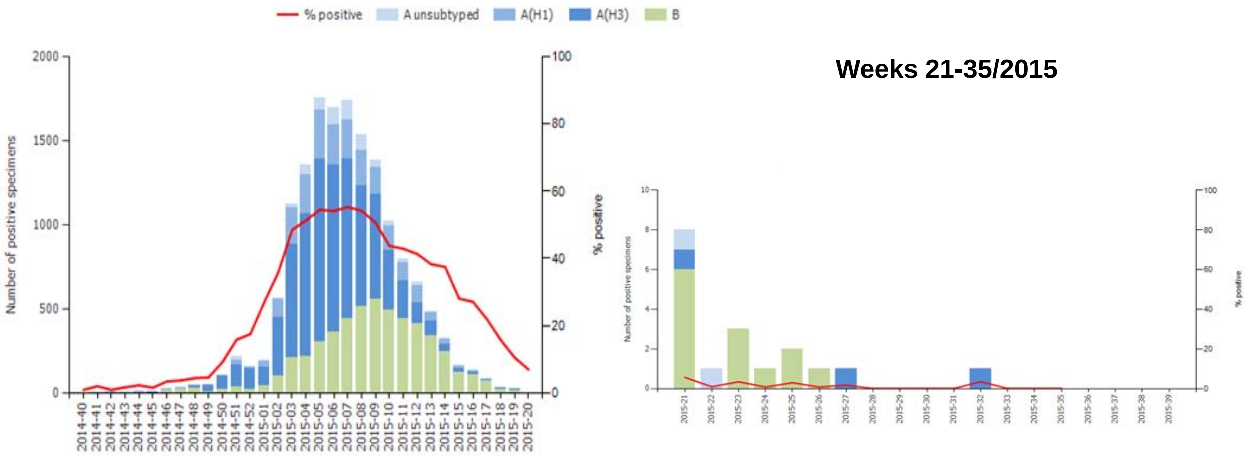
Excess all-cause mortality among the elderly (aged ≥65 years), concomitant with increased influenza activity and the predominance of A(H3N2) viruses, was observed. Across most countries, a pooled analysis showed a higher level of mortality among elderly people than in the four previous seasons (see the European project for monitoring excess mortality for public health action (EuroMOMO – http://www.euromomo.eu).

Figure E1. Influenza Activity in the WHO European Region: 2014-2015 Season

A. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



B. Sentinel Influenza Detections



C. Non-Sentinel Influenza Detections

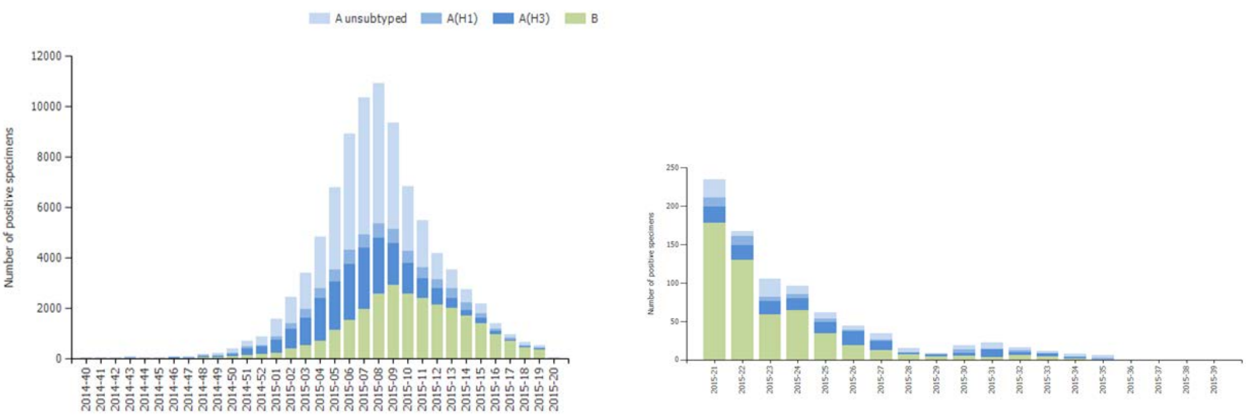


Figure E2. The Situation in EU/EEA Countries in the Week of Peak Number of Influenza Detections (week 08/2015) – *data compiled in Tessy (2015-09-01)*

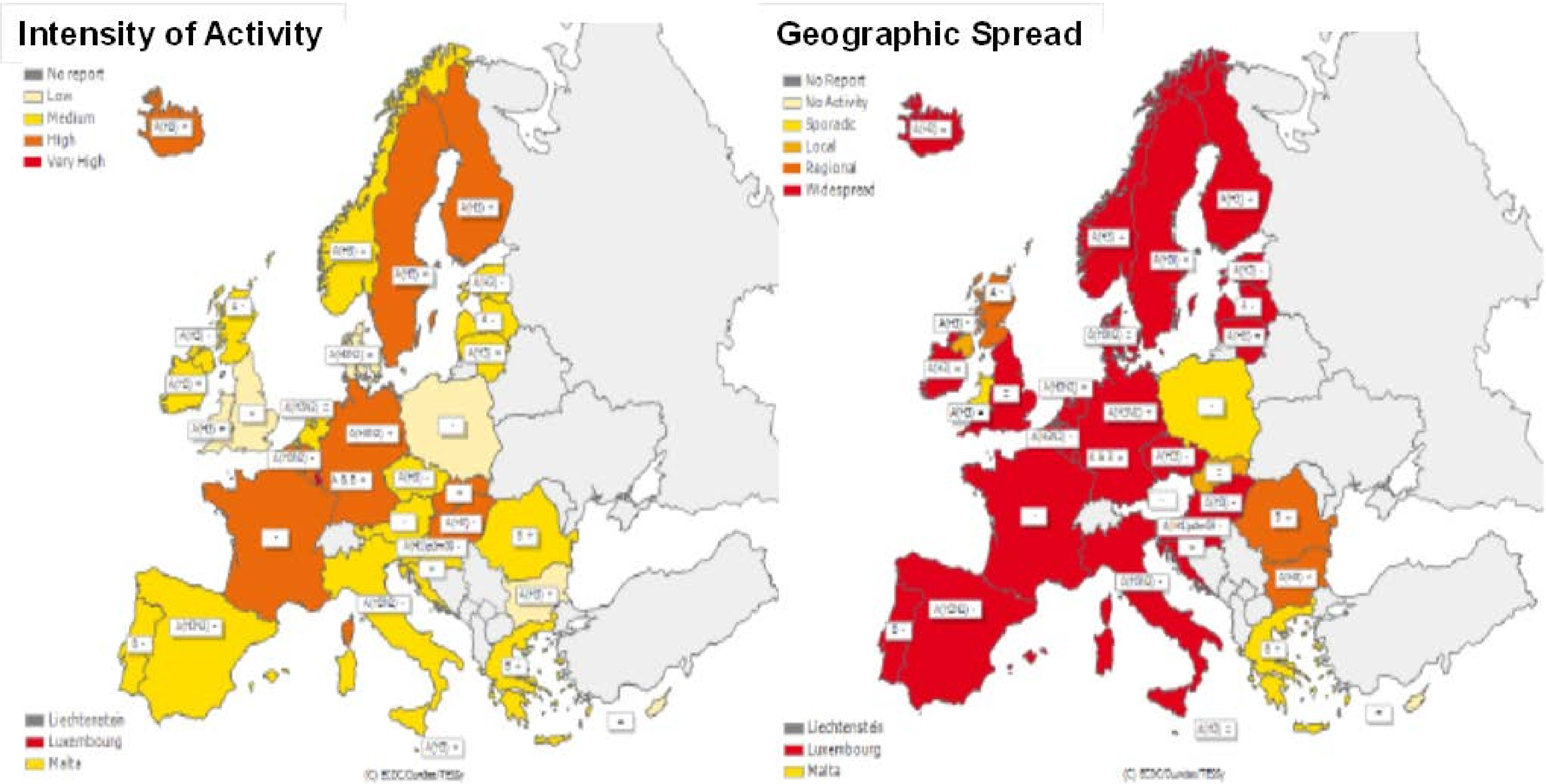


Figure E3. Development of the 2014-2015 Influenza Season in the WHO European Region

Intensity of Influenza Activity and Dominant Virus (sub)Type

WEST

Year	2014													2015																			
Country / Week	40	41	42	43	44	45	46	47	48	49	53	51	52	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Iceland									-							AH3	AH3	AH3	AH3	-	AH3												
Ireland															AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	A/B	A/B	B	B	B	B			
Portugal														B	B	B	A/B	B	A/B	A	B	B											
Spain								AH3	AH3	A	AH3	AH3	A	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	B	B	B	B	B	B	B			
United Kingdom (Northern Ireland)	AH3	B		A/B	B	B	B		AH3	B	AH3	-	A	AH3	AH3	AH3	AH3	AH3	AH3	A	AH3	A	AH3	A/B	B/AH3	A/B	A/B	B	B	B	B	B	B
United Kingdom (Wales)	-		AH3					-		AH3	AH3		-	AH3	AH3	AH3		AH3	-	AH3	AH3	AH3	AH3	B	B	AH3		B	B	B	B	B	B
United Kingdom (Scotland)		A				A	A	A	A/B	A	AH3	AH3	A	AH3	AH3	A	AH3	A	A	A	A	A	A	A	A/B	B	B	B	B	B	B	B	B
United Kingdom (England)	-	-						-																									
France																																	
Belgium												AH3		AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	A/B	B	B	B	B						
Netherlands				A/B						AH3	AH3	AH3	AH3	AH3	AH3	AH3	A	A	A	A	AH3	AH3	AH3	B	B	B	B	B	B	B	B	B	
Luxembourg												B/AH1				AH3	AH3	AH3	AH3	AH3	A/B	AH3	A/B	A/B	B	B	B	B	B				
Switzerland	A/B			A		B	B/AH3	A	A/B	A/B	A/B	A	A	AB	AH3	AH3	AH3	AH3	AH3	AH3/B	AH3	AH3	B/AH3	B/AH3	A/B	B	B	B	B				
Germany												AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3											
Denmark														A	A	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH1/H3	AH1/H3	AH1/H3	AH1/H3	AH1/H3	B/AH1/H3	B/AH1/H3	B		
Norway						A	A	AB	B	B/AH3	B/AH3	B/AH3	B/AH3	B/AH3	B/AH3	B/AH3	AH3	AH3	AH3	AH3	AH3	B/AH3	B/AH3	B/AH3	B/AH3	B	B	B	B	B	B	B	B
Italy	-	-						A		A	A	A	A	A	AH1	AH1	AH1	A	A	A	AH3	AH3	A	A/B	A/B	A/B	B	B	B				
Austria													-	-																			
Malta	-														AH3	AH3	AH3	AH3		AH3	AH3												
Slovenia												AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	AH1	B/AH1H3	B/AH1H3	B/AH3	B	B	B	B	B	B	B	B	
Sweden											AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	B/AH3	B/AH3	B	B	B	B	B	B	B	B
Croatia	-	-												AH3	AH1	A	A	A	A	A			A			B	B	B	B				
Czech Republic															B/AH3	AH3	AH3	AH3	AH3		AH3	B/AH3	B/AH3	B/AH3	B	B/AH1H3	B		B	B			
Slovakia																																	
Albania									-		AH3		-	AH3		AH3	AH3		AH3	-		AH3	AH3	AH3	AH3	AH3	AH3	AH3					
Hungary													-						AH3	AH3	AH3	AH3	AH3		B								
Poland																													B			B	
Serbia																		AH1	AH1	AH1	AH1	AH1	AH1H3	AH1H3	AH1H3	B/AH1H3							
Greece																			B	AH3/B	B	B	B	B	B	B/AH3	B			B	B	B	
Lithuania																	AH3	AH3	AH3	AH3	AH3	AH3	A	AH3		A	A	B					
Bulgaria																	AH3	AH3	AH3	AH3	AH3	AH3	B/AH1H3		AH3								
Latvia									AH1	AH1		A	A	AH3	A	AH3	AH3	AH3	AH3	A	A	A	A	B/AH1/H3	A/B	B	B	B	B	B	B	B	B
Romania													-		B	AH3	B	B	B	B	B	B	B	B	B								
Estonia																AH3	AH3	AH3	AH3		AH3	AH3	B/AH3	B/AH3	B/AH3	B/AH3	A/B	A/B					
Finland									AH3	AH3		AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3	B/AH3	AH3	B	B	AH3	B	AH3		
Belarus																AH3	AH3	AH3	AH3/B	AH3	AH3	AH3	B/AH3	B	B	B	B	B					
Republic of Moldova																			AH1/B	AH1/B	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	BAH1	B					
Ukraine																																	
Cyprus																				-													
Israel																AH3		AH3	AH3	AH3	AH3	AH3	AH3	AH3									
Turkey		-	-									B		AH3		AH1/H3					B/AH1H3	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1	B/AH1H3	B/AH1	B/AH1	BAH1	B/AH1	B
Georgia																			B	B	B	B	B										
Armenia																																	
Azerbaijan					B		B	B	B		AH3	AH3	AH3																				
Uzbekistan										AH3			AH3								B	B	B	B	B	B	B			B			
Kazakhstan																					AH3	AH3	B	B									
Russian Federation									AH3	AH3	AH3	AH3	AH3	AH3		AH3	AH3	AH3	AH3	AH3	AH3	B/AH3	B/AH3	B/AH3	B/AH3	A/B	B	A/B	B	A/B	A/B	B	B
Tajikistan																																	
Kyrgyzstan																			B		B	B/AH3	B/AH3		B	AH3	B						

EAST

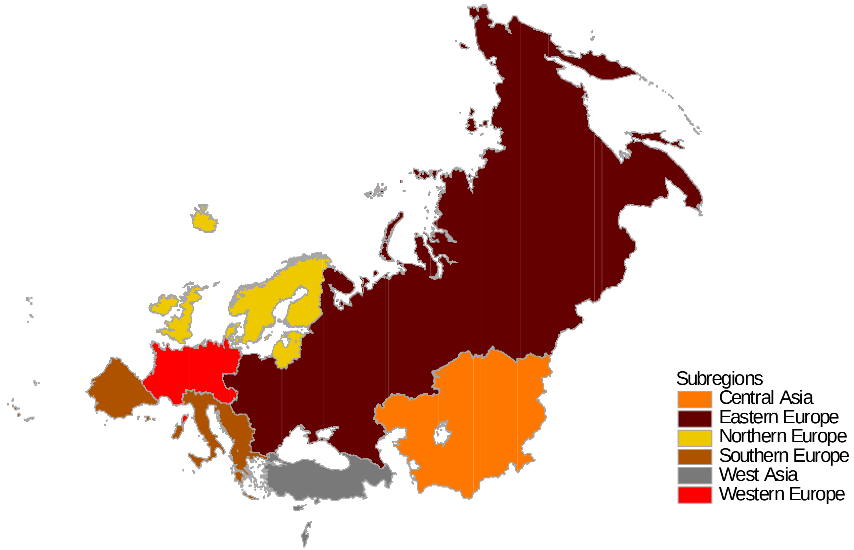
High = levels of activity higher than usual.

Medium = usual levels of activity;

Low = no activity or activity at baseline* level;

* Baseline influenza activity is the level at which clinical influenza activity remains throughout the summer and most of the winter.

Table E1. Distribution of Influenza Detection in the WHO European Region[1]: 2014-2015 Season (weeks 40-20)



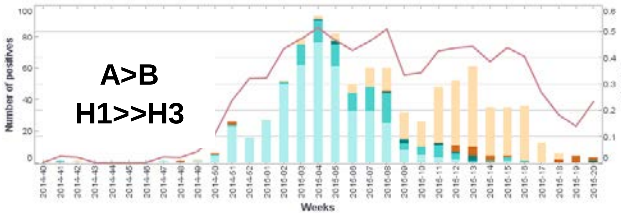
Influenza Detections*	Southern Europe		Western Europe		Eastern Europe		Northern Europe		West Asia		Central Asia		WHO European Region	
Influenza A	**12500**	**71%**	**23254**	**79%**	**9330**	**58%**	**27056**	**67%**	**1746**	**58%**	**458**	**60%**	**74344**	**69%**
Influenza A subtyped	8430		8950		8582		11057		1311		458		38788	
A (H1N1)pdm09	*2818*	*33%*	*1956*	*22%*	*1268*	*15%*	*2064*	*19%*	*864*	*66%*	*20*	*4%*	*8990*	*23%*
A (H3N2)	*5612*	*67%*	*6994*	*78%*	*7314*	*85%*	*8993*	*81%*	*447*	*34%*	*438*	*96%*	*29798*	*77%*
Influenza B	**5084**	**29%**	**6166**	**21%**	**6757**	**42%**	**13555**	**33%**	**1243**	**42%**	**310**	**40%**	**33115**	**31%**
B lineage determined	811		1092		150		1685		0		9		3747	
B/Yamagata lineage	*808*	*100%*	*1070*	*98%*	*144*	*96%*	*1648*	*98%*	*0*		*9*	*100%*	*3679*	*98%*
B/Victoria lineage	*3*	*0%*	*22*	*2%*	*6*	*4%*	*37*	*2%*	*0*		*0*	*0%*	*68*	*2%*
Total	***17584***		***29420***		***16087***		***40611***		***2989***		***768***		***107459***	

*combined ILI, ARI sentinel and non-sentinel respiratory specimens positive for influenza

[1] WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM

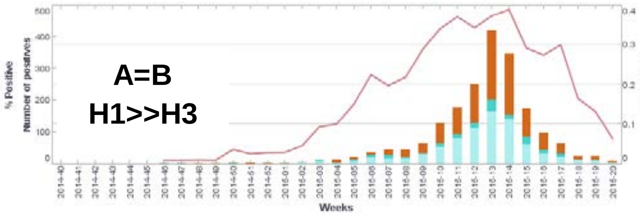
Figure E4. Patterns of Influenza Type and Subtype Circulation in the WHO European Region

Slovenia



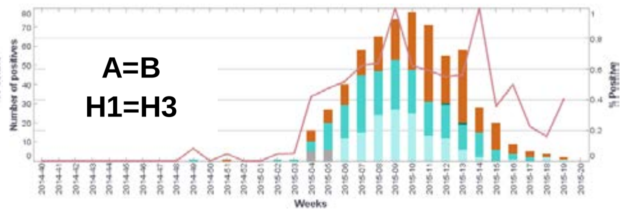
Source: combined sentinel and non-sentinel

Turkey



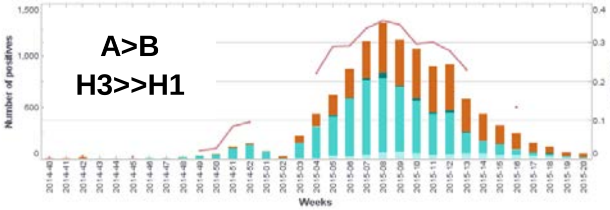
Source: combined sentinel and non-sentinel

Romania



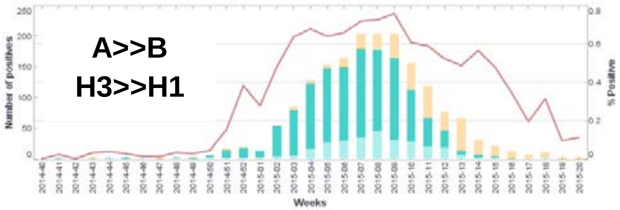
Source: combined sentinel and non-sentinel

Russian Federation



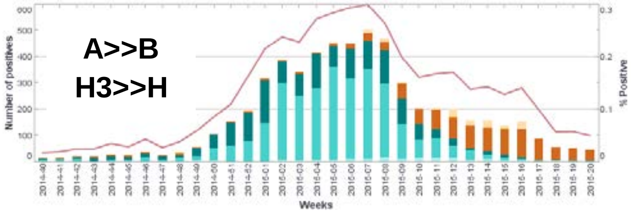
Source: combined sentinel and non-sentinel

Germany



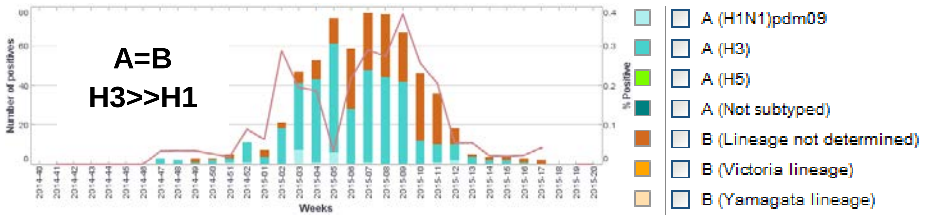
Source: combined sentinel and non-sentinel

United Kingdom



Source: combined sentinel and non-sentinel

Kazakhstan



Source: combined sentinel and non-sentinel

= represents similar ratio; > represents a ratio of at least 1.5:1; >> represents a ratio of ≥5

Figure E5. Laboratory-confirmed hospitalized influenza cases

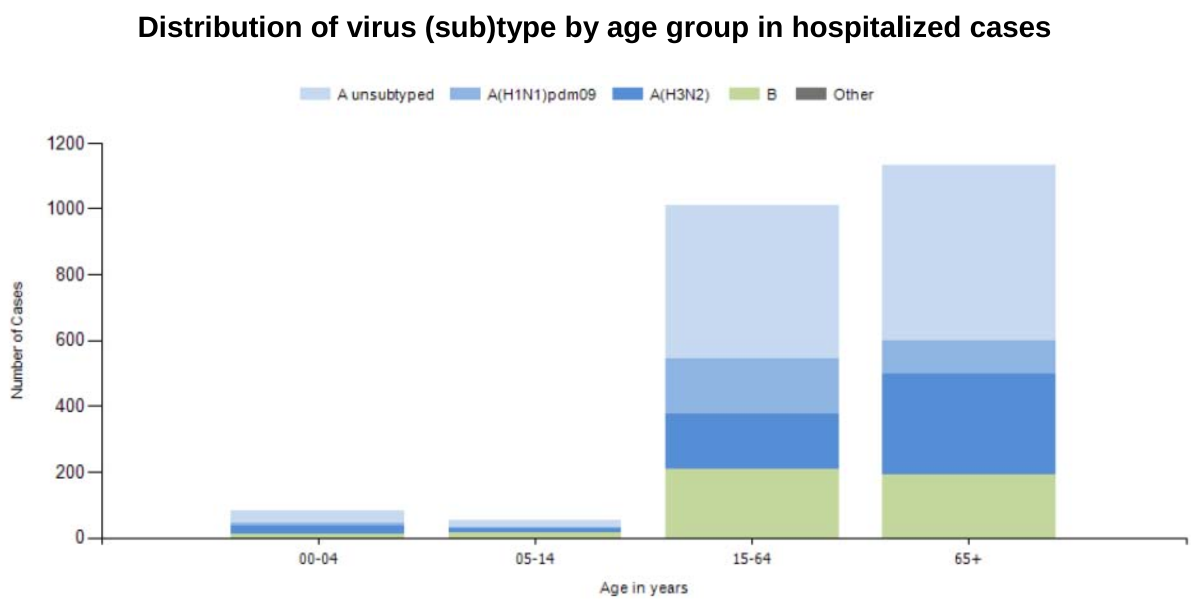


Table E2. Cumulative Numbers of Influenza Detections in the WHO Europe Region: 2014-2015 season

Virus type / subtype / lineage	Weeks 40/2014-20/2015			Weeks 40/2014-35/2015			Weeks 21-35/2015		
	Sentinel	Non-Sentinel	Combined	Sentinel	Non-Sentinel	Combined	Sentinel	Non-Sentinel	Combined
Total detections	15950	91509	107459	15968	92358	108326	18	839	867
A(H1N1)pdm09	2316	6674	8990	2316	6729	9045	0	55	55
A(H3N2)	7687	22111	29798	7690	22258	29948	3	147	150
Not subtyped	656	34900	35556	658	35021	35679	2	121	123
Total A	10659	63685	74344	10664	64008	74672	5	313	328
B/Victoria	31	37	68	32	39	71	1	2	3
B/Yamagata	1334	2345	3679	1340	2479	3819	6	134	140
B no lineage	3926	25442	29368	3932	25832	29764	6	390	396
Total B	5291	27824	33115	5304	28350	33654	13	526	539

Data compiled from Tessy (2015-09-03)

Influenza specimens received at CRICK WIC, Mill Hill Laboratory

Samples, viruses or clinical specimens, with collection dates after 2015-01-31 have been received from 39 countries in Europe, Africa, the Middle East, South America and the Far East. Two thirds were type A viruses, with A(H3N2) and A(H1N1)pdm09 viruses being received in similar numbers **(Table 1)**. Of the type B viruses, B/Yamagata lineage viruses predominated over those of the B/Victoria lineage by a ratio of ~16:1. These proportions of influenza types/subtypes/lineages are generally similar to those reported within the WHO European Region (**Figure E1-B, C** & **Tables E1, E2**).**Table 2** shows the isolation rates from specimens received with collection dates after 2015-01-31. Influenza A(H1N1)pdm09 and influenza B viruses were recovered efficiently as assessed by being able to perform HI assays. For A(H3N2) viruses, while good recovery was achieved as assessed by MUNANA-based neuraminidase activity, significant numbers could not be assessed by HI assay due to HA titres being too low (<4 in the presence of 20nM oseltamivir). **Table 3** summarises the number of viruses of the A(H1N1)pdm09 and A(H3N2) subtypes, and the two influenza B/lineages, that showed ≤2-, 4- and ≥8-fold reductions in HI titres with post-infection ferret antisera compared with the titres against their respective homologous vaccine/reference viruses; however, the antiserum raised against cell culture-propagated A/Switzerland/9715293/2013 gave a low homologous titre and reduced titres for the test viruses could only be defined as 2-fold or ≥4-fold reduced. For A(H3N2) viruses a summary of the results of plaque-reduction neutralisation assays is also shown.

Table 1. Summary of clinical samples and isolates received, with collection dates since 2015-01-31, by country*

2015 MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated	Number received	Number propagated[1]	Number received	Number propagated[2]	Number received	Number propagated	Number received	Number propagated[1]	Number received	Number propagated[1]
FEBRUARY													
Belarus	1					1	0 (0)						
Bulgaria	26			3	3	19	2 (15)					4	4
Cameroon	8			2	0	5	0 (0)	1	0				
Cyprus	12					11	1 (9)					1	1
Estonia	20			1	1	17	0 (17)					2	2
Finland	2					2	2 (0)						
France	1			1	in process								
Georgia	9			1	1	1	0 (1)	1	0			6	6
Ghana	1									1	1		
Greece	3					1	0 (0)	2	0				
Hungary	7			1	in process	3	in process					3	in process
Iceland	8			1	in process	5	in process	2	in process				
Iraq	11	4	0	4	4	2	1 (1)					1	1
Italy	41			12	12	15	5 (10)	1	0			13	13
Jordan	13			4	2	6	0 (6)	1	0			2	2
Macedonia	14	2	0	1	0	2	0 (0)	6	0			3	1
Madagascar	6					3	3 (0)					3	3
Mauritius	1					1	1 (0)						
Moldova	31			12	12	1	1 (0)					18	18
Mozambique	8			1	in process	7	in process						
Netherlands	11			2	2	7	5 (2)					2	2
Norway	1					1	1 (0)						
Oman	11	3	in process	8	in process								
Poland	3	2	0									1	1
Romania	9			2	2	4	3 (1)					3	3
Russia	18			7	6	4	4 (0)			2	2	5	5
Senegal	5			3	3	1	0 (0)					1	0
Serbia	8			2	0	3	1 (2)					3	3
Slovakia	15			5	5	8	7 (1)					2	2
Slovenia	12			4	4	3	2 (1)	1	0			4	4
Spain	58	5	0	1	1	39	16 (16)	2	0			11	9
Sweden	13			5	5	6	6 (0)					2	2
Ukraine	50			5	4	2	2 (0)	3	0			40	40
Zambia	3					3	in process						
MARCH													
Belarus	5							5	0				
Bulgaria	3					2	0 (2)					1	1
Cameroon	20	1	0	5	3	11	1 (0)	3	0				
Estonia	4					2	0 (2)					2	2
Finland	1									1	1		
France	20			5	in process	6	in process			3	in process	6	in process
French Guiana	5			2	in process	2	in process					1	in process
Georgia	10			3	3	3	1 (2)					4	4
Ghana	13			5	4	6	2 (3)					2	1
Greece	4			1	0	2	0 (0)	1	0				
Iceland	7					3	in process	4	in process				
Iraq	1			1	1								
Italy	1							1	0				
Jordan	18			11	11	2	0 (2)					5	5
Madagascar	11			1	1	1	1 (0)					9	9
Mauritius	4			1	1	3	3 (0)						
Mozambique	3			2	in process	1	in process						
Netherlands	2					1	0 (1)					1	1
Norway	11			4	in process	4	2 (1)			1	1	2	in process
Oman	10	1	in process	9	in process								
Poland	10	1	0	3	3	4	0 (0)					2	2
Romania	16			5	5	1	1 (0)					10	10
Russia	18			5	4	8	6 (1)					5	3
Slovakia	6			1	1							5	5
Slovenia	10			5	5	2	1 (1)					3	3
Senegal	3					1	0					2	2
Serbia	11			4	3	4	1 (2)	2	0			1	1
South Africa	2			1	1	1	0 (1)						
Sweden	4			3	3	1	1 (0)						
Turkey	6			1	in process	2	in process	3	in process				
Ukraine	61			24	21	7	3 (4)	1	0			29	29
Zambia	2	2	in process										

2015 MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated	Number received	Number propagated[1]	Number received	Number propagated[2]	Number received	Number propagated	Number received	Number propagated[1]	Number received	Number propagated[1]
APRIL													
Belarus	1											1	1
Cameroon	14	2	0	4	4	5	0 (0)	3	0				
Estonia	5					2	0 (2)					3	3
Finland	4					2	1 (1)			1	1	1	1
France	5			1	in process	1	in process					3	in process
French Guiana	1			1	in process								
Ghana	14			11	5	1	0 (1)					2	2
Greece	5			2	0			2	0			1	1
Jordan	38			32	in process	2	in process	4	in process				
Madagascar	4			4	4								
Mauritius	3					1	0 (0)					2	2
Norway	14			5	in process	2	0 (1)			2	2	5	in process
Oman	4			4	in process								
Poland	2					2	0 (1)						
Russia	6			2	2	1	1 (0)			1	1	2	2
Senegal	3					1	0 (1)					2	2
Slovakia	1											1	1
Slovenia	10					2	2 (0)	1	0			7	7
Serbia	4			1	1							3	3
South Africa	4			2	2	1	0 (1)			1	1		
Turkey	3			2	in process	1	in process						
Ukraine	17			9	8	4	2 (1)	3	0			1	1
MAY													
Cameroon	11	2	0	7	4	2	0 (1)						
Ghana	1			1	0								
Hong Kong	1									1	1		
Iceland	4					2	in process	2	in process				
Jordan	6			5	in process			1	in process				
Mauritius	1			1	1								
Mozambique	1					1	in process						
Norway	11			4	in process	3	in process			2	in process	2	in process
Poland	1					1	0						
Senegal	2											2	2
South Africa	11			5	in process	6	in process						
Ukraine	1			1	0								
Zambia	3			2	in process			1	in process				
JUNE													
Iceland	1					1	in process						
Jordan	1							1	in process				
Hong Kong	9			2	2	3	1 (2)			1	1	3	3
Mozambique	3							3	in process				
Norway	17			2	in process	10	in process					5	in process
South Africa	40			11	in process	23	in process					6	in process
Zambia	8	6	in process	2	in process								
JULY													
Iceland	2					2	in process						
Jordan	2			1	in process	1	in process						
Norway	5					4	in process					1	in process
South Africa	3					3	1 (2)						
Zambia	3	2	in process					1	in process				
AUGUST													
Zambia	1	1	in process										
39 Countries	1053	34		314	170	353	94 (118)	62		17	12	273	231
					29.8%		33.5%				1.6%		25.9%
		66.6%						33.4%					

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Numbers in parentheses indicate viruses recovered but with insufficient HA titre to permit HI assay

* As of 2015-09-08

Table 2. Isolation rates for specimens collected after 2015-01-31

	Clinical sample			Virus isolate		
	Number processed	Number propagated[1]	Percent recovered	Number processed	Number propagated[1]	Percent recovered
H1	73	46	63%	129	124	96%
H3	75	7 30[2]	49% (9%)[3]	189	87 88[2]	93% (46%)
B Victoria	2	2	100%	10	10	100%
B Yamagata	32	29	91%	207	202	98%

1. Propagated to sufficient HA titre to perform HI assay
2. Viruses recovered but with insufficient HA titre to permit HI assay
3. The percentages for viruses that were assayed by HI are shown in parentheses

Table 3. Summary of isolates with collection dates after 2015-01-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction microneutralisation assays

<table>
<tr><th></th><th>Number tested</th><th>≤2-fold[a]</th><th>4-fold[a]</th><th>≥8-fold[a]</th></tr>
<tr><td>H1[1]</td><td>183</td><td>180 (98.4%)</td><td>2 (1.1%)</td><td>1 (0.5%)</td></tr>
<tr><td>H3[2a]</td><td>106</td><td>1 (1%)</td><td>1 (1%)</td><td>104 (98%)</td></tr>
<tr><td>H3[2b]</td><td>106</td><td>53 (50.0%)</td><td colspan="2">53 (50%)[5]</td></tr>
<tr><td>H3[2c]</td><td>106</td><td>0</td><td>7 (6.6%)</td><td>99 (93.4%)</td></tr>
<tr><td>H3[2d]</td><td>106</td><td>80 (75.5%)</td><td>19 (17.9%)</td><td>7 (6.6%)</td></tr>
<tr><td>H3[d1]</td><td>93</td><td>13 (14.0%)</td><td>40 (43.0%)</td><td>40 (43.0%)</td></tr>
<tr><td>H3[d2]</td><td>93</td><td>52 (55.9)</td><td>32 (34.4)</td><td>9 (9.7%)</td></tr>
<tr><td>H3[d3]</td><td>93</td><td>67 (72.0%)</td><td>21 (22.6%)</td><td>5 (5.4%)</td></tr>
<tr><td>H3[d4]</td><td>93</td><td>11 (11.8%)</td><td>47 (50.6%)</td><td>35 (37.6%)</td></tr>
<tr><td>H3[d5]</td><td>93</td><td>19 (20.4%)</td><td>1 (1.1%)</td><td>73 (78.5%)</td></tr>
<tr><td>B/Victoria[3]</td><td>12</td><td>12 (100%)</td><td>0</td><td>0</td></tr>
<tr><td>B/Yamagata[4]</td><td>236</td><td>223 (94.5%)</td><td>12 (5.1%)</td><td>1 (0.4%)</td></tr>
</table>

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Compared to homologous titres for ferret antisera raised against vaccine virus A/California/7/2009

2a. Compared to homologous titres for ferret antisera raised against egg-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a)
2b. Compared to homologous titres for ferret antisera raised against cell-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a)
2c. Compared to homologous titres for ferret antisera raised against egg-propagated A/Texas/50/2012 (genetic subgroup 3C.1)
2d. Compared to homologous titres for ferret antisera raised against cell-propagated A/Hong Kong/5738/2014 (genetic subgroup 3C.2a)
d1. Compared to homologous titres for ferret antisera raised against egg-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a) in plaque-reduction neutralisation assays
d2. Compared to homologous titres for ferret antisera raised against cell-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a) in plaque-reduction neutralisation assays
d3. Compared to homologous titres for ferret antisera raised against cell-propagated A/Hong Kong/7295/2014 (genetic subgroup 3C.2a [NYT]) in plaque-reduction neutralisation assays
d4. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/4801/2014 (genetic subgroup 3C.2a [NYK]) in plaque-reduction neutralisation assays
d5. Compared to homologous titres for ferret antisera raised against egg-propagated A/Netherlands/525/2014 (genetic subgroup 3C.3b) in plaque-reduction neutralisation assays

3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

4. Compared to homologous titres for ferret antisera raised against vaccine virus B/Phuket/3073/2013

5. Unable to differentiate fold drop in titre for most viruses as it falls outside the dynamic range of the antiserum

Influenza A(H1N1)pdm09 virus analysis.

Influenza A(H1N1)pdm09 viruses were received from NICs in Europe, Africa (Cameroon, Ghana, Senegal and South Africa), the Middle East (Iraq, Jordan and Oman), the Indian Ocean (Madagascar and Mauritius) and Hong Kong SAR. **Table 4** summarises the antigenic (HI) results for viruses that were propagated successfully at the Francis Crick Institute.

The vast majority of H1N1 viruses collected after 2015-01-31 were antigenically similar to the vaccine virus **(Table 4)**. **Tables 5-1** to **5-11** show the results of HI assays of H1N1 viruses carried out since our February 2015 report. Test viruses in each table are sorted by date of collection and viruses with collection dates after 2015-01-31 are shown below red lines in **Tables 5-3** to **5-11**. All viruses but one, A/Castilla La Mancha/666/2015, were recognised by the panel of antisera at titres within 4-fold of the titres for the homologous viruses, including that raised against the A/California/7/2009 vaccine virus, with the exception of the antiserum raised against A/Christchurch/16/2010. This antiserum recognised just over half (52%) of the test viruses collected since 2015-01-31 at a titre within 4-fold of the titre for the homologous virus. A/Georgia/567/2015 was recognised by the antiserum raised against A/California/7/2009 at a titre 4-fold reduced over the homologous titre but was recognised by three of the antisera poorly. Note also that in most tests none of the antisera raised against reference viruses collected since 2011 recognised the egg-propagated vaccine virus A/California/7/2009 at a titre less than 4-fold reduced compared to the titres of the antisera with their homologous viruses. A summary of the HI results of viruses collected since 2015-01-31 is shown in **Table 5-12**.

Figure 1 shows a phylogenetic tree for the HA genes of representative H1N1 viruses. Over the last five years the HA genes have evolved and eight genetic groups have been designated, with A/California/7/2009 representing group 1. Viruses collected after 2015-01-31 fell into genetic subgroup 6B **(Figure 1).** Subgroup 6B is defined by the following amino acid substitutions in **HA1** and **HA2**:

- **D97N**, **S185T, S203T K163Q, A256T** and **K283E** in **HA1**, and **E47K, S124N** and **E172K** in **HA2** compared with A/California/7/2009, e.g. A/South Africa/3626/2013.

Several minor groupings are detected in subgroup 6B. These include a group represented by viruses form South Africa with the amino acid substitutions **S69P, T120A, K208R** and **E235D** in **HA1**; a geographically widespread group carrying the amino acid substitution **S84N**; and a group also geographically widespread carrying **N129D, S190R, K308E** in **HA1** and **E164G** in **HA2**.

An amino acid sequence alignment of the HAs for representative viruses is shown in **Figure 2**; the vaccine virus is shown in red, reference viruses are coloured purple, glycosylation motifs are highlighted and the genetic group is shown in green.

Figure 3 illustrates the location of amino acid residues, on an H1-HA structure, defining genetic subgroup 6B and those defining several of the minor groupings within subgroup 6B.

Figure 4 shows a phylogenetic tree for NA genes of representative H1N1 viruses and **Figure 5** shows a NA amino acid alignment for a selected group of these viruses. The HA and NA phylogenetic trees are generally congruent.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on just over 200 viruses with collection dates after 2015-01-31. All viruses tested showed normal inhibition (NI) to both neuraminidase inhibitors. One virus, A/Norway/2227/2015, not tested phenotypically at Mill Hill at the time of preparation of this report was sequenced and confirmed to carry NA H275Y amino acid substitution which is associated with highly reduced inhibition by oseltamivir **(Table 15)**.

Conclusion.

The A(H1N1)pdm09 viruses received were antigenically similar to the vaccine virus, A/California/7/2009, and retain susceptibility to oseltamivir and zanamivir.

Table 4. Summary of influenza A(H1N1)pdm09 viruses analysed, collected after 2015-01-31

MONTH	H1N1pdm09				
			Fold reduction in HI titre		
Country	Number received[1]	Number propagated[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
FEBRUARY					
Belarus					
Bulgaria	3	3	3		
Estonia	1	1	1		
Georgia	1	1	1		
Iraq	4	4	3	1	
Italy	12	12	12		
Jordan	4	2	2		
Moldova	12	12	12		
Netherlands	2	2	2		
Romania	2	2	2		
Russia	7	6	6		
Senegal	3	3	3		
Slovakia	5	5	5		
Slovenia	4	4	4		
Spain	1	1			1
Sweden	5	5	5		
Ukraine	5	4	4		
MARCH					
Cameroon	5	3	3		
Georgia	3	3	3		
Ghana	5	4	4		
Iraq	1	1	1		
Jordan	11	11	11		
Madagascar	1	1	1		
Mauritius	1	1	1		
Norway	4	2	2		
Poland	3	3	3		
Romania	5	5	5		
Russia	5	4	4		
Slovakia	1	1	1		
Slovenia	5	5	5		
Serbia	4	3	3		
South Africa	1	1	1		
Sweden	3	3	3		
Ukraine	24	21	21		
APRIL					
Belarus					
Cameroon	4	4	4		
Ghana	11	5	5		
Greece	2	0			
Madagascar	4	4	4		
Norway	5	2	2		
Russia	2	2	2		
Serbia	1	1	1		
South Africa	2	2	2		
Ukraine	9	8	8		
MAY					
Cameroon	7	4	4		
Mauritius	1	1	1		
South Africa	5	3	3		
JUNE					
Hong Kong	2	2	1	1	
South Africa	13	5	5		
	221	182	179	2	1
28 Countries			98.4%	1.1%	0.5%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assays

3. Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

Table 5-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-02-18 + 2015-02-25 + 2015-03-03)

				Haemagglutination inhibition titre										
				Post-infection ferret antisera										
Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St P 27/11 F23/11	A/St P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
							4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E3	640	640	640	160	80	160	160	320	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	320	80	40	40	80	80	80	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	1280	320	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	1280	2560	1280	5120	2560	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	160	320	640	640	640	1280	640	320	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	320	320	640	640	1280	1280	2560	1280	640	
A/St Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	640	1280	640	2560	1280	640	
A/St Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	1280	640	1280	2560	2560	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	160	640	640	640	1280	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	320	640	640	640	1280	640	640	
TEST VIRUSES														
A/Latvia/01-003839/2014	6B	2014-01-03	C1/MDCK1	640	640	640	640	1280	1280	1280	2560	1280	640	
A/Hormozgan/85977/2014		2014-06-15	MDCK2/MDCK1	640	320	320	640	1280	1280	1280	2560	1280	1280	
A/Ghana/DILI-14-0567/2014	6C	2014-06-16	C3/MDCK1	320	320	320	320	160	160	320	640	320	160	G155G>E
A/Ghana/DILI-14-0582/2014	6C	2014-06-23	C2/MDCK1	1280	640	640	1280	1280	1280	1280	5120	2560	1280	
A/Ghana/DILI-14-0618/2014		2014-07-04	C2/MDCK1	640	320	320	640	1280	1280	1280	2560	1280	640	
A/Ghana/DILI-14-0620/2014	6C	2014-07-07	C2/MDCK1	320	160	160	160	320	320	320	640	320	160	V30A, I116M, A186T, V234I, M257V
A/Khorasan/85136/2014	6B	2014-11-06	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	640	
A/Lorestan/89914/2014	6B	2014-11-12	MDCK1/MDCK1	640	320	320	320	640	640	1280	2560	1280	640	
A/Ghana/FS-14-0983/2014	6C	2014-11-20	MDCK1	1280	640	640	640	1280	2560	2560	5120	2560	1280	
A/Luxembourg/14064674/2014	6B	2014-11-21	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	12580	
A/Slovenia/2206/2014	6B	2014-11-23	MDCKx/MDCK1	640	640	640	1280	1280	1280	1280	5120	2560	1280	
A/Latvia/11-058122/2014	6B	2014-11-25	C1/MDCK1	320	320	320	320	640	640	1280	2560	1280	640	
A/Ghana/FS-14-1025/2014	6C	2014-11-27	MDCK1	640	320	640	640	1280	1280	1280	2560	1280	640	
A/Slovenia/2418/2014		2014-12-03	MDCKx/MDCK1	640	320	640	640	1280	640	1280	2560	1280	1280	
A/Slovenia/2585/2014		2014-12-11	MDCKx/MDCK1	640	320	320	640	1280	640	640	2560	1280	640	
A/Slovenia/2634/2014		2014-12-12	MDCK1/MDCK1	640	320	320	640	1280	640	1280	2560	1280	640	
A/Slovenia/2731/2014	6B	2014-12-12	MDCK1	640	320	320	640	1280	640	1280	2560	1280	640	
A/Norway/3154/2014	6B	2014-12-13	MDCK1	320	160	160	160	640	320	320	1280	640	640	
A/Slovenia/2743/2014		2014-12-13	MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Slovenia/2749/2014	6B	2014-12-13	MDCK1	320	160	320	320	640	320	640	1280	640	640	
A/Norway/3244/2014		2014-12-16	MDCK2	320	320	320	320	1280	640	640	2560	1280	640	
A/Croatia/1994/2014	6B	2014-12-29	E1/E2	640	320	640	640	640	1280	1280	2560	1280	640	
A/Netherlands/576/2014	6B	2014-12-29	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	1280	1280	
A/Bayern/4/2015		2015-01-12	C2/MDCK1	2560	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Brandenburg/2/2015	6B	2015-01-12	C2/MDCK1	1280	1280	1280	5120	1280	2560	1280	2560	2560	1280	
A/Sachsen/1/2015	6B	2015-01-14	C2/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	2560	
A/Albania/5742/2015	6B	2015-01-15	MDCK2	1280	640	640	640	1280	1280	1280	5120	2560	1280	
A/Bayern/8/2015	6B	2015-01-19	C2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	
A/Hessen/1/2015	6B	2015-01-20	C2/MDCK1	1280	640	640	640	1280	1280	1280	5120	2560	1280	
A/Greece/414/2015	6B	2015-01-25	MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	

Vaccine

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-03-11 + 2015-03-17 + 2015-03-25 + 2015-04-01 + 2015-04-09)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre Post infection ferret antisera A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St P 27/11 F23/11	A/St P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
							4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E3	640	640	1280	160	160	160	160	320	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	320	640	640	160	40	80	160	80	160	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	320	1280	1280	320	80	80	160	160	160	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	1280	2560	1280	5120	2560	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	160	320	640	1280	1280	640	2560	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	640	640	640	1280	1280	1280	1280	1280	640	
A/St Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	1280	1280	1280	2560	2560	1280	
A/St Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	1280	1280	1280	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	160	640	640	640	1280	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	320	640	640	320	640	640	640	1280	640	640	
TEST VIRUSES														
A/Istanbul/1584/2014	6B	2014-11-03	MDCK1	640	320	640	640	1280	1280	1280	1280	1280	1280	
A/Catalonia/7497S/2014	6B	2014-12-02	C0/MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	2560	
A/Catalonia/7536S/2014	6B	2014-12-16	C0/MDCK1	1280	640	640	1280	2560	1280	1280	5120	2560	2560	
A/Greece/1/2015	6B	2015-01-01	MDCK1	640	640	640	640	2560	1280	1280	2560	2560	1280	
A/Greece/42/2015	6B	2015-01-05	MDCK1	2560	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Catalonia/2234975NS/2015	6B	2015-01-07	C0/MDCK1	640	640	640	640	2560	1280	1280	2560	2560	1280	
A/Ankara/20/2015	6B	2015-01-07	MDCK2	640	320	640	640	640	1280	1280	2560	1280	640	
A/Catalonia/2235933NS/2015	6B	2015-01-10	C0/MDCK1	640	320	320	640	2560	1280	1280	2560	1280	1280	
A/Slovenia/187/2015	6B	2015-01-12	SIAT1	1280	1280	640	1280	2560	2560	2560	5120	5120	2560	
A/Catalonia/2236624NS/2015	6B	2015-01-13	C0/MDCK1	640	320	320	640	1280	1280	1280	2560	1280	1280	
A/Konya/61/2015	6B	2015-01-13	MDCK2	1280	640	640	1280	2560	2560	2560	5120	5120	2560	
A/Catalonia/2237143NS/2015	6B	2015-01-14	C0/MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	2560	
A/Catalonia/2237251NS/2015	6B	2015-01-14	C0/MDCK1	640	640	640	640	2560	1280	1280	2560	2560	1280	
A/Catalonia/2239209NS/2015	6B	2015-01-22	C0/MDCK1	1280	640	640	640	2560	1280	1280	2560	2560	1280	
A/Greece/364/2015	6B	2015-01-22	MDCK1	1280	1280	2560	2560	5120	5120	5120	5120	5120	2560	
A/Hong Kong/8111529/2015	6B	2015-01-23	MDCK1	1280	640	1280	1280	1280	2560	2560	5120	2560	1280	
A/Catalonia/2239980NS/2015	6B	2015-01-25	MDCK1	1280	640	640	1280	1280	1280	1280	2560	2560	1280	
A/Catalonia/2240116NS/2015	6B	2015-01-25	C0/MDCK3	1280	640	640	1280	1280	2560	2560	5120	2560	1280	
A/Hong Kong/8112398/2015	6B	2015-01-26	MDCK1	640	640	640	640	1280	1280	1280	2560	2560	1280	

Vaccine

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-04-22)

				Haemagglutination inhibition titre										
				Post infection ferret antisera										
Viruses	Genetic group	Collection date	Passage History	New A/Cal 7/09 F01/15	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St P 27/11 F23/11	A/St P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
							4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E3	1280	1280	2560	320	320	320	320	640	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	640	640	160	80	80	160	160	160	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	1280	320	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	2560	2560	2560	5120	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	640	640	1280	640	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/St Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	1280	1280	1280	2560	2560	1280	
A/St Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	160	160	160	320	640	640	640	2560	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	640	640	1280	640	2560	1280	1280	
TEST VIRUSES														
A/Iraq/361/2014	6B	2014-07-27	C2/MDCK1	640	640	640	1280	1280	2560	2560	5120	2560	1280	
A/Iraq/478/2014	6B	2014-09-22	C1/MDCK1	640	640	640	1280	2560	2560	5120	5120	5120	2560	
A/Nevsehir/1118/2014	6B	2014-10-16	MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Sivas/1327/2014	6B	2014-10-24	MDCK1	1280	1280	1280	1280	2560	5120	2560	5120	5120	2560	
A/Corum/1664/2014		2014-11-07	MDCK1	640	640	640	1280	2560	2560	2560	5120	5120	2560	
A/Iraq/835/2014	6B	2014-12-18	C1/MDCK1	640	320	320	640	1280	2560	2560	5120	2560	1280	
A/Samsun/80/2015	6B	2015-01-15	MDCK1	160	320	320	320	640	640	640	2560	1280	640	R259K
A/Madrid/777/2015	6B	2015-01-23	SIAT1/MDCK1	1280	640	1280	1280	2560	2560	5120	5120	2560	1280	
A/Madrid/777/2015	6B	2015-01-23	E1/E2	640	640	1280	1280	1280	1280	1280	2560	2560	1280	
A/Andalucia/803/2015	6B	2015-01-27	SIAT2/MDCK1	1280	640	1280	2560	2560	5120	5120	5120	5120	2560	
A/Iraq/469/2014	6B	2015-02-01	SIAT1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	
A/Iraq/474/2014	6B	2015-02-01	SIAT1	320	160	320	640	1280	1280	1280	5120	2560	1280	
A/Castilla La Mancha/666/2015	6B	2015-02-02	SIAT1/MDCK1	160	320	320	320	640	640	320	2560	640	640	P271S
A/Iraq/476/2014	6B	2015-02-05	SIAT1	640	320	640	1280	2560	2560	2560	5120	2560	2560	
A/Iraq/481/2014	6B	2015-02-08	SIAT1	640	320	640	1280	2560	2560	2560	5120	5120	2560	
A/Bulgaria/377/2015	6B	2015-02-18	SIAT2/MDCK1	1280	640	1280	1280	2560	2560	5120	5120	5120	2560	
A/Bulgaria/431/2015	6B	2015-02-24	SIAT2/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	1280	
A/Bulgaria/456/2015	6B	2015-02-24	SIAT2/MDCK1	1280	1280	1280	2560	2560	5120	5120	5120	5120	2560	

Sequences in phylogenetic tree

Vaccine

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-04-29 + 2015-05-06 + 2015-05-13)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post-infection ferret antisera										
				A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St P 27/11 F23/11	A/St P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
	Genetic group						4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E3	640	1280	1280	320	160	320	320	320	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	320	80	40	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	1280	320	80	160	160	160	320	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	1280	2560	2560	5120	2560	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	320	320	320	640	640	640	1280	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	1280	1280	1280	5120	2560	1280	
A/St Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	640	640	1280	1280	1280	2560	2560	1280	
A/St Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	1280	1280	2560	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	320	640	640	640	1280	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	320	640	640	640	1280	1280	640	
TEST VIRUSES														
A/Kocaeli/1135/2014	6B	2014-10-17	MDCK2	320	160	160	160	320	320	320	1280	640	320	
A/Iraq/57/2015	6B	2015-01-15	MDCK1	1280	640	640	640	1280	1280	1280	2560	2560	1280	
A/Thessaloniki/93/2015	6B	2015-01-23	Ex/E2	320	320	320	160	320	320	320	640	320	320	K153E>K
A/Jordan/30039/2015	6B	2015-01-28	MDCK2	1280	640	640	640	1280	1280	1280	2560	2560	1280	
A/Italy/FVG-60/2015	6B	2015-01-28	Cx/MDCK1	1280	320	640	640	ND	1280	1280	2560	2560	1280	
A/Milano/163/2015		2015-01-30	MDCK2/MDCK1	2560	640	640	1280	ND	2560	1280	5120	2560	1280	
A/Perugia/30/2015		2015-01-30	MDCK2/MDCK1	1280	320	320	640	ND	1280	1280	2560	1280	1280	
A/Firenze/7/2015		2015-02-02	MDCK2/MDCK1	1280	640	640	640	ND	1280	2560	5120	2560	1280	
A/Perugia/41/2015	6B	2015-02-02	MDCK2/MDCK1	320	160	320	320	ND	640	640	1280	640	640	D127E, D222G>D
A/Parma/179/2015		2015-02-03	MDCK2/MDCK1	1280	640	640	640	ND	1280	2560	5120	2560	1280	
A/Jordan/10220/2015	6B	2015-02-05	MDCK2	1280	640	1280	640	1280	1280	1280	2560	2560	1280	
A/Firenze/8/2015	6B	2015-02-05	MDCK2/MDCK1	640	320	320	640	ND	640	1280	2560	1280	1280	
A/Perugia/34/2015	6B	2015-02-05	MDCK1/MDCK1	1280	640	640	640	ND	1280	1280	2560	2560	1280	
A/Parma/244/2015		2015-02-05	MDCK2/MDCK1	1280	640	640	640	ND	1280	1280	2560	2560	1280	
A/Parma/270/2015		2015-02-05	MDCK2/MDCK1	1280	640	640	1280	ND	2560	2560	5120	2560	1280	
A/Parma/257/2015	6B	2015-02-10	MDCK3/MDCK1	1280	320	320	640	ND	1280	1280	2560	2560	1280	
A/Milano/236/2015	6B	2015-02-11	MDCK2/MDCK1	1280	640	640	640	ND	2560	1280	2560	2560	1280	
A/Milano/237/2015		2015-02-11	MDCK2/MDCK1	1280	320	320	640	ND	1280	1280	2560	2560	1280	
A/Jordan/20163/2015	6B	2015-02-12	MDCK1	1280	320	640	640	1280	1280	1280	2560	2560	1280	
A/Italy/FVG-93/2015	6B	2015-02-19	Cx/MDCK1	1280	640	640	640	ND	1280	1280	2560	2560	1280	
A/Milano/264/2015	6B	2015-02-20	MDCK2/MDCK1	1280	640	640	1280	ND	2560	2560	5120	2560	1280	
A/Georgia/382/2015	6B	2015-02-25	MDCK1	2560	1280	1280	1280	2560	2560	1280	5120	2560	2560	
A/Jordan/20184/2015	6B	2015-03-01	MDCK1	1280	320	640	640	1280	1280	1280	2560	2560	1280	
A/Iraq/1070/2014	6B	2015-03-03	MDCK1	1280	320	640	1280	1280	2560	2560	5120	5120	2560	
A/Jordan/30070/2015	6B	2015-03-03	MDCK1	1280	640	640	640	1280	1280	1280	5120	2560	1280	
A/Jordan/30080/2015	6B	2015-03-04	MDCK1	640	160	320	320	640	640	640	1280	1280	640	
A/Jordan/20187/2015	6B	2015-03-05	MDCK1	640	320	640	320	640	1280	1280	2560	1280	1280	
A/Jordan/20208/2015	6B	2015-03-07	MDCK1	1280	320	640	640	1280	1280	1280	2560	2560	1280	
A/Jordan/10498/2015	6B	2015-03-08	MDCK1	1280	640	640	640	1280	1280	1280	2560	2560	1280	
A/Georgia/545/2015	6B	2015-03-09	MDCK1	640	640	640	640	1280	1280	1280	2560	1280	1280	
A/Jordan/20216/2015	6B	2015-03-10	MDCK1	1280	320	320	320	1280	1280	1280	2560	1280	1280	
A/Georgia/528/2015	6B	2015-03-13	MDCK1	640	320	320	320	640	640	1280	2560	1280	640	
A/Jordan/10589/2015	6B	2015-03-16	MDCK1	1280	320	640	640	1280	1280	1280	5120	2560	1280	
A/Georgia/567/2015	6B	2015-03-17	MDCK1	320	320	640	160	160	160	160	160	160	320	N156S
A/Jordan/10628/2015	6B	2015-03-17	MDCK1	640	320	320	320	640	640	640	2560	1280	1280	
A/Jordan/20241/2015	6B	2015-03-22	MDCK1	640	320	320	320	640	640	640	2560	1280	640	
A/Jordan/10863/2015	6B	2015-03-29	MDCK2	1280	320	640	640	1280	1280	1280	2560	1280	1280	

ND = Not Done

Sequences in phylogenetic tree

Vaccine

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-05-20 + 2015-06-03)

Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/Astrak 1/11 F22/13	A/St P 27/11 F23/11	A/St P 100/11 F24/11	A/HK 5659/12 F30/12	New A/HK 5659/12 F17/15	A/Sth Afr 3626/13 F3/14	
				Haemagglutination inhibition titre – Post-infection ferret antisera										
	Genetic group						4	5	6	7	6A	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E3	1280	1280	1280	320	320	320	320	320	40	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	640	320	160	80	160	80	80	40	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	1280	320	160	160	160	640	80	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	2560	2560	2560	5120	2560	2560	5120	5120	1280	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	2560	1280	5120	2560	1280	1280	
A/St Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	1280	1280	5120	2560	1280	1280	
A/St Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	1280	1280	2560	2560	1280	640	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	160	160	320	640	1280	2560	1280	640	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	1280	1280	1280	640	1280	1280	2560	2560	640	1280	
TEST VIRUSES														
A/Serbia/NS-938/2015		2015-01-12	MDCK2	1280	640	1280	1280	2560	2560	5120	2560	ND	2560	
A/Iasi/175973/2015		2015-01-13	MDCK2/MDCK 1	1280	640	640	1280	2560	1280	5120	2560	2560	1280	
A/Serbia/NS-941/2015	6B	2015-01-16	MDCK1	2560	640	1280	2560	2560	2560	5120	5120	ND	2560	
A/Bari/5/2015		2015-01-19	MDCK2/MDCK1	2560	1280	1280	2560	5120	5120	5120	5120	ND	5120	
A/Iasi/176380/2015		2015-01-21	MDCK1/MDCK 1	1280	320	640	640	1280	1280	2560	1280	1280	1280	
A/Milano/106/2015		2015-01-23	MDCK2/MDCK1	1280	640	640	1280	2560	2560	5120	2560	ND	2560	
A/Italy/FVG-57/2015		2015-01-26	Cx/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	ND	5120	
A/Mures/176671/2015	6B	2015-01-26	SIAT1/MDCK 1	640	320	320	320	640	1280	2560	1280	1280	640	
A/Serbia/NS-957/2015	6B	2015-01-27	MDCK1	1280	640	1280	1280	2560	2560	5120	5120	ND	2560	
A/Iasi/176656/2015		2015-01-27	MDCK1/MDCK 1	1280	640	640	1280	2560	2560	2560	2560	2560	1280	
A/Bucuresti/177751/2015	6B	2015-02-17	MDCK1/MDCK 1	1280	640	640	1280	2560	2560	5120	2560	2560	2560	
A/Valcea/178103/2015		2015-02-23	SIAT1/MDCK 1	640	320	320	320	1280	1280	2560	1280	1280	1280	
A/Serbia/NS-1168/2015	6B	2015-03-04	MDCK1	2560	1280	1280	1280	5120	2560	5120	5120	ND	2560	
A/Neamt/178811/2015	6B	2015-03-04	MDCK1/MDCK 1	2560	1280	1280	1280	2560	2560	5120	2560	2560	2560	
A/Sibiu/178761/2015	6B	2015-03-05	MDCK1/MDCK 1	2560	640	1280	1280	2560	2560	5120	2560	2560	2560	
A/Serbia/NS-1194/2015	6B	2015-03-06	MDCK1	2560	640	1280	1280	2560	2560	5120	5120	ND	2560	
A/Iasi/178819/2015	6B	2015-03-06	MDCK1/MDCK 1	2560	640	640	1280	2560	2560	5120	2560	2560	2560	
A/Constanta/179080/2015	6B	2015-03-11	MDCK2/MDCK 1	2560	640	1280	1280	2560	2560	5120	2560	2560	2560	
A/Constanta/179081/2015	6B	2015-03-11	MDCK1/MDCK 1	2560	640	1280	1280	2560	2560	5120	2560	2560	2560	
A/Serbia/NS-1265/2015	6B	2015-03-26	MDCK2	1280	640	640	640	1280	1280	5120	2560	ND	1280	
A/Serbia/NS-1305/2015		2015-04-08	SIAT1	2560	1280	2560	2560	5120	5120	5120	5120	ND	5120	

ND = Not Done

Vaccine

Sequences in phylogenetic tree

Table 5-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-06-10 + 2015-06-17 + 2015-06-24)

Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14 4	A/Astrak 1/11 F22/13 5	A/St P 27/11 F23/11 6	A/St P 100/11 F24/11 7	A/HK 5659/12 F30/12 6A	A/Sth Afr 3626/13 F3/14 6B	
				Haemagglutination inhibition titre – Post-infection ferret antisera									
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E3	1280	1280	1280	320	320	320	320	160	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	320	160	80	160	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	1280	320	80	160	160	160	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	2560	2560	2560	5120	2560	2560	5120	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	1280	1280	2560	2560	1280	
A/St Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	640	640	1280	1280	2560	1280	1280	
A/St Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	640	640	1280	1280	5120	1280	640	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	160	160	320	640	640	2560	640	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	1280	640	640	640	1280	1280	2560	1280	1280	
TEST VIRUSES													
A/St Petersburg/01/2015	6B	2015-01-21	E2/E1	640	320	320	320	1280	640	1280	640	640	Q223R
A/Finland/491/2015	6B	2015-01-28	MDCK1/MDCK1	1280	320	640	640	1280	1280	2560	1280	1280	
A/St Petersburg/44/2015		2015-01-30	E1/E1	640	640	640	640	1280	640	2560	1280	1280	
A/Odessa/232/2015	6B	2015-02-09	SIAT2/SIAT1	2560	640	1280	1280	2560	2560	5120	5120	2560	
A/Ukraine/6342/2015		2015-02-16	C1/MDCK1	1280	640	640	640	2560	1280	2560	2560	1280	
A/Ukraine/6392/2015	6B	2015-02-18	C1/MDCK1	2560	1280	2560	2560	5120	2560	5120	5120	2560	
A/Ukraine/6393/2015	6B	2015-02-18	C2/MDCK1	2560	1280	1280	1280	2560	5120	5120	5120	2560	
A/Moldova/151.08/2015		2015-02-18	MDCK1/MDCK1	2560	1280	1280	1280	2560	2560	5120	5120	2560	
A/Moldova/158.08/2015	6B	2015-02-19	MDCK1/MDCK1	2560	640	1280	1280	2560	2560	5120	5120	2560	
A/St Petersburg/122/2015	6B	2015-02-26	E2/E1	1280	1280	1280	640	1280	1280	2560	2560	2560	
A/Ukraine/6394/2015	6B	2015-03-02	C2/MDCK1	640	320	320	320	640	640	1280	640	640	
A/Ukraine/160/2015	6B	2015-03-02	SIAT2/SIAT1	1280	320	640	640	1280	1280	2560	2560	1280	
A/Ukraine/6403/2015	6B	2015-03-09	C0/MDCK1	640	320	320	320	640	640	2560	1280	640	
A/Ukraine/6459/2015	6B	2015-03-09	C2/MDCK1	2560	640	640	1280	2560	2560	5120	5120	1280	
A/Ukraine/294/2015		2015-03-11	SIAT1/SIAT1	2560	1280	1280	1280	2560	2560	5120	2560	1280	
A/Ukraine/289/2015		2015-03-13	SIAT1/SIAT1	2560	1280	2560	2560	5120	5120	5120	5120	2560	
A/Ukraine/6395/2015	6B	2015-03-15	C1/MDCK1	2560	640	1280	1280	2560	2560	5120	5120	2560	
A/Ukraine/6786/2015	6B	2015-03-15	C0/MDCK2	1280	640	640	640	1280	1280	2560	2560	1280	
A/Ukraine/6807/2015		2015-03-15	C1/MDCK1	640	640	640	640	1280	1280	2560	1280	1280	
A/Ukraine/277/2015		2015-03-16	SIAT1/SIAT1	2560	640	1280	1280	2560	2560	5120	2560	1280	
A/Norway/1690/2015	6B	2015-03-17	MDCK1/MDCK3	2560	640	640	1280	2560	1280	5120	2560	2560	
A/Dnipro/454/2015	6B	2015-03-18	SIAT2/MDCK1	1280	640	640	640	1280	1280	2560	2560	1280	
A/Ukraine/377/2015	6B	2015-03-19	SIAT2/MDCK1	1280	640	640	1280	1280	2560	5120	2560	2560	
A/Norway/1637/2015	6B	2015-03-22	MDCK1	640	320	320	320	640	640	1280	640	640	N129D, A139S
A/Ukraine/369/2015		2015-03-22	SIAT1/MDCK1	1280	640	640	1280	1280	2560	5120	2560	1280	
A/Ukraine/433/2015	6B	2015-03-23	SIAT2/MDCK1	640	320	320	320	1280	640	1280	1280	640	S84N, R205K
A/Tsaralalana/1278/2015	6B	2015-03-23	MDCK1/MDCK1	1280	640	1280	1280	1280	1280	5120	2560	1280	
A/Dnipro/451/2015		2015-03-24	SIAT2/MDCK3	640	320	320	640	640	1280	2560	2560	1280	
A/Dnipro/453/2015		2015-03-24	SIAT2/MDCK3	1280	1280	1280	1280	1280	2560	5120	2560	2560	
A/Khmelnitsky/471/2015	6B	2015-03-25	SIAT2/SIAT1	2560	640	1280	1280	2560	2560	5120	2560	1280	
A/Ukraine/6781/2015	6B	2015-03-26	C0/MDCK1	2560	640	1280	1280	2560	2560	5120	5120	2560	
A/Ukraine/6782/2015		2015-03-30	C0/SIAT1	1280	640	640	640	1280	1280	2560	2560	1280	
A/Dnipro/455/2015		2015-03-30	SIAT2/MDCK3	1280	640	640	1280	1280	1280	5120	2560	1280	
A/Dnipro/456/2015		2015-03-31	SIAT2/MDCK3	1280	640	640	1280	1280	1280	2560	2560	1280	
A/Ukraine/6785/2015		2015-04-01	C0/SIAT1	2560	640	640	1280	2560	2560	5120	2560	1280	
A/Ukraine/399/2015	6B	2015-04-01	SIAT1/MDCK1	1280	640	640	640	1280	1280	2560	2560	1280	
A/Ternopil/411/2015	6B	2015-04-02	SIAT2/MDCK2	640	320	320	640	640	640	1280	1280	640	
A/Dnipro/447/2015		2015-04-03	SIAT2/MDCK3	1280	640	640	640	1280	1280	2560	2560	1280	
A/Dnipro/446/2015	6B	2015-04-07	SIAT2/MDCK3	1280	640	640	1280	2560	2560	5120	2560	2560	
A/Dnipro/458/2015		2015-04-09	SIAT1/MDCK3	2560	640	1280	1280	2560	2560	5120	5120	2560	
A/Norway/1951/2015	6B	2015-04-10	MDCK1	1280	640	640	640	1280	1280	2560	1280	1280	
A/Ukraine/427/2015	6B	2015-04-11	SIAT1/MDCK1	1280	640	640	640	1280	1280	2560	1280	1280	
A/Norway/1917/2015	6B	2015-04-13	MDCK1	1280	320	640	640	1280	640	2560	1280	1280	
A/Madagascar/1566/2015	6B	2015-04-15	MDCK1	640	320	320	320	640	640	1280	1280	640	
A/Madagascar/1600/2015		2015-04-20	MDCK2	1280	640	640	1280	2560	1280	5120	2560	2560	
A/Dnipro/448/2015	6B	2015-04-22	SIAT2/MDCK3	640	320	320	640	1280	1280	2560	1280	1280	

Sequences in phylogenetic tree

Vaccine

Table 5-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-06-30 + 2015-07-08)

Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/Astrak 1/11 F22/13	A/St P 27/11 F23/11	A/St P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
				Haemagglutination inhibition titre									
				Post-infection ferret antisera									
	Genetic group						4	5	6	7	6A	6B	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E3	1280	1280	2560	320	320	320	640	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	640	320	80	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	2560	320	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	2560	2560	2560	5120	2560	2560	5120	5120	2560	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	2560	2560	5120	2560	1280	
A/St Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	640	640	1280	1280	2560	2560	1280	
A/St Petersburg/100/2011	7	2011-03-14	E1/E3	2560	1280	1280	640	2560	2560	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	160	320	320	640	640	1280	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	1280	640	1280	640	1280	1280	2560	1280	1280	
TEST VIRUSES													
A/Sweden/74/2014		2014-12-03	MDCK2/MDCK1	1280	640	640	1280	2560	2560	5120	2560	2560	
A/Stockholm/37/2014	6B	2014-12-08	MDCK2/MDCK1	1280	640	640	640	1280	1280	2560	2560	1280	
A/Stockholm/6/2015	6B	2015-01-06	MDCK0/MDCK1	2560	640	640	1280	2560	2560	5120	2560	2560	
A/Sweden/6/2015	6B	2015-01-07	MDCK2/MDCK1	1280	640	640	1280	2560	1280	5120	2560	1280	
A/Stockholm/5/2015		2015-01-19	MDCK0/MDCK1	2560	640	640	1280	2560	2560	5120	2560	2560	
A/Moldova/050.05/2015	6B	2015-01-28	MDCK1/MDCK1	2560	1280	1280	1280	2560	2560	5120	2560	2560	
A/Moldova/085.07/2015		2015-02-06	MDCK3/MDCK2	2560	640	640	1280	2560	2560	5120	2560	2560	
A/Moldova/086.07/2015	6B	2015-02-06	MDCK2/MDCK1	1280	640	640	1280	2560	1280	2560	2560	1280	
A/Moldova/090.07/2015		2015-02-06	MDCK2/MDCK1	1280	1280	1280	1280	2560	2560	5120	2560	2560	
A/Stockholm/11/2015	6B	2015-02-06	MDCK2/MDCK1	2560	640	1280	1280	2560	2560	5120	2560	2560	
A/Moldova/078.07/2015	6B	2015-02-09	MDCK2/MDCK2	640	320	320	320	1280	640	1280	1280	640	D222G, L2133I, G264G=D
A/Moldova/107.07/2015	6B	2015-02-09	MDCK3/MDCK1	1280	640	640	1280	2560	1280	5120	2560	1280	
A/Moldova/109.07/2015		2015-02-09	MDCK1/MDCK1	2560	640	1280	1280	2560	2560	5120	5120	2560	
A/Moldova/117.07/2015		2015-02-11	MDCK3/MDCK1	2560	640	640	1280	2560	2560	5120	2560	2560	
A/Moldova/114.07/2015		2015-02-12	MDCK3/MDCK1	1280	640	640	640	2560	1280	5120	2560	1280	
A/Moldova/125.07/2015		2015-02-13	MDCK2/MDCK1	2560	1280	1280	1280	2560	2560	5120	5120	2560	
A/Moldova/150.08/2015	6B	2015-02-16	MDCK1/MDCK1	2560	1280	1280	1280	2560	2560	5120	5120	2560	
A/Stockholm/21/2015	6B	2015-02-18	MDCK0/MDCK1	1280	640	640	640	2560	1280	2560	2560	1280	
A/Sweden/18/2015	6B	2015-02-19	MDCK0/MDCK1	2560	640	1280	1280	2560	1280	5120	2560	1280	
A/Stockholm/25/2015	6B	2015-02-21	MDCK2/MDCK1	1280	640	640	1280	2560	1280	5120	2560	1280	
A/Stockholm/23/2015	6B	2015-02-23	MDCK0/MDCK1	2560	640	640	1280	2560	2560	5120	2560	1280	
A/Stockholm/28/2015	6B	2015-03-06	MDCK2/MDCK1	1280	640	640	640	1280	1280	2560	2560	1280	
A/Stockholm/27/2015	6B	2015-03-12	MDCK0/MDCK1	1280	640	640	1280	2560	2560	5120	2560	2560	
A/Ghana/DILI-15-0234/2015	6B	2015-03-18	Cx/MDCK1	640	160	320	320	640	640	1280	640	640	
A/Ghana/DILI-15-0255/2015	6B	2015-03-25	Cx/MDCK1	640	160	160	320	640	640	1280	640	640	
A/Ghana/FIPI-15-0019/2015	6B	2015-03-25	C1/MDCK1	640	320	320	320	1280	1280	2560	1280	1280	
A/Ghana/DARI-15-077/2015	6B	2015-03-30	C3/MDCK1	2560	320	320	640	640	1280	2560	1280	1280	
A/Stockholm/34/2015	6B	2015-03-30	MDCK0/MDCK1	1280	640	640	1280	1280	1280	2560	2560	1280	
A/Madagascar/1409/2015		2015-04-01	MDCK3	1280	320	640	640	1280	1280	2560	2560	1280	
A/Madagascar/1413/2015		2015-04-01	MDCK3	1280	640	640	640	1280	1280	2560	2560	1280	
A/Ghana/DILI-15-0303/2015		2015-04-02	Cx/MDCK2	640	160	320	320	640	640	1280	1280	640	
A/Ghana/FIPI-15-0023/2015	6B	2015-04-14	MDCK1	5120	640	640	640	2560	2560	5120	2560	2560	
A/Ghana/FIPI-15-0024/2015	6B	2015-04-14	MDCK1	640	320	320	320	1280	1280	2560	1280	1280	
A/Ghana/FIPI-15-0206/2015		2015-04-16	MDCK1	2560	320	320	640	1280	1280	2560	1280	1280	
A/Ghana/FS-15-242/2015		2015-04-19	SIAT1	2560	640	640	1280	2560	2560	5120	2560	2560	

Sequences in phylogenetic tree

Vaccine

Table 5-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-07-15)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre Post-infection ferret antisera A/Cal 7/09 F05/14	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/Astrak 1/11 F22/13	A/St P 27/11 F26/14	A/St P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
	Genetic group						4	5	6	7	6A	6B	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	EP1/E3	1280	1280	1280	320	320	320	640	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	320	320	80	80	40	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	1280	1280	320	160	160	160	640	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	2560	640	2560	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	1280	640	2560	2560	1280	
A/St Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	1280	640	2560	2560	1280	
A/St Petersburg/100/2011	7	2011-03-14	E1/E3	2560	1280	1280	1280	2560	1280	5120	5120	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	160	640	320	1280	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	1280	1280	1280	640	1280	1280	2560	1280	2560	
TEST VIRUSES													
A/Levice/97/2015		2015-01-09	MDCK1/MDCK1	1280	1280	1280	1280	2560	1280	5120	2560	2560	
A/Bratislava/88/2015		2015-01-09	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	
A/Nove Zamky/105/2015		2015-01-14	MDCK1/MDCK1	1280	1280	1280	1280	2560	1280	5120	2560	1280	
A/Moldova/012.04/2015		2015-01-17	MDCK2/MDCK1	2560	1280	1280	1280	2560	1280	5120	5120	2560	
A/Moldova/024.04/2015		2015-01-19	MDCK1/MDCK1	1280	1280	1280	1280	2560	1280	5120	2560	2560	
A/Bratislava/127/2015		2015-01-19	MDCK1/MDCK1	1280	1280	1280	1280	2560	1280	5120	2560	2560	
A/Kosice/163/2015		2015-01-23	MDCK1/MDCK1	1280	640	640	1280	2560	1280	5120	2560	2560	
A/Netherlands/1037/2015	6B	2015-02-02	MDCK2/MDCK1	2560	1280	1280	1280	2560	1280	5120	5120	2560	
A/Galanta/208/2015	6B	2015-02-03	MDCKx/MDCK1	1280	640	640	1280	2560	1280	5120	2560	2560	
A/Sobotiste/218/2015	6B	2015-02-03	MDCKx/MDCK1	2560	1280	1280	1280	2560	1280	5120	5120	2560	
A/Trencin/298/2015	6B	2015-02-06	MDCKx/MDCK1	1280	640	640	640	1280	1280	5120	2560	1280	
A/Netherlands/1417/2015	6B	2015-02-10	MDCK2/MDCK1	640	320	320	640	1280	640	2560	1280	1280	
A/Sastin-Straze/441/2015	6B	2015-02-18	MDCKx/MDCK1	1280	640	640	1280	1280	1280	5120	2560	2560	
A/St Petersburg/122/2015		2015-02-26	C1/MDCK1	1280	640	640	2560	2560	1280	5120	2560	2560	
A/IIV-Moscow/111/2015	6B	2015-02-26	MDCK3/MDCK1	1280	640	640	1280	2560	1280	5120	2560	2560	
A/Nitra/682/2015	6B	2015-02-28	MDCKx/MDCK1	1280	640	1280	1280	2560	1280	5120	5120	2560	
A/IIV-Moscow/110/2015		2015-02-28	MDCK3/MDCK1	640	320	320	640	1280	640	2560	1280	1280	
A/IIV-Moscow/112/2015	6B	2015-03-03	MDCK2/MDCK1	2560	640	1280	1280	2560	1280	5120	5120	2560	
A/IIV-Moscow/142/2015		2015-03-06	MDCK1/MDCK1	1280	320	320	640	1280	1280	5120	2560	1280	
A/IIV-Moscow/93/2015	6B	2015-03-10	MDCK1/MDCK1	640	320	320	640	640	640	2560	1280	1280	S84I, E172K, V272F
A/IIV-Moscow/94/2015	6B	2015-03-12	MDCK1/MDCK1	1280	640	640	1280	2560	1280	5120	2560	2560	
A/Malacky/831/2015	6B	2015-03-18	MDCKx/MDCK1	1280	640	640	1280	2560	1280	2560	2560	1280	
A/IIV-Moscow/139/2015	6B	2015-04-09	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	
A/IIV-Moscow/145/2015	6B	2015-04-16	MDCK1/MDCK1	1280	640	640	1280	2560	1280	5120	2560	2560	

Sequences in phylogenetic tree

Vaccine

Table 5-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-07-22 + 2015-07-29)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre, Post-infection ferret antisera: A/Cal 7/09 F05/14	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14 (4)	A/Astrak 1/11 F22/13 (5)	A/St P 27/11 F26/14 (6)	A/St P 100/11 F24/11 (7)	A/HK 5659/12 F30/12 (6A)	A/Sth Afr 3626/13 F3/14 (6B)	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E3	640	640	640	160	160	160	320	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	320	80	40	40	40	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	320	640	1280	160	80	80	80	160	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	2560	640	2560	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	320	320	640	1280	640	2560	1280	640	
A/St Petersburg/27/2011	6	2011-02-14	E1/E3	640	640	640	320	1280	640	2560	1280	640	
A/St Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	640	2560	1280	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	160	80	160	160	320	320	640	1280	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	640	640	320	640	640	1280	640	2560	
TEST VIRUSES													
A/Dakar/01/2015	6B	2015-01-12	P1/MDCK1	1280	640	640	640	2560	640	5120	2560	1280	
A/Dakar/03/2015	6B	2015-01-27	P1/MDCK1	1280	640	640	1280	2560	1280	5120	2560	2560	
A/Dakar/04/2015		2015-01-30	P1/MDCK1	1280	640	640	1280	2560	1280	5120	2560	2560	
A/Dakar/05/2015	6B	2015-02-03	P1/MDCK1	1280	320	640	640	1280	640	2560	1280	1280	
A/Dakar/06/2015	6B	2015-02-09	P0/MDCK1	1280	1280	1280	1280	2560	1280	5120	5120	2560	
A/Vladivostok/30/2015	6B	2015-02-11	MDCK3/MDCK2	640	320	320	320	640	320	1280	1280	640	
A/Dakar/07/2015	6B	2015-02-11	P0/MDCK1	1280	640	1280	1280	2560	1280	5120	2560	2560	
A/Yaroslavl/2/2015	6B	2015-02-12	MDCK2/MDCK2	640	320	320	640	1280	320	2560	1280	1280	
A/Slovenia/963/2015	6B	2015-02-13	MDCK1/SIAT1	2560	1280	1280	1280	2560	1280	5120	5120	2560	
A/Cameroon/15V-2691/2015	6B	2015-03-27	SIAT2	640	640	640	640	1280	640	2560	2560	1280	
A/Cameroon/15V-2682/2015	6B	2015-03-27	SIAT2	1280	640	640	640	2560	1280	5120	2560	1280	
A/Cameroon/15V-2860/2015	6B	2015-04-06	SIAT1	1280	640	640	1280	1280	640	2560	2560	1280	
A/Cameroon/15V-3038/2015	6B	2015-04-08	SIAT2	1280	640	640	640	1280	640	2560	2560	1280	
A/Cameroon/15V-3044/2015	6B	2015-04-14	SIAT2	1280	640	640	1280	2560	1280	5120	5120	2560	
A/Cameroon/15V-3360/2015	6B	2015-04-17	SIAT2	1280	640	640	1280	2560	1280	5120	5120	2560	
A/Cameroon/15V-3714/2015	6B	2015-05-04	SIAT1	640	320	640	640	1280	640	2560	2560	1280	
A/Cameroon/15V-3719/2015	6B	2015-05-07	SIAT1	640	320	320	640	1280	640	2560	2560	1280	
A/Cameroon/15V-3814/2015	6B	2015-05-07	SIAT2	640	320	320	640	1280	640	2560	2560	1280	
A/Cameroon/15V-4100/2015	6B	2015-05-26	MDCK1	640	640	640	640	1280	640	2560	2560	1280	
A/Hong Kong/11688/2015	6B	2015-06-08	MDCK1/MDCK1	1280	640	640	1280	1280	1280	5120	2560	1280	
A/Hong Kong/12243/2015	6B	2015-06-14	MDCK2/MDCK1	320	320	320	80	320	320	1280	640	640	S157T

Sequences in phylogenetic tree

Vaccine

Table 5-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-08-04)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post-infection ferret antisera: A/Cal 7/09 F05/14	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/Astrak 1/11 F22/13	A/St. P 27/11 F26/14	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
	Genetic group						4	5	6	7	6A	6B	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E3	640	640	640	160	160	160	320	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	80	320	160	80	40	40	40	40	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	640	1280	320	80	80	80	320	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	2560	1280	640	2560	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	320	320	640	1280	640	2560	1280	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	2560	640	5120	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	640	2560	1280	5120	25620	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	160	80	80	80	320	160	640	640	160	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	640	640	320	640	640	1280	640	1280	
TEST VIRUSES													
A/Slovenia/153/2015		2015-01-11	MDCKx/MDCK1	640	640	320	640	1280	640	2560	1280	1280	
A/Slovenia/367/2015		2015-01-20	MDCK1/MDCK1	640	320	320	320	1280	640	1280	1280	1280	
A/Mauritius/I-87/2015	6B	2015-01-21	MDCK2/MDCK1	320	160	320	320	320	320	640	640	640	N129D
A/Dakar/02/2015	6B	2015-01-23	C1/MDCK1	640	320	320	640	1280	640	2560	1280	1280	
A/Slovenia/596/2015		2015-01-26	MDCK1/MDCK1	640	320	320	320	640	320	1280	1280	640	
A/Slovenia/839/15	6B	2015-02-05	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	2560	1280	
A/Slovenia/826/15	6B	2015-02-06	MDCK1/MDCK1	640	640	640	640	1280	640	2560	1280	1280	
A/Slovenia/1079/15	6B	2015-02-20	MDCK1/MDCK1	640	320	320	320	640	640	2560	1280	640	
A/Slovenia/1267/2015	6B	2015-03-03	MDCK1/MDCK1	640	320	320	320	1280	640	2560	1280	640	
A/Slovenia/1314/15	6B	2015-03-05	MDCK1/MDCK1	640	320	320	320	640	640	2560	1280	640	
A/Slovenia/1413/15	6B	2015-03-11	MDCKx/MDCK1	640	640	320	640	1280	640	2560	2560	1280	
A/Slovenia/1541/15	6B	2015-03-17	MDCK1/MDCK1	640	320	320	640	1280	640	2560	1280	1280	
A/Slovenia/1550/15	6B	2015-03-17	MDCK1/MDCK1	640	640	320	640	1280	640	2560	1280	1280	
A/Mauritius/I-274/2015	6B	2015-03-18	MDCK2/MDCK1	640	320	320	320	640	320	1280	1280	640	
A/Cameroon/15V-2449/2015	6B	2015-03-23	SIAT2	640	320	320	640	1280	640	2560	1280	1280	
A/Mauritius/I-463/2015	6B	2015-05-18	MDCK2/MDCK1	640	320	320	320	1280	640	2560	1280	1280	

Sequences in phylogenetic tree

Vaccine

Table 5-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-08-12)

Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F05/14	A/Bayern 69/09 F09/15	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/Astrak 1/11 F22/13	A/St. P 27/11 F26/14	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
				Haemagglutination inhibition titre – Post-infection ferret antisera									
							4	5	6	7	6A	6B	
REFERENCE VIRUSES													
A/California/7/2009		2009-04-09	E1/E3	1280	640	320	160	320	160	320	160	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK1	160	640	320	80	80	40	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	320	1280	1280	160	160	80	160	160	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	1280	2560	2560	640	2560	2560	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	2560	640	2560	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	640	640	1280	640	2560	1280	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	640	640	2560	640	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	160	640	640	1280	640	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E3	640	640	640	640	1280	640	1280	1280	1280	
TEST VIRUSES													
A/Estonia/91956/2015	6B	2015-02-11	SIAT1/SIAT1	1280	640	640	640	2560	640	2560	2560	1280	
A/Poland/2453/2015	6B	2015-03-10	MDCK1/MDCK1	1280	640	640	640	2560	640	2560	2560	1280	
A/Poland/2456/2015	6B	2015-03-10	MDCK1/MDCK1	1280	1280	640	640	1280	640	2560	2560	1280	
A/Poland/2487/2015	6B	2015-03-19	MDCK1/MDCK1	1280	1280	640	1280	2560	1280	2560	2560	1280	
A/South Africa/R1101/2015	6B	2015-03-21	MDCK3/MDCK1	1280	1280	640	640	2560	640	2560	2560	1280	
A/South Africa/R1585/2015	6B	2015-04-16	MDCK1/MDCK1	1280	1280	640	640	2560	640	2560	2560	1280	
A/South Africa/R1730/2015		2015-04-16	MDCK2/MDCK1	1280	1280	640	640	2560	1280	2560	2560	1280	
A/South Africa/R2441/2015	6B	2015-05-18	MDCK1/MDCK1	1280	640	640	640	1280	640	2560	1280	1280	
A/South Africa/R2487/2015		2015-05-22	MDCK2/MDCK1	1280	1280	640	640	2560	1280	2560	2560	1280	
A/South Africa/R2491/2015	6B	2015-05-22	MDCK2/MDCK1	1280	1280	640	640	2560	640	2560	2560	1280	
A/South Africa/R2977/2015	6B	2015-06-05	MDCK1/MDCK1	1280	1280	640	640	2560	640	2560	2560	1280	
A/South Africa/R3127/2015	6B	2015-06-08	MDCK1/MDCK1	1280	640	640	640	2560	640	2560	2560	1280	
A/South Africa/R3425/2015	6B	2015-06-20	MDCK1/MDCK1	1280	1280	1280	1280	2560	1280	5120	2560	2560	
A/South Africa/R3474/2015	6B	2015-06-22	MDCK1/MDCK1	1280	1280	640	1280	2560	1280	5120	2560	2560	
A/South Africa/R3723/2015	6B	2015-06-29	MDCK1/MDCK1	1280	1280	640	640	2560	640	2560	2560	1280	
A/South Africa/R2574/2015	6B		MDCK1/MDCK1	320	320	320	160	80	80	80	80	160	N156S

Sequences in phylogenetic tree

Vaccine

Table 5-12. Summary of A(H1N1)pdm09 test virus HI results

Antiserum to Virus	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/HK* 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St P 27/11 F23/11	A/St P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14
Host	Egg	Cell	Cell	Egg	Cell	Cell	Egg	Egg	Cell	Egg
Genetic group	1	1	1	4	3	5	6	7	6A	6B
	Number of test viruses reacting within 2- or 4-fold of the antiserum homologous titre									
	Numbers within 2-fold									
	141	141	95	6	25	138	141	131	142	141
	Percentage within 2-fold									
	97.9	97.9	66.0	4.2	96.2	95.8	97.9	91.0	98.6	97.9
	Numbers within 4-fold									
	143	144	138	73	26	142	142	142	142	143
	Percentage within 4-fold									
	99.3	100.0	95.8	50.7	100.0	98.6	98.6	98.6	98.6	99.3
	Vaccine									

The HA sequences of 144 test viruses with collection dates after 2015-01-31 were determined: all were genetic group 6B

* This antiserum was tested against only 26 test viruses

Figure 1. Phylogenetic comparison of A(H1N1)pdm09 HA genes

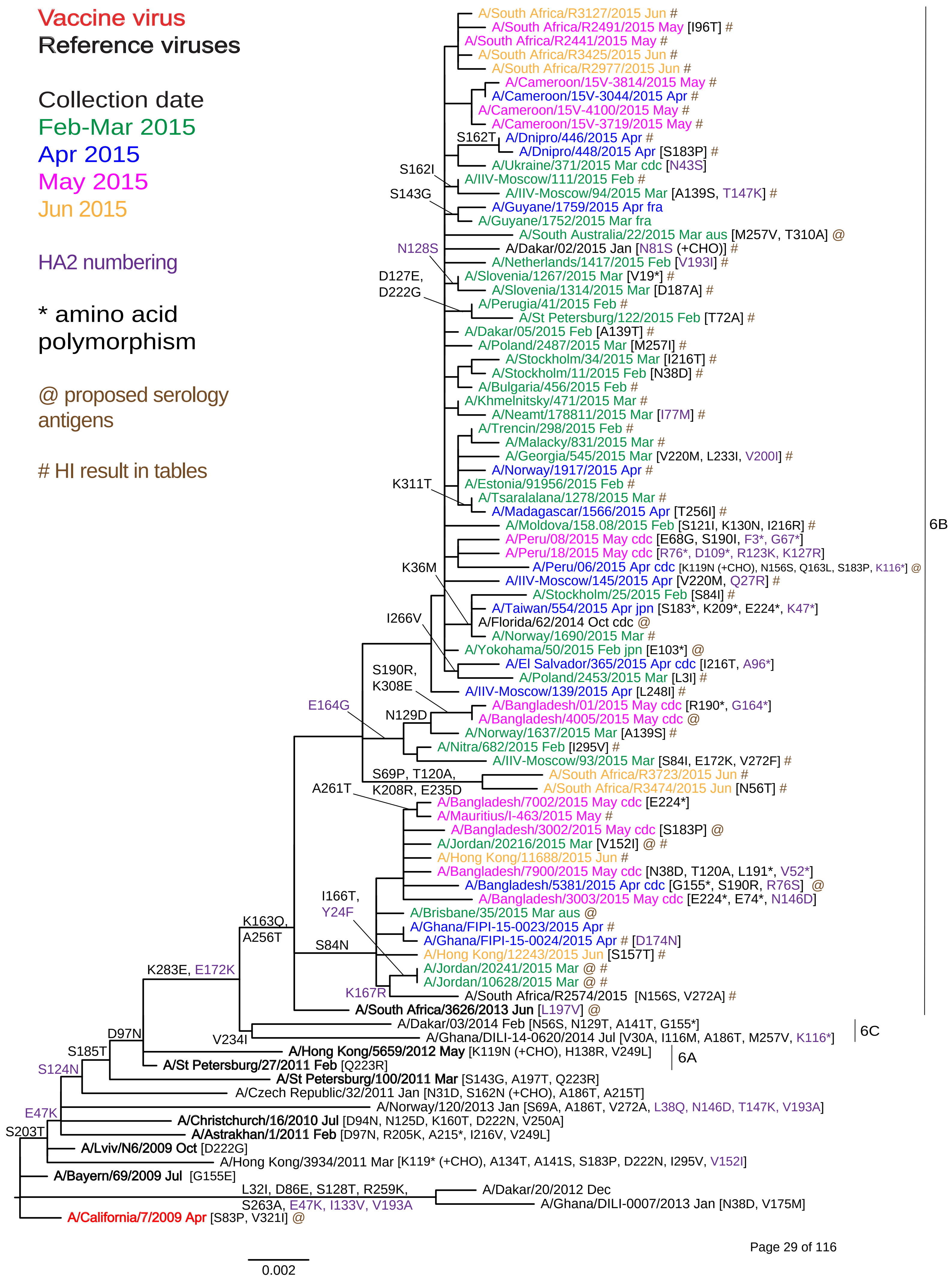


Figure 2. HA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                  10        20        30        40        50        60        70        80        90       100
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009      DTLCIGYHANNSTDTVDTVLEKNVTVTHSVNLLEDKHNGKLCKLRGVAPLHLGKCNIAGWILGNPECESLSTASSWSYIVETPSSDNGTCYPGDFIDYEE 1
A/Bayern/69/2009         ..................................................................................S................. 1
A/Lviv/N6/2009           ..................................................................................S................. 1
A/Dakar/20/2012          ...............................I..................................................S..E.............. W Africa
A/Czech Republic/32/2011 ..............................D...................................................S................. 2
A/Hong Kong/3934/2011    ..................................................................................S................. 3
A/Christchurch/16/2010   ..................................................................................S..........N...... 4
A/Astrakhan/1/2011       ..................................................................................S.............N... 5
A/Norway/120/2013        ....................................................................A.............S................. 8
A/St Petersburg/100/2011 ..................................................................................S................. 7
A/Hong Kong/5659/2012    ..................................................................................S.............N... 6A
A/St Petersburg/27/2011  ..................................................................................S.............N... 6A
A/South Africa/3626/2013 ..................................................................................S.............N... 6B
A/Hong Kong/12243/2015   ..................................................................................SN............N... 6B
A/Jordan/20241/2015      ..................................................................................SN............N... 6B
A/Bangladesh/3003/2015   ..................................................................................SN............N... 6B
A/Mauritius/I-463/2015   ..................................................................................SN............N... 6B
A/IIV-Moscow/93/2015     ..................................................................................SI............N... 6B
A/Bangladesh/01/2015     ..................................................................................S.............N... 6B
A/South Africa/R3723/2015 ...................................................................P.............S.............N... 6B
A/St Petersburg/122/2015 .......................................................................A..........S.............N... 6B
A/Madagascar/1566/2015   ..................................................................................S.............N... 6B
A/Slovenia/1314/2015     ..................................................................................S.............N... 6B
A/Guyane/1759/2015       ..................................................................................S.............N... 6B
A/Norway/1690/2015       ...................................M..............................................S.............N... 6B
A/IIV-Moscow/94/2015     ..................................................................................S.............N... 6B
A/Cameroon/15V-3814/2015 ..................................................................................S.............N... 6B
A/South Africa/R2977/2015 .................................................................................S.............N... 6B
A/Ghana/DILI-14-0620/2014 .............................A....................................................S.............N... 6C
A/Dakar/03/2014          .......................................................S..........................S.............N... 6C

                                 110       120       130       140       150       160       170       180       190       200
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009      LREQLSSVSSFERFEIFPKTSSWPNHDSNKGVTAACPHAGAKSFYKNLIWLVKKGNSYPKLSKSYINDKGKEVLVLWGIHHPSTSADQQSLYQNADAYVF
A/Bayern/69/2009         ......................................................E.............................................
A/Lviv/N6/2009           ....................................................................................................
A/Dakar/20/2012          ...........................T........................................................................
A/Czech Republic/32/2011 .............................................................N.......................T..............
A/Hong Kong/3934/2011    ..................X..............T......S.........................................P.................
A/Christchurch/16/2010   ........................D..................................T........................................
A/Astrakhan/1/2011       ....................................................................................................
A/Norway/120/2013        .....................................................................................T..............
A/St Petersburg/100/2011 ..........................................G.........................................T...........T...
A/Hong Kong/5659/2012    ..................N.................R...............................................T...............
A/St Petersburg/27/2011  ....................................................................................T...............
A/South Africa/3626/2013 ..............................................................Q.....................T...............
A/Hong Kong/12243/2015   ........................................................T.....Q.....................T...............
A/Jordan/20241/2015      ..............................................................Q..T..................T...............
A/Bangladesh/3003/2015   ..............................................................Q.....................T...............
A/Mauritius/I-463/2015   ..............................................................Q.....................T...............
A/IIV-Moscow/93/2015     ..............................................................Q........K...........T...............
A/Bangladesh/01/2015     ............................D.................................Q.....................T....X..........
A/South Africa/R3723/2015 ...................A..........................................Q.....................T...............
A/St Petersburg/122/2015 ..........................E...................................Q.....................T...............
A/Madagascar/1566/2015   ..............................................................Q.....................T...............
A/Slovenia/1314/2015     ..............................................................Q.....................T.A.............
A/Guyane/1759/2015       ..........................................G...................Q.....................T...............
A/Norway/1690/2015       ..............................................................Q.....................T...............
A/IIV-Moscow/94/2015     ......................................S......................IQ.....................T...............
A/Cameroon/15V-3814/2015 ..............................................................Q.....................T...............
A/South Africa/R2977/2015 .............................................................Q.....................T...............
A/Ghana/DILI-14-0620/2014 ..............M.....................................................................TT..............
A/Dakar/03/2014          ............................T...........T.............X.............................T...............

                                 210       220       230       240       250       260       270       280       290       300
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009      VGSSRYSKKFKPEIAIRPKVRDQEGRMNYYWTLVEPGDKITFEATGNLVVPRYAFAMERNAGSGIIISDTPVHDCNTTCQTPKGAINTSLPFQNIHPITI
A/Bayern/69/2009         ....................................................................................................
A/Lviv/N6/2009           ..T..................G..............................................................................
A/Dakar/20/2012          ..........................................................K...A.....................................
A/Czech Republic/32/2011 ..T...........T.....................................................................................
A/Hong Kong/3934/2011    ..T..................N..........................................................................V.....
A/Christchurch/16/2010   ..T..................N...........................A..................................................
A/Astrakhan/1/2011       ..T.K.........XV................................L...................................................
A/Norway/120/2013        ..T....................................................................A............................
A/St Petersburg/100/2011 ..T...................R.............................................................................
A/Hong Kong/5659/2012    ..T.............................................L...................................................
A/St Petersburg/27/2011  ..T...................R.............................................................................
A/South Africa/3626/2013 ..T....................................................T..........................E.................
A/Hong Kong/12243/2015   ..T....................................................T..........................E.................
A/Jordan/20241/2015      ..T....................................................T..........................E.................
A/Bangladesh/3003/2015   ..T....................X...............................T..........................E.................
A/Mauritius/I-463/2015   ..T....................................................T....T.....................E.................
A/IIV-Moscow/93/2015     ..T....................................................T...............F..........E.................
A/Bangladesh/01/2015     ..T....................................................T..........................E.................
A/South Africa/R3723/2015 ..T....R..........................D.....................T..........................E.................
A/St Petersburg/122/2015 ..T..................G..................................T..........................E.................
A/Madagascar/1566/2015   ..T....................................................I..........................E.................
A/Slovenia/1314/2015     ..T....................................................T..........................E.................
A/Guyane/1759/2015       ..T....................................................T..........................E.................
A/Norway/1690/2015       ..T....................................................T..........................E.................
A/IIV-Moscow/94/2015     ..T....................................................T..........................E.................
A/Cameroon/15V-3814/2015 ..T....................................................T..........................E.................
A/South Africa/R2977/2015 ..T...................................................T..........................E.................
A/Ghana/DILI-14-0620/2014 ..T..............................I......................V.........................E.................
A/Dakar/03/2014          ..T..............................I...............................................E.................
```

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                                310       320       330       340       350       360       370       380       390       400
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009        GKCPKYVKSTKLRLATGLRNIPSIQSRGLFGAIAGFIEGGWTGMVDGWYGYHHQNEQGSGYAADLKSTQNAIDEITNKVNSVIEKMNTQFTAVGKEFNHL
A/Bayern/69/2009           ....................V...............................................................................
A/Lviv/N6/2009             ....................V...............................................................................
A/Dakar/20/2012            ....................V....................................................K..........................
A/Czech Republic/32/2011   ....................V....................................................K..........................
A/Hong Kong/3934/2011      ....................V...............................................................................
A/Christchurch/16/2010     ....................V....................................................K..........................
A/Astrakhan/1/2011         ....................V....................................................K..........................
A/Norway/120/2013          ....................V...........................................Q........K..........................
A/St Petersburg/100/2011   ....................V....................................................K..........................
A/Hong Kong/5659/2012      ....................V....................................................K..........................
A/St Petersburg/27/2011    ....................V....................................................K..........................
A/South Africa/3626/2013   ....................V....................................................K..........................
A/Hong Kong/12243/2015     ....................V....................................................K..........................
A/Jordan/20241/2015        ....................V.............................F......................K..........................
A/Bangladesh/3003/2015     ....................V....................................................K..........................
A/Mauritius/I-463/2015     ....................V....................................................K..........................
A/IIV-Moscow/93/2015       ....................V....................................................K..........................
A/Bangladesh/01/2015       .......E............V....................................................K..........................
A/South Africa/R3723/2015  ....................V....................................................K..........................
A/St Petersburg/122/2015   ....................V....................................................K..........................
A/Madagascar/1566/2015     ..........T.........V....................................................K..........................
A/Slovenia/1314/2015       ....................V....................................................K..........................
A/Guyane/1759/2015         ....................V....................................................K..........................
A/Norway/1690/2015         ....................V....................................................K..........................
A/IIV-Moscow/94/2015       ....................V....................................................K..........................
A/Cameroon/15V-3814/2015   ....................V....................................................K..........................
A/South Africa/R2977/2015  ....................V....................................................K..........................
A/Ghana/DILI-14-0620/2014  ....................V....................................................K..........................
A/Dakar/03/2014            ....................V....................................................K..........................

                                410       420       430       440       450       460       470       480       490       500
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009        EKRIENLNKKVDDGFLDIWTYNAELLVLLENERTLDYHDSNVKNLYEKVRSQLKNNAKEIGNGCFEFYHKCDNTCMESVKNGTYDYPKYSEEAKLNREEI
A/Bayern/69/2009           ....................................................................................................
A/Lviv/N6/2009             ....................................................................................................
A/Dakar/20/2012            ...........................................................V........................................
A/Czech Republic/32/2011   ..................................................N.................................................
A/Hong Kong/3934/2011      ..............................................................................I.....................
A/Christchurch/16/2010     ....................................................................................................
A/Astrakhan/1/2011         ....................................................................................................
A/Norway/120/2013          ........................................................................DK..........................
A/St Petersburg/100/2011   ..................................................N.................................................
A/Hong Kong/5659/2012      ..................................................N.................................................
A/St Petersburg/27/2011    ..................................................N.................................................
A/South Africa/3626/2013   ..................................................N...............................................K.
A/Hong Kong/12243/2015     ..................................................N...............................................K.
A/Jordan/20241/2015        ..................................................N..........................................R....K.
A/Bangladesh/3003/2015     X.......................X.........................N.....................D.........................K.
A/Mauritius/I-463/2015     ..................................................N...............................................K.
A/IIV-Moscow/93/2015       ..................................................N.......................................G.......K.
A/Bangladesh/01/2015       ..................................................N.......................................X.......K.
A/South Africa/R3723/2015  ..................................................N...............................................K.
A/St Petersburg/122/2015   ..................................................N...............................................K.
A/Madagascar/1566/2015     ..................................................N...............................................K.
A/Slovenia/1314/2015       ..................................................N...S...........................................K.
A/Guyane/1759/2015         ..................................................N...............................................K.
A/Norway/1690/2015         ..................................................N...............................................K.
A/IIV-Moscow/94/2015       ..................................................N......................K........................K.
A/Cameroon/15V-3814/2015   ..................................................N...............................................K.
A/South Africa/R2977/2015  ..................................................N...............................................K.
A/Ghana/DILI-14-0620/2014  ..........................................X.......N...............................................K.
A/Dakar/03/2014            ..................................................N...............................................K.

                                510       520       530       540
                           ....|....|....|....|....|....|....|....|....|....
A/California/7/2009        DGVKLESTRIYQILAIYSTVASSLVLVVSLGAISFWMCSNGSLQCRICI
A/Bayern/69/2009           .................................................
A/Lviv/N6/2009             .................................................
A/Dakar/20/2012            ...................A.............................
A/Czech Republic/32/2011   .................................................
A/Hong Kong/3934/2011      .................................................
A/Christchurch/16/2010     .................................................
A/Astrakhan/1/2011         .................................................
A/Norway/120/2013          ...................A.............................
A/St Petersburg/100/2011   .................................................
A/Hong Kong/5659/2012      .................................................
A/St Petersburg/27/2011    .................................................
A/South Africa/3626/2013   .......................V.........................
A/Hong Kong/12243/2015     .................................................
A/Jordan/20241/2015        .................................................
A/Bangladesh/3003/2015     .................................................
A/Mauritius/I-463/2015     .................................................
A/IIV-Moscow/93/2015       .................................................
A/Bangladesh/01/2015       .................................................
A/South Africa/R3723/2015  .................................................
A/St Petersburg/122/2015   .................................................
A/Madagascar/1566/2015     .................................................
A/Slovenia/1314/2015       .................................................
A/Guyane/1759/2015         .................................................
A/Norway/1690/2015         .................................................
A/IIV-Moscow/94/2015       .................................................
A/Cameroon/15V-3814/2015   .................................................
A/South Africa/R2977/2015  .................................................
A/Ghana/DILI-14-0620/2014  .................................................
A/Dakar/03/2014            .................................................
```

Vaccine virus: **Reference virus**: **Genetic group**: **HA1/HA2 boundary**: **Potential N-linked glycosylation site**

Figure 3. H1 HA locations of amino acid substitutions that define genetic subgroup 6B and some minor groupings within subgroup 6B.

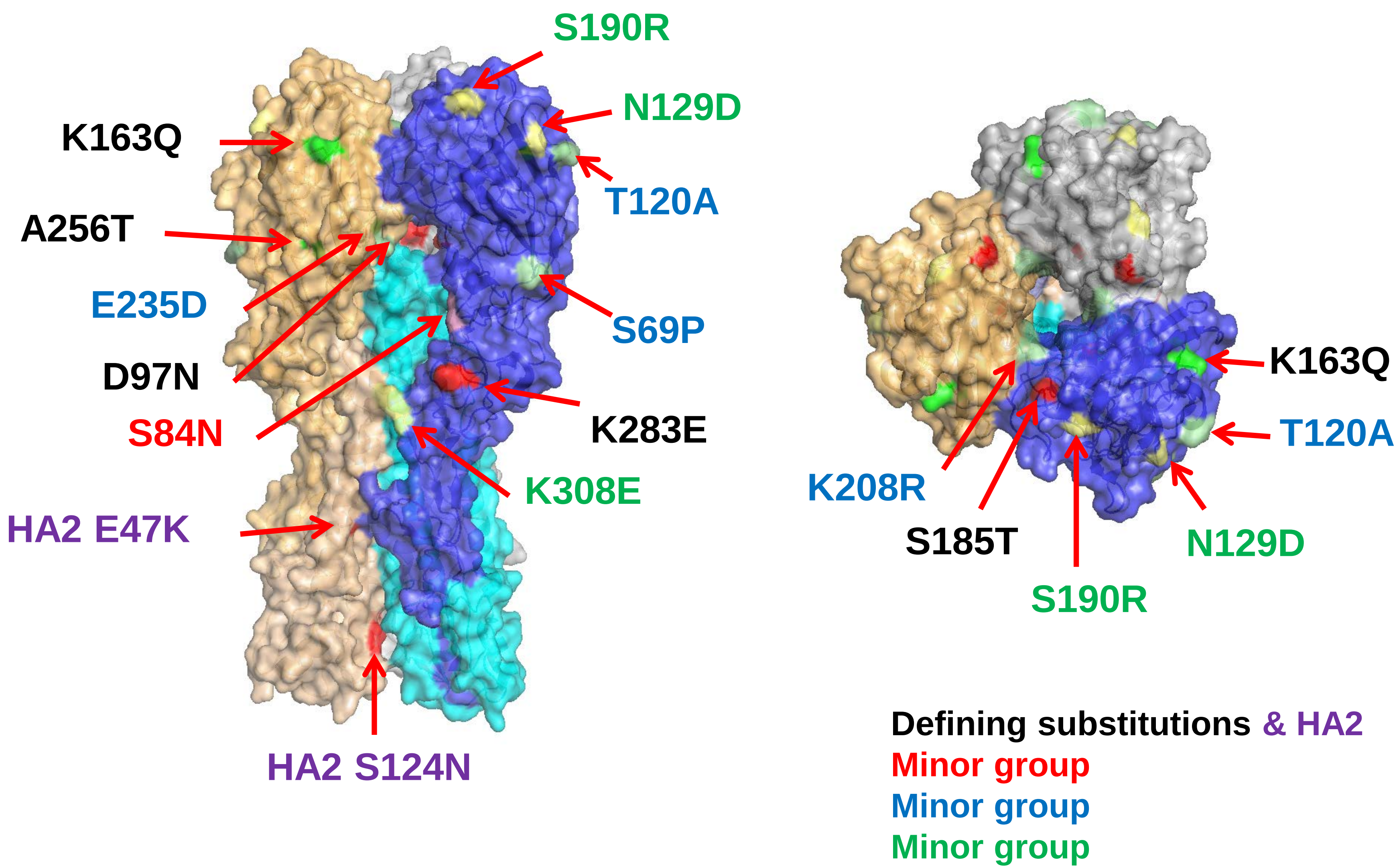

Figure 4. Phylogenetic comparison of A(H1N1)pdm09 NA genes

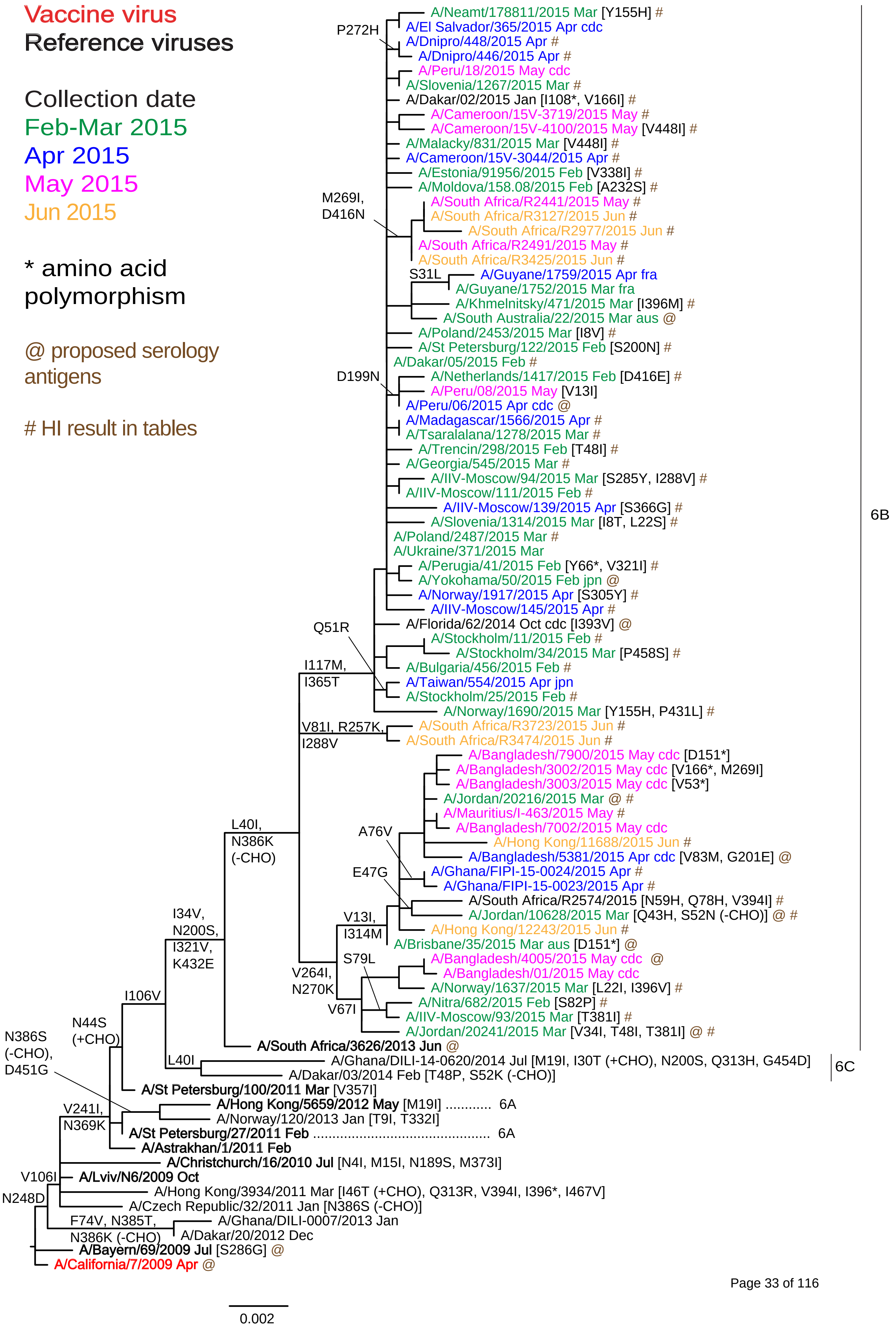

Figure 5. NA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                   10        20        30        40        50        60        70        80        90       100
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009       MNPNQKIITIGSVCMTIGMANLILQIGNIISIWISHSIQLGNQNQIETCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLAGNSSLCPVSGWAIY 1
A/Bayern/69/2009          .................................................................................................... 1
A/Lviv/N6/2009            .................................................................................................... 1
A/Dakar/20/2012           .........................................................................V.......................... W Africa
A/Czech Republic/32/2011  .................................................................................................... 2
A/Hong Kong/3934/2011     .............................................T...................................................... 3
A/Christchurch/16/2010    ...I..........I..................................................................................... 4
A/Astrakhan/1/2011        .................................................................................................... 5
A/Norway/120/2013         ........I........................................................................................... 8
A/St Petersburg/100/2011  ...........................................S........................................................ 7
A/Hong Kong/5659/2012     ..................I................................................................................. 6A
A/St Petersburg/27/2011   .................................................................................................... 6A
A/South Africa/3626/2013  .................................V.........S........................................................ 6B
A/Hong Kong/12243/2015    ............I....................V.....I...S........................................................ 6B
A/Jordan/20241/2015       .................................V.....I...S...I..................I................................. 6B
A/Bangladesh/3003/2015    ............I....................V.....I...S........X............................................... 6B
A/Mauritius/I-463/2015    ............I....................V.....I...S........................................................ 6B
A/IIV-Moscow/93/2015      .................................V.....I...S......................I...........L..................... 6B
A/Bangladesh/01/2015      .................................V.....I...S......................I................................. 6B
A/South Africa/R3723/2015 .................................V.....I...S....................................I................... 6B
A/St Petersburg/122/2015  .................................V.....I...S........................................................ 6B
A/Madagascar/1566/2015    .................................V.....I...S........................................................ 6B
A/Slovenia/1314/2015      .......T.............S...........V.....I...S........................................................ 6B
A/Guyane/1759/2015        ..............................L..V.....I...S........................................................ 6B
A/Norway/1690/2015        .................................V.....I...S........................................................ 6B
A/IIV-Moscow/94/2015      .................................V.....I...S........................................................ 6B
A/Cameroon/15V-3719/2015  .................................V.....I...S........................................................ 6B
A/South Africa/R2977/2015 .................................V.....I...S........................................................ 6B
A/Ghana/DILI-14-0620/2014 ..................I..........T.........I...S........................................................ 6C
A/Dakar/03/2014           .......................................I...S...P...K................................................ 6C

                                  110       120       130       140       150       160       170       180       190       200
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009       SKDNSVRIGSKGDVFVIREPFISCSPLECRTFFLTQGALLNDKHSNGTIKDRSPYRTLMSCPIGEVPSPYNSRFESVAWSASACHDGINWLTIGISGPDN
A/Bayern/69/2009          ....................................................................................................
A/Lviv/N6/2009            .....I..............................................................................................
A/Dakar/20/2012           ....................................................................................................
A/Czech Republic/32/2011  .....I..............................................................................................
A/Hong Kong/3934/2011     .....I..............................................................................................
A/Christchurch/16/2010    .....I..................................................................................S...........
A/Astrakhan/1/2011        .....I..............................................................................................
A/Norway/120/2013         .....I..............................................................................................
A/St Petersburg/100/2011  .....I..............................................................................................
A/Hong Kong/5659/2012     .....I..............................................................................................
A/St Petersburg/27/2011   .....I..............................................................................................
A/South Africa/3626/2013  ...................................................................................................S
A/Hong Kong/12243/2015    ...................................................................................................S
A/Jordan/20241/2015       ...................................................................................................S
A/Bangladesh/3003/2015    ...................................................................................................S
A/Mauritius/I-463/2015    ...................................................................................................S
A/IIV-Moscow/93/2015      ...................................................................................................S
A/Bangladesh/01/2015      ...................................................................................................S
A/South Africa/R3723/2015 ...................................................................................................S
A/St Petersburg/122/2015  ................M...................................................................................
A/Madagascar/1566/2015    ................M..................................................................................S
A/Slovenia/1314/2015      ................M..................................................................................S
A/Guyane/1759/2015        ................M..................................................................................S
A/Norway/1690/2015        ................M.....................................H............................................S
A/IIV-Moscow/94/2015      ................M..................................................................................S
A/Cameroon/15V-3719/2015  ................M..................................................................................S
A/South Africa/R2977/2015 ................M..................................................................................S
A/Ghana/DILI-14-0620/2014 ...................................................................................................S
A/Dakar/03/2014           ....................................................................................................

                                  210       220       230       240       250       260       270       280       290       300
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009       GAVAVLKYNGIITDTIKSWRNNILRTQESECACVNGSCFTVMTDGPSNGQASYKIFRIEKGKIVKSVEMNAPNYHYEECSCYPDSSEITCVCRDNWHGSN
A/Bayern/69/2009          .....................................................................................G..............
A/Lviv/N6/2009            ...............................................D....................................................
A/Dakar/20/2012           ...............................................D....................................................
A/Czech Republic/32/2011  ...............................................D....................................................
A/Hong Kong/3934/2011     ...............................................D....................................................
A/Christchurch/16/2010    ...............................................D....................................................
A/Astrakhan/1/2011        ........................................I......D....................................................
A/Norway/120/2013         ........................................I......D....................................................
A/St Petersburg/100/2011  ........................................I......D....................................................
A/Hong Kong/5659/2012     ........................................I......D....................................................
A/St Petersburg/27/2011   ........................................I......D....................................................
A/South Africa/3626/2013  ........................................I......D....................................................
A/Hong Kong/12243/2015    ........................................I......D...............I.....K..............................
A/Jordan/20241/2015       ........................................I......D...............I.....K..............................
A/Bangladesh/3003/2015    ........................................I......D...............I.....K..............................
A/Mauritius/I-463/2015    ........................................I......D...............I.....K..............................
A/IIV-Moscow/93/2015      ........................................I......D...............I.....K..............................
A/Bangladesh/01/2015      ........................................I......D...............I.....K..............................
A/South Africa/R3723/2015 ........................................I......D........K..............................V............
A/St Petersburg/122/2015  ........................................I......D....................................................
A/Madagascar/1566/2015    ........................................I......D....................................................
A/Slovenia/1314/2015      ........................................I......D....................................................
A/Guyane/1759/2015        ........................................I......D....................................................
A/Norway/1690/2015        ........................................I......D....................................................
A/IIV-Moscow/94/2015      ........................................I......D....................................Y..V............
A/Cameroon/15V-3719/2015  ........................................I......D....................................................
A/South Africa/R2977/2015 ........................................I......D....................I...............................
A/Ghana/DILI-14-0620/2014 ........................................I......D....................................................
A/Dakar/03/2014           ........................................I......D....................................................
```

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                                    310       320       330       340       350       360       370       380       390       400
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009        RPWVSFNQNLEYQIGYICSGIFGDNPRPNDKTGSCGPVSSNGANGVKGFSFKYGNGVWIGRTKSISSRNGFEMIWDPNGWTGTDNNFSIKQDIVGINEWS
A/Bayern/69/2009           ....................................................................................................
A/Lviv/N6/2009             ....................................................................................................
A/Dakar/20/2012            ....................................................................................TK..............
A/Czech Republic/32/2011   .....................................................................................S..............
A/Hong Kong/3934/2011      ............R................................................................................I.X....
A/Christchurch/16/2010     ........................................................................I...........................
A/Astrakhan/1/2011         ....................................................................K...............................
A/Norway/120/2013          ...............................I....................................K................S..............
A/St Petersburg/100/2011   ........................................................I...........K...............................
A/Hong Kong/5659/2012      ....................................................................K................S..............
A/St Petersburg/27/2011    ....................................................................K...............................
A/South Africa/3626/2013   ....................V...............................................K...............................
A/Hong Kong/12243/2015     .............M......V...............................................K................K..............
A/Jordan/20241/2015        ....................V...............................................K...........I....K..............
A/Bangladesh/3003/2015     .............M......V...............................................K................K..............
A/Mauritius/I-463/2015     .............M......V...............................................K................K..............
A/IIV-Moscow/93/2015       ....................V...............................................K...........I....K..............
A/Bangladesh/01/2015       ....................V...............................................K................K..............
A/South Africa/R3723/2015  ....................V...............................................K................K..............
A/St Petersburg/122/2015   ....................V...........................................T...K................K..............
A/Madagascar/1566/2015     ....................V...........................................T...K................K..............
A/Slovenia/1314/2015       ....................V...........................................T...K................K..............
A/Guyane/1759/2015         ....................V...........................................T...K................K..............
A/Norway/1690/2015         ....................V...........................................T...K................K..............
A/IIV-Moscow/94/2015       ....................V...........................................T...K................K..............
A/Cameroon/15V-3719/2015   ....................V...........................................T...K................K..............
A/South Africa/R2977/2015  ....................V...........................................T...K................K..............
A/Ghana/DILI-14-0620/2014  ............H.......................................................K...............................
A/Dakar/03/2014            ....................................................................K...............................

                                    410       420       430       440       450       460
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/California/7/2009        GYSGSFVQHPELTGLDCIRPCFWVELIRGRPKENTIWTSGSSISFCGVNSDTVGWSWPDGAELPFTIDK
A/Bayern/69/2009           .....................................................................
A/Lviv/N6/2009             .....................................................................
A/Dakar/20/2012            .....................................................................
A/Czech Republic/32/2011   .....................................................................
A/Hong Kong/3934/2011      ..................................................................V..
A/Christchurch/16/2010     .....................................................................
A/Astrakhan/1/2011         .....................................................................
A/Norway/120/2013          ..................................................G..................
A/St Petersburg/100/2011   .....................................................................
A/Hong Kong/5659/2012      ..................................................G..................
A/St Petersburg/27/2011    .....................................................................
A/South Africa/3626/2013   ...............................E.....................................
A/Hong Kong/12243/2015     ...............................E.....................................
A/Jordan/20241/2015        ...............................E.....................................
A/Bangladesh/3003/2015     ...............................E.....................................
A/Mauritius/I-463/2015     ...............................E.....................................
A/IIV-Moscow/93/2015       ...............................E.....................................
A/Bangladesh/01/2015       ...............................E.....................................
A/South Africa/R3723/2015  ...............................E.....................................
A/St Petersburg/122/2015   ...............................E.....................................
A/Madagascar/1566/2015     ...............................E.....................................
A/Slovenia/1314/2015       ...............................E.....................................
A/Guyane/1759/2015         ...............................E.....................................
A/Norway/1690/2015         ..............................LE.....................................
A/IIV-Moscow/94/2015       ...............................E.....................................
A/Cameroon/15V-3719/2015   ...............................E.....................................
A/South Africa/R2977/2015  ...............N...............E.....................................
A/Ghana/DILI-14-0620/2014  .....................................................D...............
A/Dakar/03/2014            .....................................................................
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2015-01-31 have been received from NICs in Europe, Africa (Ghana and South Africa), the Indian Ocean (Madagascar and Mauritius), the Middle East (Iraq) and from Hong Kong SAR. Influenza A(H3N2) viruses have become increasingly difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pig, turkey and humans or the loss of the ability of viruses to agglutinate any of these RBCs. Of the successfully propagated viruses collected since 2015-01-31, just under 40% had sufficient HA titre to be analysed by HI assays conducted using guinea pig RBCs in the presence of 20nM oseltamivir, added to circumvent any NA-mediated binding to the RBCs **(Figure 6, Table 6)**. The remainder were unable to agglutinate guinea pig RBCs at all (44% of viruses collected since 2015-01-31) or were unable to agglutinate RBCs in the presence of 20nM oseltamivir (17% of viruses). The majority (81%) of viruses in these last two categories belonged to genetic subgroup 3C.2a. Over half (53%) of the viruses in genetic subgroup 3C.2a, collected since 2015-01-31, that were able to bind guinea pig RBCs in the presence of oseltamivir showed either loss of, or polymorphism at, the 158/160 glycosylation motif in HA1 acquired by viruses in genetic subgroup 3C.2a **(Tables 8-1 to 8-14)**.

HI analyses.

A total of 101 viruses could be analysed by HI assay and summaries of the HI results is shown in **Tables 3 and 7**. Individual HI results are shown in **Tables 8-1** to **8-14**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are viruses with collection dates after 2015-01-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees presented here are highlighted. Notable substitutions in the HA glycoprotein are also shown. Two sets of HI results for egg-propagated cultivars of viruses from Australia are shown in **Tables 8-2 and 8-12**.

A summary of the HI results for all viruses, tested since our report for the February 2015 consultation, in genetic subgroups 3C.2a, 3C.3a and 3C.3b is shown in **Table 8-15**; viruses in genetic group 3C.3 were not included in this summary table and represented only 11% of the viruses tested and sequenced (only 9% of those collected since 2015-01-31). All test viruses, except the egg-propagated viruses with results shown in **Tables 8-2 and 8-12,** were propagated in MDCK-SIAT1 cells.

Post-infection ferret antiserum raised against egg-propagated A/Texas/50/2012, a former vaccine virus, recognised the vast majority of cell culture-propagated test viruses poorly in HI assays (93% of those collected since 2015-01-31, and 95% of those tested since our report for the February 2015 consultation, for which gene sequencing had been done) showing ≥8-fold decreased titres compared with the titre of the antiserum with the homologous virus. However, of the 37 test viruses falling in genetic subgroup 3C.3b, determined by HA gene sequencing, 24% were recognised at titres within 4-fold of the homologous titre by this antiserum.

Antiserum raised against the currently recommended vaccine virus, egg-propagated A/Switzerland/9715293/2013, recognised even smaller proportions (~2%) of test viruses, both those collected since 2015-01-31 and all those analysed since our report for the February 2015 consultation, at titres within 4-fold of the titre for the homologous virus (**Table 8-15**).

A summary of the HI results for viruses collected since 2015-01-31 with an antiserum raised against the cell culture-propagated cultivar of A/Switzerland/9715293/2013 is shown in **Table 7**. This antiserum recognised just over 50% of viruses at titres within 2-fold of the titre of the antiserum with the homologous virus. Due to the low homologous titre of the antiserum it was not possible, in some assays, to differentiate between titres of the antiserum for viruses with 4-fold or greater reductions (**Table 7**). Of the viruses analysed genetically and subjected to HI assay since our report for the February 2015 consultation the antiserum raised against cell culture-propagated A/Switzerland/9715293/2013 recognised only 27% of the genetic subgroup 3C.2a viruses at titres within 4-fold of the titre for the homologous virus. Recognition of viruses in genetic subgroups 3C.3a and 3C.3b was better at 91% and 68%, respectively (**Table 8-15**).

Compared to antiserum raised against the egg-propagated cultivar of A/Switzerland/9715293/2013, antiserum raised against egg-propagated A/Stockholm/6/2014 recognised substantially more (80%) of the test viruses at titres within 4-fold of the titre for the homologous virus. For viruses analysed since our report for the February 2015 consultation: 66% of the 3C.2a viruses, 89% of the 3C.3a and 97% of the 3C.3b viruses were recognised at titres within 4-fold of the titre for the homologous virus (**Table 8-15**).

In addition, antiserum raised against the cell-propagated cultivar of A/Stockholm/6/2014 recognised the test viruses better than the antiserum raised against cell-propagated A/Switzerland/9715293/2013: 87% of the 3C.2a viruses, 96% of 3C.3a viruses and 97% of 3C.3b viruses were recognised at titres within 4-fold of the titre for the homologous virus (**Table 8-15)**.

An antiserum raised against the egg-propagated cultivar of A/Hong Kong/5738/2014 (3C.2a) recognised none of the genetically characterised cell culture-propagated test viruses at titres within 4-fold of the titre for the homologous virus.

An antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/5738/2014 recognised 88% of the 3C.2a viruses analysed, 88% of the 3C.3a viruses and 97% of the 3C.3b viruses at titres within 4-fold of the titre for the homologous virus.

An antiserum raised against the egg-propagated cultivar of A/Hong Kong/4801/2014 was used in **Tables 8-11 to 8-14** and recognised all but one of the15 cell-propagated 3C.2a viruses analysed and all seven of the 3C.3b viruses at titres within 4-fold of the titre for the homologous virus (**Table 8-15**).

An antiserum raised against a cell-culture propagated reference virus in genetic subgroup 3C.3b, A/Netherlands/525/2014, was used in **Tables 8-9 to 8-14**. It recognised all 23 of the viruses in genetic subgroup 3C.3b at titres within 4-fold of the titre for the homologous virus but none of the viruses in the other genetic subgroups (38 in group 3C.2a and six in 3C.3a) (**Table 8-15**). Currently, we have no antisera raised against egg-propagated viruses in genetic subgroup 3C.3b.

Virus Neutralisation.

Plaque reduction neutralisation assays (PRNA) were used to complement HI tests and allowed antigenic characterisation of viruses that could not be assayed by HI.

The results of virus neutralisation assays done in MDCK-SIAT1 cells are shown in **Tables 9-1 to 9-7**. The genetic groups and noteworthy sequence features in the HA are marked. **Table 9-1** shows the titres of the reference viruses and their corresponding post-infection ferret antisera from which a sub-set was selected for virus neutralisation assays on test viruses that had circulated recently. The selected antisera were raised against a cell culture-propagated 3C.2a virus, A/Hong Kong/7295/2014, that retained the glycosylation motif at residues 158-160 of HA1; an egg-propagated 3C.2a virus, A/Hong Kong/4801/2014 that had lost the glycosylation motif at residues 158-160 in HA1 and had acquired egg-adaptive changes of N96S and L194P in HA1; both the cell culture-propagated and egg-propagated cultivars of A/Switzerland/9715293/2013 in genetic subgroup 3C.3a; and cell culture-propagated A/Netherlands/525/2014, belonging to genetic subgroup 3C.3b. A summary table of the results is shown in **Table 9-8**. A total of 86 test viruses were analysed.

The antiserum raised against the currently recommended vaccine virus, egg-propagated A/Switzerland/9715293/2013, had a low homologous titre of between 160 and 320. It recognised 28 of 57 (49%) 3C.2a viruses, 9 of 15 (60%) 3C.3a viruses and 12 of 14 (86%) 3C.3b test viruses at titres within 4-fold of the titre for the homologous virus (summarised in **Table 9-8**).

Somewhat higher reactivity, relative to that for the homologous virus, was seen with the antiserum raised against the cell culture-propagated cultivar of A/Switzerland/9715293/2013 which recognised 86% of the 3C.2a viruses tested, and all of the viruses in genetic subgroups 3C.3a and 3C.3b at titres within 4-fold of the titre for the homologous virus (**Table 9-8**).

The antiserum raised against the egg-propagated cultivar of A/Hong Kong/4801/2014 (3C.2a) recognised 80% of the 3C.2a test viruses at titres within 4-

fold of the titre for the homologous virus, but only 20% of the 15 3C.3a and 50% of the 14 3C.3b test viruses were recognised at titres within 4-fold of the titre for the homologous virus (**Table 9-8**).

The antiserum raised against the cell culture-propagated cultivar of A/Hong Kong/7295/2014 recognised all 57 3C.2a test viruses at titres within 4-fold of the titre for the homologous virus and 55 (96%) within 2-fold, 12 of 15 (80%) 3C.3a test viruses at titres within 4-fold of the titre for the homologous virus, three (20%) within 2-fold, and it recognised 13 of the 14 (93%) 3C.3b test viruses at titres within 4-fold of the titre for the homologous virus and 6 (43%) within 2-fold of the titre for the homologous virus (**Table 9-8**).

The antiserum raised against A/Netherlands/525/2014 recognised all 14 viruses from genetic subgroup 3C.3b very well, at titres within 2-fold of the titre of the antiserum for the homologous virus but test viruses from genetic subgroups 3C.2a and 3C.3a were recognised poorly (**Table 9-8**).

A higher proportion of viruses in genetic subgroup 3C.2a were recognised by antiserum raised against A/Hong Kong/4801/2014 at titres within 4-fold of the titre for the homologous virus, compared to the antiserum raised against A/Switzerland/9715293/2013. Moreover, a higher proportion of viruses in genetic subgroup 3C.2a were recognised better by the antiserum raised against cell culture-propagated A/Hong Kong/7295/2014 than they were by the antiserum raised against cell culture-propagated A/Switzerland/9715293/2013.

A non-parametric statistical analysis of the results from virus neutralisation assays confirmed that cell culture-propagated viruses in genetic subgroup 3C.2a were recognised better by the antiserum raised against cell culture-propagated A/Hong Kong/7295/2014 than they were by the antiserum raised against cell culture-propagated A/Switzerland/9715293/2013, relative to the homologous titres of the antisera. The analyses also indicated that these cell culture-propagated 3C.2a test viruses were recognised equally by the antiserum raised against egg-propagated A/Hong Kong/4801/2014 and the antiserum raised against egg-propagated A/Switzerland/9715293/2013 when assessed relative to the homologous titres of the antisera. However, they were recognised better in absolute titres by the antiserum raised against egg-propagated A/Hong Kong/4801/2014 than by the antiserum raised against egg-propagated A/Switzerland/9715293/2013, with the antiserum raised against egg-propagated A/Hong Kong/4801/2014 having the higher homologous titre.

Genetic analyses.

Phylogenetic analysis of the HA genes of representative recently circulating H3N2 viruses is shown in **Figure 7**. The HA genes fell within genetic group 3C. This group has three subdivisions: 3C.1, 3C.2 and 3C.3; further subdivisions have emerged in genetic group subdivisions 3C.2 and 3C.3 yielding subgroups designated as 3C.2a, 3C.3a and 3C.3b respectively (**Figure 7**).

The previously used vaccine virus A/Texas/50/2012 belongs to genetic subdivision 3C.1. Most of the viruses received with collection dates since 2015-01-31 that have been sequenced fall into subgroups 3C.2a (61%), 3C.3b (20%), 3C.3a (10%) and genetic group subdivision 3C.3 (9%). Amino acid substitutions that define these new subgroups compared with A/Texas/50/2012 are:

(**3C.2a**) **L3I**, **N144S** (resulting in the loss of a potential glycosylation site), **N145S, F159Y**, **K160T** (resulting in the gain of a potential glycosylation site), **V186G**[@], **N225D** and **Q311H** in **HA1**, e.g. A/Hong Kong/5738/2014.

(**3C.3a**) **T128A** (resulting in the loss of a potential glycosylation site), **A138S, R142G, N145S, F159S, V186G**[@] and **N225D** in **HA1**, e.g. A/Switzerland/9715293/2013.

(**3C.3b**) **E62K, K83R, N122D** (resulting in the loss of a potential glycosylation site), **T128A** (resulting in the loss of a potential glycosylation site), **R142G, N145S, L157S, V186G**[@] and **R261Q** in **HA1** with **M18K** in **HA2**, e.g. A/Netherlands/525/2014.

[@]Note: the **G186V** substitution in HA1 occurred during adaptation of A/Texas/50/2012 to propagation in hens' eggs.

Figure 8 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny **(Figure 7)**. The vaccine viruses A/Texas/50/2012 and A/Switzerland/9715293/2013 are shown in red, glycosylation motifs are highlighted and the reference viruses are coloured purple. The genetic group to which each of the sequences belongs is also shown.

Figure 9 shows the location on a 2005 H3 HA structure of amino acid substitutions that define genetic subgroups 3C.2a **(Figure 9A)**, 3C.3a **(Figure 9B)** and 3C.3b **(Figure 9C)** each compared with the vaccine virus A/Texas/50/2012.

Figure 10 shows a NA phylogenetic tree for representative viruses with the HA genetic groups indicated. **Figure 11** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured purple and the genetic group to which the corresponding HA gene sequences belong is also shown.

The NA gene sequences of viruses with HA genes in genetic group 3C do not cluster in the same manner as the HA sequences. NA genes of viruses with HA genes in the 3C.2a subgroup fall into at least two distinct clusters within one of the major groups. Both main clusters share the NA amino acid substitution **E221D**; one cluster is defined by the NA substitution **I392T**, and is represented by the reference virus

A/Hong Kong/5738/2014, the other main cluster is defined by the amino acid substitutions **T267K** and **I380V,** some also carrying **S245N**, which results in the addition of a glycosylation site, e.g. A/Hong Kong/7127/2014**.** Viruses in genetic subgroup 3C.3a fall into one main cluster; defined by the substitutions **E221D** and **I392T,** e.g. A/Switzerland/9715293/2013. **Figure 12** indicates the positions of the amino acid substitutions on the structure of an N2 NA for four reference viruses in the three main genetic clusters.

Antiviral resistance

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 265 viruses recovered from samples with collection dates after 2015-01-31. All showed normal inhibition (NI) by both neuraminidase inhibitors (**Table 14**).

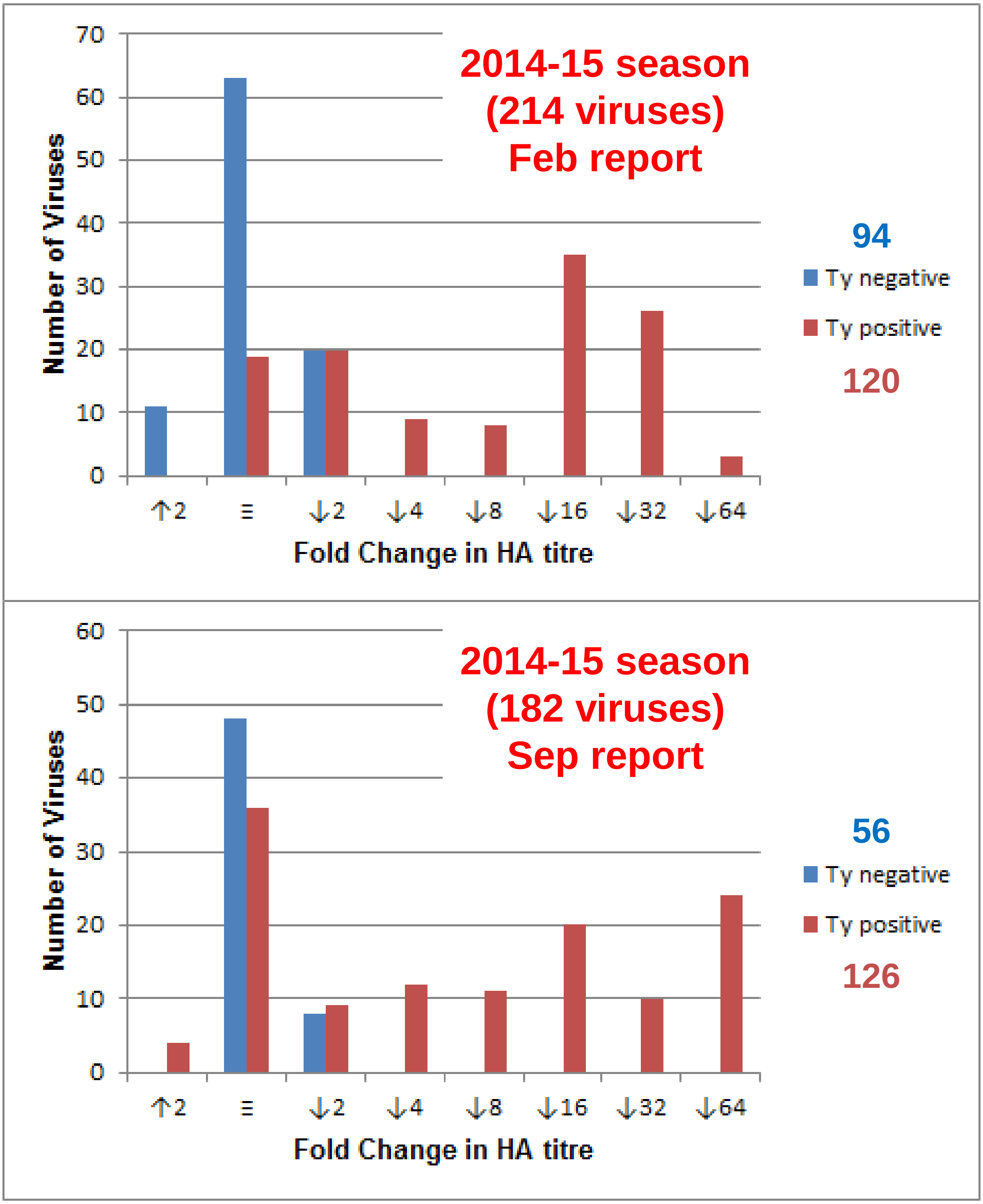
Conclusion.

Most recently collected test viruses analysed showed poor reactivity with antisera raised against the egg-propagated vaccine virus A/Texas/50/2012 in HI assays. Viruses in genetic subgroup 3C.2a were the most common. Many viruses in subgroup 3C.2a did not agglutinate RBCs and could not therefore be characterised by HI assay. Of the viruses in genetic subgroup 3C.2a that were able to agglutinate RBCs in the presence of oseltamivir over half (53%) became a mixed population during culture resulting in the loss or partial loss of the glycosylation site motif at resides 158 to 160 in HA1.

In HI assays antisera raised against most egg-propagated viruses recognised the cell-propagated test viruses poorly. Antisera raised against a 3C.3b virus recognised only test viruses in this genetic group. The antiserum raised against a cell culture-propagated 3C.2a virus (polymorphic for the glycosylation motif at residues 158 to 160 in HA1) recognised viruses in genetic subgroups 3C.2a, 3C.3a and 3C.3b similarly, as did the antiserum raised against the cell culture-propagated cultivar of A/Stockholm/6/2014 (3C.3a), but the antiserum raised against the cell culture-propagated cultivar of A/Switzerland/9715293/2013 (3C.3a) recognised only a minority of test viruses in genetic subgroup 3C.2a.

In PRNA subgroup 3C.2a viruses were recognised similarly by antisera raised against egg-propagated A/Switzerland/9715293/2013 (3C.3a) and egg-propagated A/Hong Kong/4801/2014 (3C.2a), although a higher proportion of viruses in genetic subgroup 3C.2a were recognised well by antiserum raised against a cell culture-propagated reference virus in genetic subgroup 3C.2a, A/Hong Kong/7295/2014, than by the antiserum raised against the cell culture-propagated reference virus A/Switzerland/9715293/2013 (3C.3a). The egg-adaptive changes of viruses in genetic subgroups 3C.2a and 3C.3a resulted in the generation of antisera with poorer recognition of cell-propagated test viruses compared to the homologous viruses.

Figure 6. Analysis of H3N2 Influenza Virus Agglutination of Guinea Pig RBCs and the Effect of 20nM Oseltamivir



Fold shift in HA activity in the presence of 20nM oseltamivir:
[Enhancement (↑2) Equivalence (≡) Reduction (↓2-↓64)]

Table 6. A(H3N2) viruses recovered from specimens collected after 2015-01-31, as assessed by NA activity (MUNANA-based assay)

RBC binding phenotype[1]			HI assay possible	Number of isolates	Number sequenced[2]	Number in clade 3C.3b	Number in clade 3C.3a	Number in clade 3C.2a	Number in clade 3.3
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir							
+	+	+	Yes	65	43 (66%)	11	9	21	2
-	+	+	Yes	62	44 (71%)	17	9	11	7
-	-	+	Yes	1	1 (100%)	1	0	0	0
+	-	+	Yes	0					
			Totals	128 (39%)	88 (69%)	29	18	32	9
+	+	-	No	46	41 (89%)	1	0	40	0
+	-	-	No	0	0	0	0	0	0
-	+	-	No	10	5 (50%)	0	0	5	0
-	-	-	No	143	61 (43%)	10	1	42	8
			Totals	199 (61%)	107 (54%)	11	1	87	8
			Totals	327	195 (60%)	40	19	119	17

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4
2. As of 2015-08-31
* These viruses showed either loss [NYK] or polymorphism [K/NYT or NYK/T] at the 158/160 glycosylation motif [NYT] in HA1

Table 7. Summary of influenza A(H3N2) viruses analysed, collected after 2015-01-31

MONTH Country	H3N2 Number received[1]	Number propagated[2]	≤2 fold[3]	≥4 fold[3,4]
FEBRUARY				
Bulgaria	19	2	1	1
Cyprus	11	1		1
Finland	2	2		2
Iraq	2	1		1
Italy	15	5	5	
Madagascar	3	3	3	
Mauritius	1	1	1	
Moldova	1	1		1
Netherlands	7	5	4	1
Norway	1	1		1
Romania	4	3	2	1
Russia	4	4	4	
Serbia	3	1		1
Slovakia	8	7	5	2
Slovenia	3	2	2	
Spain	39	16	13	3
Sweden	6	6	3	3
Ukraine	2	2		2
MARCH				
Cameroon	11	1		1
Georgia	3	1		1
Ghana	6	2	1	1
Madagascar	1	1	1	
Mauritius	3	3	2	1
Norway	4	2		2
Romania	1	1		1
Russia	8	6	1	5
Slovenia	2	1	1	
Serbia	4	1		1
Sweden	1	1	1	
Ukraine	7	3		3
APRIL				
Finland	2	1		1
Russia	1	1		1
Slovenia	2	2	1	1
Ukraine	4	2	1	1
JUNE				
Hong Kong	3	1		1
South Africa	21	7	4	3
JULY				
South Africa	3	1		1
	218	101	56	45
23 Countries			55.4%	44.6%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir
3. Compared to homologous titres for ferret antisera raised against tissue culture propagated virus A/Switzerland/9715293/2013
4. Unable to differentiate fold drop in titre for most viruses as it falls outside the dynamic range of the antiserum

Table 8-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-02-13 + 2015-02-17 + 2015-02-20

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F25/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	
				Haemagglutination inhibition titre[1] — Post-infection ferret antisera											
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E2	320	80	80	80	160	40	80	<	80	<	<	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	320	640	320	640	320	160	320	160	40	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	2560	1280	640	320	640	80	640	160	80	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	320	1280	1280	640	640	160	640	320	80	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	640	640	1280	1280	160	640	80	640	320	80	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	40	320	160	640	160	160	160	160	40	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	320	320	80	640	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	80	320	160	640	160	160	160	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	80	160	320	160	320	80	640	160	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	40	80	80	320	160	320	160	80	160	320	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	80	40	80	320	160	320	160	80	40	320	1280	
TEST VIRUSES															
A/Dakar/18/2014	3C.3a	2014-07-17	C1/SIAT1	<	40	40	160	80	640	160	80	80	80	<	
A/Dakar/19/2014		2014-07-28	C2/SIAT1	<	40	40	160	80	320	160	40	80	80	<	
A/Dakar/21/2014		2014-08-04	C2/SIAT1	<	80	80	160	160	640	160	160	160	160	40	
A/Dakar/32/2014	3C.3a	2014-08-08	C1/SIAT1	<	40	40	160	80	320	160	80	80	80	<	
A/Dakar/22/2014		2014-08-18	C1/SIAT1	<	40	40	160	80	640	160	80	80	160	<	
A/Dakar/24/2014	3C.3a	2014-08-25	C2/SIAT1	<	40	40	160	160	640	160	160	160	160	<	
A/Dakar/25/2014		2014-08-30	C1/SIAT1	<	40	40	160	80	640	160	80	80	80	<	
A/Dakar/26/2014	3C.3a	2014-09-18	C1/SIAT1	<	40	40	80	80	320	80	80	80	80	<	
A/Dakar/27/2014	3C.3a	2014-09-23	C2/SIAT1	<	80	80	160	160	640	160	80	80	160	<	
A/Paris/1971/2014	3C.2a	2014-10-02	MDCK2/SIAT2	<	40	<	80	40	160	40	40	40	80	<	+cho
A/Niedersachsen/8/2014		2014-10-09	SIAT2/SIAT1	<	80	40	160	80	640	160	80	80	80	<	
A/Madrid/15042/2014	3C.3a	2014-10-24	SIAT2/SIAT1	<	80	40	160	80	640	160	80	80	80	<	
A/Aragón/2103/2014	3C.3b	2014-12-02	SIAT2/SIAT1	80	160	80	320	160	320	160	40	80	80	<	Q197H
A/Astrahkan/26/2014	3C.2a	2014-12-17	C2/SIAT2	<	80	40	40	40	80	40	40	40	40	<	+cho
A/Glasgow/400097/2015	3C.2a	2015-01-02	SIAT1	<	80	40	160	80	320	80	<	40	160	<	±cho
A/England/20/2015	3C.3	2015-01-05	SIAT1/SIAT2	160	160	80	320	320	320	320	80	160	160	<	
A/Castilla La Mancha/13/2015		2015-01-05	SIAT2/SIAT1	<	40	40	160	80	640	160	80	80	160	<	
A/Castilla La Mancha/14/2015		2015-01-05	SIAT2/SIAT1	80	160	160	320	160	320	160	40	80	160	<	
A/Glasgow/401678/2015	3C.3a	2015-01-07	SIAT1	<	40	40	80	80	640	160	80	80	80	<	
A/Saarland/2/2015	3C.3	2015-01-12	SIAT2/SIAT1	40	160	80	320	160	320	160	40	80	80	<	
A/Nordrhein-Westfalen/12/2015		2015-01-19	SIAT2/SIAT1	<	80	<	80	80	320	80	<	40	160	<	
A/Rheinland-Pfalz/2/2015	3C.3b	2015-01-19	SIAT2/SIAT1	160	160	160	640	640	640	320	80	160	160	<	Q197H
A/Thuringen/7/2015	3C.2a	2015-01-19	SIAT2/SIAT1	<	<	<	80	80	320	40	<	40	160	<	-cho
A/Berlin/40/2015	3C.3b	2015-01-21	SIAT2/SIAT1	40	80	80	160	80	160	80	<	40	40	<	Q197H
A/Baden-Wurttemberg/9/2015	3C.2a		SIAT2/SIAT1	<	<	40	80	80	160	80	<	40	80	<	-cho
						Vaccine NH2014-15						Vaccine SH2015 NH2015-16			

1. < = <40

PRNA performed

Table 8-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-02-24 + 2015-03-06 + 2015-03-10

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	
				Haemagglutination inhibition titre[1] — Post-infection ferret antisera											
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E2	320	80	80	80	160	40	80	<	40	<	<	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	320	640	320	640	160	80	160	160	40	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	640	1280	640	640	160	640	40	640	80	40	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	320	320	640	640	640	320	80	320	320	40	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	320	640	640	2560	160	640	40	640	320	40	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	<	80	40	160	80	320	80	80	80	80	40	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	160	160	80	320	160	320	80	640	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	40	160	80	640	160	80	80	80	<	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	160	160	320	320	320	80	640	160	<	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	80	80	160	160	320	80	40	80	160	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E2 clone121	40	40	80	160	160	160	80	40	40	320	640	
TEST VIRUSES															
A/Mauritius/483/2014	3C.2a	2014-06-02	MDCK2/SIAT1	<	<	40	80	40	160	40	<	40	80	<	-cho
A/Cameroon/14V-5950/2014		2014-07-09	MDCK2	<	80	80	160	160	640	320	80	80	160	<	
A/Ghana/DILI-14-0657/2014		2014-07-22	C1/SIAT1	<	80	40	160	160	640	160	80	40	80	<	
A/Ghana/DILI-14-0659/2014	3C.3a	2014-07-22	C1/SIAT1	<	40	40	160	160	320	160	80	40	160	<	
A/Victoria/265/2014	3C.2a	2014-08-11	E4/E1	<	40	40	80	80	160	80	40	<	160	320	-cho
IVR-178 (A/New Caledonia/71/2014)	3C.2a	2014-08-13	E5/E1	<	<	<	80	40	80	40	<	<	160	320	-cho
A/Dakar/01/2014	3C.3a	2014-08-23	C2/MDCK1	<	80	40	160	160	640	160	80	80	80	<	
A/Madagascar/2395/2014	3C.3a	2014-08-25	MDCK2/SIAT1	<	80	80	160	160	640	320	160	80	160	40	
A/Victoria/5006/2014	3C.2a	2014-08-25	E4/E1	<	40	40	160	80	160	80	40	<	160	320	-cho
A/Victoria/673/2014	3C.2a	2014-09-07	E4/E1	<	<	<	80	40	80	40	<	<	160	320	-cho
A/Malta/MX000757/2014		2014-10-01	SIAT1	40	80	80	160	80	320	80	<	40	40	<	
A/Denmark/44/2014	3C.3a	2014-10-03	SIAT3/SIAT2	<	40	40	160	80	320	160	80	40	80	<	
A/Brisbane/341/2014	3C.2a	2014-10-03	E4/E1	<	<	<	40	40	80	40	<	<	160	320	-cho
A/Finland/462/2014	3C.2a	2014-10-08	SIAT1/SIAT1	<	40	40	80	80	160	80	<	<	80	<	-cho
A/Algiers/08/2014	3C.3a	2014-10-12	C0/SIAT1	<	40	40	80	80	640	160	80	80	80	<	
A/Denmark/45/2014	3C.3a	2014-10-20	SIAT5/SIAT2	<	40	40	160	160	320	160	80	40	80	40	
A/Malta/MX001178/2014	3C.3b	2014-10-21	SIAT1	40	80	40	160	80	160	80	<	40	40	<	Q197H
A/Malta/MX001204/2014		2014-10-22	SIAT1	40	80	80	320	160	320	80	<	80	80	<	
A/Ghana/DILI-14-1010/2014	3C.3a	2014-10-30	SIAT1	<	40	40	80	80	320	160	<	40	80	<	
A/Ghana/DILI-14-1014/2014	3C.3a	2014-11-04	SIAT1	<	40	40	80	80	320	160	<	40	80	<	
A/Israel/O-6286/2014	3C.3	2014-11-10	SIATx/SIAT1	40	80	80	160	160	320	160	<	80	80	<	
A/Switzerland/656/2014		2014-11-14	SIAT1	160	160	160	320	160	320	160	<	80	80	<	
A/Denmark/46/2014		2014-11-20	SIAT2/SIAT2	<	40	40	160	160	320	160	80	40	80	<	
A/Ghana/DILI-14-1152/2014	3C.3a	2014-12-10	SIAT1	<	80	40	80	160	640	160	80	40	80	<	
A/Leon/256/2014	3C.3a	2014-12-17	SIAT1	<	<	<	<	<	160	80	40	<	40	<	
A/Netherlands/507/14	3C.2a	2014-12-20	MDCK1/SIAT1	<	40	40	160	80	160	80	<	<	160	<	-cho
A/Jordan/30485/2014		2014-12-23	SIAT2	<	80	40	80	160	640	160	80	40	80	<	
A/Czech Republic/20/2014	3C.3a	2014-12-31	MDCK2/SIAT1	<	80	40	160	160	640	160	80	80	160	<	
A/Catalonia/0684834S/2015		2015-01-07	C0/MDCK2	<	40	40	80	80	320	80	40	<	160	<	
A/Catalonia/2235376NS/2015	3C.2a	2015-01-08	C0/SIAT2	<	40	40	80	40	160	40	<	<	80	<	+cho
A/Catalonia/2235968NS/2015	3C.3a	2015-01-10	C0/SIAT1	<	<	<	<	<	160	40	<	<	<	<	
						Vaccine NH2014-15						Vaccine SH2015 NH2015-16			

1. < = <40

Sequences in phylogenetic tree

Table 8-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-03-13 + 2015-03-17 + 2015-03-20 + 2015-03-24

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F08/14 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	A/HK 7127/14 F11/15 3C.2a New	A/HK 4801/14 F12/15 3C.2a New	
				Haemagglutination inhibition titre[1] — Post-infection ferret antisera													
REFERENCE VIRUSES																	
A/Perth/16/2009		2009-07-04	E3/E2	320	80	80	80	160	40	80	<	40	<	<	ND	ND	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/MDCK2/SIAT1	80	160	320	320	160	320	80	40	80	160	<	ND	ND	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	640	640	160	640	40	320	160	80	ND	ND	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	320	1280	1280	640	640	80	640	320	80	ND	ND	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	320	640	640	2560	80	640	40	640	320	40	ND	ND	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	80	160	160	640	160	160	80	160	40	320	320	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	160	160	80	160	160	320	80	320	160	40	160	160	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	80	40	160	160	320	160	160	80	160	40	160	160	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E2 clone 123	40	320	320	320	320	640	640	160	1280	320	40	160	160	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	40	160	160	320	320	640	320	80	80	320	80	640	320	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E2 clone121	40	80	80	320	160	320	160	80	80	320	1280	2560	640	
A/Hong Kong/7127/2014	3C.2a	2014-07-29	E5	40	40	40	160	80	160	80	40	40	160	640	1280	160	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E5 isolate 1	40	40	40	160	80	160	160	40	40	160	1280	1280	320	
TEST VIRUSES																	
A/Greece/2263/2014	3C.3a		SIAT1	<	40	40	80	160	320	160	80	40	160	<	ND	ND	
A/Cameroon/14V-5949/2014		2014-07-02	MDCK3	<	40	40	160	80	640	160	80	40	80	40	ND	ND	
A/Ghana/DILI-14-0658/2014		2014-07-22	C1/SIAT2	<	80	80	320	160	640	320	160	80	160	40	ND	ND	
A/Zambia/06-036/2014		2014-08-10	MDCK2/SIAT1	<	80	80	160	160	320	160	40	40	80	<	ND	ND	
A/Zambia/04-230/2014		2014-09-15	MDCK2/SIAT1	<	80	80	160	80	640	160	80	40	80	<	ND	ND	
A/Zambia/06-023/2014		2014-09-15	MDCK2/SIAT1	160	320	320	640	640	640	320	160	160	320	80	ND	ND	
A/Zambia/06-011/2014		2014-09-30	MDCK2/SIAT1	40	80	160	320	160	320	160	40	80	160	<	ND	ND	
A/Zambia/02-346/2014		2014-10-25	MDCK2/SIAT1	<	40	40	160	160	640	160	80	40	160	<	ND	ND	
A/Kazakhstan/3784/2014	3C.3a	2014-11-25	SIAT1	<	80	80	320	160	640	320	160	80	320	40	ND	ND	
A/Kazakhstan/3781/2014		2014-11-28	SIAT1	<	160	80	320	160	640	320	160	80	320	40	ND	ND	
A/Kazakhstan/3785/2014		2014-12-02	SIAT1	<	160	80	160	160	640	320	160	80	160	40	ND	ND	
A/Kazakhstan/3782/2014	3C.3a	2014-12-09	SIAT1	<	160	80	320	160	640	320	160	80	320	40	ND	ND	
A/Kazakhstan/3793/2014		2014-12-23	SIAT1	<	80	80	160	160	640	320	160	80	160	40	80	80	
A/Kazakhstan/3790/2014		2014-12-24	SIAT1	<	160	160	320	320	1280	320	160	160	320	80	320	160	
A/Kazakhstan/3791/2014	3C.3a	2014-12-24	SIAT1	<	80	80	160	160	640	320	160	80	160	40	80	40	
A/Greece/17/2015	3C.3	2015-01-02	SIAT1	40	160	160	320	320	320	160	40	80	160	<	ND	ND	
A/Kazakhstan/3798/2014		2014-01-03	SIAT1	<	80	80	320	160	640	320	160	80	160	40	ND	ND	
A/Kazakhstan/3797/2015		2015-01-03	SIAT1	<	160	80	320	160	640	320	160	160	160	80	80	80	
A/Greece/51/2015		2015-01-08	SIAT1	40	160	160	320	160	320	160	<	40	160	<	ND	ND	
A/Estonia/91229/2015	3C.2a	2015-01-21	SIAT1/SIAT2	<	40	40	160	40	320	80	40	<	80	40	ND	ND	+cho
A/Greece/327/2015	3C.3	2015-01-21	SIAT1	40	80	80	160	80	160	40	<	<	40	<	ND	ND	
A/Estonia/91299/2015	3C.2a	2015-01-23	SIAT2	<	40	40	80	80	160	80	40	<	80	<	ND	ND	+cho
A/Cyprus/F41 /2015		2015-01-28	SIAT1	40	80	40	160	80	160	40	<	<	40	<	ND	ND	
A/Cyprus/F52/2015	3C.3	2015-02-02	SIAT1	40	80	80	160	80	160	80	<	40	40	<	ND	ND	
						Vaccine NH2014-15						Vaccine SH2015 NH2015-16					

1. < = <40; ND = Not Done

Table 8-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-03-27 + 2015-04-01 + 2015-04-02

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F05/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	
				Haemagglutination inhibition titre[1] — Post-infection ferret antisera											
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E2	320	80	320	80	160	40	80	<	40	<	<	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/MDCK2/SIAT1	160	320	640	640	320	320	160	80	160	80	40	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	5120	1280	1280	320	640	80	640	160	80	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	1280	1280	1280	640	640	160	640	320	80	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	640	160	1280	2560	160	640	80	640	640	80	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	40	80	320	320	160	640	160	160	160	160	80	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	160	320	80	160	320	320	320	160	640	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	40	40	320	160	80	640	160	80	80	80	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	80	320	160	160	640	320	640	160	640	320	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	80	320	160	160	320	160	40	40	160	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E2 clone121	40	80	80	320	160	320	160	80	80	320	1280	
TEST VIRUSES															
A/Zambia/06-041/2014			MDCK2/SIAT1	40	160	320	320	160	320	160	40	40	160	<	
A/Zambia/06-027/2014	3C.3		MDCK2/SIAT2	40	80	320	160	160	320	80	<	40	80	<	
A/Zambia/04-215/2014		2014-01-09	MDCK2/SIAT1	40	160	320	320	320	320	160	40	80	160	<	
A/Madagascar/269/2014	3C.3	2014-01-27	SIAT1	40	80	320	320	160	320	160	<	40	80	<	
A/Madagascar/431/2014		2014-02-06	SIAT1	<	80	320	160	160	320	160	<	40	80	<	
A/Madagascar/460/2014		2014-02-10	SIAT1	<	80	160	320	160	160	160	<	40	80	<	
A/Madagascar/493/2014	3C.3	2014-02-18	SIAT1	40	160	320	320	160	320	160	<	40	80	<	
A/Madagascar/582/2014		2014-02-19	SIAT1	40	160	320	320	160	320	160	<	40	80	<	
A/Madagascar/801/2014		2014-02-24	SIAT1	40	160	320	320	160	320	160	<	40	160	<	
A/Madagascar/897/2014	3C.3	2014-03-07	SIAT1	40	160	320	320	160	320	160	<	40	80	<	
A/Zambia/04-278/2014		2014-03-11	MDCK2/SIAT1	80	160	320	320	320	640	160	40	80	160	40	
A/Madagascar/2576/2014	3C.3a	2014-09-08	SIAT1	<	40	80	160	160	640	160	80	80	80	<	
A/Madagascar/2572/2014	3C.3a	2014-09-11	MDCK1/SIAT1	40	160	160	320	160	640	320	160	80	160	40	
A/Madagascar/2583/2014		2014-09-11	MDCK2/SIAT1	<	80	160	320	160	640	160	160	80	160	40	
A/Madagascar/2586/2014		2014-09-12	MDCK1/SIAT1	<	80	160	320	160	640	320	160	80	160	40	
A/Madagascar/2611/2014		2014-09-12	MDCK2/SIAT1	<	80	160	320	160	640	320	160	80	160	40	
A/Madagascar/2614/2014		2014-09-15	MDCK2/SIAT1	<	80	160	320	160	640	320	160	80	160	40	
A/Madagascar/2636/2014		2014-09-15	MDCK1/SIAT1	<	80	160	320	160	640	320	160	160	160	40	
A/Madagascar/2616/2014		2014-09-16	MDCK2/SIAT1	<	80	160	320	160	640	320	160	80	160	40	
A/Madagascar/2618/2014		2014-09-17	MDCK2/SIAT1	<	80	160	320	160	1280	320	160	80	160	40	
A/Madagascar/2632/2014	3C.3a	2014-09-17	SIAT1	<	80	160	320	160	640	320	160	80	160	40	
A/Madagascar/2639/2014		2014-09-17	MDCK1/SIAT1	<	80	160	320	160	640	320	160	80	160	40	
A/Madagascar/2634/2014		2014-09-18	MDCK2/SIAT1	<	80	160	320	160	640	320	160	80	160	40	
A/Madagascar/2679/2014		2014-09-19	MDCK2/SIAT1	<	80	160	320	160	640	320	160	80	160	40	
A/Madagascar/2730/2014		2014-09-30	SIAT1	<	40	80	160	160	640	160	160	80	160	40	
A/Antalya/1183/2014	3C.2a	2014-10-21	SIAT1	<	<	40	80	80	160	<	<	<	160	<	+cho
A/Gumushane/1211/2014	3C.2a	2014-10-22	SIAT1	<	<	80	160	80	160	80	<	<	160	<	±cho
A/Istanbul/1628/2014	3C.2a	2014-11-04	MDCK1	<	<	80	40	40	80	40	<	<	40	<	±cho
A/Hatay/2314/2014	3C.3a	2014-11-13	MDCK2	<	<	40	80	80	320	80	40	<	80	<	
A/Erzurum/2568/2014	3C.3a	2014-12-09	MDCK1	<	40	80	160	80	320	160	80	40	160	<	
A/Cyprus/F7/2015	3C.3	2015-01-16	SIAT1	40	80	160	160	80	160	80	<	40	40	<	
A/Hatay/141/2015	3C.2a	2015-01-17	MDCK2	<	<	40	40	40	160	40	<	<	40	<	±cho
A/Cyprus/F16/2015		2015-01-19	SIAT1	80	160	320	320	160	320	160	40	80	80	<	
A/Greece/261/2015	3C.3	2015-01-20	SIAT3	80	320	640	320	320	320	320	40	80	160	40	
A/Cyprus/F19/2015	3C.2a	2015-01-22	SIAT1	<	<	<	40	40	40	<	<	<	40	<	±cho
A/Cyprus/F34/2015		2015-01-22	SIAT1	80	160	320	320	160	320	160	40	80	80	<	
A/Cyprus/F33/2015	3C.3	2015-01-23	SIAT1	80	160	320	160	160	320	160	40	40	80	<	
						Vaccine NH2014-15						Vaccine SH2015 NH2015-16			

1. < = <40

Table 8-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)
2015-04-21 + 2015-04-24 + 2015-04-28 + 2015-05-01 + 2015-05-05

Haemagglutination inhibition titre[1] — Post-infection ferret antisera

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F05/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E2	640	160	320	160	160	40	80	<	40	<	<	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/MDCK2/SIAT1	40	160	160	160	80	160	160	40	40	40	40	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	640	2560	320	320	160	320	40	320	80	40	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	640	1280	1280	1280	640	640	160	640	320	160	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	320	160	640	1280	80	640	40	640	320	80	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	160	320	160	640	160	160	80	160	80	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	40	160	160	80	160	160	320	80	320	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	80	160	80	320	160	80	80	80	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E2 clone 123	40	160	320	160	320	160	320	80	640	160	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	40	80	160	80	160	80	<	40	160	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	<	<	160	80	80	160	40	40	40	160	640	
TEST VIRUSES															
A/Madagascar/2286/2014		2014-08-11	MDCK2/SIAT2	<	40	80	160	80	320	160	80	40	80	40	
A/Madagascar/2457/2014		2014-09-01	MDCK2/SIAT2	<	80	80	160	160	640	160	80	80	160	40	
A/Madagascar/2456/2014		2014-09-03	MDCK2/SIAT2	<	40	80	80	80	320	160	80	40	160	<	
A/Madagascar/2580/2014		2014-09-10	MDCK1/SIAT2	<	40	40	80	40	320	160	40	40	80	<	
A/Madagascar/2584/2014		2014-09-11	MDCK2/SIAT2	<	40	40	80	80	320	160	80	40	80	<	
A/Jordan/20028/2014		2014-12-28	SIAT2	<	<	40	80	80	320	80	40	<	80	<	
A/Bulgaria/252/2015	3C.2a	2015-01-26	SIAT2/MDCK3	<	40	160	80	80	320	80	<	<	80	<	±cho
A/Bulgaria/252/2015	3C.2a	2015-01-26	SIAT2/MDCK4	<	40	80	80	80	320	80	40	40	80	<	±cho
A/Bulgaria/253/2015	3C.2a	2015-01-27	SIAT2/MDCK3	<	80	160	160	80	320	80	40	40	160	40	±cho
A/Castilla La Mancha/513/2015	3C.2a	2015-01-27	SIAT1/SIAT1	<	<	40	80	80	160	40	<	<	80	<	±cho
A/Navarra/508/2015	3C.3a	2015-01-28	SIAT2/SIAT1	<	<	<	<	<	160	40	40	<	40	<	
A/Castilla La Mancha/523/2015		2015-01-30	SIAT1/SIAT1	<	40	80	160	160	640	160	80	40	160	<	
A/Iraq/463/2014	3C.2a	2015-02-01	SIAT1	<	40	80	80	80	320	80	<	<	160	40	-cho
A/Castilla La Mancha/532/2015		2015-02-02	SIAT1/SIAT1	<	80	160	160	160	1280	320	80	40	160	<	
A/Bulgaria/243/2015	3C.2a	2015-02-02	SIAT2/SIAT3	<	40	80	80	80	160	80	40	40	80	<	±cho
A/Extremadura/729/2015	3C.3b	2015-02-03	SIAT2/SIAT2	40	40	160	80	80	160	80	<	40	40	<	Q197H
A/Extremadura/731/2015	3C.3a	2015-02-03	SIAT2/SIAT1	<	40	160	160	80	640	160	80	40	160	40	
A/Castilla La Mancha/660/2015	3C.3b	2015-02-04	SIAT1/SIAT1	320	160	640	640	320	640	160	40	80	160	<	Q197H
A/Castilla La Mancha/658/2015		2015-02-04	SIAT1/SIAT1	<	80	160	160	80	320	80	<	<	80	<	
A/Pais Vasco/722/2015		2015-02-05	SIAT2/SIAT1	<	40	80	80	80	640	160	40	40	160	<	
A/Castilla La Mancha/659/2015	3C.3a	2015-02-05	SIAT1/SIAT1	<	40	80	80	80	640	160	80	40	160	<	
A/Castilla La Mancha/646/2015	3C.3a	2015-02-05	SIAT1/SIAT1	<	40	40	80	80	320	80	40	<	80	<	
A/Navarra/745/2015	3C.3b	2015-02-06	SIAT2/SIAT1	160	160	640	1280	640	320	320	80	320	320	40	Q197H
A/Navarra/642/2015	3C.3a	2015-02-06	SIAT2/SIAT1	<	40	80	80	80	640	160	80	40	160	<	
A/Bulgaria/353/2015	3C.2a	2015-02-10	SIAT2/SIAT2	<	<	<	<	<	<	<	<	<	<	<	+cho
A/Castilla La Mancha/718/2015	3C.3a	2015-02-10	SIAT1/SIAT1	<	<	<	<	<	160	40	40	40	<	<	
A/Navarra/797/2015	3C.2a	2015-02-13	SIAT2/SIAT1	40	40	320	320	160	320	320	40	40	320	40	-cho
A/Georgia/532/2015	3C.2a	2015-03-09	SIAT1	<	160	80	80	80	320	160	<	40	160	40	±cho, Q197R

Vaccine NH2014-15 (A/Texas/50/2012 Egg column)

Vaccine SH2015 NH2015-16 (A/Switz/9715293/13 Egg column)

1. < = <40

Sequences in phylogenetic tree | PRNA performed

Table 8-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

2015-05-12 + 2015-05-19 + 2015-05-22 + 2015-05-29

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F05/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F10/15 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	
				Haemagglutination inhibition titre[1] — Post-infection ferret antisera											
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	640	160	160	40	160	<	80	<	40	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/MDCK2/SIAT1	80	160	320	160	80	320	160	40	80	80	40	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	2560	640	640	160	640	40	640	160	80	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	1280	1280	640	640	640	160	640	320	80	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	640	1280	1280	1280	160	640	80	640	320	80	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT4	<	40	80	160	160	640	160	80	80	80	40	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	160	320	320	160	80	160	320	160	640	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	80	160	80	640	160	160	80	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	80	320	320	160	160	320	320	160	1280	320	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	40	80	80	80	320	80	40	40	160	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	40	320	160	40	160	80	40	40	320	640	
TEST VIRUSES															
A/Iraq/470/2014		2014-07-30	C1/SIAT1	40	80	320	320	80	320	160	40	80	80	<	
A/Iraq/821/2014		2014-12-17	C1/SIAT1	<	40	160	160	80	320	160	<	40	160	<	
A/Galicia/756/2015		2014-12-17	SIAT2/SIAT3	40	80	160	160	80	80	160	<	40	80	<	
A/Leon/256/2014	3C.3a	2014-12-18	MDCK1/SIAT1	<	<	<	<	<	40	160	40	40	40	<	
A/Navarra/618/2015	3C.3	2015-01-13	SIAT2/SIAT2	80	160	320	320	160	160	320	40	160	160	40	
A/Navarra/616/2015		2015-01-15	SIAT2/SIAT1	40	80	160	160	40	160	80	<	40	80	<	
A/Galicia/760/2015		2015-01-20	SIAT2/SIAT1	80	160	320	320	160	320	160	40	80	160	<	
A/Serbia/NS-949/2015	3C.2a	2015-01-20	MDCK2	320	40	40	40	<	80	40	<	<	40	<	+cho
A/Castilla La Mancha/527/2015	3C.3a	2015-01-24	SIAT1/SIAT1	<	40	80	160	40	640	160	80	80	160	<	
A/Milano/119/2015	3C.3a	2015-01-26	MDCK2/SIAT1	40	160	320	320	160	320	160	40	40	80	<	
A/Milano/144/2015	3C.3	2015-01-26	MDCK2/SIAT1	40	80	160	160	80	1280	80	40	40	40	<	
A/Roma/7/2015	3C.2a	2015-01-27	Cx/MDCK3	<	40	80	80	40	160	40	<	<	80	<	+cho
A/Roma/14/2015	3C.2a	2015-01-28	Cx/MDCK2	<	<	<	40	<	<	<	<	<	80	<	+cho
A/Milano/191/2015	3C.3a	2015-01-30	MDCK2/SIAT1	40	160	320	320	160	1280	320	160	160	320	80	
A/Sassari/9/2015	3C.3a	2015-02-03	Cx/SIAT1	40	160	320	320	160	1280	320	160	160	320	80	
A/Perugia/45/2015	3C.3a	2015-02-06	MDCK1/SIAT1	<	80	160	160	160	640	320	160	80	160	40	
A/Perugia/46/2015	3C.3a	2015-02-06	MDCK1/SIAT1	40	160	320	320	160	1280	320	320	160	320	80	
A/Leon/201/2015	3C.3a	2015-02-06	MDCK1/SIAT1	<	80	160	160	80	640	320	160	160	160	40	
A/Soria/198/2015	3C.3	2015-02-06	MDCK1/SIAT1	80	80	320	160	80	80	80	<	40	40	<	
A/Valladolid/205/2015	3C.3	2015-02-06	MDCK1/SIAT1	80	160	320	320	160	160	320	80	80	160	<	
A/Milano/210/2015	3C.3	2015-02-10	MDCK2/SIAT1	80	160	320	320	80	320	160	80	80	80	<	
A/Burgos/207/2015	3C.3a	2015-02-10	MDCK1/SIAT1	<	160	160	320	160	640	320	160	160	160	40	
A/Perugia/52/2015	3C.3a	2015-02-12	MDCK1/SIAT1	40	160	320	320	160	1280	320	320	160	320	80	
A/Serbia/NS-1107/2015	3C.3b	2015-02-23	SIAT1	<	40	160	160	80	40	160	<	40	40	<	Q197H
A/Serbia/NS-1256/2015		2015-03-24	MDCK1	40	160	320	160	80	80	160	<	40	80	<	

1. < = <40

Vaccine NH2014-15 (A/Texas 50/12 Egg F05/13)

Vaccine SH2015 NH2015-16 (A/Switz 9715293/13 Egg F32/14)

PRNA performed

Table 8-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-06-02 + 2015-06-05 + 2015-06-09

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F05/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F10/15 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	
				Haemagglutination inhibition titre[1] — Post-infection ferret antisera											
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	640	80	320	80	80	40	80	<	40	<	<	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/MDCK2/SIAT1	80	160	640	320	160	320	160	40	80	80	40	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	640	2560	640	320	160	640	40	320	160	80	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	320	640	640	320	320	320	80	320	320	40	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	320	1280	640	1280	80	640	40	640	320	40	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	40	80	160	80	320	160	80	80	80	40	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	320	320	80	80	160	320	80	640	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	80	160	40	320	160	80	80	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	320	160	80	160	320	80	640	320	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3	40	40	160	160	80	320	80	40	80	160	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	40	160	160	40	160	80	40	40	320	640	
TEST VIRUSES															
A/Iasi/176654/2015	3C.3a	2015-01-27	MDCK 1/SIAT1	<	<	<	<	<	80	40	40	40	40	<	
A/Iasi/177049/2015	3C.3a	2015-02-04	MDCK 1/SIAT1	<	<	<	<	<	80	40	40	40	40	<	
A/Iasi/177050/2015	3C.3a	2015-02-04	MDCK 1/SIAT1	<	<	<	<	<	80	40	40	40	<	<	
A/Finland/494/2015	3C.3	2015-02-11	SIAT2/SIAT1	40	40	160	80	40	160	40	<	<	40	<	
A/Timis/177582/2015	3C.3	2015-02-12	MDCK 1/SIAT1	40	80	160	80	80	80	40	<	40	<	<	
A/Ukraine/6456/2015	3C.2a	2015-02-28	C2/SIAT1	<	40	40	40	40	160	40	<	<	80	<	+cho
A/Ukraine/165/2015	3C.3b	2015-02-28	SIAT1/SIAT1	80	80	320	160	80	320	80	<	40	80	<	Q197H
A/Bucuresti/179471/2015	3C.3b	2015-03-18	SIAT1/SIAT1	80	80	160	160	80	160	80	<	40	40	<	
A/Khmelnitsky/466/2015	3C.2a	2015-03-26	SIAT2/SIAT1	<	<	40	40	<	40	<	<	<	80	<	+cho, S144R
A/Khmelnitsky/467/2015	3C.3b	2015-03-27	SIAT2/SIAT1	40	40	80	80	40	80	40	<	<	40	<	Q197H
A/Ukraine/6784/2015	3C.2a	2015-03-31	C0/SIAT1	<	<	<	<	<	40	<	<	<	80	<	+cho, S144R
A/Ukraine/6808/2015	3C.3b	2015-04-12	C1/SIAT1	<	160	320	320	160	320	160	40	40	80	<	Q197H
A/Ukraine/6809/2015	3C.2a	2015-04-13	C1/SIAT1	<	40	80	40	40	160	40	<	<	80	<	+cho, S144R
A/Finland/525/2015	3C.3	2015-04-13	SIAT1/SIAT1	40	80	160	80	80	160	40	<	<	40	<	

1. < = <40

Vaccine NH2014-15 (A/Texas/50/12 Egg F05/13)

Vaccine SH2015 NH2015-16 (A/Switz/9715293/13 Egg F32/14)

Sequences in phylogenetic tree | PRNA performed

Table 8-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-06-12

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post-infection ferret antisera A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F05/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F10/15 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	640	160	320	80	160	40	160	<	40	<	<	
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	640	640	320	640	160	80	80	160	40	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	2560	640	640	320	640	80	320	160	80	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	1280	1280	640	640	320	80	320	320	80	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	640	2560	640	1280	80	640	40	640	320	40	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT5	<	40	80	160	80	320	80	80	40	80	<	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	160	320	80	80	160	320	80	640	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	<	40	80	40	320	80	80	40	80	<	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	320	160	80	320	320	80	640	320	<	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3	<	40	80	160	80	320	80	40	40	160	<	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E2 clone121	40	40	320	160	40	160	80	40	40	320	640	
TEST VIRUSES															
A/Norway/3172/2014		2014-12-09	MDCK2	<	40	160	80	80	160	80	<	40	160	<	
A/Norway/353/2015		2015-01-04	MDCK1	40	80	320	160	160	160	80	<	40	80	<	
A/Manjakaray/13/2015		2015-01-05	MDCK1	<	40	80	160	40	320	80	40	40	80	<	
A/Moramanga/50/2015	3C.3a	2015-01-08	MDCK1	<	<	80	80	40	320	80	40	40	80	<	
A/Norway/217/2015		2015-01-12	MDCK1	40	80	320	160	80	160	80	<	<	40	<	
A/Maevatanana/172/2015	3C.3a	2015-01-19	MDCK1	<	<	80	80	40	320	80	40	<	80	<	
A/Toamasina/180/2015		2015-01-19	MDCK2	<	40	80	160	80	640	160	80	40	160	<	
A/Nosy Be/284/2015		2015-01-19	MDCK1	<	40	80	80	80	320	80	40	40	80	<	
A/Norway/923/2015		2015-01-20	SIAT2	40	80	160	160	80	160	80	<	<	40	<	
A/Tsiroanomandidy/278/2015		2015-01-22	MDCK2	<	40	80	160	80	640	160	80	40	160	<	
A/Manjakaray/215/15		2015-01-23	MDCK1	<	40	80	160	40	640	160	80	40	160	<	
A/Antsirabe/289/2015		2015-01-28	MDCK1	<	40	80	80	40	320	160	40	40	80	<	
A/Finland/506/2015	3C.3b	2015-02-02	SIAT1	80	80	160	160	80	160	80	<	40	80	<	Q197H
A/Tsiroanomandidy/362/2015		2015-02-02	MDCK1	<	40	80	80	40	320	160	40	40	160	<	
A/Behoririka/355/2015	3C.3a	2015-02-04	MDCK1	<	40	40	80	40	320	80	40	<	40	<	
A/Moldova/111.07/2015	3C.2a	2015-02-10	MDCK2	<	40	80	80	40	160	80	<	<	80	<	±cho, S144R
A/Manjakaray/612/2015	3C.3a	2015-02-23	MDCK1	<	40	80	80	40	320	80	40	40	80	<	
A/Norway/1151/2015	3C.3	2015-02-25	SIAT1	<	40	160	80	40	80	40	<	<	40	<	
A/Norway/1656/2015	3C.3b	2015-03-21	SIAT1	80	80	160	160	80	160	80	<	40	80	<	
A/Tsiroanomandidy/1384/2015	3C.3a	2015-03-30	MDCK1	<	40	80	80	80	320	80	80	40	80	<	
						Vaccine NH2014-15						Vaccine SH2015 NH2015-16			

1. < = <40

Sequences in phylogenetic tree

PRNA performed

Table 8-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-06-19 + 2015-06-23 + 2015-06-25 + 2015-07-01

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Texas 50/12 Egg F05/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F10/15 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	A/Neth 525/14 F23/15 3C.3b New	A/Eng 527/14 F21/15 3C.3b New	
				Haemagglutination inhibition titre[1] — Post-infection ferret antisera												
REFERENCE VIRUSES																
A/Perth/16/2009		2009-07-04	E3/E3	640	320	160	160	40	160	<	40	<	40	80	320	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	2560	640	320	160	640	40	320	160	80	320	320	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	1280	1280	640	640	640	80	320	320	80	320	320	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	1280	640	640	80	640	40	320	320	80	320	1280	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	<	80	160	80	320	160	80	80	80	80	80	80	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	160	80	80	160	320	80	640	160	80	80	80	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	40	160	40	320	160	80	80	80	40	40	ND	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	160	80	160	320	80	640	320	40	80	80	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3	<	160	160	80	320	160	40	80	160	40	40	80	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	160	80	40	80	80	40	40	320	640	80	80	
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT2	80	320	320	80	320	160	40	80	80	<	640	640	
TEST VIRUSES																
A/Stockholm/31/2014	3C.2a	2014-11-26	MDCK1/SIAT1	<	40	<	<	80	40	<	<	40	<	<	ND	±cho
A/Stockholm/28/2014	3C.3b	2014-12-02	MDCK1/SIAT1	80	320	320	80	160	80	40	40	80	<	640	1280	
A/Stockholm/36/2014	3C.2a	2014-12-02	MDCK2/SIAT1	<	80	40	<	160	80	<	<	80	<	<	ND	±cho
A/Stockholm/30/2014	3C.2a	2014-12-04	MDCK1/SIAT1	<	80	80	40	160	80	<	<	80	<	<	40	±cho
A/Stockholm/32/2014	3C.3a	2014-12-09	MDCK1/SIAT1	<	80	80	40	320	160	40	40	80	<	<	40	
A/Stockholm/38/2014	3C.2a	2014-12-27	MDCK1/SIAT1	<	80	80	<	80	80	<	<	80	<	<	<	+cho
A/Mahajanga/122/2015		2015-01-05	MDCK2/SIAT1	<	80	80	<	320	160	80	40	80	<	<	<	
A/Stockholm/1/2015	3C.2a	2015-01-12	MDCK1/SIAT1	<	160	80	40	160	80	<	<	160	40	<	<	+cho
A/Stockholm/7/2015	3C.2a	2015-01-13	MDCK1/SIAT1	<	80	80	<	160	80	<	<	80	<	<	40	+cho
A/Stockholm/8/2015	3C.2a	2015-01-25	MDCK1/SIAT1	<	80	80	40	160	80	<	<	80	<	<	40	+cho
A/Stockholm/12/2015	3C.3	2015-02-10	MDCK1/SIAT1	<	80	40	<	<	<	<	<	<	<	160	160	
A/Stockholm/13/2015	3C.2a	2015-02-10	MDCK0/SIAT1	<	80	80	40	160	80	<	<	80	<	<	<	+cho
A/Stockholm/18/2015	3C.3b	2015-02-11	MDCK0/SIAT1	80	320	320	80	320	160	80	40	80	40	320	1280	
A/Stockholm/20/2015	3C.3b	2015-02-24	MDCK0/SIAT1	80	320	320	80	320	160	160	40	80	<	640	1280	Q197H
A/Stockholm/19/2015	3C.3b	2015-02-25	MDCK0/SIAT1	80	320	160	80	320	80	80	40	80	<	320	1280	
A/Sweden/15/2015	3C.2a	2015-02-27	MDCK2/SIAT3	<	40	40	<	80	40	<	<	40	<	<	ND	
A/Sweden/16/2015	3C.3b	2015-03-05	MDCK1/SIAT1	80	320	320	80	320	160	80	40	80	<	320	640	
A/Ghana/DILI-15-0233/2015	3C.2a	2015-03-16	Cx/SIAT2	<	80	160	80	320	80	80	<	160	40	<	ND	+cho
A/Ghana/FS-15-0185/2015	3C.2a	2015-03-20	C2/SIAT2	<	80	160	80	320	80	<	<	160	<	<	ND	+cho
A/Norway/1800/2015		2015-03-25	SIAT3	<	80	80	40	80	40	<	<	40	<	160	ND	
					Vaccine NH2014-15						Vaccine SH2015 NH2015-16					

1. < = <40; ND = Not Done

Sequences in phylogenetic tree | PRNA performed

Table 8-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-07-06 + 2015-07-10

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post-infection ferret antisera A/Perth 16/09 F18/11	A/Texas 50/12 Egg F36/12 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F10/15 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	A/Neth 525/14 F23/15 3C.3b	
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	640	320	80	80	40	160	<	40	<	<	80	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	5120	640	640	320	1280	40	640	160	80	640	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	1280	640	640	640	640	160	320	320	80	640	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	2560	640	1280	160	640	40	640	320	80	320	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	<	80	160	80	320	160	160	80	80	40	40	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	640	80	40	160	320	80	640	160	40	80	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	80	160	40	320	160	80	80	160	40	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E2 clone 123	40	160	160	80	320	320	80	640	320	80	80	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3	<	160	160	80	320	160	80	80	160	40	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E2 clone121	40	160	160	40	160	160	80	40	320	1280	80	
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT2	40	320	160	80	160	80	40	40	80	<	640	
TEST VIRUSES															
A/Netherlands/773/2014		2014-08-10	MDCK2/SIAT1	40	160	160	40	160	160	<	<	40	<	160	
A/Netherlands/778/2014		2014-11-25	MDCK2/SIAT1	40	320	320	80	320	160	40	40	80	<	320	
A/Netherlands/779/2014		2014-12-02	MDCK2/SIAT1	80	320	160	80	320	160	40	40	80	<	320	
A/Khabarovsk/1/2015		2014-12-29	C1/SIAT1	<	160	160	80	640	160	160	80	160	40	40	
A/Levice/84/2015	3C.2a	2015-01-07	MDCK1/SIAT1	<	80	80	40	160	40	<	<	160	<	<	±cho
A/Kosice/128/2015	3C.3b	2015-01-15	MDCK1/SIAT1	80	320	320	80	320	160	40	80	80	<	640	Q197H
A/Khabarovsk/3/2015	3C.3a	2015-01-16	C3/SIAT1	<	160	160	40	640	320	160	80	160	40	40	
A/Khabarovsk/12/2015	3C.3a	2015-01-19	C1/SIAT1	<	80	40	<	40	80	<	<	<	<	<	
A/St Petersburg/14/2015		2015-01-20	C2/SIAT1	<	80	80	80	160	80	<	<	80	<	<	
A/Netherlands/1018/2015	3C.3b	2015-01-20	MDCK2/SIAT1	80	640	320	80	320	160	40	40	80	<	640	Q197H
A/Partizanske/131/2015	3C.3b	2015-01-21	MDCK1/SIAT1	160	1280	640	160	320	320	40	160	160	40	1280	Q197H
A/Kaliningrad/4/2015	3C.2a	2015-01-23	C1/SIAT1	<	80	160	40	160	80	<	40	80	40	<	-cho
A/St Petersburg/21/2015		2015-01-26	C1/SIAT1	<	160	160	80	640	320	160	80	320	40	40	
A/Trnava/176/2015	3C.2a	2015-01-29	MDCK2/SIAT1	<	80	80	40	160	80	<	<	80	<	<	±cho
A/Netherlands/1034/2015	3C.3b	2015-02-02	MDCK2/SIAT1	80	640	320	80	320	160	40	40	80	<	640	
A/Nove Zamky/201/2015	3C.2a	2015-02-02	MDCK1/SIAT1	<	80	80	40	160	160	<	<	160	<	<	±cho
A/Nitra/205/2015	3C.3b	2015-02-02	MDCKx/SIAT1	160	1280	320	160	320	320	40	40	80	<	1280	
A/Poprad/237/2015	3C.3b	2015-02-02	MDCKx/SIAT1	80	640	320	80	320	160	40	80	80	<	640	Q197H
A/St Petersburg/119/2015	3C.3a	2015-02-03	C2/SIAT1	<	80	160	80	640	160	160	80	160	40	40	
A/St Petersburg/134/2015	3C.2a	2015-02-09	C2/SIAT2	<	80	160	40	160	80	40	<	160	<	<	±cho, S144R
A/Bratislava/329/2015		2015-02-11	MDCKx/SIAT1	80	640	320	160	160	160	40	80	80	<	320	
A/St Petersburg/57/2015	3C.3a	2015-02-12	C1/SIAT1	<	80	80	<	320	160	80	40	80	<	40	±cho, S144R
A/Netherlands/1497/2015	3C.3b	2015-02-12	MDCK2/SIAT2	40	160	160	40	160	80	<	<	40	<	320	
A/Netherlands/1726/2015	3C.3b	2015-02-16	MDCK2/SIAT1	160	640	640	160	640	320	80	160	160	40	1280	Q197H
A/Bratislava/437/2015	3C.3b	2015-02-17	MDCKx/SIAT1	80	640	320	160	320	160	40	80	80	<	640	Q197H
A/Trencin/558/2015	3C.2a	2015-02-23	MDCKx/SIAT1	<	80	80	40	160	80	<	<	80	<	<	±cho
A/Netherlands/1903/2015	3C.2a	2015-02-24	MDCK2/SIAT1	<	80	160	40	320	80	40	40	160	<	<	±cho
A/Netherlands/1916/2015	3C.3b	2015-02-25	MDCK2/SIAT1	80	640	320	160	320	160	80	80	80	<	1280	Q197H
A/Komarno/618/2015	3C.3b	2015-02-25	MDCKx/SIAT1	160	640	320	80	320	160	40	40	80	<	640	
A/St Petersburg/28/2015	3C.2a	2015-02-26	C2/SIAT1	<	80	160	80	160	160	40	40	160	40	<	±cho, S144R
					Vaccine NH2014-15						Vaccine SH2015 NH2015-16				

1. < = <40

Sequences in phylogenetic tree | PRNA performed

Table 8-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-07-14 + 2015-07-17

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Texas 50/12 Egg F36/12 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F10/15 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	A/Neth 525/14 F23/15 3C.3b	A/HK 4801/14 F12/15 3C.2a	X-263B HK/4801/14 F26/15 3C.2a	A/HK 7127/14 F11/15 3C.2a	X-261 HK/7127/14 F25/15 3C.2a	
				Haemagglutination inhibition titre[1] — Post-infection ferret antisera															
REFERENCE VIRUSES																			
A/Perth/16/2009		2009-07-04	E3/E3	320	320	80	80	40	80	<	<	<	<	40	ND	ND	ND	ND	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	5120	640	320	320	640	40	320	160	80	320	ND	ND	ND	ND	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	1280	640	320	640	320	80	320	160	80	320	ND	ND	ND	ND	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	2560	640	640	160	640	40	320	320	40	320	ND	ND	ND	ND	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	<	80	160	80	320	160	80	80	80	40	40	ND	ND	ND	ND	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	640	80	40	160	320	80	320	160	<	80	ND	ND	ND	ND	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	40	80	40	160	80	40	40	80	<	<	ND	ND	ND	ND	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E2 clone 123	<	640	160	80	160	320	80	320	160	<	40	40	<	80	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3	<	80	160	80	160	80	40	40	160	40	<	ND	ND	ND	ND	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	80	80	<	160	80	40	<	160	640	40	ND	ND	ND	ND	
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT3	80	320	160	80	160	80	<	40	40	<	640	80	80	80	40	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6 isolate 1	<	40	40	<	80	40	<	<	160	640	40	160	1280	640	320	
NYMC X-263B (A/HK/4801/2014)	3C.2a	2014-02-26	Ex	<	40	40	<	40	40	40	<	160	640	40	160	1280	640	640	
A/Hong Kong/7127/2014	3C.2a	2014-07-29	E5	40	40	80	<	80	80	40	<	160	320	40	160	1280	640	320	
NYMC X-261 (A/HK/7127/2014)	3C.2a	2014-07-29	Ex	40	40	80	40	80	80	40	40	160	320	40	160	640	640	640	
TEST VIRUSES																			
A/N. Novgorod/303/2014	3C.2a	2014-12-19	MDCK3/SIAT1	<	40	80	40	160	80	<	<	80	<	<	ND	ND	ND	ND	+cho, S144R
A/Vladimir/135/2015	3C.2a	2015-03-13	MDCK1/SIAT1	<	40	40	40	160	40	<	<	80	<	<	ND	ND	ND	ND	+cho, S144R
A/Vladivostok/36/2015	3C.3a	2015-03-14	MDCK2/SIAT2	<	80	80	40	320	160	80	40	80	<	<	40	40	80	40	
A/Moscow/103/2015	3C.2a	2015-03-15	MDCK1/SIAT2	<	40	40	<	80	40	<	<	<	<	<	<	<	80	<	+cho, S144R
A/Tomsk/5/2015	3C.2a	2015-03-16	MDCK1/SIAT1	<	80	80	40	160	80	<	<	80	<	<	ND	ND	ND	ND	+cho
A/Moscow/101/2015	3C.3b	2015-03-18	MDCK1/SIAT1	40	160	80	40	160	80	<	<	40	<	320	ND	ND	ND	ND	
A/Moscow/100/2015	3C.2a	2015-03-19	MDCK1/SIAT1	<	80	80	40	160	80	<	<	80	<	<	ND	ND	ND	ND	+cho, S144R
A/Moscow/133/2015	3C.2a	2015-04-09	MDCK1/SIAT1	<	80	160	80	320	80	<	40	80	<	<	ND	ND	ND	ND	+cho
					Vaccine NH2014-15						Vaccine SH2015 NH2015-16								

1. < = <40; ND = Not Done

Sequences in phylogenetic tree — PRNA performed

Table 8-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-07-24 + 2015-07-28

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] / Post-infection ferret antisera: A/Perth 16/09 F18/11	A/Texas 50/12 Egg F36/12 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F10/15 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121	A/Neth 525/14 F23/15 3c.3b	A/HK 4801/14 F12/15 3C.3a	NYMC X-263B F26/15 3C.3a	A/HK 7127/14 F11/15 3C.3a	A/HK 7127/14 NIB F52/15 3C.3a	NYMC X-261 F25/15 3C.3a	NIBSC NIB-93 NIB F53/15 3C.3a
REFERENCE VIRUSES																				
A/Perth/16/2009		2009-07-04	E3/E3	640	320	80	80	40	80	<	<	<	<	80	<	40	160	ND	40	ND
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	2560	640	320	160	640	40	320	160	80	320	80	40	320	ND	160	ND
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	1280	640	320	640	320	80	320	160	80	320	ND	ND	ND	ND	ND	ND
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	2560	640	640	80	640	40	320	320	40	320	80	40	160	ND	80	ND
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	<	80	160	80	320	160	160	80	160	40	80	160	80	160	ND	80	ND
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	160	640	80	80	160	320	80	640	160	<	80	80	40	80	ND	80	ND
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	80	40	160	80	80	40	80	40	<	40	40	80	ND	40	ND
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E2 clone 123	40	640	160	80	160	320	80	640	160	40	80	80	40	80	ND	80	ND
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	80	160	80	160	80	40	40	160	40	<	80	40	160	ND	40	ND
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	<	40	80	<	80	80	40	40	160	640	40	320	1280	1280	ND	640	ND
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT3	40	320	160	80	160	80	40	80	40	40	640	80	80	80	ND	40	ND
TEST VIRUSES																				
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6 isolate 1	<	40	80	<	80	80	40	40	160	640	40	160	1280	1280	ND	640	ND
NYMC X-263B (A/Hong Kong/4801/2014)	3C.2a	2014-02-26	Ex	<	40	40	<	40	40	<	<	80	640	40	160	640	640	ND	640	ND
A/Hong Kong/7127/2014	3C.2a	2014-07-29	E5	40	40	80	<	80	80	40	40	160	320	40	160	640	640	640	640	640
NYMC X-261 (A/Hong Kong/7127/2014)	3C.2a	2014-07-29	Ex	<	40	80	<	80	80	40	40	160	320	40	160	640	640	320	640	320
NIB-93 (A/Hong Kong/7127/2014)	3C.2a	2014-07-29	E7	ND	ND	ND	ND	80	80	40	40	80	640	ND	ND	ND	1280	640	640	640
A/South Australia/9/2015	3C.2a	2015-02-12	E5/E1	<	40	40	<	<	40	<	<	40	320	<	80	640	320	ND	320	ND
A/Fiji/2/2015	3C.2a	2015-03-03	E5/E1	<	40	40	<	80	80	<	<	80	320	40	80	640	640	ND	320	ND
A/Brisbane/47/2015	3C.2a	2015-03-18	E4/E1	<	40	40	<	80	40	<	40	80	320	40	80	640	640	ND	320	ND
A/Victoria/503/2015	3C.2a	2015-03-31	E6/E1	<	40	40	<	40	40	<	<	80	320	<	80	640	320	ND	320	ND
A/Singapore/KK943/2014	3C.2a	2014-07-10	E5/E1	<	40	80	<	80	80	40	40	160	640	40	80	640	640	ND	320	ND
					Vaccine NH2014-15						Vaccine SH2015 NH2015-16									

1. < = <40; ND =Not Done

Sequences in phylogenetic tree

Table 8-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-07-28 + 2015-07-31

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Texas 50/12 Egg F36/12 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F10/15 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 T/C NIB F13/14 3C.3a	A/Switz 9715293/13 Egg F32/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 4801/14 F12/15 3C.2a	A/Neth 525/14 F23/15 3c.3b	
				Haemagglutination inhibition titre[1] – Post-infection ferret antisera											
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	320	320	80	80	40	80	<	40	<	<	80	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	5120	640	320	160	640	40	640	160	80	320	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT2	160	1280	640	320	640	320	160	320	320	160	320	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	2560	640	1280	80	640	40	640	320	160	320	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	<	80	160	80	320	160	160	160	160	160	80	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	640	160	80	160	320	160	640	160	40	80	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT4	<	160	80	80	640	160	160	160	160	160	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E2 clone 123	40	640	160	80	160	320	80	1280	160	80	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	160	160	80	160	160	40	80	160	160	40	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6 isolate 1	<	40	80	<	80	80	40	40	160	320	80	
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT3	80	320	320	160	160	160	80	160	80	160	1280	
TEST VIRUSES															
A/Slovenia/314/2015	3C.2a	2015-01-16	MDCKx/SIAT1	<	40	40	40	160	40	<	80	80	80	<	+cho
A/Slovenia/369/2015	3C.2a	2015-01-20	MDCK1/SIAT1	<	40	40	<	40	40	<	40	40	40	<	+cho
A/Slovenia/438/2015	3C.3b	2015-01-21	MDCKx/SIAT1	160	640	320	160	320	160	80	<	160	80	1280	Q197H
A/Slovenia/1066/15	3C.3b	2015-02-19	MDCK1/SIAT1	160	1280	320	160	320	160	80	160	160	160	1280	Q197H
A/Slovenia/1077/2015	3C.3b	2015-02-19	MDCKx/SIAT1	160	1280	320	160	320	160	80	160	160	160	1280	Q197H
A/Mauritius/I-203/2015	3C.3b	2015-02-25	MDCK2/SIAT1	80	320	320	80	320	160	80	80	80	80	640	
A/Mauritius/I-238/2015	3C.2a	2015-03-06	MDCK3/SIAT1	40	40	160	80	320	160	80	80	160	160	40	+cho
A/Mauritius/I-240/2015	3C.2a	2015-03-09	MDCK2/SIAT1	40	160	160	80	640	160	80	80	160	160	40	+cho
A/Slovenia/1411/2015	3C.3b	2015-03-11	MDCK1/SIAT1	80	640	320	10	320	160	80	160	80	160	1280	Q197H
A/Cameroon/15V-2465/2015	3C.2a	2015-03-23	SIAT1	<	80	160	80	160	80	40	40	160	80	<	+cho, S144R
A/Mauritius/I-288/2015	3C.2a	2015-03-23	MDCK2/SIAT1	<	80	160	80	320	160	40	80	160	160	40	+cho
A/Slovenia/2049/2015	3C.3b	2015-04-10	MDCK1/SIAT1	80	1280	320	160	320	160	40	80	40	160	1280	
A/Slovenia/2069/2015	3C.3b	2015-04-13	MDCKx/SIAT1	80	1280	320	160	320	160	80	160	80	160	1280	
A/Hong Kong/11743/2015	3C.2a	2015-06-09	MDCK1/SIAT2	<	80	80	40	320	80	40	40	80	80	<	+cho
					Vaccine NH2014-15						Vaccine SH2015 NH2015-16				

1. < = <40

Sequences in phylogenetic tree | PRNA performed

Table 8-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-08-11 + 2015-08-14

				Haemagglutination inhibition titre[1]											
				Post-infection ferret antisera											
Viruses		Collection Date	Passage History	A/Perth 16/09 F18/11	A/Texas 50/12 Egg F36/12	A/Samara 73/13 F24/13	A/HK 146/13 F10/15	A/Stock 6/14 T/C F14/14	A/Stock 6/14 Egg F20/14	A/Switz 9715293/13 T/C NIB F13/14	A/Switz 9715293/13 Egg F32/14	A/HK 5738/14 T/C F30/14	A/HK 4801/14 F12/15	A/Neth 525/14 F23/15	
	Genetic group				3C.1	3C.3	3C.2	3C.3a	3C.3a isolate 2	3C.3a	3C.3a cl123	3C.2a	3C.2a	3c.3b	
REFERENCE VIRUSES															
A/Perth/16/2009		2009-07-04	E3/E3	640	320	80	80	40	80	<	40	<	<	80	
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	320	5120	640	320	160	640	80	640	160	80	320	
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	1280	640	320	640	320	160	320	320	160	320	
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	2560	640	640	80	640	80	640	160	160	320	
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	160	80	640	160	160	160	160	80	40	
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	40	320	80	40	80	320	80	320	80	40	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	<	80	80	40	320	160	80	80	80	80	40	
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E2 clone 123	40	640	80	80	160	320	80	640	160	40	40	
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3	40	160	160	80	320	160	80	80	160	160	40	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6/E1 isolate 1	40	40	80	<	80	80	40	40	160	320	40	
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT3	160	640	640	160	320	320	160	160	160	160	1280	
TEST VIRUSES															
A/South Africa/R2490/2015	3C.2a	2015-05-22	MDCK2/SIAT2	<	160	160	80	160	80	40	40	160	160	40	±cho, Q197R
A/South Africa/R2825/2015	3C.2a	2015-06-01	MDCK2/SIAT1	<	80	160	80	320	160	40	80	160	160	<	-cho, Q197R
A/South Africa/R3402/2015	3C.2a	2015-06-16	MDCK1/SIAT1	<	160	160	80	320	160	40	80	320	160	40	-cho, Q197R
A/South Africa/R3512/2015	3C.2a	2015-06-19	MDCK2/SIAT2	<	80	80	40	160	80	<	40	80	80	<	±cho
A/South Africa/R3457/2015	3C.2a	2015-06-22	MDCK1/SIAT1	<	160	160	80	320	160	40	80	160	160	<	-cho, Q197R
A/South Africa/R3720/2015	3C.2a	2015-06-29	MDCK1/SIAT1	<	160	160	80	320	160	<	80	160	160	40	±cho, Q197R
A/South Africa/R3778/2015	3C.2a	2015-06-29	MDCK2/SIAT2	<	80	80	40	160	80	<	40	160	80	<	±cho, Q197R
A/South Africa/R3944/2015	3C.2a	2015-07-06	MDCK2/SIAT1	<	80	160	80	320	80	<	40	80	160	<	±cho, Q197R
					Vaccine NH2014-15						Vaccine SH2015 NH2015-16				

1. < = <40

Sequences in phylogenetic tree | PRNA performed

Table 8-15. Summary of HI reactivity of cell-propagated A(H3N2) Test viruses by clade

HI Table	Test Virus genetic group	No of Test Viruses by genetic group		Number of test viruses reacting within 4-fold of the antiserum homologous titre													
			Antiserum to Virus	A/Perth 16/09	A/Vic 361/11	A/Texas 50/12	A/Samara 73/13	A/HK 146/13	A/Stock 6/14	A/Stock 6/14	A/Switz 9715293/13	A/Switz 9715293/13	A/HK 5738/14	A/HK 5738/14	A/HK 4801/14	A/Neth 525/14	A/Eng 527/14
			Host	Egg	Vic	Egg	Cell	Egg	Cell	Egg	Cell	Egg	Cell	Egg	Egg	Cell	Cell
			Genetic group		3C.1	3C.1	3C.3	3C.2	3C.3a	3C.3a	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a	3C.3b	3C.3b
X-1	3C.2a	5		0	2	0	0	0	4	2	2	0	4	0			
	3C.3a	7		0	2	0	0	0	7	7	7	1	7	0			
	3C.3b	3		2	3	0	2	1	3	3	1	1	2	0			
X-2	3C.2a	4		0	0	0	1	0	4	3	0	0	4	0			
	3C.3a	12		0	4	0	6	0	12	11	9	0	8	0	1 (0)		
	3C.3b	1		0	1	0	1	0	1	1	0	0	1	0			
X-3	3C.2a	2		0	2	0	0	0	2	2	0	0	0	0			
	3C.3a	4		0	4	0	2	0	4	4	4	0	4	0			
	3C.3b																
X-4	3C.2a	5		0	0	0	0	0	3	1	0	0	2	0			
	3C.3a	5		0	2	0	2	0	5	5	5	0	5	0			
	3C.3b																
X-5	3C.2a	9		0	7	0	1	0	8	7	4	0	8	0			
	3C.3a	5		0	3	0	0	0	5	3	5	0	4	0			
	3C.3b	3		2	3	3	2	2	3	3	2	1	3	0			
X-6	3C.2a	3		1	2	0	0	0	1	0	0	0	2	0			
	3C.3a	10		0	9	0	6	0	9	10	10	0	10	0			
	3C.3b	1		0	1	0	0	0	0	1	0	0	1	0			
X-7	3C.2a	4		0	1	0	0	0	2	0	0	0	4	0			
	3C.3a	3		0	0	0	0	0	3	0	3	0	2	0			
	3C.3b	4		0	3	0	3	0	4	3	1	0	4	0			
X-8	3C.2a	1		0	0	0	0	0	1	1	1	0	1	0			
	3C.3a	5		0	0	0	0	0	5	5	4	0	5	0			
	3C.3b	2		0	2	0	0	0	2	2	0	0	2	0			
X-9	3C.2a	10		0		0	0	0	10	9	1	0	10	0		0	
	3C.3a	1		0		0	0	0	1	1	1	0	1	0		0	
	3C.3b	4		0		0	3	0	4	4	4	0	4	0		4	4
X-10	3C.2a	8		0		0	4	0	8	7	3	0	8	0		0	
	3C.3a	4		0		0	2	0	3	4	3	0	3	0		0	
	3C.3b	11		4		2	11	0	11	11	10	1	11	0		11	
X-11	3C.2a	6		0		0	1	0	6	4	0	0	5	0	1 (0)	0	
	3C.3a	1		0		0	0	0	1	1	1	0	1	0	1 (1)	0	
	3C.3b	1		0		0	0	0	1	1	0	0	1	0		1	
X-12*	3C.2a	5		0		0	0	0	3	2	1	0	5	5	5	0	
	3C.3a																
	3C.3b																
X-13	3C.2a	7		0		0	4	0	7	5	5	0	7		6	0	
	3C.3a																
	3C.3b	7		7		4	7	0	7	7	7	0	7		7	7	
X-14	3C.2a	8		0		0	6	0	8	8	4	0	8		8	0	
	3C.3a																
	3C.3b																
	Totals			Percentage within 4-fold													
	3C.2a	77		1.3	42.4	0.0	22.1	0.0	87.0	66.2	27.3	0.0	88.3	100.0	90.5	0.0	
	3C.3a	57		0.0	47.1	0.0	31.6	0.0	96.5	89.5	91.2	1.8	87.7	0.0	50.0	0.0	
	3C.3b	37		40.5	92.9	24.3	78.4	8.1	97.3	97.3	67.6	8.1	97.3	0.0	100.0	100.0	100.0
						Vaccine NH2014-15						Vaccine SH2015 NH2015-16					

* Egg-propagated viruses

Table 9-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2015-07-09

Viruses	Genetic group	Collection Date	Passage History	A/HK 7295/14 T/C F02/15 3C.2a		A/HK 7127/14 Egg F11/15 3C.2a		A/HK 4801/14 Egg F12/15 3C.2a		A/HK 4244/14 Egg F24/15 3C.2a		A/Stock 6/2014 T/CF14/14 3C.3a		NYMC X-251 Egg F20/15 3C.3a		A/Switz 9715293/13 T/C NIB F13/14 3C.3a		A/Switz 9715293/13 Egg CI123 F32/14 3C.3a		A/Eng 527/14 T/C F21/15 3C.3b		A/Neth 525/14 T/C F23/15 3C.3b		Egg adaptation HA1 substitutions compared to the corresponding cell isolate
				Neutralisation titre[1] — Post-infection ferret antisera																				
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																								
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	160	214	160	162	80	110	80	114	160	141	40	39	80	74	40	45	40	54	40	38	
A/Hong Kong/7127/2014	3C.2a	2014-07-29	E6	160	150	640	947	320	285	80	82	80	101	20	24	80	62	20	22	80	83	40	53	T160K, L194P, D225G
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6	160	196	640	965	320	359	80	84	80	80	<	0	40	56	20	29	40	53	80	69	N96S, T160K, L194P
A/Hong Kong/4244/2014	3C.2a	2014-02-06	E5	160-320	240	160	224	160	130	1280	1541	80	88	320	279	80	80	640	679	1280	1836	320	410	T160K, G186V, S219F, D225G
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT2	40	33	20	29	20	26	40	31	80	106	20	28	80	66	20	28	20	29	40	31	
NYMC X-251 (A/Stockholm/6/2014)	3C.3a	2014-02-06	E6	40	47	80	101	20	26	160	196	80	101	320	318	40	55	320	300	20	26	20	25	G186V, D225N
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	20	21	20	22	20	21	20	25	80	69	20	21	160	153	20	23	20	20	20	23	
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	E4/E2	80	77	80	68	40	35	160	188	160	204	160	127	40	40	320	318	40	32	40	36	R156,V186
A/England/527/2014	3C.3b	2014-11-18	SIAT1/SIAT1	80	77	80	96	40	53	160	185	160	142	40	34	80	74	40	34	1280	989	1280	1806	
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	79	80	103	80	68	160	148	160	130	40	55	80	66	80	72	1280	1826	1280	1622	
																		Vaccine SH 2015 NH 2015-16						

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation; < = <10

Table 9-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2015-07-16

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post-infection ferret antisera A/HK 7295/14 T/C F02/15 3C.2a 2-fold	Read	A/HK 4801/14 Egg F12/15 3C.2a 2-fold	Read	A/Switz 9715293/13 T/C NIB F13/14 3C.3a 2-fold	Read	A/Switz 9715293/13 Egg Cl123 F25/14 3C.3a 2-fold	Read	A/Neth 525/14 T/C F23/15 3C.3b 2-fold	Read	Egg adaptation HA1 substitutions compared to the corresponding cell isolate
REFERENCE VIRUSES														
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	320	250	80	77	80	74	80	111	40	38	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6	160	126	320	350	40	56	20	20	40	49	N96S, T160K, L194P
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	20	21	20	21	160	153	20	23	20	23	
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	E4/E2	80	70	<	1	40	40	160	132	10	10	R156,V186
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	61	40	49	80	66	80	76	1280	1023	
TEST VIRUSES														Additional HA information*
														L3I, Q33R, N144S, N145S, F159Y, K160T, *N225D* , N278K, Q311H
A/Pais Vasco/635/2015	3C.2a	2015-02-02	SIAT2/SIAT1	640	526	160	188	320	284	160	149	40	54	Q197R, R208G
A/Milano/256/2015	3C.2a	2015-02-03	MDCK2/SIAT1	160	184	80	97	80	91	80	71	40	48	V88I
A/Pais Vasco/724/2015	3C.2a	2015-02-05	SIAT2/SIAT1	320	289	80	64	80	78	80	75	<	0	
A/Jordan/30029/2015	3C.2a	2015-02-10	SIAT1	640	901	160-320	240	160	128	160-320	240	80	66	
A/Parma/224/2015	3C.2a	2015-02-10	MDCK1/SIAT1	640	544	80	99	80	67	40-80	60	40	59	
A/Italy/FVG-84/2015	3C.2a	2015-02-11	Cx/SIAT1	160	215	80	103	80	76	80	72	40	42	S198P
A/Jordan/30052/2015	3C.2a	2015-02-14	SIAT1	640	480	80	61	80	61	80	63	<	0	
A/Jordan/10702/2015	3C.2a	2015-03-22	SIAT1	320	262	80	76	80	69	80	74	80	66	R142K, N171K
														Q33R, E62K, K83R, N122D, T128A, R142G, N145S, L157S, R261Q, N278K
A/Castilla La Mancha/660/2015	3C.3b	2015-02-04	SIAT1/SIAT1	80	73	20	20	40	41	80	115	1280	1131	N144K, Q197H

Vaccine SH 2015 NH 2015-16

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation; < = <10

* Amino acid substitutions relative to cell-propagated A/Victoria/361/2011 that define genetic groups are shown together with those observed in individual viruses

Sequences in phylogenetic tree | HI assay performed

Table 9-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2015-07-29

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre[1]										Egg adaptation HA1 substitutions compared to the corresponding cell isolate
				Post-infection ferret antisera										
				A/HK 7295/14 T/C F02/15 3C.2a		A/HK 4801/14 Egg F12/15 3C.2a		A/Switz 9715293/13 T/C NIB F13/14 3C.3a		A/Switz 9715293/13 Egg Cl123 F25/14 3C.3a		A/Neth 525/14 T/C F23/15 3C.3b		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	320	255	80	77	80	74	80	111	40	38	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6	640	819	320	462	40	40	40	48	80	67	N96S, T160K, L194P
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	20	21	20	21	160	153	20	23	20	23	
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	E4/E2	80	77	40	35	40	40	320	318	40	36	R156,V186
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	61	40	49	80	66	80	76	1280	1023	
TEST VIRUSES														Additional HA information*
														L3I, Q33R, N144S, N145S, F159Y, K160T, *N225D* , N278K, Q311H
A/Sassari/8/2015	3C.2a	2015-02-02	Cx/SIAT2	640	729	80	116	40	57	40	54	40	40	
A/Parma/220/2015	3C.2a	2015-02-03	MDCK2/SIAT1	1280	1187	320	274	160	145	160	139	80	115	
A/Parma/234/2015	3C.2a	2015-02-03	MDCK3/SIAT1	640	496	80	108	80	67	80	63	40	40	
A/Parma/224/2015	3C.2a	2015-02-10	MDCK1/SIAT1	640	544	80	99	80	67	40-80	60	40	59	
A/Italy/FVG-89/2015	3C.2a	2015-02-13	Cx/SIAT1	1280	1092	160	154	80	106	80	105	80	89	S114T
A/Italy/FVG-92/2015	3C.2a	2015-02-17	Cx/SIAT1	1280	1226	320	400	160	138	160	128	20	20	
														Q33R, T128A, A138S, R142G, N145S, F159S, *N225D* , N278K
A/Sassari/9/2015	3C.3a	2015-02-03	Cx/SIAT1	80	75	40	46	80	109	80	67	80	100	Y94H, L157S
A/Leon/201/2015	3C.3a	2015-02-06	MDCK1/SIAT1	80	64	20	20	80	97	40	44	40	40	L3F, Y94H
A/Burgos/207/2015	3C.3a	2015-02-10	MDCK1/SIAT1	80	72	40	52	80	105	80	72	80	64	Y94H, P169S
														Q33R, T128A, R142G, N145S, L157S, N278K
A/Soria/198/2015	3C.3	2015-02-06	MDCK1/SIAT1	160	155	80	68	80	70	80	113	2560	3271	D53N, I214T
A/Serbia/NS-1014/2015	3C.3	2015-02-10	SIAT1	320	253	80	96	80	80	160	176	2560	3474	G5E, N31S, E62K, N122D
A/Serbia/NS-1241/2015	3C.3	2015-03-19	SIAT1	160	138	40-80	60	40	47	80	72	2560	2088	G5E, N31S, E62K, N122D, S312N

Vaccine SH 2015 NH 2015-16

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

* Amino acid substitutions relative to cell-propagated A/Victoria/361/2011 that define genetic groups are shown together with those observed in individual viruses

Sequences in phylogenetic tree | HI assay performed

Table 9-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2015-08-03

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post-infection ferret antisera A/HK 7295/14 T/C F02/15 3C.2a		A/HK 4801/14 Egg F12/15 3C.2a		A/Switz 9715293/13 T/C NIB F13/14 3C.3a		A/Switz 9715293/13 Egg Cl123 F32/14 3C.3a		A/Neth 525/14 T/C F23/15 3C.3b		Egg adaptation HA1 substitutions compared to the corresponding cell isolate
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	320	355	80	77	80	74	80	111	40	38	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6	640	819	320	462	40	40	40	48	80	67	N96S, T160K, L194P
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	20	21	20	21	160	153	20	23	20	23	
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	E4/E2	80	77	40	35	40	40	320	318	40	36	R156,V186
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	61	40	49	80	66	80	76	1280	1023	
TEST VIRUSES														Additional HA information*
														L3I, Q33R, N144S, N145S, F159Y, K160T, *N225D* , N278K, Q311H
A/Castilla La Mancha/530/2015	3c.2a	2015-02-05	SIAT1/SIAT1	640	567	80	118	80	84	40	53	40	40	
A/Serbia/NS-1261/2015	3c.2a	2015-03-15	SIAT1	320	442	40	48	20	20	20	20	<	0	
A/Finland/526/2015	3c.2a	2015-04-13	SIAT1/SIAT1	320-640	480	80	90	80	72	80	66	40	51	
														Q33R, T128A, A138S, R142G, N145S, F159S, *N225D* , N278K
A/Extremadura/731/2015	3c.3a	2015-02-03	SIAT2/SIAT1	160	125	80	66	160	125	80	94	160	170	Y94H
A/Iasi/177049/2015	3c.3a	2015-02-04	MDCK1/SIAT1	80	66	40	40	80	84	80	64	40	59	Y94H, K276N
A/Castilla La Mancha/646/2015	3c.3a	2015-02-05	SIAT1/SIAT1	40	40	20	20	80	80	40	40	40	40	Y94H
A/Castilla La Mancha/659/2015	3c.3a	2015-02-05	SIAT1/SIAT1	80	115	40	59	160	123	80	74	160	148	Y94H
A/Perugia/46/2015	3c.3a	2015-02-06	MDCK1/SIAT1	160	183	80	77	80	102	80	76	20	20	Y94H
A/Castilla La Mancha/718/2015	3c.3a	2015-02-10	SIAT1/SIAT1	40	40	10	10	40	54	40	40	20	20	Y94H, R208K, K276N
														Q33R, E62K, K83R, N122D, T128A, R142G, N145S, L157S, R261Q, N278K
A/Aragon/2103/2014	3c.3b	2014-12-02	SIAT2/SIAT1	80	104	40	53	40-80	60	80	116	2560	3657	Q197H
A/Navarra/745/2015	3c.3b	2015-02-06	SIAT2/SIAT1	160	158	80	99	80	74	320	196	2560	3268	N144S, Q197H
A/Ukraine/165/2015	3c.3b	2015-02-28	SIAT1/SIAT1	80	65	40	44	40	40	40	59	1280	1136	V112I, Q197H
A/Norway/1825/2015	3c.3b	2015-03-30	SIAT1	80	119	40	51	40	45	80	71	2560	2425	D122N
														Q33R, T128A, R142G, N145S, L157S, N278K
A/Finland/494/2015	3c.3	2015-02-11	SIAT2/SIAT1	80	97	40	48	40	50	80	75	1280	1250	G5E, N31S, N122D
A/Norway/1656/2015	3c.3	2015-03-21	SIAT1/SIAT1	80	92	40	53	40	50	80	87	2560	2176	E62K, K83R, N122D, Q197H, R261Q

Vaccine SH 2015 NH 2015-16

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation; < = <10

* Amino acid substitutions relative to cell-propagated A/Victoria/361/2011 that define genetic groups are shown together with those observed in individual viruses

Sequences in phylogenetic tree | HI assay performed

Table 9-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2015-08-13

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post-infection ferret antisera A/HK 7295/14 T/C F02/15 3C.2a		A/HK 4801/14 Egg F12/15 3C.2a		A/Switz 9715293/13 T/C NIB F13/14 3C.3a		A/Switz 9715293/13 Egg Cl123 F32/14 3C.3a		A/Neth 525/14 T/C F23/15 3C.3b		Egg adaptation HA1 substitutions compared to the corresponding cell isolate
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	320	355	80	77	80	74	80	111	40	38	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6	640	819	320	462	40	40	40	48	80	67	N96S, T160K, L194P
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	20	21	20	21	160	153	20	23	20	23	
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	E4/E2	80	77	40	35	40	40	320	318	40	36	R156,V186
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	61	40	49	80	66	80	76	1280	1023	
TEST VIRUSES														Additional HA information*
														L3I, Q33R, N144S, N145S, F159Y, K160T, *N225D* , N278K, Q311H
A/Stockholm/1/2015	3C.2a	2015-01-12	MDCK1/SIAT1	160	175	80	83	40	57	40	52	40	40	T160K[2]
A/Stockholm/7/2015	3C.2a	2015-01-13	MDCK1/SIAT1	160	160	80	78	40	55	40	56	40	48	T160K
A/Stockholm/8/2015	3C.2a	2015-01-25	MDCK1/SIAT1	320	242	160	143	80	75	80	70	40-80	60	T160K
A/Stockholm/9/2015	3C.2a	2015-01-27	MDCK1/SIAT1	160	223	80	90	40	58	40	59	40	40	S114T
A/Stockholm/13/2015	3C.2a	2015-02-10	MDCK0/SIAT1	320	240	160	124	80	71	80	61	40	40	T160K
A/Moldova/111.07/2015	3C.2a	2015-02-10	MDCK2/SIAT1	320	279	80	92	40	57	40	46	40	43	S144R
A/Sweden/15/2015	3C.2a	2015-02-27	MDCK2/SIAT1	160	226	80	75	40	56	40	50	40	40	
														Q33R, T128A, A138S, R142G, N145S, F159S, *N225D* , N278K
A/Stockholm/32/2014	3C.3a	2014-12-09	MDCK1/SIAT1	40	49	40	40	40	46	40	46	40	42	S159X, S312N, K326R
A/Iasi/177050/2015	3C.3a	2015-02-04	MDCK1/SIAT1	80	66	40	40	80	84	80	64	40	59	Y94H, K276N
A/Behoririka/355/2015	3C.3a	2015-02-04	MDCK1/SIAT1	80	98	40	40	80	71	40	58	80	66	T135A, K326R
A/Manjakaray/612/2015	3C.3a	2015-02-23	MDCK1/SIAT1	160	222	80	75	80	97	80	76	80	68	S198P, K326R
														Q33R, E62K, K83R, N122D, T128A, R142G, N145S, L157S, R261Q, N278K
A/Stockholm/18/2015	3C.3b	2015-02-11	MDCK0/SIAT1	160	176	80	113	80	75	80	77	1280	1638	
A/Stockholm/20/2015	3C.3b	2015-02-24	MDCK0/SIAT1	160	124	80	68	40-80	60	80	71	1280	1246	Q197H
A/Stockholm/19/2015	3C.3b	2015-02-25	MDCK0/SIAT1	320	244	80	109	80	69	80	106	1280	1280	
A/Sweden/16/2015	3C.3b	2015-03-05	MDCK1/SIAT1	320	264	160	168	80	107	80	85	640	711	
														Q33R, T128A, R142G, N145S, L157S, N278K
A/Stockholm/12/2015	3C.3	2015-02-10	MDCK1/SIAT1	80	98	40	48	40	47	40	40	320	320	D53N, I214T, *N225D*

Vaccine SH 2015 NH 2015-16

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation; < = <10

* Amino acid substitutions relative to cell-propagated A/Victoria/361/2011 that define genetic groups are shown together with those observed in individual viruses

[2] Polymorhism/substitution resulting in loss of a potential glycosylation site (158-160)

Sequences in phylogenetic tree | HI assay performed

Table 9-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2015-08-17

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post-infection ferret antisera A/HK 7295/14 T/C F02/15 3C.2a		A/HK 4801/14 Egg F12/15 3C.2a		A/Switz 9715293/13 T/C NIB F13/14 3C.3a		A/Switz 9715293/13 Egg Cl123 F32/14 3C.3a		A/Neth 525/14 T/C F23/15 3C.3b		Egg adaptation HA1 substitutions compared to the corresponding cell isolate
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	320	355	80	77	80	74	80	111	40	38	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6	640	819	320	462	40	40	40	48	80	67	N96S, T160K, L194P
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	20	21	20	21	160	153	20	23	20	23	
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	E4/E2	80	77	40	35	40	40	320	318	40	36	R156,V186
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	61	40	49	80	66	80	76	1280	1023	
TEST VIRUSES														Additional HA information*
														L3I, Q33R, N144S, N145S, F159Y, K160T, *N225D* , N278K, Q311H
A/South Africa/R0740/2015	3C.2a	2015-03-02	MDCK3/SIAT1	160	153	40	51	40	52	40	52	40	46	Q197R, P221L
A/South Africa/R1734/2015	3C.2a	2015-04-27	MDCK1/SIAT1	320	411	80	80	80	60	80	63	40	40	Q197R
A/South Africa/R2490/2015	3C.2a	2015-05-22	MDCK2/SIAT1	320	272	80	111	80	67	80	65	40	59	T160X[2], Q197R
A/South Africa/R2665/2015	3C.2a	2015-05-25	MDCK2/SIAT1	320	400	80	63	40	53	40	58	40	40	R142K, Q197R
A/South Africa/R2651/2015	3C.2a	2015-05-25	MDCK2/SIAT1	320	347	80	73	80	66	80	60	40	40	Q197R, I214T
A/South Africa/R2582/2015	3C.2a	2015-05-26	MDCK2/SIAT1	320	382	80	111	80	91	80	116	80	64	
A/South Africa/R2991/2015	3C.2a	2015-06-05	MDCK2/SIAT1	160	232	160	130	80	60	80	64	80	63	T160X, Q197R
A/South Africa/R3280/2015		2015-06-09	MDCK2/SIAT1	80	105	20	20	20	20	<	5	<	1	SCI[3]
A/South Africa/R3276/2015	3C.2a	2015-06-13	MDCK3/SIAT1	160	133	40	40	20	20	20	20	20	20	Q197R SCI
A/South Africa/R4157/2015	3C.2a	2015-06-13	MDCK2/SIAT1	160	173	40	44	40	40	20	20	20	20	Q197R SCI
A/South Africa/R3663/2015	3C.2a	2015-06-14	MDCK2/SIAT1	160	187	40	50	40	43	40	40	10	10	
A/South Africa/R3659/2015	3C.2a	2015-06-17	MDCK2/SIAT1	160	197	80	62	40	42	40	40	40	40	S114X, N158X
A/South Africa/R3512/2015	3C.2a	2015-06-19	MDCK2/SIAT1	80	66	20	20	<	1	<	1	<	0	N158X, T160X SCI
A/South Africa/R3445/2015	3C.2a	2015-06-20	MDCK2/SIAT1	160	160	80	67	40	47	40	50	40	47	N158X, Q197R, P221L
A/South Africa/R3664/2015	3C.2a	2015-06-22	MDCK2/SIAT1	160	211	40	55	40	49	40	42	40	40	
A/South Africa/R3910/2015	3C.2a	2015-06-25	MDCK2/SIAT1	160	222	80	67	80	73	40	58	40	40	I48T
A/South Africa/R3777/2015	3C.2a	2015-06-26	MDCK2/SIAT1	320	251	80	87	80	79	80	83	80	77	N158X, Q197R
A/South Africa/R3803/2015	3C.2a	2015-06-29	MDCK2/SIAT1	160	230	80	61	40	44	80	65	40	59	K92R, Q197R
A/South Africa/R3778/2015	3C.2a	2015-06-29	MDCK2/SIAT1	160	189	40	57	20	20	20	20	10	10	N158X, Q197R
A/South Africa/R3989/2015	3C.2a	2015-07-06	MDCK2/SIAT1	80	99	<	5	<	3	<	3	10	10	V182I, Q197R SCI
A/South Africa/R3797/2015	3C.2a		MDCK2/SIAT1	160	137	40	40	20	20	20	20	10	10	N158X, Q197R SCI

Vaccine SH 2015 NH 2015-16

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation; < = <10

* Amino acid substitutions relative to cell-propagated A/Victoria/361/2011 that define genetic groups are shown together with those observed in individual viruses

[2] Polymorhism/substitution resulting in loss of a potential glycosylation site (158-160)

[3] SCI = single cell infection

Sequences in phylogenetic tree | HI assay performed

Table 9-7. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2015-08-20

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre[1] Post-infection ferret antisera: A/HK 7295/14 T/C F02/15 3C.2a 2-fold	Read	A/HK 4801/14 Egg F12/15 3C.2a 2-fold	Read	A/Switz 9715293/13 T/C NIB F13/14 3C.3a 2-fold	Read	A/Switz 9715293/13 Egg CI123 F32/14 3C.3a 2-fold	Read	A/Neth 525/14 T/C F23/15 3C.3b 2-fold	Read	Egg adaptation HA1 substitutions compared to the corresponding cell isolate
REFERENCE VIRUSES														
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	320	355	80	77	80	74	80	111	40	38	
A/Hong Kong/4801/2014	3C.2a	2014-02-26	E6	640	819	320	462	40	40	40	48	80	67	N96S, T160K, L194P
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	20	21	20	21	160	153	20	23	20	23	
A/Switzerland/ 9715293/2013	3C.3a	2013-12-06	E4/E2	80	77	40	35	40	40	320	318	40	36	R156,V186
A/Netherlands/525/2014	3C.3b	2014-12-17	SIAT2/SIAT1	80	61	40	49	80	66	80	76	1280	1023	
TEST VIRUSES														Additional HA information*
														L3I, Q33R, N144S, N145S, F159Y, K160T, *N225D* , N278K, Q311H
A/Estonia/91651/2015	3C.2a	2015-02-02	SIAT2/SIAT1	320	436	80	75	80	60	40	42	20	20	
A/Estonia/92115/2015	3C.2a	2015-02-12	SIAT2/SIAT1	640	577	80	118	80	66	40	53	40	40	
A/Nitra/563/2015	3C.2a	2015-02-19	MDCKx/SIAT1	640	525	160	152	80	80	80	64	20	20	I140M
A/Netherlands/1908/2015	3C.2a	2015-02-23	MDCK2/SIAT1	320	400	80	105	80	68	80	64	40	56	
A/Netherlands/1903/2015	3C.2a	2015-02-24	MDCK2/SIAT1	320	384	160	135	80	89	40	47	20	20	N158X[2]
A/Mauritius/I-238/2015	3C.2a	2015-03-06	MDCK3/SIAT1	320	309	80	107	80	74	80	63	40	50	N171K
A/Mauritius/I-240/2015	3C.2a	2015-03-09	MDCK2/SIAT1	320	276	80	67	20	20	<	5	20	20	N171K
A/Moscow/100/2015	3C.2a	2015-03-19	MDCK1/SIAT1	320	480	40	58	10	10	40	40	<	1	S144R
A/Moscow/133/2015	3C.2a	2015-04-09	MDCK1/SIAT1	320	378	80	62	40	40	80	63	<	1	I140R
A/Dakar/10/2015	3C.2a	2015-04-12	C1/SIAT1	320	240	80	77	40	57	40	53	40	40	N158X, L316M, R329K
A/Hong Kong/11743/2015	3C.2a	2015-06-09	MDCK1/SIAT1	640	505	80	83	40	53	40	40	40	40	D7N, S114T, T128N
A/Hong Kong/11706/2015	3C.2a	2015-06-09	MDCK2/SIAT1	320	311	80	84	40	47	20	20	20	20	S114T
A/Hong Kong/11713/2015	3C.2a	2015-06-10	MDCK1/SIAT1	640	480	80	83	80	63	80	60	40	43	S54N, R261L
														Q33R, T128A, A138S, R142G, N145S, F159S, *N225D* , N278K
A/St Petersburg/119/2015	3C.3a	2015-02-03	C2/SIAT1	40	40	20	20	80	68	20	20	20	20	A212S, K326R, SCI[3]
A/Vladivostok/36/2015	3C.3a	2015-03-14	MDCK2/SIAT1	80	64	40	40	80	98	80	76	40	40	A212S, K326R
														Q33R, E62K, K83R, N122D, T128A, R142G, N145S, L157S, R261Q, N278K
A/Bratislava/437/2015	3C.3b	2015-02-17	MDCKx/SIAT1	80	77	40	55	40	58	80	74	1280	1262	Q197H
A/Slovenia/1066/15	3C.3b	2015-02-19	MDCK1/SIAT1	80	102	80	72	80	74	80	78	1280	1165	S124R, Q197H
A/Slovenia/1077/2015	3C.3b	2015-02-19	MDCKx/SIAT1	160	126	80	63	80	64	80	68	2560	2425	Q197H
A/Mauritius/I-203/2015	3C.3b	2015-02-25	MDCK2/SIAT1	40	58	40	40	80	40	40	40	1280	1499	
A/Moscow/101/2015	3C.3b	2015-03-18	MDCK1/SIAT1	80	78	40	43	40	57	40	49	1280	1060	

Vaccine SH 2015 NH 2015-16 (A/Switz 9715293/13 Egg CI123 F32/14)

[1] Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation; < = <10

* Amino acid substitutions relative to cell-propagated A/Victoria/361/2011 that define genetic groups are shown together with those observed in individual viruses

[2] Polymorhism/substitution resulting in loss of a potential glycosylation site (158-160)

[3] SCI = single cell infection

Sequences in phylogenetic tree | HI assay performed

Table 9-8. Influenza A(H3N2) viruses - Summary of Plaque Reduction Neutralisation Assays (MDCK-SIAT)

Assay Date	Antiserum to Virus	Number of test viruses reacting within 4-fold and 2-fold of the antiserum homologous titre									
		A/HK 7295/14 T/C F02/15		A/HK 4801/14 Egg F12/15		A/Switz 9715293/13 T/C NIB F13/14		A/Switz 9715293/13 Egg Cl123 F32/14		A/Neth 525/14 T/C F23/15	
Virus Genetic Group		3C.2a		3C.2a		3C.3a		3C.3a		3C.3b	
	Number of Viruses	4-fold	2-fold	4-fold	2-fold	4-fold	2-fold	4-fold	2-fold	4-fold	2-fold
16-Jul											
3C.2a	8	8	8	8	2	8	2	8	8	0	0
3C.3a	0	0	0	0	0	0	0	0	0	0	0
3C.3b	1	1	0	0	0	1	0	1	1	1	1
29-Jul											
3C.2a	6	6	6	6	3	6	5	5	2	0	0
3C.3a	3	3	0	0	0	3	3	2	0	0	0
3C.3b	0	0	0	0	0	0	0	0	0	0	0
03-Aug											
3C.2a	3	3	3	2	0	2	2	1	0	0	0
3C.3a	6	5	2	2	0	6	5	4	0	0	0
3C.3b	4	4	1	1	0	4	2	3	1	4	4
13-Aug											
3C.2a	7	7	7	7	2	7	2	2	0	0	0
3C.3a	4	3	1	1	0	4	3	2	0	0	0
3C.3b	4	4	4	4	1	4	4	4	0	4	4
17-Aug											
3C.2a	20	20	18	11	1	15	7	7	0	0	0
3C.3a	0	0	0	0	0	0	0	0	0	0	0
3C.3b	0	0	0	0	0	0	0	0	0	0	0
20-Aug											
3C.2a	13	13	13	12	2	11	7	5	0	0	0
3C.3a	2	1	0	0	0	2	2	1	0	0	0
3C.3b	5	4	1	2	0	5	2	4	0	5	5
Summary - Numbers											
3C.2a	57	57	55	46	10	49	25	28	10	0	0
3C.3a	15	12	3	3	0	15	13	9	0	0	0
3C.3b	14	13	6	7	1	14	8	12	2	14	14
Summary - Percentage											
3C.2a		100.0	96.5	80.7	17.5	86.0	43.9	49.1	17.5	0.0	0.0
3C.3a		80.0	20.0	20.0	0.0	100.0	86.7	60.0	0.0	0.0	0.0
3C.3b		92.9	42.9	50.0	7.1	100.0	57.1	85.7	14.3	100.0	100.0

Vaccine SH2015 NH2015-16

Figure 7. Phylogenetic comparison of A(H3N2) HA genes

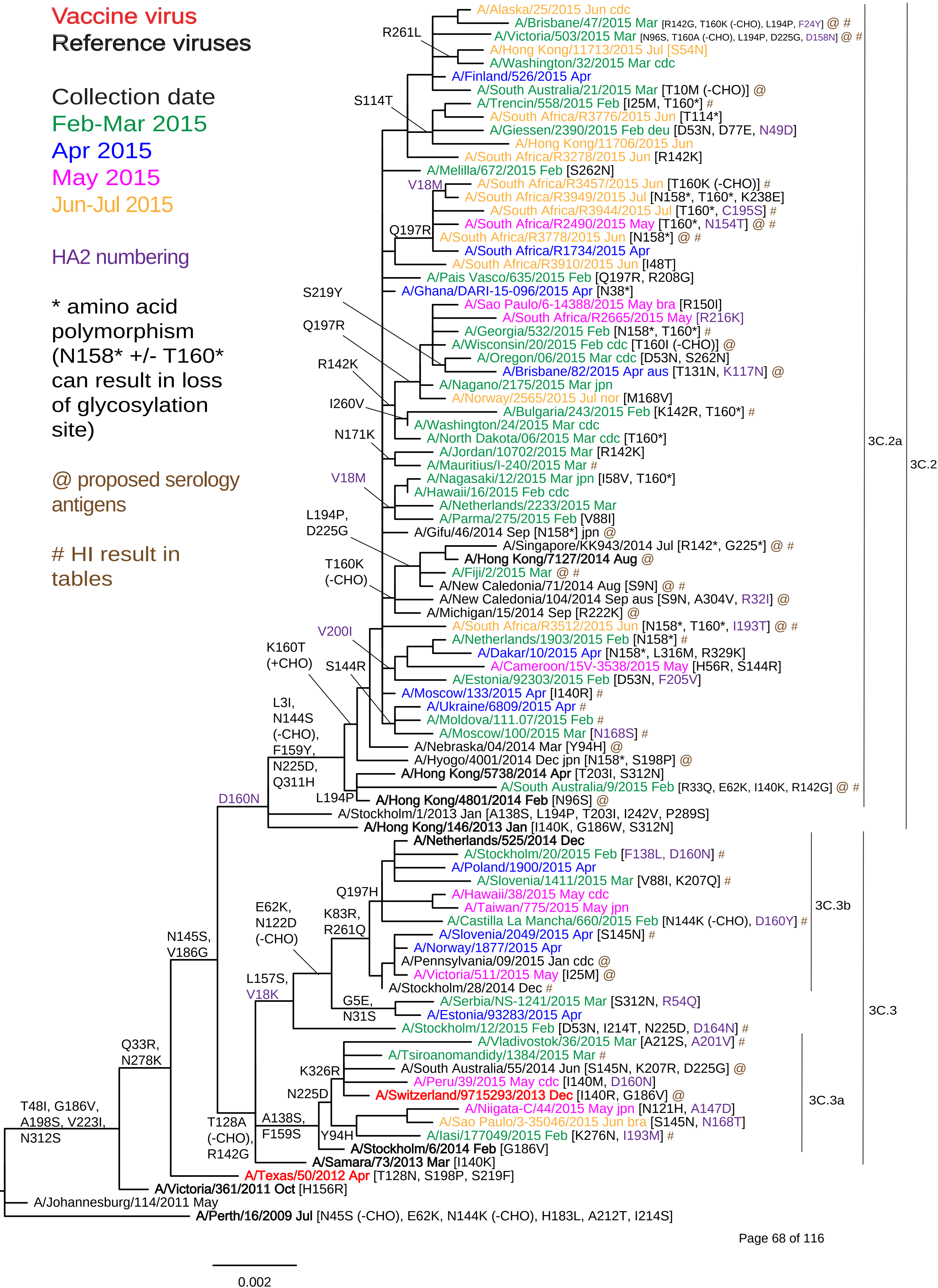


Figure 8. HA protein alignment for A(H3N2) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                     10        20        30        40        50        60        70        80        90       100
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             QKLPGNDNSTATLCLGHHAVPNGTIVKTITNDQIEVTNATELVQSSSTGEICDSPHQILDGKNCTLIDALLGDPQCDGFQNKKWDLFVERSKAYSNCYPY 1
A/Johannesburg/114/2011     ............................................N................E...................................... 7
A/Victoria/361/2011         ............................................N..I.............E...................................... 3C
A/Texas/50/2012             ................................R...........N..I.............E...................................... 3C.1
A/Samara/73/2013            ................................R...........N..I.............E...................................... 3C.3
A/Stockholm/6/2014          ................................R...........N..I.............E...................................... 3C.3a
A/Sao Paulo/3-35046/2015    ................................R...........N..I.............E...............................H...... 3C.3a
A/Switzerland/9715293/2013  ................................R...........N..I.............E...................................... 3C.3a
A/Vladivostok/36/2015       ................................R...........N..I.............E...................................... 3C.3a
A/Estonia/93283/2015        ....E.........................S.R...........N..I.................................................... 3C.3
A/Stockholm/28/2014         ................................R...........N..I..................................R................. 3C.3b
A/Poland/1900/2015          ................................R...........N..I..................................R................. 3C.3b
A/Netherlands/525/2014      ................................R...........N..I..................................R................. 3C.3b
A/Hong Kong/146/2013        ................................R...........N..I.............E...................................... 3C.2
A/Hong Kong/4801/2014       ..I.............................R...........N..I.............E.................................S.... 3C.2a
A/Hong Kong/5738/2014       ..I.............................R...........N..I.............E...................................... 3C.2a
A/Moscow/100/2015           ..I.............................R...........N..I.............E...................................... 3C.2a
A/Ukraine/6809/2015         ..I.............................R...........N..I.............E...................................... 3C.2a
A/Cameroon/15V-3538/2015    ..I.............................R...........N..I.......R.....E...................................... 3C.2a
A/South Africa/R3512/2015   ..I.............................R...........N..I.............E...................................... 3C.2a
A/Nagasaki/12/2015          ..I.............................R...........N..I.........V...E...................................... 3C.2a
A/Mauritius/I-240/2015      ..I.............................R...........N..I.............E...................................... 3C.2a
A/Bulgaria/243/2015         ..I.............................R...........N..I.............E...................................... 3C.2a
A/Norway/2565/2015          ..I.............................R...........N..I.............E...................................... 3C.2a
A/Brisbane/82/2015          ..I.............................R...........N..I.............E...................................... 3C.2a
A/South Africa/R2665/2015   ..I.............................R...........N..I.............E...................................... 3C.2a
A/South Africa/R2490/2015   ..I.............................R...........N..I.............E...................................... 3C.2a
A/South Africa/R3949/2015   ..I.............................R...........N..I.............E...................................... 3C.2a
A/South Africa/R3776/2015   ..I.............................R...........N..I.............E...................................... 3C.2a
A/Hong Kong/11713/2015      ..I.............................R...........N..I.....N.......E...................................... 3C.2a

                                    110       120       130       140       150       160       170       180       190       200
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             DVPDYASLRSLVASSGTLEFNNESFNWTGVTQNGTSSACIRRSKNSFFSRLNWLTHLNFKYPALNVTMPNNEQFDKLYIWGVLHPGTDKDQIFLYAQASG
A/Johannesburg/114/2011     ...........................................N......................................H.................
A/Victoria/361/2011         ...........................................N...........R..........................H..V...........S..
A/Texas/50/2012             ...........................N...............N......................................H..V...........P..
A/Samara/73/2013            ...........................A...........K.G.NS.....................................H..............S..
A/Stockholm/6/2014          ...........................A.........S...G.NS.............S.......................H..V...........S..
A/Sao Paulo/3-35046/2015    ...........................A.........S...G.N..............S.......................H..............S..
A/Switzerland/9715293/2013  ...........................A.........S.R.G.NS.............S.......................H..V...........S..
A/Vladivostok/36/2015       ...........................A.........S...G.NS.............S.......................H..............S..
A/Estonia/93283/2015        .....................D.....A.............G.NS...........S.........................H..............S..
A/Stockholm/28/2014         .....................D.....A.............G.NS...........S.........................H..............S..
A/Poland/1900/2015          .....................D.....A.............G.NS...........S.........................H.............HS..
A/Netherlands/525/2014      .....................D.....A.............G.NS...........S.........................H.............HS..
A/Hong Kong/146/2013        .......................................K...NS.....................................H..W...........S..
A/Hong Kong/4801/2014       ...........................................SS.............Y.......................H..........P...S..
A/Hong Kong/5738/2014       ...........................................SS.............Y.......................H..........P...S..
A/Moscow/100/2015           ...........................................RS.............YT......................H..............S..
A/Ukraine/6809/2015         ...........................................RS.............YT......................H..............S..
A/Cameroon/15V-3538/2015    ...........................................RS.............YT......................H..............S..
A/South Africa/R3512/2015   ...........................................SS............XYX......................H..............S..
A/Nagasaki/12/2015          ...........................................SS.............YX......................H..............S..
A/Mauritius/I-240/2015      ...........................................SS.............YT..........K...........H..............S..
A/Bulgaria/243/2015         ...........................................SS.............YX......................H..............S..
A/Norway/2565/2015          .........................................K.SS.............YT.......V..............H.............RS..
A/Brisbane/82/2015          ..............................N..........K.SS.............YT......................H.............RS..
A/South Africa/R2665/2015   .........................................K.SS.............YT......................H.............RS..
A/South Africa/R2490/2015   ...........................................SS.............YX......................H.............RS..
A/South Africa/R3949/2015   ...........................................SS............XYX......................H.............RS..
A/South Africa/R3776/2015   .............X.............................SS.............YT......................H..............S..
A/Hong Kong/11713/2015      ...........................................SS.............YT......................H..............S..

                                    210       220       230       240       250       260       270       280       290       300
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             RITVSTKRSQQTVSPNIGSRPRVRNIPSRISIYWTIVKPGDILLINSTGNLIAPRGYFKIRSGKSSIMRSDAPIGKCNSECITPNGSIPNDKPFQNVNRI
A/Johannesburg/114/2011     ...........A.I......................................................................................
A/Victoria/361/2011         ...........A.I........I.............................................................................
A/Texas/50/2012             ...........A.I....F...I......................................................K......................
A/Samara/73/2013            ...........A.I........I......................................................K......................
A/Stockholm/6/2014          ...........A.I........I......................................................K......................
A/Sao Paulo/3-35046/2015    ...........A.I........I.D....................................................K......................
A/Switzerland/9715293/2013  ...........A.I........I.D....................................................K......................
A/Vladivostok/36/2015       ...........S.I........I.D....................................................K......................
A/Estonia/93283/2015        ...........A.I........I......................................................K......................
A/Stockholm/28/2014         ...........A.I........I.....................................Q................K......................
A/Poland/1900/2015          ...........A.I........I.....................................Q................K......................
A/Netherlands/525/2014      ...........A.I........I.....................................Q................K......................
A/Hong Kong/146/2013        ...........A.I........I......................................................K......................
A/Hong Kong/4801/2014       ...........A.I........I.D....................................................K......................
A/Hong Kong/5738/2014       ..I........A.I........I.D....................................................K......................
A/Moscow/100/2015           ...........A.I........I.D....................................................K......................
A/Ukraine/6809/2015         ...........A.I........I.D....................................................K......................
A/Cameroon/15V-3538/2015    ...........A.I........I.D....................................................K......................
A/South Africa/R3512/2015   ...........A.I........I.D....................................................K......................
A/Nagasaki/12/2015          ...........A.I........I.D....................................................K......................
A/Mauritius/I-240/2015      ...........A.I........I.D....................................................K......................
A/Bulgaria/243/2015         ...........A.I........I.D..................................V.................K......................
A/Norway/2565/2015          ...........A.I........I.D....................................................K......................
A/Brisbane/82/2015          ...........A.I....Y...I.D....................................................K......................
A/South Africa/R2665/2015   ...........A.I........I.D....................................................K......................
A/South Africa/R2490/2015   ...........A.I........I.D....................................................K......................
A/South Africa/R3949/2015   ...........A.I........I.D............E.......................................K......................
A/South Africa/R3776/2015   ...........A.I........I.D....................................................K......................
A/Hong Kong/11713/2015      ...........A.I........I.D...................................L................K......................
```

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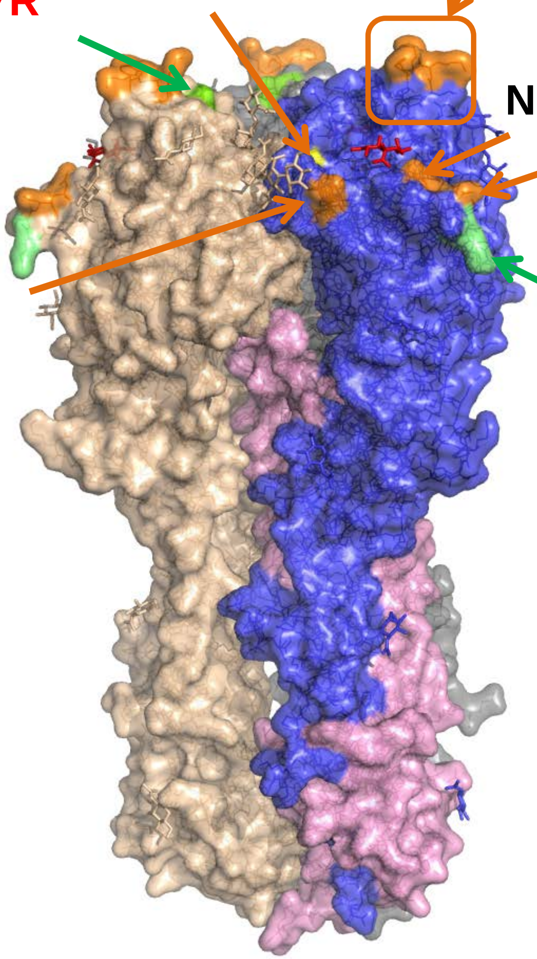
                                    310       320       330       340       350       360       370       380       390       400
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             TYGACPRYVKQNTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNEKFHQIEKEFS
A/Johannesburg/114/2011     ....................................................................................................
A/Victoria/361/2011         ...........S........................................................................................
A/Texas/50/2012             ...........S........................................................................................
A/Samara/73/2013            ...........S........................................................................................
A/Stockholm/6/2014          ...........S........................................................................................
A/Sao Paulo/3-35046/2015    ...........S........................................................................................
A/Switzerland/9715293/2013  ...........S.............R..........................................................................
A/Vladivostok/36/2015       ...........S.............R..........................................................................
A/Estonia/93283/2015        ...........S..................................K.....................................................
A/Stockholm/28/2014         ...........S..................................K.....................................................
A/Poland/1900/2015          ...........S..................................K.....................................................
A/Netherlands/525/2014      ...........S..................................K.....................................................
A/Hong Kong/146/2013        ....................................................................................................
A/Hong Kong/4801/2014       ..........HS........................................................................................
A/Hong Kong/5738/2014       ..........H.........................................................................................
A/Moscow/100/2015           ..........HS........................................................................................
A/Ukraine/6809/2015         ..........HS........................................................................................
A/Cameroon/15V-3538/2015    ..........HS........................................................................................
A/South Africa/R3512/2015   ..........HS........................................................................................
A/Nagasaki/12/2015          ..........HS..................................M.....................................................
A/Mauritius/I-240/2015      ..........HS........................................................................................
A/Bulgaria/243/2015         ..........HS........................................................................................
A/Norway/2565/2015          ..........HS........................................................................................
A/Brisbane/82/2015          ..........HS........................................................................................
A/South Africa/R2665/2015   ..........HS........................................................................................
A/South Africa/R2490/2015   ..........HS........................................................................................
A/South Africa/R3949/2015   ..........HS..................................M.....................................................
A/South Africa/R3776/2015   ..........HS........................................................................................
A/Hong Kong/11713/2015      ..........HS........................................................................................

                                    410       420       430       440       450       460       470       480       490       500
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSEMNKLFEKTKKQLRENAEDMGNGCFKIYHKCDNACIGSIRNGTYDHDVYRDEALNNRF
A/Johannesburg/114/2011     ....................................................................................................
A/Victoria/361/2011         ....................................................................................................
A/Texas/50/2012             ....................................................................................................
A/Samara/73/2013            ....................................................................................................
A/Stockholm/6/2014          ....................................................................................................
A/Sao Paulo/3-35046/2015    ................................................................................................T...
A/Switzerland/9715293/2013  ....................................................................................................
A/Vladivostok/36/2015       ....................................................................................................
A/Estonia/93283/2015        ....................................................................................................
A/Stockholm/28/2014         ....................................................................................................
A/Poland/1900/2015          ....................................................................................................
A/Netherlands/525/2014      ....................................................................................................
A/Hong Kong/146/2013        ........................................................................................N...........
A/Hong Kong/4801/2014       ........................................................................................N...........
A/Hong Kong/5738/2014       ........................................................................................N...........
A/Moscow/100/2015           ........................................................................................N.......S...
A/Ukraine/6809/2015         ........................................................................................N...........
A/Cameroon/15V-3538/2015    ........................................................................................N...........
A/South Africa/R3512/2015   ........................................................................................N...........
A/Nagasaki/12/2015          ........................................................................................N...........
A/Mauritius/I-240/2015      ........................................................................................N...........
A/Bulgaria/243/2015         ........................................................................................N...........
A/Norway/2565/2015          ........................................................................................N...........
A/Brisbane/82/2015          .............................................N..........................................N...........
A/South Africa/R2665/2015   ........................................................................................N...........
A/South Africa/R2490/2015   ..................................................................................T.....N...........
A/South Africa/R3949/2015   ........................................................................................N...........
A/South Africa/R3776/2015   ........................................................................................N...........
A/Hong Kong/11713/2015      ........................................................................................N...........

                                    510       520       530       540       550
                            ....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI
A/Johannesburg/114/2011     ..................................................
A/Victoria/361/2011         ..................................................
A/Texas/50/2012             ..................................................
A/Samara/73/2013            ..................................................
A/Stockholm/6/2014          ..................................................
A/Sao Paulo/3-35046/2015    ..................................................
A/Switzerland/9715293/2013  ..................................................
A/Vladivostok/36/2015       .............................V....................
A/Estonia/93283/2015        ..................................................
A/Stockholm/28/2014         ..................................................
A/Poland/1900/2015          ..................................................
A/Netherlands/525/2014      ..................................................
A/Hong Kong/146/2013        ..................................................
A/Hong Kong/4801/2014       ..................................................
A/Hong Kong/5738/2014       ..................................................
A/Moscow/100/2015           ..................................................
A/Ukraine/6809/2015         ..................................................
A/Cameroon/15V-3538/2015    ............................I.....................
A/South Africa/R3512/2015   .....................T............................
A/Nagasaki/12/2015          ..................................................
A/Mauritius/I-240/2015      ..................................................
A/Bulgaria/243/2015         ..................................................
A/Norway/2565/2015          ..................................................
A/Brisbane/82/2015          ..................................................
A/South Africa/R2665/2015   ............................................K.....
A/South Africa/R2490/2015   ..................................................
A/South Africa/R3949/2015   ..................................................
A/South Africa/R3776/2015   ..................................................
A/Hong Kong/11713/2015      ..................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 9A. Amino acid substitutions associated with the HA of 3C.2a viruses relative to A/Texas/50/2012



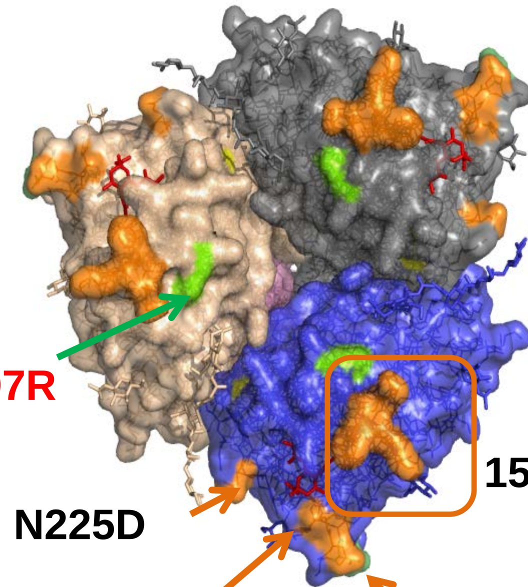


Figure 9B. Amino acid substitutions associated with the HA of 3C.3a viruses relative to A/Texas/50/2012

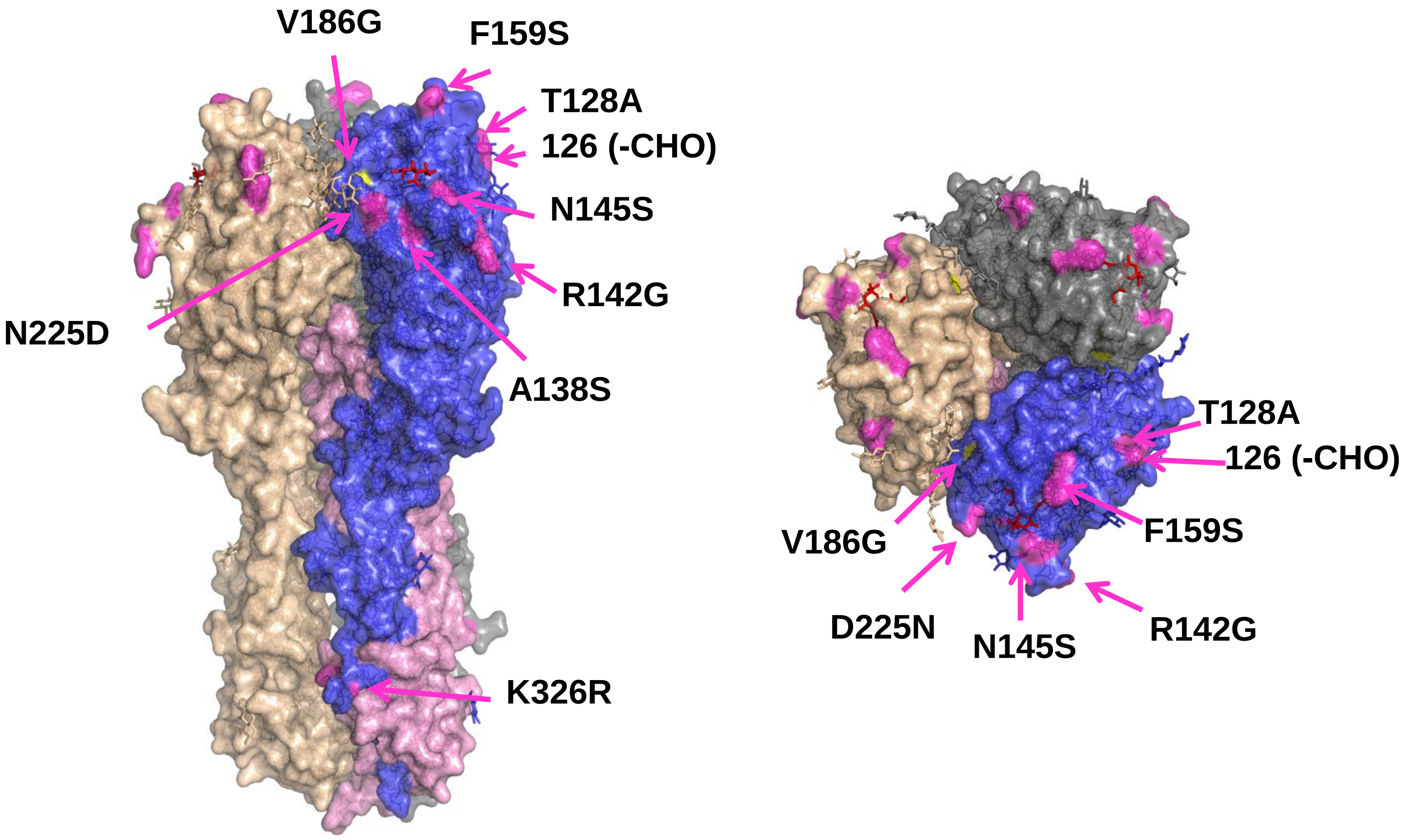
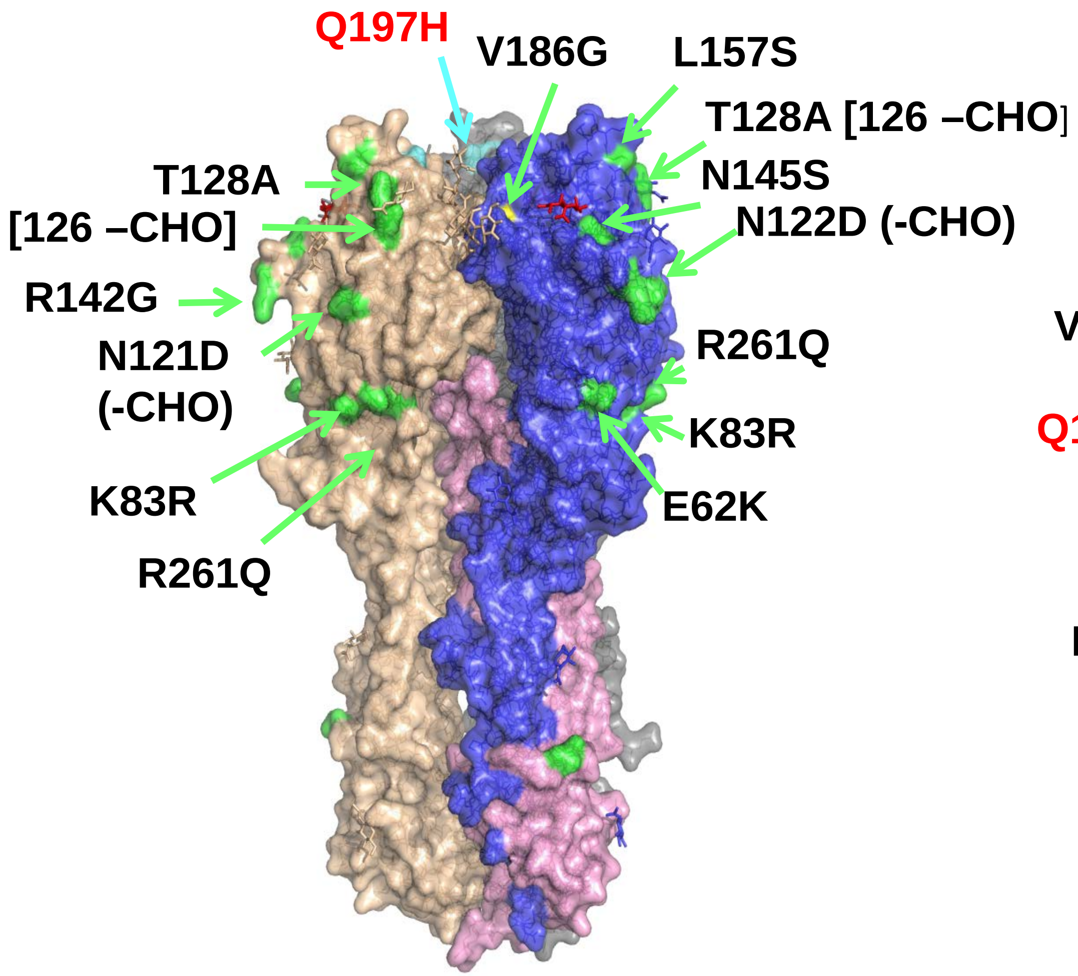


Figure 9C. Amino acid substitutions associated with the HA of 3C.3b viruses relative to A/Texas/50/2012



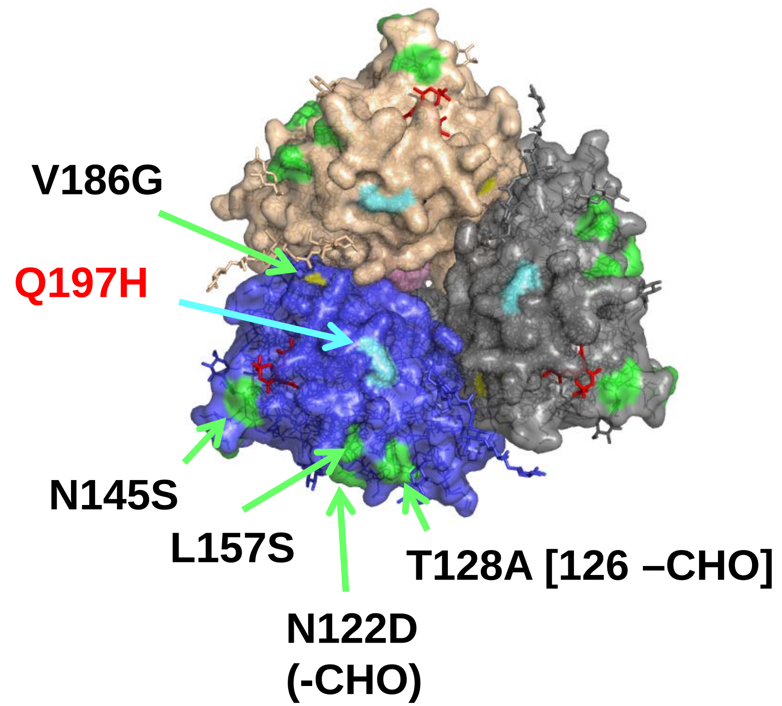


Figure 10. Phylogenetic comparison of A(H3N2) NA genes

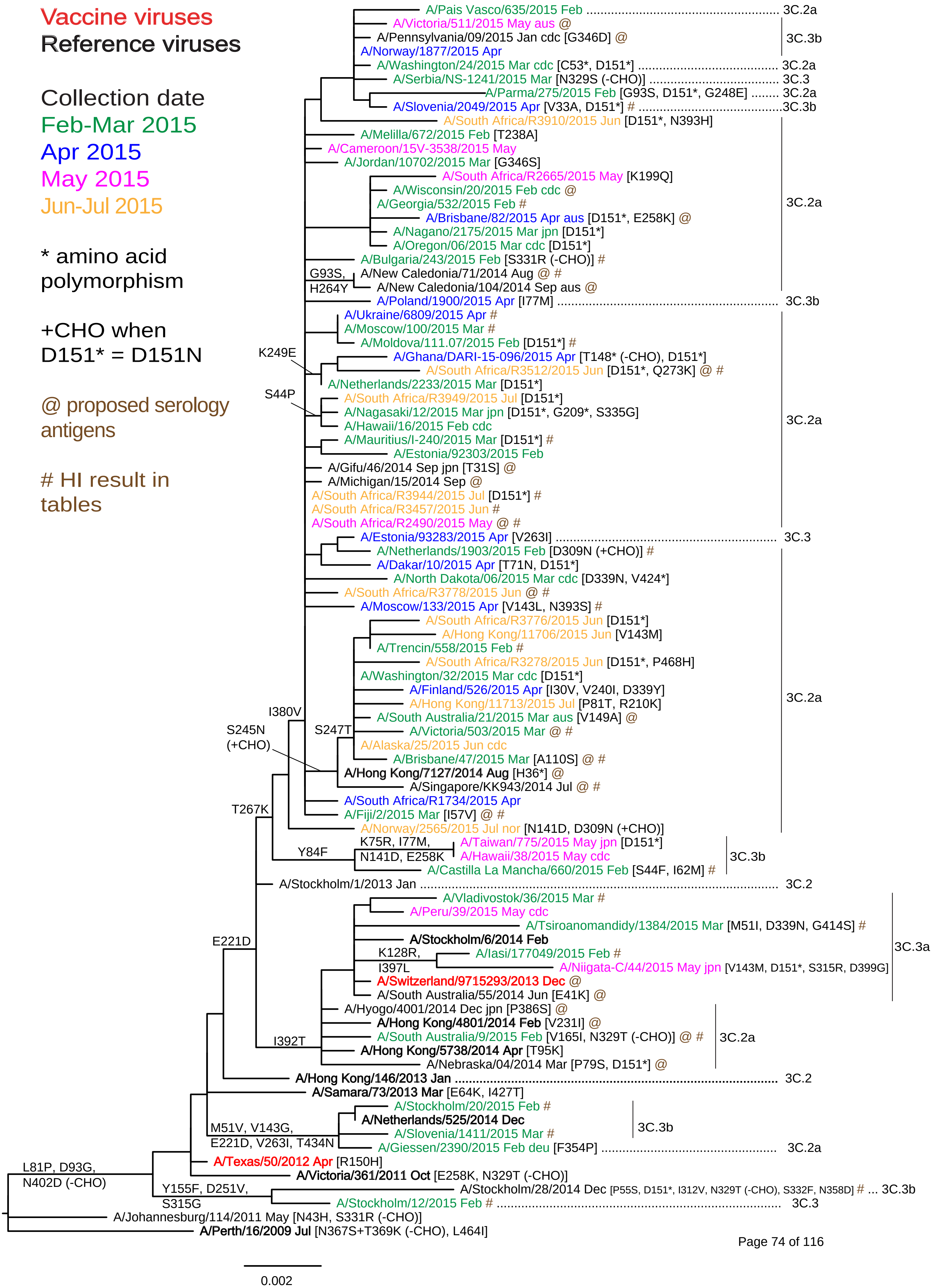


Figure 11. NA protein alignment for A(H3N2) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                    10        20        30        40        50        60        70        80        90       100
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             MNPNQKIITIGSVSLTISTICFFMQIAILITTVTLHFKQYEFNSPPNNQVMLCEPTIIERNITEIVYLTNTTIEKEICPKLAEYRNWSKPQCDITGFAPF 1
A/Johannesburg/114/2011     ..........................................H......................................................... 7
A/Victoria/361/2011         ................................................................................P...........G....... 3C
A/Texas/50/2012             ................................................................................P...........G....... 3C.1
A/Samara/73/2013            ...............................................................K................P...........G....... 3C.3
A/Stockholm/6/2014          ................................................................................P...........G....... 3C.3a
A/Niigata-C/44/2015         ................................................................................P...........G....... 3C.3a
A/Switzerland/9715293/2013  ................................................................................P...........G....... 3C.3a
A/Vladivostok/36/2015       ................................................................................P...........G....... 3C.3a
A/Estonia/93283/2015        ................................................................................P...........G....... 3C.3
A/Victoria/511/2015         ................................................................................P...........G....... 3C.3b
A/Poland/1900/2015          ............................................................................M...P...........G....... 3C.3b
A/Netherlands/525/2014      ..................................................V.............................P...........G....... 3C.3b
A/Hong Kong/146/2013        ................................................................................P...........G....... 3C.2
A/Hong Kong/4801/2014       ................................................................................P...........G....... 3C.2a
A/Hong Kong/5738/2014       ................................................................................P...........G.K..... 3C.2a
A/Moscow/100/2015           ................................................................................P...........G....... 3C.2a
A/Ukraine/6809/2015         ................................................................................P...........G....... 3C.2a
A/Cameroon/15V-3538/2015    ................................................................................P...........G....... 3C.2a
A/South Africa/R3512/2015   ................................................................................P...........G....... 3C.2a
A/Nagasaki/12/2015          ...........................................P....................................P...........G....... 3C.2a
A/Mauritius/I-240/2015      ................................................................................P...........G....... 3C.2a
A/Bulgaria/243/2015         ................................................................................P...........G....... 3C.2a
A/Norway/2565/2015          ................................................................................P...........G....... 3C.2a
A/Brisbane/82/2015          ................................................................................P...........G....... 3C.2a
A/South Africa/R2665/2015   ................................................................................P...........G....... 3C.2a
A/South Africa/R2490/2015   ................................................................................P...........G....... 3C.2a
A/South Africa/R3949/2015   ...........................................P....................................P...........G....... 3C.2a
A/South Africa/R3776/2015   ................................................................................P...........G....... 3C.2a
A/Hong Kong/11713/2015      ................................................................................T...........G....... 3C.2a

                                   110       120       130       140       150       160       170       180       190       200
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             SKDNSIRLSAGGDIWVTREPYVSCDPDKCYQFALGQGTTLNNVHSNNTVRDRTPYRTLLMNELGVPFHLGTKQVCIAWSSSSCHDGKAWLHVCITGDDKN
A/Johannesburg/114/2011     ....................................................................................................
A/Victoria/361/2011         ........................................................X...........................................
A/Texas/50/2012             .................................................H..................................................
A/Samara/73/2013            ....................................................................................................
A/Stockholm/6/2014          ....................................................................................................
A/Niigata-C/44/2015         ...........................R..............M.......X.................................................
A/Switzerland/9715293/2013  ....................................................................................................
A/Vladivostok/36/2015       ....................................................................................................
A/Estonia/93283/2015        ....................................................................................................
A/Victoria/511/2015         ....................................................................................................
A/Poland/1900/2015          ....................................................................................................
A/Netherlands/525/2014      ..........................................G.........................................................
A/Hong Kong/146/2013        ....................................................................................................
A/Hong Kong/4801/2014       ....................................................................................................
A/Hong Kong/5738/2014       ....................................................................................................
A/Moscow/100/2015           ....................................................................................................
A/Ukraine/6809/2015         ....................................................................................................
A/Cameroon/15V-3538/2015    ....................................................................................................
A/South Africa/R3512/2015   ..................................................X.................................................
A/Nagasaki/12/2015          ..................................................X.................................................
A/Mauritius/I-240/2015      ..................................................X.................................................
A/Bulgaria/243/2015         ....................................................................................................
A/Norway/2565/2015          ........................................D...........................................................
A/Brisbane/82/2015          ..................................................X.................................................
A/South Africa/R2665/2015   ..................................................................................................Q.
A/South Africa/R2490/2015   ....................................................................................................
A/South Africa/R3949/2015   ..................................................X.................................................
A/South Africa/R3776/2015   ..................................................X.................................................
A/Hong Kong/11713/2015      ....................................................................................................

                                   210       220       230       240       250       260       270       280       290       300
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             ATASFIYNGRLVDSVVSWSKEILRTQESECVCINGTCTVVMTDGSASGKADTKILFIEEGKIVHTSTLSGSAQHVEECSCYPRYPGVRCVCRDNWKGSNR
A/Johannesburg/114/2011     ....................................................................................................
A/Victoria/361/2011         .........................................................K..........................................
A/Texas/50/2012             ....................................................................................................
A/Samara/73/2013            ....................................................................................................
A/Stockholm/6/2014          ....................D...............................................................................
A/Niigata-C/44/2015         ....................D...............................................................................
A/Switzerland/9715293/2013  ....................D...............................................................................
A/Vladivostok/36/2015       ....................D...............................................................................
A/Estonia/93283/2015        ....................D.........................................I...K.................................
A/Victoria/511/2015         ....................D.............................................K.................................
A/Poland/1900/2015          ....................D.............................................K.................................
A/Netherlands/525/2014      ....................D.........................................I.....................................
A/Hong Kong/146/2013        ....................................................................................................
A/Hong Kong/4801/2014       ....................D.........I.....................................................................
A/Hong Kong/5738/2014       ....................D...............................................................................
A/Moscow/100/2015           ....................D.............................................K.................................
A/Ukraine/6809/2015         ....................D.............................................K.................................
A/Cameroon/15V-3538/2015    ....................D.............................................K.................................
A/South Africa/R3512/2015   ....................D...........................E.................K.....K...........................
A/Nagasaki/12/2015          ........X...........D.............................................K.................................
A/Mauritius/I-240/2015      ....................D.............................................K.................................
A/Bulgaria/243/2015         ....................D.............................................K.................................
A/Norway/2565/2015          ....................D.............................................K.................................
A/Brisbane/82/2015          ....................D....................................K........K.................................
A/South Africa/R2665/2015   ....................D.............................................K.................................
A/South Africa/R2490/2015   ....................D.............................................K.................................
A/South Africa/R3949/2015   ....................D.............................................K.................................
A/South Africa/R3776/2015   ....................D.......................N.T...................K.................................
A/Hong Kong/11713/2015      .........K..........D.......................N.T...................K.................................
```

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                                    310       320       330       340       350       360       370       380       390       400
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009             PIVDINIKDHSIVSSYVCSGLVGDTPRKNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNDVWMGRTISEKSRLGYETFKVIEGWSNPKSKLQINRQVIVDR
A/Johannesburg/114/2011     ..............................R...................................N.T...............................
A/Victoria/361/2011         ............................T.....................................N.T...............................
A/Texas/50/2012             ..................................................................N.T...............................
A/Samara/73/2013            ..................................................................N.T...............................
A/Stockholm/6/2014          ..................................................................N.T......................T........
A/Niigata-C/44/2015         ..............R...................................................N.T......................T....L.G.
A/Switzerland/9715293/2013  ..................................................................N.T......................T........
A/Vladivostok/36/2015       ..................................................................N.T......................T........
A/Estonia/93283/2015        ..................................................................N.T..........V....................
A/Victoria/511/2015         ..................................................................N.T..........V....................
A/Poland/1900/2015          ..................................................................N.T..........V....................
A/Netherlands/525/2014      ..................................................................N.T...............................
A/Hong Kong/146/2013        ..................................................................N.T...............................
A/Hong Kong/4801/2014       ..................................................................N.T......................T........
A/Hong Kong/5738/2014       ..................................................................N.T......................T........
A/Moscow/100/2015           ..................................................................N.T..........V....................
A/Ukraine/6809/2015         ..................................................................N.T..........V....................
A/Cameroon/15V-3538/2015    ..................................................................N.T..........V....................
A/South Africa/R3512/2015   ..................................................................N.T..........V....................
A/Nagasaki/12/2015          ..................................G...............................N.T..........V....................
A/Mauritius/I-240/2015      ..................................................................N.T..........V....................
A/Bulgaria/243/2015         ..............................R...................................N.T..........V....................
A/Norway/2565/2015          ........N.........................................................N.T...............................
A/Brisbane/82/2015          ..................................................................N.T..........V....................
A/South Africa/R2665/2015   ..................................................................N.T..........V....................
A/South Africa/R2490/2015   ..................................................................N.T..........V....................
A/South Africa/R3949/2015   ..................................................................N.T..........V....................
A/South Africa/R3776/2015   ..................................................................N.T..........V....................
A/Hong Kong/11713/2015      ..................................................................N.T..........V....................

                                    410       420       430       440       450       460
                            ....|....|....|....|....|....|....|....
A/Perth/16/2009             GNRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTSGTYGTGSWPDGADINLMPI
A/Johannesburg/114/2011     ...............................................................L.....
A/Victoria/361/2011         .D.............................................................L.....
A/Texas/50/2012             .D.............................................................L.....
A/Samara/73/2013            .D........................T....................................L.....
A/Stockholm/6/2014          .D.............................................................L.....
A/Niigata-C/44/2015         .D.............................................................L.....
A/Switzerland/9715293/2013  .D.............................................................L.....
A/Vladivostok/36/2015       .D.............................................................L.....
A/Estonia/93283/2015        .D.............................................................L.....
A/Victoria/511/2015         .D.............................................................L.....
A/Poland/1900/2015          .D.............................................................L.....
A/Netherlands/525/2014      .D...............................N.............................L.....
A/Hong Kong/146/2013        .D.............................................................L.....
A/Hong Kong/4801/2014       .D.............................................................L.....
A/Hong Kong/5738/2014       .D.............................................................L.....
A/Moscow/100/2015           .D.............................................................L.....
A/Ukraine/6809/2015         .D.............................................................L.....
A/Cameroon/15V-3538/2015    .D.............................................................L.....
A/South Africa/R3512/2015   .D.............................................................L.....
A/Nagasaki/12/2015          .D.............................................................L.....
A/Mauritius/I-240/2015      .D.............................................................L.....
A/Bulgaria/243/2015         .D.............................................................L.....
A/Norway/2565/2015          .D.............................................................L.....
A/Brisbane/82/2015          .D.............................................................L.....
A/South Africa/R2665/2015   .D.............................................................L.....
A/South Africa/R2490/2015   .D.............................................................L.....
A/South Africa/R3949/2015   .D.............................................................L.....
A/South Africa/R3776/2015   .D.............................................................L.....
A/Hong Kong/11713/2015      .D.............................................................L.....
```
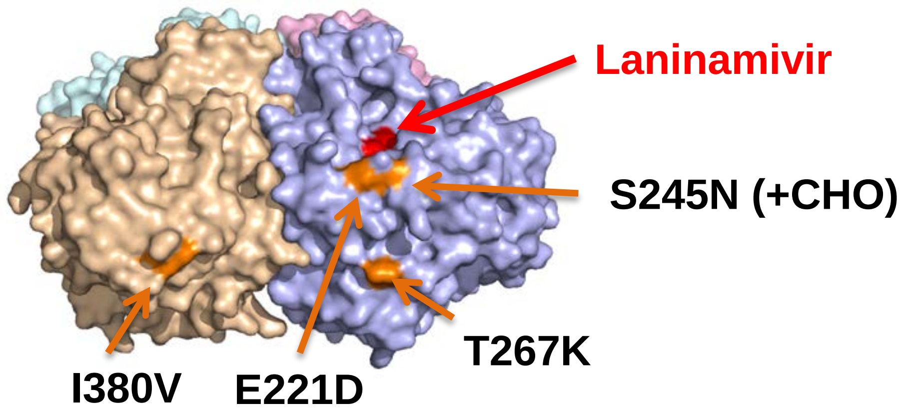
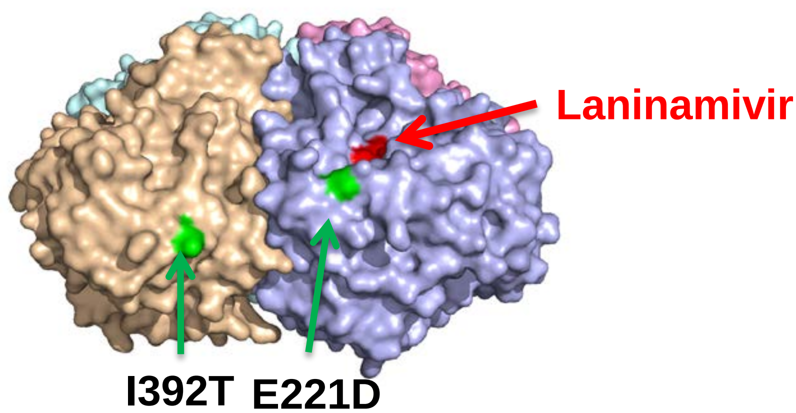

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

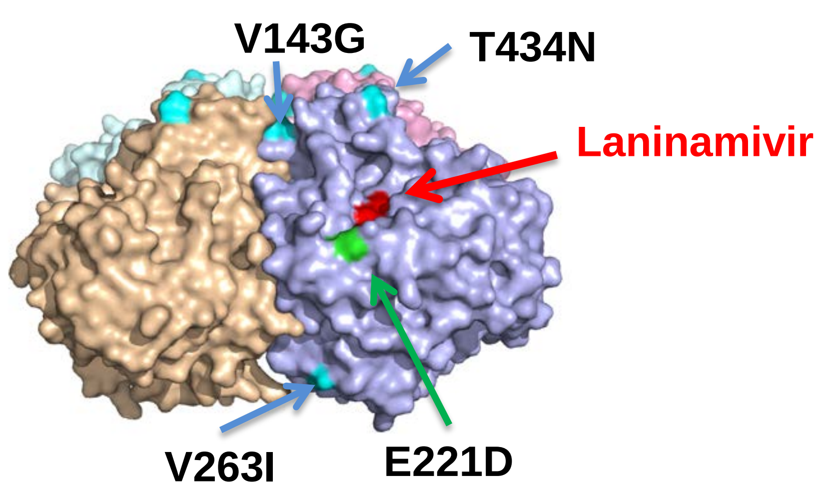
Figure 12. N2 NA amino acid substitutions in selected reference viruses.



HA group 3C.2a: A/Hong Kong/7127/2014



HA group 3C.3a: A/Switzerland/9715293/2013
HA group 3C.2a: A/Hong Kong/4801/2014



HA group 3C.3b: A/Netherlands/525/2014

Influenza B virus analyses.

Influenza B viruses represented one third of the samples received with collection dates after 2015-01-31 (**Table 1**). B/Yamagata lineage viruses predominated over those of the B/Victoria lineage at a ratio of approximately 16:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from Finland, France, Ghana, Norway, Russia, South Africa and Hong Kong SAR. The collection details and summary results are shown in **Table 10**.

HI results for the propagated viruses are shown in **Tables 11-1 to 11-3**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 11-2** and **11-3** are those with collection dates after 2015-01-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees presented here are highlighted. The results for two egg-propagated viruses received from Australia are shown in **Table 11-3**. All cell-propagated test viruses showed ≥8-fold reductions in HI titres compared to titres with the homologous virus with post-infection ferret antisera raised against the recommended vaccine virus for quadrivalent vaccines, B/Brisbane/60/2008. Similarly, the cell-propagated test viruses were poorly recognised by post-infection ferret antisera raised against egg-propagated reference viruses: B/Malta/636714/2011, B/Johannesburg/3964/2012 and B/South Australia/81/2012. However, a group of six viruses from Cameroon and Ghana collected in November and December 2014 were recognised well by the antiserum raised against egg-propagated B/South Australia/81/2012. In contrast, almost all of the test viruses showed reactivity within 4-fold of the titres with the corresponding homologous virus for antisera raised against viruses genetically closely related to B/Brisbane/60/2008 but propagated in cells. These antisera were raised against B/Paris/1762/2009, B/Hong Kong/514/2009, B/Odessa/3886/2010 and B/Formosa/V2367/2012, viruses that are considered to be surrogate cell-propagated antigens representing the egg-propagated B/Brisbane/60/2008 prototype strain. All antisera recognised all the test viruses at titres within 4-fold of the titre for the homologous virus, with the exception that the antiserum raised against B/Formosa/V2367/2012 failed to recognise four viruses within 4-fold of the homologous titre. Moreover, it is notable that a group of viruses from Cameroon collected in July, August, September and November 2014 and a virus from Zambia collected in September 2014 showed generally poorer recognition by these antisera raised against cell-propagated viruses.

Phylogenetic analysis of the HA gene of representative B/Victoria lineage viruses is shown in **Figure 13** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 14**. The HA and NA genes of the analysed viruses fell into genetic group 1A, the B/Brisbane/60/2008 genetic group. **Figure 18** shows a phylogenetic tree for the NA gene and **Figure 19** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins.

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from countries in Europe, Africa (Ghana, Senegal and South Africa), the Indian Ocean (Madagascar, Mauritius), the Middle East (Iraq and Jordan) and Hong Kong SAR. The collection details and results are summarised in **Table 12**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 13-1** to **13-17**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 13-4** to **13-17** are viruses with collection dates after 2015-01-31. The clade into which the HA falls is shown, where known, and viruses for which gene sequences are included in phylogenetic trees are highlighted. A summary of the results is shown in **Table 3-18**.

Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013, recommended for use in the Southern Hemisphere for 2015 and in the Northern Hemisphere for 2015/2016, recognised approximately 99% of viruses collected after 2015-01-31 at titres within 4-fold of the titre with the homologous virus, 94% within 2-fold. However, the ferret antiserum raised against a cell-propagated cultivar of B/Phuket/3073/2013 recognised only 63% of viruses collected since 2015-01-31 at titres within 4-fold of its titre with the homologous virus, only 37% within 2-fold of the homologous titre. These antisera recognised the test viruses better than an antiserum raised against the previously recommended vaccine virus, egg-propagated B/Massachusetts/02/2012, which only recognised 40% of the test viruses at titres within 4-fold of the titre of the antiserum for the homologous virus. However, the antiserum raised against cell culture-propagated B/Massachusetts/02/2012 recognised a similar proportion of the test viruses at a titre within 4-fold and within 2-fold of its homologous titre as did the antiserum raised against the cell culture-propagated cultivar of B/Phuket/3073/2013 compared with its homologous titre. Test viruses were recognised somewhat better by an antiserum raised against cell culture-propagated B/Estonia/55669/2011, another virus from the B/Massachusetts/02/2012 clade (clade 2), with 76% of viruses being recognised by the antiserum at a titre within 4-fold of the titre of the antiserum for the homologous virus (51% within 2-fold of the homologous titre).

Antisera raised against other viruses from the B/Phuket/3073/2013 clade (clade 3) propagated in eggs, those raised against B/Wisconsin/1/2010, B/Stockholm/11/2011 and B/Hong Kong/3417/2014, also recognised the test viruses well, recognising 99%, 90% and 100%, respectively, of the test viruses at titres within 4-fold of the titres of the antisera for their homologous viruses, and 90%, 89% and 100%, respectively, within 2-fold of the homologous titres.

Figure 17 shows a phylogenetic analysis of the HA gene of representative B/Yamagata lineage viruses and an amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 18**. **Figure 19** shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 20** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. All HA and NA genes of recently collected viruses fell into the B/Phuket/3073/2013 clade (clade 3).

Antiviral Analysis.

Sensitivity to the neuraminidase inhibitors zanamivir and oseltamivir was assessed for 271 influenza B viruses collected after 2015-01-31 (**Table 14**). One virus, B/Ukraine/6295/2015 showed reduced inhibition to oseltamivir associated with D197N amino acid substitution in the NA. Another virus showed D197G amino acid substitution in NA, B/Extremadura/728/2015, but it was inhibited within the normal range by both oseltamivir and zanamivir.

Conclusion.

Viruses of the B/Yamagata lineage have predominated. Most viruses of the B/Victoria lineage were antigenically similar to viruses closely related to B/Brisbane/60/2008. Viruses of the B/Yamagata lineage fell into genetic clade 3, the B/Phuket/3073/2013 genetic clade, and were recognised well by antisera raised against egg-propagated vaccine/reference viruses from genetic clade 3.

Table 10. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2015-01-31

MONTH / Country	B Victoria lineage				
	Number received[1]	Number propagated[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
FEBRUARY					
Ghana	1	1	1		
Russia	2	2	2		
MARCH					
Finland	1	1	1		
France	2	2	2		
Norway	1	1	1		
APRIL					
Finland	1	1	1		
Norway	2	2	2		
Russia	1	1	1		
South Africa	1	1	1		
MAY					
Hong Kong	1	1	1		
JUNE					
Hong Kong	1	1	1		
6 Countries	14	14	14	0	0
			100.0%	0.0%	0.0%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay

3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

Table 11-1. Antigenic analyses of influenza B/Victoria lineage viruses (2015-02-18 + 2015-03-04 + 2015-03-11)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post-infection ferret antisera									
				B/Bris[1,3] 60/08 Sh 522	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F22/12	B/Paris[2] 1762/09 F07/11	B/HK[2] 514/09 F19/13	B/Odessa[2] 3886/10 F19/11	B/Malta[2] 636714/11 F33/11	B/Jhb[2] 3964/12 F01/13	B/For[4] V2367/12 F04/13	B/Sth Aus[2] 81/12 F41/13
	Genetic group			1A		1A	1A	1B	1B	1A	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	640	640	40	<	20	<	80	160	40	160
B/Brisbane/60/2008	1A	2008-08-04	E4/E5	5120	160	640	80	160	80	320	640	320	1280
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	<	10	80	80	80	10	40	40	80
B/Hong Kong/514/2009	1B	2009-10-11	MDCK3	2560	<	10	80	160	160	10	20	40	80
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK2	2560	<	10	40	40	80	10	40	40	80
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	160	40	80	40	320	320	160	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	160	320	160	1280	1280	1280	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	40	160	80	80	40	160	160	160	640
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	80	320	80	80	80	320	320	320	640
TEST VIRUSES													
B/Ghana/DILI-14-0572/2014	1A	2014-06-16	C1/MDCK1	2560	10	<	160	160	80	20	40	80	160
B/Cameroon/14V-5323 GVFI/2014	1A	2014-07-11	MDCK1/MDCK1	640	<	10	10	40	10	10	<	10	20
B/Cameroon/14V-5325 GVFI/2014		2014-07-12	MDCK1/MDCK1	640	<	10	10	40	10	<	<	10	20
B/Cameroon/14V-5326 GVFI/2014		2014-07-14	MDCK1/MDCK1	640	<	10	10	40	10	<	<	20	20
B/Cameroon/14V-5328 GVFI/2014	1A	2014-07-15	MDCK1/SIAT1	320	<	10	<	80	10	<	<	10	40
B/Cameroon/14V-5329 GVFI/2014		2014-07-15	MDCK1/MDCK1	640	<	10	10	40	10	<	<	20	20
B/Cameroon/14V-5330 GVFI/2014		2014-07-15	MDCK1/MDCK1	640	<	10	10	40	10	10	<	20	20
B/Cameroon/14V-5327 GVFI/2014	1A	2014-07-16	MDCK1/MDCK1	1280	<	10	80	80	80	10	40	40	80
B/Cameroon/14V-5097/2014	1A	2014-07-24	MDCK2/MDCK1	640	<	10	10	40	10	10	<	20	20
B/Cameroon/14V-5847 GVFI/2014		2014-08-16	MDCK1/MDCK2	640	<	10	10	40	10	<	<	10	10
B/Cameroon/14V-5846 GVFI/2014	1A	2014-08-18	MDCK1/MDCK1	640	<	10	<	40	10	<	<	10	10
B/Cameroon/14V-6241 GVFI/2014		2014-08-25	MDCK1/MDCK1	1280	<	10	40	40	40	10	20	40	40
B/Ghana/DILI-14-0818/2014	1A	2014-09-10	C2/MDCK1	1280	10	<	80	80	80	20	<	40	80
B/Ghana/DILI-14-0932/2014		2014-09-10	C2/MDCK1	1280	10	20	80	160	80	20	40	80	80
B/Zambia/01-133/2014	1A	2014-09-17	MDCK2/MDCK1	640	<	10	<	80	20	<	<	20	20
B/Ghana/DILI-14-0948/2014		2014-09-25	C2/MDCK1	1280	<	10	160	160	80	20	40	40	80
B/Ghana/FS-14-0820/2014	1A	2014-09-29	C2/MDCK1	1280	<	10	40	160	80	10	20	40	80
B/Cameroon/14V-7666/2014		2014-10-23	MDCK1	640	<	20	40	160	80	10	40	40	80
B/Cameroon/14V-7601/2014		2014-10-28	MDCK1	640	<	10	40	80	80	10	20	40	80
B/Cameroon/14V-7878/2014	1A	2014-11-03	MDCK1	1280	<	20	80	160	160	80	40	80	80
B/Cameroon/14V-7876/2014		2014-11-03	MDCK1	320	<	10	<	40	10	<	<	20	20
B/Cameroon/14V-8169/2014	1A	2014-11-11	MDCK1	1280	<	40	10	320	40	10	40	40	40
B/Cameroon/14V-8134/2014		2014-11-11	MDCK1	640	<	20	40	160	80	10	40	40	80
B/Cameroon/14V-8130/2014		2014-11-11	MDCK1	2560	<	20	40	160	80	10	40	40	80
B/Cameroon/14V-8349/2014		2014-11-12	MDCK1	1280	<	40	80	320	160	40	80	80	160
B/Cameroon/14V-8351/2014		2014-11-13	MDCK1	1280	<	40	80	160	160	20	80	80	160
B/Ghana/FS-14-1023/2014	1A	2014-11-26	MDCK1	1280	10	20	40	160	80	20	160	40	320
B/Ghana/DILI-14-1147/2014	1A	2014-12-11	MDCK2	1280	<	20	40	160	80	20	160	40	320
B/Ghana/FS-14-1049/2014	1A	2014-12-12	MDCK2	1280	<	20	40	80	80	20	160	40	320
B/Ghana/FS-14-1053/2014		2014-12-14	MDCK1	1280	<	10	40	80	80	10	80	40	160

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. <=<20

Vaccine*

Sequences in phylogenetic tree

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 11-2. Antigenic analyses of influenza B/Victoria lineage viruses
(2015-06-10 + 2015-06-17 + 2015-06-24 + 2015-06-30 + 2015-07-01 + 2015-07-15 + 2015-07-29)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre Post-infection ferret antisera B/Bris[1,3] 60/08 Sh 522 1A	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F22/12 1A	B/Paris[2] 1762/09 F07/11 1A	B/HK[2] 514/09 F19/13 1B	B/Odessa[2] 3886/10 F13/15 1B	B/Malta[2] 636714/11 F33/11 1A	B/Jhb[2] 3964/12 F01/13 1A	B/For[4] V2367/12 F04/13 1A	B/Sth Aus[2] 81/12 F41/13 1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	640	640	80	<	20	10	80	160	80	160
B/Brisbane/60/2008	1A	2008-08-04	E4/E5	1280	160	640	80	160	160	640	640	320	1280
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	1280	10	20	80	160	160	20	40	80	160
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	640	<	20	80	80	160	10	40	40	80
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	1280	10	20	80	80	160	20	40	80	160
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	80	160	40	80	80	320	320	160	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	160	320	160	1280	1280	1280	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	80	320	80	80	160	160	160	320	1280
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	80	640	80	160	160	320	320	320	1280
TEST VIRUSES													
B/Moldova/049.05/2015	1A	2015-01-28	MDCK1/MDCK1	1280	10	<	80	80	320	20	40	80	80
B/Ghana/DILI-15-093/2015	1A	2015-01-29	C2/MDCK1	320	20	40	40	40	80	40	80	80	160
B/Krasnoyarsk/1/2015	1A	2015-02-16	C1/MDCK1	640	10	<	80	80	80	<	<	40	40
B/Ghana/DILI-15-174/2015	1A	2015-02-26	Cx/MDCK1	160	<	10	40	40	80	10	10	40	80
B/Ryazan/2/2015	1A	2015-02-26	C1/MDCK1	1280	20	<	80	80	160	10	<	80	160
B/Norway/1570/2015	1A	2015-03-06	MDCK1/MDCK3	1280	<	<	80	80	80	<	<	<	40
B/Finland/507/2015	1A	2015-03-31	MDCK2/MDCK1	1280	20	20	80	80	320	20	40	80	80
B/Moscow/113/2015	1A	2015-04-08	MDCK1/MDCK1	640	10	<	80	80	80	10	<	40	160
B/Finland/530/2015	1A	2015-04-20	MDCK1/MDCK1	1280	10	20	160	160	320	20	40	80	80
B/Norway/2102/2015	1A	2015-04-20	MDCK1	640	<	20	80	80	80	10	<	80	80
B/Norway/2179/2015	1A	2015-04-30	MDCK1	1280	20	40	160	320	160	40	80	80	160
B/Hong Kong/1183/2015	1A	2015-05-28	MDCK1/MDCK1	1280	20	10	80	160	160	20	<	80	40
B/Hong Kong/1337/2015	1A	2015-06-11	MDCK1/MDCK1	1280	20	10	80	160	160	20	<	80	40

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. <=<20

Vaccine*

Sequences in phylogenetic tree

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 11-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2015-08-11 + 2015-09-09

Viruses	Genetic group	Collection date	Passage History	B/Bris[1,3] 60/08 Sh 522	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F22/12	B/Paris[2] 1762/09 F07/11	B/HK[2] 514/09 F19/13	B/Odessa[2] 3886/10 F13/15	B/Malta[2] 636714/11 F33/11	B/Jhb[2] 3964/12 F01/13	B/For[4] V2367/12 F04/13	B/Sth Aus[2] 81/12 F41/13
				Haemagglutination inhibition titre									
				Post-infection ferret antisera									
	Genetic group			1A		1A	1A	1B	1B	1A	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	640	640	40	<	10	10	80	160	40	160
B/Brisbane/60/2008	1A	2008-08-04	E4/E5	1280	80	320	40	80	160	320	320	320	640
B/Paris/1762/2009	1A	2009-02-09	C2/MDCK2	1280	10	40	160	160	320	20	40	160	160
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	640	<	10	80	80	160	10	20	40	80
B/Odessa/3886/2010	1B	2010-03-19	C2/MDCK2	1280	10	20	80	80	160	20	40	80	160
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	80	160	40	80	80	320	320	160	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	2560	320	640	80	160	160	640	640	640	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	40	160	40	40	160	160	160	320	320
B/South Australia/81/2012	1A	2012-11-28	E4/E1	1280	80	320	40	80	160	320	320	320	640
TEST VIRUSES													
B/Alsace/1443/2015	1A	2015-03-09	MDCK1/MDCK1	1280	<	<	80	80	320	20	40	80	80
B/Paris/1566/2015	1A	2015-03-23	MDCK1/MDCK1	1280	<	20	80	80	320	20	40	80	80
B/Victoria/502/2015	1A	2015-03-31	MDCK1/MDCK1	640	<	<	80	40	160	<	<	80	40
B/Victoria/502/2015	1A	2015-03-31	E3/E1	320	40	80	40	40	40	80	160	80	320
B/Brisbane/46/2015	1A	2015-04-28	E4/E1	1280	80	320	80	160	160	320	320	320	1280
B/South Africa/R1588/2015	1A	2015-04-30	MDCK1/MDCK1	1280	10	10	160	80	320	20	40	160	160

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. <=<20

Vaccine*

Sequences in phylogenetic tree

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Figure 13. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

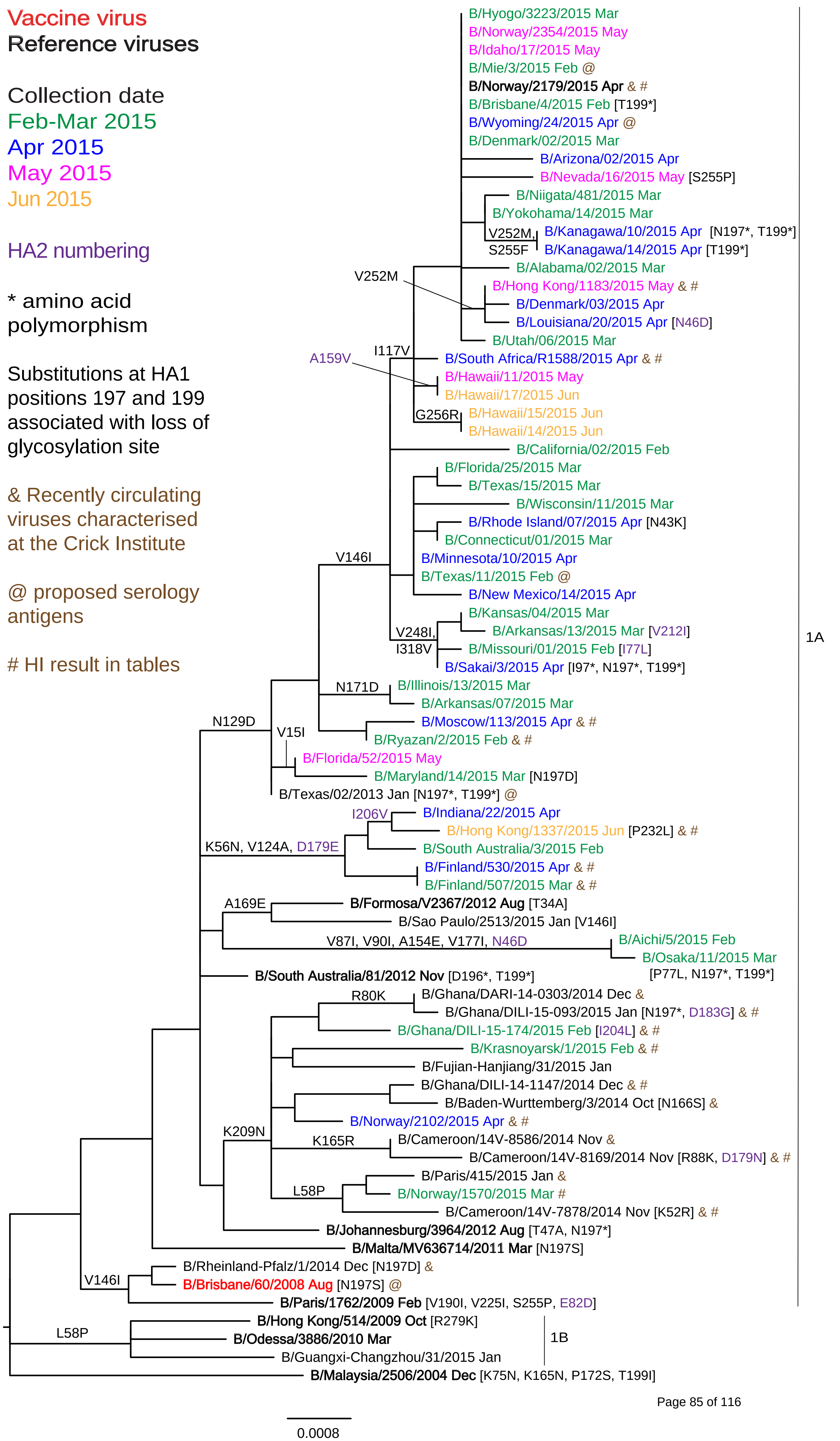


Figure 14. HA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                      10        20        30        40        50        60        70        80        90       100
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSHFANLKGTETRGKLCPKCLNCTDLDVALGRPKCTGNIPSARVSILHEVRPVTSGCFPIMHD
B/Hong Kong/514/2009          .........................................................P................K......................... 1B
B/Odessa/3886/2010            .........................................................P................K......................... 1B
B/Guangxi-Changzhou/31/2015   .........................................................P................K......................... 1B
B/Brisbane/60/2008            ..........................................................................K......................... 1A
B/Paris/1762/2009             ..........................................................................K......................... 1A
B/Malta/MV636714/2011         ..........................................................................K......................... 1A
B/Johannesburg/3964/2012      ..............................................A...........................K......................... 1A
B/Norway/1570/2015            .........................................................P................K......................... 1A
B/Cameroon/14V-8169/2014      ..........................................................................K............K............ 1A
B/Norway/2102/2015            ..........................................................................K......................... 1A
B/Krasnoyarsk/1/2015          ..........................................................................K.....................M... 1A
B/Ghana/DILI-15-093/2015      ..........................................................................K....K.................... 1A
B/South Australia/81/2012     ..........................................................................K......................... 1A
B/Formosa/V2367/2012          .................................A........................................K......................... 1A
B/Osaka/11/2015               ..........................................................................K.L.........I..I.......... 1A
B/Finland/530/2015            .......................................................N..................K......................... 1A
B/South Australia/3/2015      .......................................................N..................K......................... 1A
B/Hong Kong/1337/2015         .......................................................N..................K......................... 1A
B/Texas/02/2013               ..........................................................................K......................... 1A
B/Florida/52/2015             ..............I...........................................................K......................... 1A
B/Moscow/113/2015             ..........................................................................K......................... 1A
B/Arkansas/07/2015            ..........................................................................K......................... 1A
B/Sakai/3/2015                ..........................................................................K.....................X... 1A
B/Rhode Island/07/2015        ..........................................K...............................K......................... 1A
B/Hawaii/15/2015              ..........................................................................K......................... 1A
B/Hawaii/11/2015              ..........................................................................K......................... 1A
B/Hong Kong/1183/2015         ..........................................................................K......................... 1A
B/Kanagawa/14/2015            ..........................................................................K......................... 1A
B/Norway/2179/2015            ..........................................................................K......................... 1A

                                     110       120       130       140       150       160       170       180       190       200
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          RTKIRQLPNLLRGYEHIRLSTHNVINAENAPGGPYKIGTSGSCPNVTNGNGFFATMAWAVPKNDNNKTATNSLTIEVPYICTEGEDQITVWGFHSDNEIQ
B/Hong Kong/514/2009          ................................................................K......P..........................T.
B/Odessa/3886/2010            ................................................................K......P..........................T.
B/Guangxi-Changzhou/31/2015   ................................................................K......P..........................T.
B/Brisbane/60/2008            .............................................I..................K......P........................S.T.
B/Paris/1762/2009             .............................................I..................K......P.................I........T.
B/Malta/MV636714/2011         ................................................................K......P........................S.T.
B/Johannesburg/3964/2012      ................................................................K......P........................X.T.
B/Norway/1570/2015            ................................................................K......P..........................T.
B/Cameroon/14V-8169/2014      ................................................................R......P..........................T.
B/Norway/2102/2015            ................................................................K......P..........................T.
B/Krasnoyarsk/1/2015          ................................................................K......P..........................T.
B/Ghana/DILI-15-093/2015      ................................................................K......P........................X.T.
B/South Australia/81/2012     ................................................................K......P.......................X..X.
B/Formosa/V2367/2012          ................................................................K...E..P..........................T.
B/Osaka/11/2015               .....................................................E..........K......P....I...................X.X.
B/Finland/530/2015            .......................A........................................K......P..........................T.
B/South Australia/3/2015      .......................A........................................K......P..........................T.
B/Hong Kong/1337/2015         .......................A........................................K......P..........................T.
B/Texas/02/2013               ............................D...................................K......P........................X.X.
B/Florida/52/2015             ............................D...................................K......P..........................T.
B/Moscow/113/2015             ............................D...................................K......P..........................T.
B/Arkansas/07/2015            ............................D...................................K.....DP..........................T.
B/Sakai/3/2015                ............................D................I..................K......P........................X.X.
B/Rhode Island/07/2015        ............................D................I..................K......P..........................T.
B/Hawaii/15/2015              ................V...........D................I..................K......P..........................T.
B/Hawaii/11/2015              ................V...........D................I..................K......P..........................T.
B/Hong Kong/1183/2015         ................V...........D................I..................K......P..........................T.
B/Kanagawa/14/2015            ................V...........D................I..................K......P..........................X.
B/Norway/2179/2015            ................V...........D................I..................K......P..........................T.

                                     210       220       230       240       250       260       270       280       290       300
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          MAKLYGDSKPQKFTSSANGVTTHYVSQIGGFPNQTEDGGLPQSGRIVVDYMVQKSGKTGTITYQRGILLPQKVWCASGRSKVIKGSLPLIGEADCLHEKY
B/Hong Kong/514/2009          ..............................................................................K.....................
B/Odessa/3886/2010            ....................................................................................................
B/Guangxi-Changzhou/31/2015   ....................................................................................................
B/Brisbane/60/2008            ....................................................................................................
B/Paris/1762/2009             ........................I.............................P.............................................
B/Malta/MV636714/2011         ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Norway/1570/2015            ........N...........................................................................................
B/Cameroon/14V-8169/2014      ........N...........................................................................................
B/Norway/2102/2015            ........N...........................................................................................
B/Krasnoyarsk/1/2015          ........N................................................A..........................................
B/Ghana/DILI-15-093/2015      ........N...........................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/Osaka/11/2015               ....................................................................................................
B/Finland/530/2015            ....................................................................................................
B/South Australia/3/2015      ....................................................................................................
B/Hong Kong/1337/2015         ...............................L....................................................................
B/Texas/02/2013               ....................................................................................................
B/Florida/52/2015             ....................................................................................................
B/Moscow/113/2015             ....................................................................................................
B/Arkansas/07/2015            ....................................................................................................
B/Sakai/3/2015                ...............................................I....................................................
B/Rhode Island/07/2015        ....................................................................................................
B/Hawaii/15/2015              .......................................................R............................................
B/Hawaii/11/2015              ....................................................................................................
B/Hong Kong/1183/2015         ...................................................M................................................
B/Kanagawa/14/2015            ...................................................M..F.............................................
B/Norway/2179/2015            ....................................................................................................
```

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                                      310       320       330       340       350       360       370       380       390       400
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         GGLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLN
B/Hong Kong/514/2009         ....................................................................................................
B/Odessa/3886/2010           ....................................................................................................
B/Guangxi-Changzhou/31/2015  ....................................................................................................
B/Brisbane/60/2008           ....................................................................................................
B/Paris/1762/2009            ....................................................................................................
B/Malta/MV636714/2011        ....................................................................................................
B/Johannesburg/3964/2012     ....................................................................................................
B/Norway/1570/2015           ....................................................................................................
B/Cameroon/14V-8169/2014     ....................................................................................................
B/Norway/2102/2015           ....................................................................................................
B/Krasnoyarsk/1/2015         ....................................................................................................
B/Ghana/DILI-15-093/2015     ....................................................................................................
B/South Australia/81/2012    ....................................................................................................
B/Formosa/V2367/2012         ....................................................................................................
B/Osaka/11/2015              ............................................................................................D.......
B/Finland/530/2015           ....................................................................................................
B/South Australia/3/2015     ....................................................................................................
B/Hong Kong/1337/2015        ....................................................................................................
B/Texas/02/2013              ....................................................................................................
B/Florida/52/2015            ....................................................................................................
B/Moscow/113/2015            ....................................................................................................
B/Arkansas/07/2015           ....................................................................................................
B/Sakai/3/2015               .................V..................................................................................
B/Rhode Island/07/2015       ....................................................................................................
B/Hawaii/15/2015             ....................................................................................................
B/Hawaii/11/2015             ....................................................................................................
B/Hong Kong/1183/2015        ....................................................................................................
B/Kanagawa/14/2015           ....................................................................................................
B/Norway/2179/2015           ....................................................................................................

                                      410       420       430       440       450       460       470       480       490       500
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         SLSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIA
B/Hong Kong/514/2009         ....................................................................................................
B/Odessa/3886/2010           ....................................................................................................
B/Guangxi-Changzhou/31/2015  ....................................................................................................
B/Brisbane/60/2008           ....................................................................................................
B/Paris/1762/2009            ............................D.......................................................................
B/Malta/MV636714/2011        ....................................................................................................
B/Johannesburg/3964/2012     ....................................................................................................
B/Norway/1570/2015           ....................................................................................................
B/Cameroon/14V-8169/2014     ....................................................................................................
B/Norway/2102/2015           ....................................................................................................
B/Krasnoyarsk/1/2015         ....................................................................................................
B/Ghana/DILI-15-093/2015     ....................................................................................................
B/South Australia/81/2012    ....................................................................................................
B/Formosa/V2367/2012         ....................................................................................................
B/Osaka/11/2015              ....................................................................................................
B/Finland/530/2015           ....................................................................................................
B/South Australia/3/2015     ....................................................................................................
B/Hong Kong/1337/2015        ....................................................................................................
B/Texas/02/2013              ....................................................................................................
B/Florida/52/2015            ....................................................................................................
B/Moscow/113/2015            ....................................................................................................
B/Arkansas/07/2015           ....................................................................................................
B/Sakai/3/2015               ............................X................................................................X......
B/Rhode Island/07/2015       ....................................................................................................
B/Hawaii/15/2015             ....................................................................................................
B/Hawaii/11/2015             ....................................................................................................
B/Hong Kong/1183/2015        ....................................................................................................
B/Kanagawa/14/2015           ....................................................................................................
B/Norway/2179/2015           ....................................................................................................

                                      510       520       530       540       550       560       570
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         AGTFDAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMIAIFVVYMVSRDNVSCSICL
B/Hong Kong/514/2009         ......................................................................
B/Odessa/3886/2010           ......................................................................
B/Guangxi-Changzhou/31/2015  ......................................................................
B/Brisbane/60/2008           ......................................................................
B/Paris/1762/2009            ......................................................................
B/Malta/MV636714/2011        ......................................................................
B/Johannesburg/3964/2012     ......................................................................
B/Norway/1570/2015           ......................................................................
B/Cameroon/14V-8169/2014     .........................N............................................
B/Norway/2102/2015           ......................................................................
B/Krasnoyarsk/1/2015         ......................................................................
B/Ghana/DILI-15-093/2015     .............................G........................................
B/South Australia/81/2012    ......................................................................
B/Formosa/V2367/2012         ......................................................................
B/Osaka/11/2015              ......................................................................
B/Finland/530/2015           .........................E............................................
B/South Australia/3/2015     .........................E............................................
B/Hong Kong/1337/2015        .........................E..........................V.................
B/Texas/02/2013              ......................................................................
B/Florida/52/2015            ......................................................................
B/Moscow/113/2015            ......................................................................
B/Arkansas/07/2015           ......................................................................
B/Sakai/3/2015               ......................................................................
B/Rhode Island/07/2015       ......................................................................
B/Hawaii/15/2015             ......................................................................
B/Hawaii/11/2015             .....V................................................................
B/Hong Kong/1183/2015        ......................................................................
B/Kanagawa/14/2015           ......................................................................
B/Norway/2179/2015           ......................................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 15. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

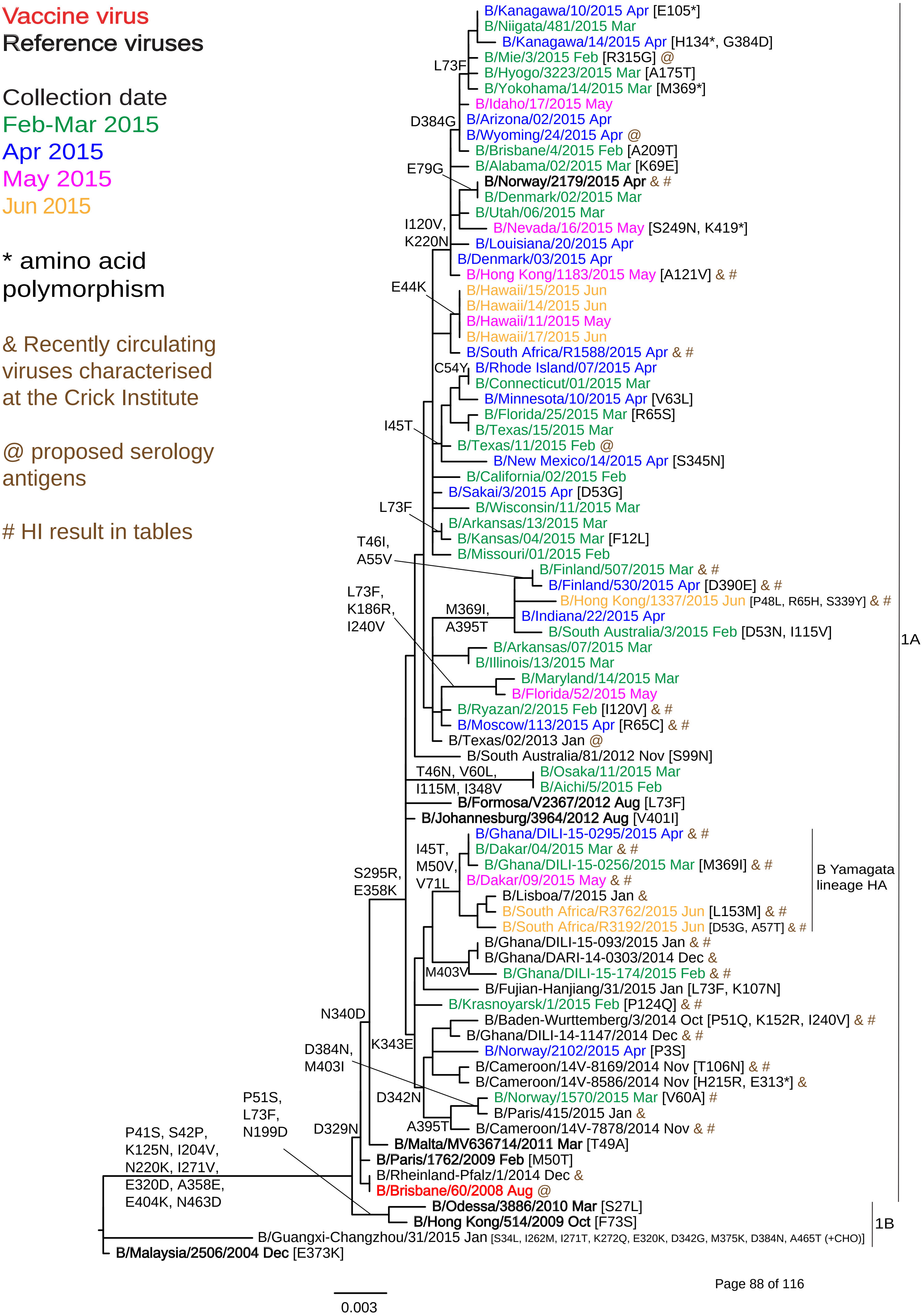

Figure 16. NA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                     10        20        30        40        50        60        70        80        90       100
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFPSTEITAPTMPLDCANASNVQAVNRSATKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Hong Kong/514/2009         ........................................SP........S.....................S........................... 1B
B/Odessa/3886/2010           ..........................L.............SP........S.....................F........................... 1B
B/Guangxi-Changzhou/31/2015  .................................L.................................................................. 1B
B/Brisbane/60/2008           ........................................SP.......................................................... 1A
B/Paris/1762/2009            ........................................SP.......T.................................................. 1A
B/Malta/MV636714/2011        ........................................SP......A................................................... 1A
B/Johannesburg/3964/2012     ........................................SP.......................................................... 1A
B/Norway/1570/2015           ........................................SP.................A........................................ 1A
B/Cameroon/14V-8169/2014     ........................................SP.......................................................... 1A
B/Norway/2102/2015           ..S.....................................SP.......................................................... 1A
B/Krasnoyarsk/1/2015         ........................................SP.......................................................... 1A
B/Ghana/DILI-15-093/2015     ........................................SP.......................................................... 1A
B/South Australia/81/2012    ........................................SP........................................................N. 1A
B/Formosa/V2367/2012         ........................................SP..............................F........................... 1A
B/Osaka/11/2015              ........................................SP...N.............L........................................ 1A
B/Finland/530/2015           ........................................SP...I........V............................................. 1A
B/South_Australia/3/2015     ........................................SP..........N............................................... 1A
B/Hong Kong/1337/2015        ........................................SP.....L................H................................... 1A
B/Texas/02/2013              ........................................SP.......................................................... 1A
B/Florida/52/2015            ........................................SP..............................F........................... 1A
B/Moscow/113/2015            ........................................SP......................C................................... 1A
B/Arkansas/07/2015           ........................................SP.......................................................... 1A
B/Sakai/3/2015               ........................................SP..........G............................................... 1A
B/Rhode Island/07/2015       ........................................SP..T........Y.............................................. 1A
B/Hawaii/15/2015             ........................................SP.K........................................................ 1A
B/Hawaii/11/2015             ........................................SP.K........................................................ 1A
B/Hong Kong/1183/2015        ........................................SP.......................................................... 1A
B/Kanagawa/14/2015           ........................................SP..............................F........................... 1A
B/Norway/2179/2015           ........................................SP....................................G..................... 1A

                                    110       120       130       140       150       160       170       180       190       200
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         HRFGETKGNSAPLIIREPFIACGPKECKHFALTHYAAQPGGYYNGTRGDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDNNA
B/Hong Kong/514/2009         ........................N.........................................................................D.
B/Odessa/3886/2010           ........................N.........................................................................D.
B/Guangxi-Changzhou/31/2015  ....................................................................................................
B/Brisbane/60/2008           ........................N...........................................................................
B/Paris/1762/2009            ........................N...........................................................................
B/Malta/MV636714/2011        ........................N...........................................................................
B/Johannesburg/3964/2012     ........................N...........................................................................
B/Norway/1570/2015           ........................N...........................................................................
B/Cameroon/14V-8169/2014     .....N..................N...........................................................................
B/Norway/2102/2015           ........................N...........................................................................
B/Krasnoyarsk/1/2015         .......................QN...........................................................................
B/Ghana/DILI-15-093/2015     ........................N...........................................................................
B/South Australia/81/2012    ........................N...........................................................................
B/Formosa/V2367/2012         ........................N...........................................................................
B/Osaka/11/2015              ..............M.........N...........................................................................
B/Finland/530/2015           ........................N...........................................................................
B/South_Australia/3/2015     ..............V.........N...........................................................................
B/Hong Kong/1337/2015        ........................N...........................................................................
B/Texas/02/2013              ........................N...........................................................................
B/Florida/52/2015            ........................N............................................................R..............
B/Moscow/113/2015            ........................N...........................................................................
B/Arkansas/07/2015           ........................N...........................................................................
B/Sakai/3/2015               ........................N...........................................................................
B/Rhode Island/07/2015       ........................N...........................................................................
B/Hawaii/15/2015             ........................N...........................................................................
B/Hawaii/11/2015             ........................N...........................................................................
B/Hong Kong/1183/2015        ...................VV...N...........................................................................
B/Kanagawa/14/2015           ...................V....N........X..................................................................
B/Norway/2179/2015           ...................V....N...........................................................................

                                    210       220       230       240       250       260       270       280       290       300
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         LLKIKYGEAYTDTYHSYANNILRTQESACNCIGGNCYLMITDGSASGVSECRFLKIREGRIIKEIFPTGRIKHTEECTCGFASNKTIECACRDNSYTAKR
B/Hong Kong/514/2009         ...V...............K..................................................V.............................
B/Odessa/3886/2010           ...V...............K..................................................V.............................
B/Guangxi-Changzhou/31/2015  .............................................................M........TQ............................
B/Brisbane/60/2008           ...V...............K..................................................V.............................
B/Paris/1762/2009            ...V...............K..................................................V.............................
B/Malta/MV636714/2011        ...V...............K..................................................V.............................
B/Johannesburg/3964/2012     ...V...............K..................................................V.......................R.....
B/Norway/1570/2015           ...V...............K..................................................V.......................R.....
B/Cameroon/14V-8169/2014     ...V...............K..................................................V.......................R.....
B/Norway/2102/2015           ...V...............K..................................................V.......................R.....
B/Krasnoyarsk/1/2015         ...V...............K..................................................V.......................R.....
B/Ghana/DILI-15-093/2015     ...V...............K..................................................V.......................R.....
B/South Australia/81/2012    ...V...............K..................................................V.......................R.....
B/Formosa/V2367/2012         ...V...............K..................................................V.......................R.....
B/Osaka/11/2015              ...V...............K..................................................V.......................R.....
B/Finland/530/2015           ...V...............K..................................................V.......................R.....
B/South_Australia/3/2015     ...V...............K..................................................V.......................R.....
B/Hong Kong/1337/2015        ...V...............K..................................................V.......................R.....
B/Texas/02/2013              ...V...............K..................................................V.......................R.....
B/Florida/52/2015            ...V...............K...................V..............................V.......................R.....
B/Moscow/113/2015            ...V...............K..................................................V.......................R.....
B/Arkansas/07/2015           ...V...............K..................................................V.......................R.....
B/Sakai/3/2015               ...V...............K..................................................V.......................R.....
B/Rhode Island/07/2015       ...V...............K..................................................V.......................R.....
B/Hawaii/15/2015             ...V...............K..................................................V.......................R.....
B/Hawaii/11/2015             ...V...............K..................................................V.......................R.....
B/Hong Kong/1183/2015        ...V..................................................................V.......................R.....
B/Kanagawa/14/2015           ...V..................................................................V.......................R.....
B/Norway/2179/2015           ...V..................................................................V.......................R.....
```

```
                                    310       320       330       340       350       360       370       380       390       400
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          PFVKLNVETDTAEIRLMCTETYLDTPRPDDGSITGPCESNGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWADSDALAFSGVM
B/Hong Kong/514/2009          ...................D.....................................E..............E...........................
B/Odessa/3886/2010            ...................D.....................................E..............E...........................
B/Guangxi-Changzhou/31/2015   ...................K.....................G..............................E.K........N................
B/Brisbane/60/2008            ...................D........N............................E..............E...........................
B/Paris/1762/2009             ...................D........N............................E..............E...........................
B/Malta/MV636714/2011         ...................D........N..........D.................E..............E...........................
B/Johannesburg/3964/2012      ...................D........N..........D.................K..............E...........................
B/Norway/1570/2015            ...................D........N..........D.NE..............K..............E..........N..........T.....
B/Cameroon/14V-8169/2014      ...................D........N..........D.NE..............K..............E...........................
B/Norway/2102/2015            ...................D........N..........D.NE..............K..............E...........................
B/Krasnoyarsk/1/2015          ...................D........N..........D..E..............K..............E...........................
B/Ghana/DILI-15-093/2015      ...................D........N..........D..E..............K..............E...........................
B/South Australia/81/2012     ...................D........N..........D.................K..............E...........................
B/Formosa/V2367/2012          ...................D........N..........D.................K..............E...........................
B/Osaka/11/2015               ...................D........N..........D.......V.........K..............E...........................
B/Finland/530/2015            ...................D........N..........D.................K..........I...E................E....T.....
B/South_Australia/3/2015      ...................D........N..........D.................K..........I...E.....................T.....
B/Hong Kong/1337/2015         ...................D........N.........YD.................K..........I...E.....................T.....
B/Texas/02/2013               ...................D........N..........D.................K..............E...........................
B/Florida/52/2015             ...................D........N..........D.................K..............E...........................
B/Moscow/113/2015             ...................D........N..........D.................K..............E...........................
B/Arkansas/07/2015            ...................D........N..........D.................K..............E...........................
B/Sakai/3/2015                ...................D........N..........D.................K..............E...........................
B/Rhode Island/07/2015        ...................D........N..........D.................K..............E...........................
B/Hawaii/15/2015              ...................D........N..........D.................K..............E...........................
B/Hawaii/11/2015              ...................D........N..........D.................K..............E...........................
B/Hong Kong/1183/2015         ...................D........N..........D.................K..............E...........................
B/Kanagawa/14/2015            ...................D........N..........D.................K..............E...........................
B/Norway/2179/2015            ...................D........N..........D.................K..............E...........................

                                    410       420       430       440       450       460
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Malaysia/2506/2004          VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKETWHSAATAIYCLMGSGQLLWDTVTGVNMAL
B/Hong Kong/514/2009          ...K..........................................................D...
B/Odessa/3886/2010            ...K..........................................................D...
B/Guangxi-Changzhou/31/2015   ................................................................T.
B/Brisbane/60/2008            ...K..........................................................D...
B/Paris/1762/2009             ...K..........................................................D...
B/Malta/MV636714/2011         ...K..........................................................D...
B/Johannesburg/3964/2012      I..K..........................................................D...
B/Norway/1570/2015            ..IK..........................................................D...
B/Cameroon/14V-8169/2014      ...K..........................................................D...
B/Norway/2102/2015            ...K..........................................................D...
B/Krasnoyarsk/1/2015          ...K..........................................................D...
B/Ghana/DILI-15-093/2015      ..VK..........................................................D...
B/South Australia/81/2012     ...K..........................................................D...
B/Formosa/V2367/2012          ...K..........................................................D...
B/Osaka/11/2015               ...K..........................................................D...
B/Finland/530/2015            ...K..........................................................D...
B/South_Australia/3/2015      ...K..........................................................D...
B/Hong Kong/1337/2015         ...K..........................................................D...
B/Texas/02/2013               ...K..........................................................D...
B/Florida/52/2015             ...K..........................................................D...
B/Moscow/113/2015             ...K..........................................................D...
B/Arkansas/07/2015            ...K..........................................................D...
B/Sakai/3/2015                ...K..........................................................D...
B/Rhode Island/07/2015        ...K..........................................................D...
B/Hawaii/15/2015              ...K..........................................................D...
B/Hawaii/11/2015              ...K..........................................................D...
B/Hong Kong/1183/2015         ...K..........................................................D...
B/Kanagawa/14/2015            ...K..........................................................D...
B/Norway/2179/2015            ...K..........................................................D...
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 12. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2015-01-31

MONTH	B Yamagata lineage				
Country	Number received[1]	Number propagated[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
FEBRUARY					
Bulgaria	4	4	4		
Cyprus	1	1	1		
Estonia	2	2	2		
Georgia	6	6	6		
Iraq	1	1	1		
Italy	13	13	9	4	
Jordan	2	2	1	1	
Macedonia	3	1	1		
Madagascar	3	3	3		
Moldova	18	18	18		
Netherlands	2	2	2		
Poland	1	1	1		
Romania	3	3	3		
Russia	5	5	5		
Serbia	3	3	3		
Slovakia	2	2	2		
Slovenia	4	4	4		
Spain	11	9	9		
Sweden	2	2	2		
Ukraine	40	40	37	3	
MARCH					
Bulgaria	1	1	1		
Estonia	2	2	2		
Georgia	4	4	4		
Ghana	2	1	1		
Jordan	5	5	3	2	
Madagascar	9	9	9		
Netherlands	1	1	1		
Poland	2	2	2		
Romania	10	10	10		
Russia	5	3	3		
Slovakia	5	5	5		
Slovenia	3	3	3		
Senegal	2	2	2		
Serbia	1	1	1		
Ukraine	29	29	27	2	
APRIL					
Belarus	1	1	1		
Estonia	3	3	3		
Finland	1	1	1		
Ghana	2	2	2		
Greece	1	1	1		
Mauritius	2	2	2		
Norway	5	1			1
Russia	2	2	2		
Senegal	2	2	2		
Slovakia	1	1	1		
Slovenia	7	7	7		
Serbia	3	3	3		
Ukraine	1	1	1		
MAY					
Senegal	2	2	2		
JUNE					
Hong Kong	3	3	3		
South Africa	7	5	5		
	250	237	224	12	1
30 Countries			94.5%	5.1%	0.4%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assay
3. Reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-grown vaccine virus)

Table 13-1. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-02-18)

Viruses		Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
				Haemagglutination inhibition titre										
					Post infection-ferret antisera									
	Genetic Group			1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	1280	160	640	80	1280	160	160	20	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	640	80	320	40	640	80	80	10	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	160	160	160	320	20	320	40	160	20	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	160	80	160	10	320	40	80	20	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	80	40	40	160	160	320	40	160	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	2560	320	640	160	320	80	640	160	160	20	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	320	320	640	640	160	80	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	160	160	160	320	20	320	40	160	20	320
B/Phuket/3073/2013	3	2013-11-21	M2/M2	1280	160	160	320	320	320	320	320	640	640	320
B/Hong Kong/3417/2014	3	2014-06-04	E5/E2	320	160	160	160	160	20	160	40	80	40	320
TEST VIRUSES														
B/Lisboa/29/2014	3	2014-12-18	SIAT1/SIAT1	2560	320	320	320	640	640	320	640	2560	1280	320
B/Lisboa/23/2014	3	2014-12-23	SIAT1/SIAT1	1280	160	160	320	320	320	320	320	640	160	320
B/Lisboa/20/2014	3	2014-12-23	SIAT2/SIAT1	640	160	160	160	160	80	160	80	160	160	320
B/Lisboa/26/2014	3	2014-12-31	SIAT2/SIAT1	640	80	80	160	160	40	160	80	80	1280	640
B/Lisboa/25/2014	3	2015-01-02	SIAT1/SIAT1	1280	160	160	320	320	160	320	320	160	160	320
B/Lisboa/23/2015	3	2015-01-03	SIAT2/SIAT1	640	80	80	160	160	40	160	80	80	320	320
B/Lisboa/5/2015	3	2015-01-05	SIAT1/SIAT1	1280	160	160	320	320	320	320	320	320	640	320
B/Lisboa/6/2015	3	2015-01-06	SIAT1/SIAT1	1280	160	160	160	320	160	320	320	320	80	320
B/Lisboa/17/2015	3	2015-01-06	SIAT1/SIAT1	640	80	80	160	160	40	160	40	80	1280	320
B/Lisboa/7/2015	3	2015-01-07	SIAT1/SIAT1	1280	160	160	160	320	80	320	80	320	160	320
B/Lisboa/11/2015	3	2015-01-08	SIAT1/SIAT1	640	80	160	160	160	80	160	80	160	1280	320
B/Lisboa/12/2015	3	2015-01-09	SIAT1/SIAT1	2560	320	320	320	640	320	640	640	640	80	320
B/Niedersachsen/1/2015		2015-01-09	C2/MDCK1	640	160	160	320	320	80	320	80	160	1280	320
B/Slovenia/196/2015	3	2015-01-12	MDCK1	2560	320	320	320	640	640	320	640	640	160	320
B/Lisboa/21/2015	3	2015-01-12	SIAT1/SIAT1	640	160	160	160	320	80	320	80	160	1280	320
B/Thuringen/1/2015	3	2015-01-12	C2/MDCK1	320	80	80	80	80	40	160	40	40	160	320
B/Bayern/2/2015		2015-01-12	C2/MDCK1	320	80	80	160	160	40	160	40	80	40	160
B/Berlin/1/2015	3	2015-01-12	C1/MDCK1	320	80	80	160	160	40	160	80	80	80	160
B/Bayern/3/2015	3	2015-01-19	C2/MDCK1	320	80	80	80	80	20	160	40	40	80	320
B/Berlin/3/2015	3	2015-01-23	C2/MDCK1	640	160	160	160	160	40	160	80	80	80	320
B/Greece/462/2015	3	2015-01-27	MDCK2	640	160	160	320	320	80	160	80	160	160	320
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-2. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-02-25)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre / Post-infection ferret antisera										
				B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F16/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
				1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	160	320	80	1280	160	160	40	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	2560	320	640	160	320	40	640	160	160	20	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	320	320	320	320	20	160	40	160	20	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	160	80	160	10	80	20	80	20	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	160	80	80	160	80	320	40	40	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	2560	640	640	160	320	80	640	160	160	20	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	320	320	320	640	160	40	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	160	320	320	320	20	160	40	160	20	320
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	320	320	320	320	320	160	640	640	640	320
B/Hong Kong/3417/2014	3	2014-06-04	E5/E2	320	160	160	160	160	20	80	40	80	40	320
TEST VIRUSES														
B/Mauritius/424/2014		2014-05-20	MDCK2/MDCK1	1280	160	160	320	320	320	640	160	320	160	320
B/Mauritius/454/2014	3	2014-05-26	MDCK3/MDCK1	640	80	160	160	160	80	80	40	160	160	320
B/Mauritius/627/2014	2	2014-07-23	MDCK2/MDCK1	2560	160	320	160	320	320	640	160	320	160	160
B/Tehran/69425/2014	3	2014-09-25	MDCK1/MDCK1	640	160	160	320	320	80	80	80	320	160	320
B/Isfahan/85647/2014	3	2014-10-06	MDCK2/MDCK1	640	160	160	320	320	80	80	80	160	320	320
B/Oman /5364/2014		2014-10-12	MDCK2/MDCK1	640	160	160	320	320	160	80	80	320	80	320
B/Israel/O-5791/2014		2014-10-21	MDCKx/MDCK1	640	80	160	160	320	80	80	80	160	160	320
B/Ukraine/426/2014	3	2014-10-31	MDCKx/MDCK1	640	160	160	320	320	80	80	80	160	640	320
B/Meknes/10/2015		2014-11-11	MDCK1/MDCK1	320	80	80	160	160	80	40	80	80	1280	320
B/Jordan/20512/2014	3	2014-11-25	MDCK1/MDCK1	320	80	80	160	160	40	40	40	80	160	160
B/Madrid/15047/2014	3	2014-11-27	SIAT2/MDCK2	320	80	80	160	160	80	40	80	160	160	320
B/Meknes/35/2015	3	2014-11-27	MDCK1/MDCK1	320	80	80	160	160	80	40	80	80	160	320
B/Rabat/43/2015		2014-12-03	MDCK1/MDCK1	640	160	80	160	160	80	80	80	160	160	320
B/Agadir/68/2015	3	2014-12-12	MDCK1/MDCK1	320	80	80	160	160	40	40	40	80	1280	320
B/Sale/70/2015	3	2014-12-13	MDCK1/MDCK1	1280	160	160	320	320	320	160	320	640	640	160
B/Meknes/38/2015		2014-12-15	MDCK1/MDCK1	1280	160	160	320	320	320	160	160	320	160	320
B/Lisboa/8/2015		2015-01-05	SIAT1/SIAT3	2560	160	320	320	640	640	320	640	640	640	320
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Table 13-3. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-03-04)

				Haemagglutination inhibition titre										
				Post-infection ferret antisera										
Viruses		Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F16/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F33-34/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
	Genetic Group			1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	320	640	160	320	80	640	160	320	20	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	160	320	80	160	40	320	80	80	10	80
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	160	160	160	160	20	160	40	160	20	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	80	80	80	160	10	80	40	80	20	80
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	80	40	40	160	80	320	40	40	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	2560	640	640	160	320	40	640	160	160	20	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	320	160	320	160	320	320	160	40	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	80	160	160	160	10	160	40	160	20	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	1280	160	160	160	160	160	160	160	160	160	160
B/Hong Kong/3417/2014	3	2014-06-04	E5/E2	320	80	160	160	80	10	80	40	80	80	320
TEST VIRUSES														
B/Zambia/04-214/2014	3	2014-03-09	MDCK2/MDCK1	160	40	40	40	40	20	40	40	20	80	320
B/Mauritius/487/2014	3	2014-06-03	MDCK2/MDCK2	5120	160	160	320	320	320	320	320	160	80	160
B/Mauritius/571/2014		2014-07-03	MDCK2/MDCK2	640	80	80	160	80	160	80	80	80	160	160
B/Egypt/4223/2014	3	2014-08-02	Cx/MDCK1	640	160	160	160	80	40	80	80	80	160	160
B/Egypt/4012/2014		2014-08-04	Cx/MDCK1	320	160	80	160	160	40	80	80	80	80	80
B/Egypt/4091/2014	3	2014-08-04	Cx/MDCK1	320	160	160	160	160	40	80	80	80	<	20
B/Egypt/4984/2014		2014-08-07	Cx/MDCK1	640	160	80	160	160	40	80	80	80	40	160
B/Madagascar/2340/2014	3	2014-08-18	MDCK1/MDCK1	320	80	80	80	80	40	80	80	80	40	80
B/Egypt/4253/2014	3	2014-08-18	Cx/MDCK1	1280	640	160	320	160	40	160	80	160	80	160
B/Egypt/4255/2014		2014-08-19	Cx/MDCK1	1280	160	160	320	160	80	160	80	160	320	160
B/Egypt/4257/2014	3	2014-08-20	Cx/MDCK1	320	80	80	80	80	40	40	40	80	1280	320
B/Madagascar/2383/2014		2014-08-25	MDCK1/MDCK1	640	80	80	80	80	160	80	320	80	40	160
B/Madagascar/2384/2014	3	2014-08-25	MDCK1/MDCK1	320	80	80	160	80	40	40	80	80	80	320
B/Madagascar/2441/2014		2014-08-25	MDCK1/MDCK1	320	80	80	160	160	80	80	80	80	160	160
B/Madagascar/2439/2014	2	2014-08-28	MDCK2/MDCK1	1280	160	160	160	160	160	160	320	80	80	160
B/Egypt/4395/2014	3	2014-09-10	Cx/MDCK1	160	40	40	40	40	40	40	40	40	160	320
B/Zambia/04-241/2014	3	2014-09-22	MDCK2/MDCK1	320	80	40	80	80	20	40	40	40	160	320
B/Zambia/04-242/2014			MDCK2/MDCK1	320	40	40	80	80	40	40	40	40	80	160
B/Zambia/04-245/2014		2014-09-29	MDCK2/MDCK1	<	<	<	<	<	<	<	<	<	160	160
B/Israel/O-6350/2014	3	2014-11-19	MDCKx/MDCK2	320	40	80	160	80	80	40	80	80	80	160
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40;   2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Table 13-4. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-03-11 + 2015-03-17)

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F27/13	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
				1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	1280	320	640	160	1280	160	320	40	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	2560	320	640	160	320	160	640	160	160	20	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	320	320	320	320	40	320	40	320	40	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	160	160	320	40	160	40	160	40	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	80	80	80	640	160	320	80	80	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	1280	1280	320	640	160	640	160	320	40	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	320	640	640	640	320	160	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	160	160	160	320	40	320	40	160	40	160
B/Phuket/3073/2013	3	2013-11-21	M2/M2	1280	160	160	320	320	160	320	320	320	640	320
B/Hong Kong/3417/2014	3	2014-06-04	E5/E2	320	160	80	160	160	20	160	40	160	160	160
TEST VIRUSES														
B/Egypt/4125/2014		2014-08-25	Cx/MDCK1	640	320	80	320	80	10	160	80	160	160	320
B/Catalonia/7573S/2014	3	2014-12-22	C0/MDCK1	1280	160	160	320	320	160	320	320	320	640	320
B/Catalonia/0683410S/2014	3	2014-12-22	MDCK2	160	80	<	80	160	40	40	40	80	320	320
B/Valencia/0686198S/2015	3	2015-01-12	C0/MDCK1	1280	160	160	320	320	160	320	320	640	80	160
B/Hong Kong/8113256/2015	3	2015-01-24	MDCK1	640	80	40	160	80	40	160	40	80	1280	320
B/Hong Kong/8112624/2015	3	2015-01-25	MDCK1	2560	320	320	320	320	320	320	640	640	80	160
B/Cyprus/F72/2015	3	2015-02-06	MDCK1	160	80	<	80	160	40	40	40	80	80	160
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

Haemagglutination inhibition titre: Post-infection ferret antisera

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-5. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-03-20 + 2015-04-01 + 2015-04-09)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre: Post-infection ferret antisera										
				B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia2 55669/11 F27/13	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
	Genetic Group			1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	1280	160	640	160	1280	160	320	20	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	320	80	160	80	320	80	80	<	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	320	80	80	160	160	20	80	20	80	<	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	320	160	160	320	40	160	40	160	20	320
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	<	40	40	640	40	320	40	40	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	160	320	160	640	160	160	<	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	320	160	320	320	320	320	160	40	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	160	160	160	320	40	160	40	160	40	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	1280	160	160	320	320	160	160	160	320	320	320
B/Hong Kong/3417/2014	3	2014-06-04	E5/E2	160	80	40	80	80	20	40	40	40	20	160
TEST VIRUSES														
B/Adana/208/2015	3	2015-01-23	MDCK1	320	80	<	80	80	40	40	40	80	80	160
B/Konya/173/2015	3	2015-01-22	MDCK1	320	80	<	160	160	80	80	80	80	80	320
B/Sivas/1488/2014	3	2014-10-28	MDCK1	640	80	40	160	160	80	80	80	80	160	160
B/Diyarbakir/2634/2014	3	2014-12-18	MDCK1	320	80	<	80	80	40	20	40	80	80	160
B/Kocaeli/1850/2014	3	2014-12-23	MDCK2	160	40	<	80	80	40	20	40	80	80	160
B/Greece/5/2015		2015-01-02	MDCK2	320	80	<	80	80	40	20	40	80	80	160
B/Greece/198/2015	3	2015-01-19	MDCK2	320	80	40	80	80	40	80	40	80	80	160
B/Greece/303/2015		2015-01-21	MDCK2	640	80	80	160	80	80	160	80	160	160	160
B/Greece/323/2015	3	2015-01-22	MDCK2	160	80	40	80	40	40	80	40	80	80	160
B/Greece/403/2015	3	2015-01-24	MDCK2	640	160	80	160	160	80	160	80	160	160	320
B/Greece/404/2015		2015-01-24	MDCK2	320	80	40	80	80	40	80	40	80	80	80
B/Macedonia/19/2015	3	2015-01-28	MDCK2	320	80	40	80	80	80	160	80	80	80	160
B/Macedonia/36/2015		2015-02-11	MDCK2	640	80	80	160	160	80	80	160	160	640	160
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Table 13-6. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-04-29 + 2015-05-01)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre / Post-infection ferret antisera: B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F27/13	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
	Genetic Group			1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	160	640	160	640	160	160	20	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	2560	320	320	80	320	160	640	160	160	10	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	160	160	160	640	40	160	40	160	20	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	80	320	40	80	40	80	20	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	80	80	80	640	80	320	80	40	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	320	320	80	320	160	640	160	160	10	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	320	320	160	160	320	320	320	160	40	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	160	160	160	320	40	160	40	160	40	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	320	160	320	640	320	320	640	320	640	320
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	160	80	40	80	80	20	40	40	40	10	160
TEST VIRUSES														
B/Kocaeli/1127/2014		2014-10-17	MDCK1	640	160	80	160	160	80	80	80	160	160	320
B/Diyarbakir/1151/2014		2014-10-19	MDCK2	160	80	40	80	80	40	40	40	80	80	160
B/Tokat/1148/2014	3	2014-10-20	MDCK1	320	80	40	80	80	40	40	40	80	80	160
B/Kutahya/1232/2014	3	2014-10-22	MDCK1	640	80	80	160	160	80	80	160	160	320	320
B/Diyarbakir/1268/2014		2014-10-23	MDCK1	320	80	40	80	80	40	40	80	80	80	160
B/Adana/1261/2014		2014-10-23	MDCK1	320	80	40	80	160	40	40	80	80	80	160
B/Kocaeli/1258/2014		2014-10-23	MDCK1	320	80	40	80	80	40	40	40	80	80	160
B/Istanbul/1364/2014	3	2014-10-25	MDCK1	320	80	40	80	80	40	40	40	80	80	160
B/Erzurum/2512/2014		2014-12-03	MDCK2	160	80	40	80	80	40	40	40	80	80	160
B/Kocaeli/1839/2014		2014-12-19	MDCK1	160	80	40	80	80	40	40	40	80	80	160
B/Georgia/20/2015	3	2015-01-06	MDCK1	640	80	40	160	80	40	40	40	80	80	320
B/Extremadura/740/2015	3	2015-01-28	SIAT2/SIAT1	640	80	80	160	160	80	80	80	160	160	320
B/Extremadura/732/2015	3	2015-02-02	SIAT2/SIAT1	320	80	80	160	80	80	80	80	80	160	160
B/Georgia/132/2015		2015-02-02	MDCK1	1280	320	160	320	320	160	320	320	640	640	320
B/Extremadura/728/2015	3	2015-02-03	SIAT2/SIAT1	640	160	160	160	160	160	160	320	320	640	320
B/Extremadura/725/2015		2015-02-03	SIAT2/SIAT1	640	160	160	320	320	160	160	320	320	640	320
B/Georgia/162/2015	3	2015-02-04	MDCK1	320	80	40	160	80	40	40	40	80	80	160
B/Georgia/169/2015	3	2015-02-05	MDCK1	640	160	160	320	320	160	160	320	320	640	320
B/Iraq/478/2014	3	2014-02-05	SIAT1	640	80	80	160	160	80	80	80	160	160	160
B/Georgia/198/2015		2015-02-09	MDCK1	160	80	40	80	80	40	40	40	80	80	160
B/Bulgaria/317/2015	3	2015-02-10	SIAT1/SIAT1	640	80	80	160	160	80	80	80	160	160	160
B/Bulgaria/351/2015	3	2015-02-13	SIAT2/SIAT1	320	160	80	160	160	80	80	80	160	160	320
B/Bulgaria/354/2015	3	2015-02-13	SIAT1/SIAT1	320	80	80	160	80	80	40	80	160	160	160
B/Georgia/287/2015		2015-02-13	MDCK1	640	80	40	80	80	40	20	40	80	40	160
B/Jordan/30046/2015	3	2015-02-16	MDCK1	320	80	40	80	40	80	20	80	80	80	160
B/Georgia/330/2015	3	2015-02-18	MDCK1	320	80	80	160	160	80	80	80	160	160	320
B/Bulgaria/442/2015	3	2015-02-24	SIAT2/SIAT1	320	160	80	160	160	80	80	80	160	80	320
B/Jordan/30059/2015	3	2015-02-24	MDCK1	160	80	20	80	40	40	10	40	40	40	160
B/Georgia/446/2015	3	2015-03-02	MDCK1	320	80	80	160	160	80	80	80	160	160	320
B/Jordan/20200/2015	3	2015-03-03	MDCK1	80	80	20	80	40	40	10	20	40	40	160
B/Bulgaria/523/2015	3	2015-03-06	SIAT2/SIAT1	320	80	80	80	160	80	80	80	160	160	320
B/Jordan/20212/2015	3	2015-03-09	MDCK1	160	80	20	80	40	40	20	40	40	40	160
B/Georgia/498/2015	3	2015-03-10	MDCK1	320	80	80	160	160	80	80	80	160	160	320
B/Georgia/539/2015	3	2015-03-11	MDCK1	640	160	80	160	160	80	80	80	160	160	320
B/Jordan/10585/2015	3	2015-03-16	MDCK1	320	80	80	160	80	40	40	80	80	80	320
B/Georgia/575/2015	3	2015-03-18	MDCK1	320	80	80	160	160	80	80	80	160	160	160
B/Jordan/20237/2015	3	2015-03-20	MDCK1	320	80	40	160	80	40	40	40	80	80	160
B/Jordan/20289/2015	3	2015-03-22	MDCK1	320	80	40	80	40	80	40	80	80	160	160
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-7. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-05-13 + 2015-05-20 + 2015-05-22)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre: Post-infection ferret antisera: B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F06/15	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F05/15	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
Genetic Group				1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	320	160	160	160	640	80	160	10	160
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	320	80	80	160	320	80	80	<	80
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	160	80	160	80	40	80	10	80	10	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	640	80	40	80	160	40	80	<	80	10	80
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	<	40	20	160	40	160	40	20	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	2560	640	640	160	160	160	640	160	160	10	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	320	160	160	40	320	320	320	160	40	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	160	80	160	80	40	160	10	160	20	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	1280	160	80	320	40	320	160	320	320	320	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	160	80	40	80	20	20	40	20	40	10	160
TEST VIRUSES														
B/Jordan/30007/2015	3	2015-01-19	MDCK1	1280	320	320	320	80	640	320	1280	640	1280	320
B/Jordan/20111/2015	3	2015-01-21	MDCK1	640	160	160	160	80	160	160	80	160	160	160
B/Italy/FVG-04/2015		2015-01-22	Cx/MDCK1	320	160	80	160	40	40	40	40	160	80	160
B/Jordan/20114/2015	3	2015-01-26	MDCK1	1280	160	160	160	40	160	80	160	160	320	320
B/Valladolid/135/2015		2015-01-28	MDCK1/MDCK1	320	80	80	160	40	80	40	80	160	160	160
B/Roma/20/2015	3	2015-01-29	Cx/SIAT1	2560	160	<	320	160	40	160	80	160	160	160
B/Milano/12/2015	3	2015-02-02	MDCK2/MDCK1	160	40	<	80	20	40	40	40	80	80	80
B/Perugia/3/2015	3	2015-02-03	MDCK1/MDCK1	160	80	<	80	20	20	40	10	40	40	160
B/Parma/4/2015	3	2015-02-03	MDCK1/MDCK1	640	160	160	160	40	320	160	320	320	640	160
B/Milano/11/2015		2015-02-04	MDCK2/MDCK1	640	160	80	160	40	160	160	160	320	320	160
B/Perugia/7/2015		2015-02-05	MDCK1/MDCK1	160	40	<	80	20	40	40	40	40	80	160
B/Parma/6/2015		2015-02-09	MDCK1/MDCK1	160	80	<	80	20	40	40	40	80	160	160
B/Serbia/NS-1020/2015	3	2015-02-10	MDCK1	320	80	40	160	40	40	40	40	80	80	320
B/Palencia/225/2015	3	2015-02-11	MDCK1/MDCK1	320	80	<	80	40	40	20	80	80	160	320
B/Valladolid/224/2015		2015-02-11	MDCK1/MDCK1	640	160	160	320	80	320	320	320	640	320	320
B/Salamanca/232/2015	3	2015-02-13	MDCK1/MDCK1	160	80	<	80	40	40	20	40	80	160	320
B/Milano/24/2015		2015-02-16	MDCK2/MDCK1	320	80	<	160	40	80	80	80	160	160	160
B/Perugia/16/2015		2015-02-16	MDCK1/MDCK1	160	80	<	80	20	20	40	20	80	80	160
B/Perugia/18/2015		2015-02-16	MDCK1/MDCK1	160	80	<	80	20	20	40	10	40	40	160
B/Palencia/241/2015	3	2015-02-16	MDCK1/MDCK1	320	160	160	160	40	80	40	80	80	160	320
B/Valladolid/240/2015		2015-02-16	MDCK1/MDCK2	320	80	80	160	20	80	80	80	160	320	320
B/Milano/25/2015	3	2015-02-18	MDCK2/MDCK1	640	160	80	320	40	320	160	320	320	640	160
B/Palencia/252/2015	3	2015-02-18	MDCK1/MDCK1	320	160	<	160	40	80	80	80	160	160	320
B/Serbia/NS-1065/2015		2015-02-19	MDCK2	320	80	160	80	20	40	40	40	80	80	160
B/Italy/FVG-08/2015	3	2015-02-20	Cx/MDCK1	320	40	<	80	20	80	40	80	80	160	160
B/Perugia/22/2015	3	2015-02-23	MDCK1/MDCK1	160	40	<	80	20	20	20	40	40	40	160
B/Parma/13/2015	3	2015-02-23	MDCK1/MDCK1	2560	160	160	320	80	320	160	320	640	640	160
B/Serbia/NS-1095/2015	3	2015-02-23	MDCK1	320	80	160	80	40	40	40	40	80	80	320
B/Serbia/NS-1280/2015	3	2015-03-31	MDCK1	320	80	80	80	40	40	40	40	80	80	320
B/Serbia/NS-1298/2015		2015-04-04	MDCK2	320	160	160	160	40	40	80	40	80	80	320
B/Serbia/NS-1313/2015	3	2015-04-15	MDCK1	320	160	80	160	40	40	40	40	80	80	160
B/Serbia/NS-1315/2015		2015-04-16	MDCK2	1280	320	320	320	80	320	320	320	320	320	320

Vaccine NH2014-15 (B/Mass 02/12 Egg F42/14)

Vaccine SH2015 NH2015-16 (B/Phuket 3073/13 Egg F36/14)

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-8. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-06-03)

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia2 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
				Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	160	320	40	1280	320	160	20	160
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	320	80	160	40	640	160	80	10	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	160	160	160	320	10	160	40	80	40	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	80	320	10	160	40	80	40	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	80	40	40	80	80	640	40	80	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	2560	640	640	160	320	80	1280	320	160	40	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	320	320	160	640	640	160	80	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	160	160	160	320	10	160	40	80	40	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	1280	320	320	320	320	320	320	640	320	640	320
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	80	80	40	40	40	160	40	40	20	20	160
TEST VIRUSES														
B/Valladolid/137/2015		2015-01-28	MDCK1/MDCK1	320	80	80	160	160	40	160	160	80	320	160
B/Valladolid/138/2015	3	2015-01-28	MDCK1/MDCK1	640	80	160	160	160	160	160	320	160	640	160
B/Iasi/176858/2015		2015-01-30	MDCK1/MDCK1	160	160	160	320	320	160	320	320	320	640	160
B/Bucuresti/177398/2015	3	2015-02-09	MDCK1/MDCK1	80	80	80	160	160	80	160	160	160	320	160
B/Iasi/178240/2015		2015-02-25	MDCK1/MDCK1	80	80	80	320	320	80	320	640	320	640	320
B/Calarasi/178283/2015		2015-02-25	MDCK2/MDCK1	160	160	80	320	320	80	320	640	160	1280	320
B/Galati/178448/2015		2015-03-02	MDCK2/MDCK1	320	80	80	160	160	80	160	160	160	320	80
B/Dolj/178771/2015		2015-03-05	MDCK2/MDCK1	640	80	160	160	160	80	160	160	160	320	160
B/Dolj/179263/2015	3	2015-03-07	MDCK2/MDCK1	320	80	80	80	80	80	80	160	160	320	80
B/Constanta/179082/2015	3	2015-03-11	MDCK1/MDCK1	320	80	80	160	80	40	80	80	80	160	160
B/Teleorman/179501/2015	3	2015-03-18	MDCK1/MDCK1	320	80	80	160	160	40	160	160	80	160	160
B/Bucuresti/179506/2015		2015-03-18	MDCK1/MDCK1	640	80	160	320	160	160	320	320	160	640	320
B/Calarasi/179521/2015	3	2015-03-19	MDCK1/MDCK1	320	80	80	160	160	80	160	160	160	320	160
B/Tulcea/179584/2015	3	2015-03-19	MDCK2/MDCK1	320	80	80	160	160	80	160	160	160	320	160
B/Bucuresti/179677/2015	3	2015-03-21	MDCK1/MDCK1	320	80	80	160	80	40	80	80	80	160	160
B/Iasi/179780/2015	3	2015-03-25	MDCK1/MDCK1	320	80	80	160	160	40	80	80	80	160	160
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-9. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-06-10)

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
				Haemagglutination inhibition titre – Post-infection ferret antisera										
Genetic Group				1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	160	320	80	640	320	320	40	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	320	80	160	40	640	160	80	10	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	320	160	320	320	20	160	80	160	40	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	80	160	20	160	40	160	40	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	160	80	80	80	160	80	640	160	80	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	160	320	80	1280	160	320	20	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	320	320	160	640	640	320	160	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	320	320	320	320	20	160	80	320	80	320
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	320	320	640	640	320	640	640	640	1280	640
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	160	80	40	80	80	10	40	40	40	20	160
TEST VIRUSES														
B/Finland/489/2014	3	2014-10-31	MDCKx/MDCK1/MDCK1	320	80	40	160	80	40	40	80	160	80	160
B/Finland/527/2014		2014-10-31	MDCKx/MDCK1/MDCK1	320	160	80	160	160	40	80	80	160	160	320
B/Ukraine/6387/2015	3	2015-02-18	C1/MDCK1	160	80	40	160	160	20	40	80	160	80	320
B/Ukraine/6451/2015		2015-02-19	C2/SIAT1	1280	160	160	320	640	320	320	640	640	1280	640
B/Ukraine/6359/2015	3	2015-02-21	C1/MDCK1	160	80	40	80	80	20	40	80	80	80	320
B/Ukraine/6360/2015		2015-02-21	C1/MDCK1	160	80	40	160	80	40	40	80	160	160	320
B/Ukraine/6389/2015		2015-02-21	C1/MDCK1	640	160	160	320	320	80	320	160	640	640	640
B/Ukraine/6361/2015	3	2015-02-23	C1/MDCK1	160	80	40	80	80	10	40	40	80	40	320
B/Ukraine/6363/2015		2015-02-24	C1/MDCK1	320	80	40	160	160	20	40	80	160	80	320
B/Ukraine/6391/2015	3	2015-02-24	C1/MDCK1	160	40	20	160	80	20	20	40	160	80	320
B/Ukraine/6452/2015	3	2015-02-25	C2/SIAT1	320	160	80	320	160	80	80	160	320	320	320
B/Ukraine/6453/2015	3	2015-02-26	C2/SIAT1	320	80	80	320	160	80	80	320	320	640	320
B/Ukraine/6365/2015	3	2015-03-01	C1/MDCK1	320	160	80	160	160	80	80	160	320	320	320
B/Ukraine/6367/2015		2015-03-02	C1/MDCK1	1280	320	320	640	320	320	320	1280	1280	1280	640
B/Ukraine/6368/2015	3	2015-03-02	C1/MDCK1	320	160	160	320	320	80	160	160	320	320	320
B/Ukraine/6462/2015	3	2015-03-02	C2/SIAT1	640	160	160	320	160	80	160	160	320	320	320
B/Ukraine/6463/2015	3	2015-03-02	C2/SIAT1	1280	320	320	640	320	160	320	320	640	640	640
B/Ukraine/6787/2015	3	2015-03-03	C1/MDCK1	320	80	80	160	80	40	80	80	160	160	320
B/Ukraine/6366/2015		2015-03-04	C1/MDCK1	320	160	80	320	320	40	80	160	320	320	320
B/Ukraine/6458/2015	3	2015-03-04	C2/SIAT1	320	160	80	160	160	40	80	80	160	160	320
B/Ukraine/6791/2015	3	2015-03-09	C1/MDCK1	640	160	160	320	320	80	320	640	640	640	320
B/Ukraine/6457/2015		2015-03-10	C2/MDCK1	640	160	160	320	320	40	160	640	320	160	640
B/Ukraine/6788/2015		2015-03-10	C1/MDCK1	320	80	80	160	80	20	40	80	160	160	320
B/Ukraine/6464/2015	3	2015-03-11	C3/MDCK1	640	160	160	320	320	80	160	640	160	320	640
B/Ukraine/6468/2015	3	2015-03-18	C2/MDCK1	320	80	40	80	80	20	40	40	160	80	160
B/Ukraine/6469/2015	3	2015-03-19	C2/SIAT1	640	160	80	160	160	40	80	80	160	160	320
B/Ukraine/6792/2015	3	2015-03-24	C1/MDCK1	1280	160	160	320	320	160	320	640	640	1280	640
B/Ukraine/6789/2015	3	2015-03-25	C1/MDCK1	1280	160	160	320	320	160	320	320	640	640	320
B/Ukraine/6790/2015	3	2015-03-29	C1/MDCK1	320	80	80	160	80	20	40	80	160	160	320
B/Ukraine/6461/2015	3	2015-04-01	C0/MDCK1	320	80	40	80	80	20	40	80	160	80	320
B/Finland/513/2015	3	2015-04-02	MDCK1/MDCK1	320	80	80	160	80	40	40	80	160	160	320
B/Greece/T-440/2015	3	2015-04-17	MDCK1	320	80	80	160	160	40	80	160	160	160	320
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40;   2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-10. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-06-17)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre: Post-infection ferret antisera B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
	Genetic Group			1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	1280	320	640	80	1280	320	320	40	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	2560	640	640	160	320	80	1280	320	320	20	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	320	160	320	320	20	160	80	320	40	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	320	160	160	320	20	160	80	160	40	320
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	160	80	40	80	160	80	640	160	40	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	5120	640	1280	160	640	80	1280	320	320	40	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	320	160	640	640	320	160	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	320	320	320	320	20	160	80	320	80	320
B/Phuket/3073/2013	3	2013-11-21	M2/M2	2560	320	320	640	640	320	640	640	1280	1280	320
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	160	40	40	80	80	10	40	40	80	20	160
TEST VIRUSES														
B/Ukraine/6269/2014	3	2014-12-29	C1/MDCK1	160	80	<	80	80	10	20	40	80	40	320
B/Ukraine/6270/2014		2014-12-29	C1/MDCK1	160	80	<	80	80	20	20	40	80	80	320
B/Ukraine/6271/2014		2014-12-30	C1/MDCK1	160	80	<	80	80	10	20	40	80	40	320
B/Ukraine/6291/2015		2015-01-15	C1/MDCK1	160	80	80	160	320	20	40	80	80	80	320
B/Ukraine/6272/2015		2015-01-16	C1/MDCK1	160	80	<	80	160	20	20	40	80	80	320
B/Ukraine/6273/2015	3	2015-01-19	C1/MDCK1	160	80	<	160	320	20	20	80	80	80	320
B/Ukraine/6324/2015		2015-01-26	C1/MDCK1	160	80	<	160	320	20	40	80	80	80	320
B/Ukraine/6325/2015		2015-01-27	C1/MDCK1	320	160	80	160	160	20	40	80	160	160	640
B/Ukraine/6329/2015		2015-01-27	C1/MDCK1	160	80	80	160	320	20	40	80	160	160	320
B/Ukraine/6292/2015		2015-01-28	C2/MDCK1	160	40	<	80	40	10	10	40	80	40	160
B/Ukraine/6326/2015		2015-01-28	C1/MDCK1	320	80	80	160	160	20	40	80	160	80	320
B/Ukraine/6295/2015	3	2015-01-29	C2/MDCK1	320	160	160	160	160	40	80	80	160	160	320
B/Ukraine/6332/2015	3	2015-01-29	C1/MDCK1	320	80	<	160	160	20	40	80	160	160	320
B/Norway/698/2015	3	2015-01-29	MDCK1/MDCK1	320	160	160	160	320	40	320	80	160	160	320
B/Ukraine/6454/2015		2015-02-03	C2/MDCK1	1280	160	320	320	640	320	320	640	640	1280	640
B/Ukraine/6327/2015		2015-02-04	C1/MDCK1	320	80	<	160	160	40	80	80	160	160	320
B/Ukraine/6447/2015		2015-02-05	C2/MDCK1	1280	320	320	320	640	320	320	640	1280	1280	640
B/Ukraine/6330/2015	3	2015-02-06	C1/MDCK1	640	160	80	320	320	80	160	160	320	320	640
B/Ukraine/6331/2015		2015-02-06	C1/MDCK1	640	80	160	320	160	80	160	160	320	320	320
B/Ukraine/6328/2015		2015-02-08	C1/MDCK1	640	160	160	320	320	80	160	160	320	320	640
B/Ukraine/6442/2015		2015-02-10	C2/MDCK1	2560	160	320	320	640	320	320	640	1280	1280	640
B/Ukraine/6443/2015		2015-02-10	C2/MDCK1	1280	160	160	320	320	320	320	640	640	1280	640
B/Ukraine/6455/2015		2015-02-11	C2/MDCK1	1280	320	320	320	320	320	320	640	640	1280	640
B/Ukraine/6357/2015	3	2015-02-14	C1/MDCK1	320	80	<	80	80	20	40	80	80	80	160
B/Ukraine/6341/2015	3	2015-02-15	C1/MDCK1	320	80	<	80	160	20	80	80	160	160	160
B/Ukraine/6355/2015		2015-02-15	C0/MDCK1	2560	160	320	320	640	320	320	640	640	1280	320
B/Ukraine/6358/2015		2015-02-15	C1/MDCK1	640	160	80	320	320	80	160	160	320	320	320
B/Ukraine/6362/2015		2015-02-15	C1/MDCK1	320	160	<	160	160	40	80	80	160	160	320
B/Ukraine/6445/2015		2015-02-15	C2/MDCK1	1280	160	160	320	320	80	320	320	640	640	320
B/Ukraine/6446/2015		2015-02-15	C2/MDCK1	640	160	160	320	320	160	320	320	640	640	320
B/Ukraine/6448/2015		2015-02-16	C2/MDCK1	1280	160	320	320	640	320	320	640	640	1280	640
B/Ukraine/6449/2015	3	2015-02-17	C2/MDCK1	1280	160	160	320	320	160	320	640	640	640	640
B/Ukraine/6450/2015		2015-02-17	C2/MDCK1	1280	320	320	640	640	320	320	640	640	1280	320
B/Ukraine/6364/2015		2015-02-25	C1/MDCK3	1280	160	320	320	640	320	640	640	640	640	320
B/Ukraine/6369/2015		2015-02-27	C1/MDCK3	5120	320	320	320	640	640	640	640	1280	1280	320
B/Ukraine/361/2015	3	2015-03-29	SIAT2/MDCK1	160	80	<	80	80	20	20	40	80	80	320
B/Norway/1881/2015	3	2015-04-02	MDCK1	160	80	<	40	80	10	20	40	40	40	160
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-11. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-06-24)

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
				Haemagglutination inhibition titre										
				Post-infection ferret antisera										
	Genetic Group			1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	320	640	160	320	80	640	160	320	20	160
B/Brisbane/3/2007	2	2007-09-03	E2/E3	2560	640	1280	160	320	80	1280	320	320	40	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	320	320	320	320	40	160	80	320	40	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	160	160	320	20	160	40	160	40	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	80	40	40	160	40	640	80	40	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	320	640	160	320	80	640	160	160	20	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	320	320	320	640	640	320	80	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	320	320	320	320	20	160	80	320	80	320
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	1280	320	320	640	320	320	320	640	1280	1280	320
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	80	80	40	40	40	10	40	20	40	20	160
TEST VIRUSES														
B/Antsirabe/144/2015		2015-01-07	MDCK1/MDCK1	640	160	160	160	160	80	80	160	160	320	320
B/Tsaralalana/35/2015		2015-01-08	MDCK1/MDCK1	640	160	160	320	160	80	160	160	320	160	320
B/Antsohihy/76/2015		2015-01-09	MDCK1/MDCK1	640	160	320	320	320	160	160	160	640	640	320
B/Itaosy/126/2015	3	2015-01-10	MDCK1/MDCK1	160	80	160	320	160	80	80	80	160	160	160
B/NosyBe/283/2015		2015-01-19	MDCK1/MDCK1	640	160	160	320	160	80	80	160	320	320	320
B/Mahajanga/271/2015	3	2015-01-21	MDCK1/MDCK1	80	80	80	160	80	80	80	80	160	160	160
B/Toamasina/307/2015		2015-01-27	MDCK1/MDCK1	1280	160	320	160	160	320	160	640	160	320	320
B/Odessa/222/2015		2015-01-31	SIAT2/MDCK1	320	80	80	160	160	40	80	80	160	160	320
B/Kharkov/263/2015	3	2015-02-02	SIAT2/SIAT2	2560	160	320	320	320	640	320	640	640	1280	320
B/Odessa/386/2015		2015-02-04	SIAT2/MDCK1	640	160	160	160	160	80	80	80	320	320	320
B/Odessa/234/2015	3	2015-02-08	SIAT2/MDCK1	640	160	160	160	160	40	80	80	160	160	320
B/Ukraine/6444/2015		2015-02-09	C2/MDCK3	320	80	160	160	320	80	80	160	640	640	320
B/Ukraine/108/2015	3	2015-02-09	SIAT2/MDCK1	320	40	40	80	40	20	40	40	160	160	80
B/Ukraine/143/2015		2015-02-16	SIAT1/MDCK1	640	80	160	160	160	80	80	80	320	320	160
B/Antananarivo/478/2015		2015-02-18	MDCK1/MDCK1	640	80	160	320	160	80	80	160	160	320	320
B/Moldova/133.08/2015	3	2015-02-18	MDCK1/MDCK1	320	80	160	80	80	40	40	80	160	80	320
B/Moldova/138.08/2015	3	2015-02-18	MDCK1/MDCK1	640	160	160	160	80	80	80	160	160	160	320
B/Antananarivo/552/2015		2015-02-19	MDCK1/MDCK1	1280	640	320	640	320	160	320	320	640	640	320
B/Odessa/392/2015		2015-02-20	SIAT2/MDCK1	1280	160	320	320	640	320	320	320	640	1280	320
B/Ukraine/142/2015		2015-02-21	SIAT1/MDCK1	320	80	80	80	80	40	40	80	160	160	320
B/Odessa/390/2015	3	2015-02-23	SIAT2/MDCK1	2560	320	320	320	320	640	320	640	1280	1280	320
B/Antsirabe/674/2015		2015-02-24	MDCK1/MDCK1	640	640	160	320	160	80	160	160	320	320	320
B/Ukraine/153/2015		2015-03-01	SIAT1/MDCK1	640	160	160	160	160	80	160	160	320	320	320
B/Zaporizza/338/2015	3	2015-03-03	SIAT2/MDCK1	320	80	80	80	80	40	40	40	80	80	160
B/Zaporizza/339/2015	3	2015-03-03	SIAT2/MDCK1	2560	320	320	640	640	320	640	640	1280	1280	320
B/Maevatanana/844/2015		2015-03-04	MDCK1/MDCK1	640	640	160	320	320	160	160	320	640	640	320
B/Ukraine/287/2015		2015-03-05	SIAT1/MDCK1	640	160	160	160	80	80	80	80	160	160	320
B/Zaporizza/336/2015		2015-03-05	SIAT2/MDCK1	640	160	160	160	160	40	80	80	160	160	320
B/Odessa/393/2015	3	2015-03-05	SIAT2/MDCK1	640	160	160	160	160	40	80	80	160	160	320
B/Antananarivo/839/2015		2015-03-05	MDCK1/MDCK1	640	640	160	320	160	80	160	160	320	320	320
B/Zaporizza/333/2015	3	2015-03-07	SIAT2/MDCK1	320	80	160	160	80	40	80	80	160	160	160
B/Kharkov/266/2015	3	2015-03-10	SIAT2/MDCK1	640	160	160	160	160	80	160	80	160	160	320
B/Ukraine/275/2015		2015-03-10	SIAT1/MDCK1	320	160	160	160	160	40	80	80	160	160	320
B/Zaporizza/335/2015		2015-03-10	SIAT2/MDCK1	640	160	320	160	320	80	160	160	320	160	320
B/Mahajanga/882/2015		2015-03-10	MDCK1/MDCK1	1280	640	320	320	320	320	160	320	640	1280	320
B/Dnipro/347/2015	3	2015-03-12	SIAT2/MDCK1	640	160	160	160	160	40	80	80	160	160	320
B/Tsaralalana/1003/2015		2015-03-16	MDCK1/MDCK1	1280	160	320	320	320	160	160	320	640	640	320
B/Antananarivo/1012/2015	3	2015-03-16	MDCK1/MDCK1	640	640	160	320	160	80	80	80	320	320	320
B/Antsirabe/1037/2015	3	2015-03-16	MDCK1/MDCK1	1280	640	320	640	320	320	320	320	640	640	320
B/Tsiroanomandidy/1272/2015		2015-03-23	MDCK1/MDCK1	1280	160	320	320	320	160	160	320	640	640	320
B/Tsaralalana/1283/2015	2	2015-03-24	MDCK1/MDCK1	640	160	160	80	80	160	80	320	160	160	160
B/Antananarivo/1315/2015	3	2015-03-27	MDCK1/MDCK1	2560	320	640	320	640	640	320	640	1280	1280	640
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-12. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-06-30)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre: Post-infection ferret antisera: B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[5] 12/11 F06/15	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
	Genetic Group			1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	640	320	320	80	1280	320	320	40	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	640	80	160	40	640	160	160	20	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	160	320	320	160	20	320	80	160	40	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	160	80	160	10	160	80	80	40	320
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	80	160	40	40	160	80	640	80	80	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	320	640	160	160	80	640	320	160	20	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	320	160	320	640	1280	320	160	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	160	160	160	160	20	160	80	160	40	320
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	1280	320	320	320	160	320	640	640	640	1280	640
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	160	80	80	80	40	10	80	80	80	40	320
TEST VIRUSES														
B/St Petersburg/6/2015	3	2015-01-19	E1/E2	320	40	80	80	<	10	40	40	40	10	160
B/Moldova/053.06/2015	3	2015-02-02	MDCK2/MDCK1	320	80	160	160	80	40	80	160	160	160	320
B/Moldova/057.06/2015	3	2015-02-03	MDCK1/MDCK1	640	160	160	160	80	80	160	320	320	160	640
B/Moldova/059.06/2015		2015-02-03	MDCK1/MDCK1	640	160	160	320	160	80	160	320	320	320	640
B/Moldova/060.06/2015		2015-02-03	MDCK2/MDCK1	320	80	160	160	80	40	80	160	160	160	320
B/Moldova/072.06/2015		2015-02-05	MDCK1/MDCK1	320	80	80	80	40	40	80	160	160	160	320
B/Moldova/075.06/2015		2015-02-05	MDCK1/MDCK1	320	80	80	80	80	40	80	80	160	160	320
B/Moldova/128.08/2015		2015-02-06	MDCK1/MDCK1	320	80	80	80	40	40	80	80	80	160	320
B/Moldova/081.07/2015		2015-02-09	MDCK2/MDCK1	640	80	160	160	80	80	160	160	160	320	320
B/Moldova/087.07/2015		2015-02-09	MDCK2/MDCK1	320	80	160	160	80	80	160	160	320	320	320
B/Moldova/103.07/2015		2015-02-09	MDCK2/MDCK1	320	80	160	160	80	80	160	160	160	320	320
B/Moldova/099.07/2015	3	2015-02-11	MDCK1/MDCK1	320	80	160	160	80	40	160	160	160	160	320
B/Moldova/104.07/2015		2015-02-11	MDCK2/MDCK1	320	80	160	160	80	80	160	160	160	160	320
B/Moldova/105.07/2015	3	2015-02-11	MDCK2/MDCK1	640	160	160	160	160	80	320	320	320	320	640
B/Moldova/110.07/2015	3	2015-02-11	MDCK2/MDCK1	320	80	80	160	80	40	80	160	160	160	320
B/Moldova/141.08/2015		2015-02-16	MDCK1/MDCK1	320	80	160	80	80	40	160	160	160	80	320
B/Moldova/156.08/2015		2015-02-17	MDCK1/MDCK1	320	80	80	80	80	40	80	160	160	160	320
B/Ghana/DILI-15-0256/2015	3	2015-03-25	Cx/MDCK1	640	160	160	320	80	80	160	160	160	160	640
B/Ghana/DILI-15-0295/2015	3	2015-04-01	C1/MDCK1	640	160	160	320	160	80	320	160	160	160	640
B/Ghana/DILI-15-0311/2015	3	2015-04-07	Cx/MDCK1	640	160	160	320	80	80	160	160	160	160	640

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Vaccine NH2014-15 (B/Mass 02/12 Egg F42/14)

Vaccine SH2015 NH2015-16 (B/Phuket 3073/13 Egg F36/14)

Sequences in phylogenetic tree

Table 13-13. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-07-08)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre: Post-infection ferret antisera: B/Fl[1,3] 4/06 SH479	B/Mass[1,3] 02/12 SH587	B/Phuket[1,3] 3073/13 SH614	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[1] 12/11 F06/15	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
	Genetic Group			1	NEW	NEW	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES																
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	5120	1280	640	640	320	320	80	1280	320	320	20	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	2560	2560	1280	320	640	80	160	40	1280	160	160	10	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	640	640	1280	160	160	320	160	10	640	40	160	20	160
B/Stockholm/12/2011	3	2011-03-28	EX/E2	640	640	1280	80	80	40	80	<	160	20	80	20	80
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	1280	640	40	80	20	<	80	160	320	40	40	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	2560	1280	320	640	80	160	40	1280	80	80	10	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	2560	1280	320	320	160	80	160	640	320	160	40	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	640	640	2560	160	320	160	160	10	320	40	160	20	160
B/Phuket/3073/2013	3	2013-11-21	M2/M2	1280	1280	5120	160	160	320	80	160	640	320	320	640	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	160	160	1280	80	80	40	<	10	80	20	40	10	160
TEST VIRUSES																
B/Stockholm/9/2014	3	2014-12-10	MDCK2/MDCK1	320	1280	2560	80	80	160	<	80	320	160	320	640	160
B/Stockholm/8/2014	3	2014-12-15	MDCK2/MDCK1	320	1280	2560	160	80	80	<	40	160	80	160	160	160
B/Stockholm/1/2015	3	2015-01-11	MDCK2/MDCK1	320	320	1280	160	80	80	<	10	80	20	80	20	160
B/Moldova/022.04/2015		2015-01-21	MDCK1/MDCK1	640	1280	2560	160	80	160	<	80	320	80	160	320	160
B/Stockholm/3/2015	3	2015-02-11	MDCK1/MDCK1	160	640	2560	80	80	80	<	40	160	80	160	160	160
B/Sweden/7/2015	3	2015-02-19	MDCK2/MDCK1	160	640	2560	160	80	80	<	40	160	40	80	80	160
												Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-14. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-07-15 + 2015-07-22)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre Post-infection ferret antisera B/Phuket[1,3] 3073/13 SH614	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F06/15	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/Phuket[2] 3073/13 T/C F27/15	B/HK[4] 3417/14 Egg St Judes F715/14
	Genetic Group			3	1	2	3	3	2	2	2	3	3	NEW	3
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	320	640	160	160	80	640	160	320	20	40	160
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	320	80	160	40	640	80	160	10	20	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	160	160	160	80	20	160	40	160	20	40	160
B/Stockholm/12/2011	3	2011-03-28	EX/E2	1280	160	160	80	160	10	160	40	80	20	20	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	80	160	40	40	160	80	320	80	40	40	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	320	320	80	160	40	640	80	160	10	20	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	320	320	160	160	160	640	320	160	80	80	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	160	160	160	160	20	160	40	160	40	40	160
B/Phuket/3073/2013	3	2013-11-21	M2/M2	5120	320	320	320	160	320	640	640	640	640	640	320
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	2560	80	80	80	40	20	80	40	80	20	40	320
TEST VIRUSES															
B/St Petersburg/6/2015		2015-01-19	C2/MDCK1	2560	80	160	80	40	40	160	80	160	80	80	320
B/Saratov/2/2015	3	2015-01-29	C2/MDCK1	5120	160	320	320	320	640	1280	640	1280	1280	640	320
B/Astrakhan/10/2015	3	2015-02-02	C1/MDCK1	5120	160	320	320	160	160	320	160	640	320	320	320
B/Sastin-Sraze/202/2015	3	2015-02-03	MDCK1/MDCK1	2560	160	160	160	80	40	160	80	160	160	80	320
B/Rostov-on-Don/4/2015	3	2015-02-05	C1/MDCK1	5120	160	160	160	80	80	320	160	320	320	160	320
B/Arkhangelsk/76/2015	3	2015-02-13	C1/MDCK1	5120	320	640	640	320	640	1280	640	1280	1280	640	640
B/Moscow/59/2015	3	2015-02-18	MDCK1/MDCK1	5120	160	320	320	160	320	320	320	640	1280	ND	320
B/Netherlands/1734/2015	3	2015-02-18	MDCK2/MDCK1	2560	80	80	80	40	40	80	40	160	80	80	160
B/Michalovce/587/2015	3	2015-02-18	MDCKx/MDCK1	5120	160	320	320	160	320	320	640	640	1280	320	320
B/Netherlands/1905/2015		2015-02-24	MDCK2/MDCK1	5120	160	160	160	80	80	320	160	320	320	160	320
B/Krasnoyarsk/6/2015	3	2015-02-27	C1/MDCK1	5120	160	320	320	160	160	320	160	640	320	320	320
B/Orenburg/4/2015	3	2015-03-10	MDCK3/MDCK2	5120	320	160	160	160	320	320	320	640	1280	ND	320
B/Vladimir/108/2015	3	2015-03-10	MDCK1/MDCK1	5120	320	320	320	160	320	320	320	640	640	320	320
B/Netherlands/2403/2015	3	2015-03-13	MDCK2/MDCK1	2560	80	160	160	40	40	160	80	160	80	80	320
B/Levice/844/2015	3	2015-03-19	MDCKx/MDCK1	2560	80	80	80	40	40	80	40	160	80	80	160
B/Yaroslavl/95-T/2015	3	2015-03-24	MDCK1/MDCK2	5120	160	160	320	160	320	320	320	640	640	ND	320
B/Galanta/912/2015	3	2015-03-26	MDCKx/MDCK1	2560	80	80	80	40	40	80	40	160	80	40	160
B/Trencin/913/2015		2015-03-26	MDCKx/MDCK1	2560	80	80	160	40	80	80	80	160	160	80	320
B/Nitra/920/2015	3	2015-03-26	MDCKx/MDCK1	2560	80	80	80	40	40	80	80	160	80	80	320
B/Trencin/927/2015	3	2015-03-31	MDCKx/MDCK1	2560	80	80	80	40	40	80	80	160	160	80	320
B/Moscow/110/2015	3	2015-04-07	MDCK1/MDCK1	5120	320	320	320	320	320	640	640	1280	1280	320	640
B/Moscow/112/2015	3	2015-04-08	MDCK1/MDCK1	5120	160	160	160	80	160	160	160	640	320	160	320
B/Trnava/1063/2015	3	2015-04-29	MDCKx/MDCK1	2560	80	80	80	40	40	80	40	160	80	80	160
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16			

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC; ND = Not Done

Sequences in phylogenetic tree

Table 13-15. Antigenic analyses of influenza B/Yamagata lineage viruses (2015-07-29)

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre: Post-infection ferret antisera: B/Phuket[1,3] 3073/13 SH614	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[1] 12/11 F06/15	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
	Genetic Group				1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	1280	320	320	80	1280	160	320	40	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	640	160	320	80	1280	160	160	20	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	5120	320	320	320	320	20	320	40	320	40	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	160	80	160	10	160	40	80	20	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	5120	160	320	160	80	320	320	640	640	320	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	2560	640	640	160	320	80	1280	160	320	20	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	640	320	160	160	640	320	320	80	640
B/Phuket/3073/2013	3	2013-11-21	E4/E2	5120	320	320	320	160	20	320	40	320	40	320
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	320	640	640	160	320	640	320	640	640	640
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	80	80	20	10	80	40	80	10	160
TEST VIRUSES														
B/Dakar/01/2015	3	2015-01-12	P1/MDCK1	5120	160	160	160	80	80	320	160	160	320	320
B/Dakar/05/2015	3	2015-03-11	P0/MDCK1	2560	80	80	160	<	40	160	80	160	160	160
B/Dakar/06/2015	3	2015-04-08	P0/MDCK1	5120	320	320	320	320	640	640	640	640	1280	640
B/Slovenia/2035/2015	3	2015-04-10	MDCK1/MDCK1	2560	160	160	160	<	40	160	80	160	80	320
B/Slovenia/2039/2015	3	2015-04-10	MDCK1/MDCK1	5120	160	160	160	80	80	160	80	160	160	320
B/Dakar/07/2015	3	2015-04-14	P0/MDCK1	5120	160	320	320	160	320	640	640	640	1280	320
B/Slovenia/2078/2015	3	2015-04-14	MDCKx/MDCK1	5120	160	160	320	80	160	320	320	320	320	320
B/Slovenia/2168/2015	3	2015-04-21	MDCK1/MDCK1	2560	160	160	160	<	40	160	80	160	160	320
B/Slovenia/2210/2015	3	2015-04-24	MDCKx/MDCK1	5120	160	160	160	<	40	320	80	160	160	320
B/Hong Kong/1323/2015	3	2015-06-08	MDCK1/MDCK1	5120	160	160	320	80	80	320	80	320	160	320
B/Hong Kong/1325/2015	3	2015-06-09	MDCK1/MDCK1	5120	320	640	320	160	160	640	320	640	640	640
B/Hong Kong/1327/2015	3	2015-06-11	MDCK1/MDCK1	5120	160	160	160	80	80	320	80	160	160	320
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-16. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-08-04

Viruses	Genetic Group	Collection date	Passage History	B/Phuket[1,3] 3073/13 SH614	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F06/15	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
				Haemagglutination inhibition titre										
					Post-infection ferret antisera									
Genetic Group					1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	2560	640	1280	160	320	80	1280	160	320	40	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	640	80	160	40	640	80	160	20	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	160	160	160	80	10	160	40	160	20	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	80	80	80	80	10	160	20	80	20	80
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	5120	80	160	160	80	320	320	640	320	320	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	320	640	80	160	80	1280	160	160	20	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	320	160	80	160	320	320	160	40	160
B/Phuket/3073/2013	3	2013-11-21	E4/E2	2560	160	160	160	160	20	320	40	160	40	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	320	80	320	320	320	320	640	160
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	40	40	20	10	40	20	40	10	160
TEST VIRUSES														
B/Slovenia/765/2015	3	2015-02-03	MDCK1/MDCK1	5120	80	160	160	40	40	160	80	160	160	160
B/Slovenia/935/2015		2015-02-11	MDCKx/MDCK1	2560	80	80	80	20	20	80	40	80	80	160
B/Slovenia/1069/2015	3	2015-02-19	MDCK1/MDCK1	1280	80	80	80	20	20	80	40	80	80	160
B/Slovenia/1143/2015		2015-02-24	MDCK1/MDCK1	2560	80	80	160	40	40	160	80	160	160	160
B/Slovenia/1312/2015	3	2015-03-04	MDCK1/MDCK1	2560	40	80	80	20	20	80	40	80	80	160
B/Slovenia/1736/2015	3	2015-03-25	MDCK1/MDCK1	2560	40	80	80	20	20	40	40	80	40	160
B/Slovenia/1750/2015		2015-03-25	MDCK1/MDCK1	1280	40	80	80	20	20	40	40	80	40	160
B/Dakar/04/2015	3	2015-03-28	C0/MDCK2	2560	160	160	160	40	40	160	80	160	160	160
B/Slovenia/1900/2015		2015-04-01	MDCKx/MDCK1	1280	40	40	40	20	20	40	40	80	40	160
B/Brest/1416/2015	3	2015-04-05	MDCK1	5120	160	160	160	80	160	320	160	320	320	160
B/Slovenia/2006/2015	3	2015-04-08	MDCK1/MDCK1	2560	80	80	80	40	40	80	40	160	160	160
B/Mauritius/I-388/2015	3	2015-04-17	MDCK2/MDCK1	5120	160	160	160	40	80	320	80	320	160	160
B/Mauritius/I-400/2015	3	2015-04-24	MDCK2/MDCK1	2560	160	160	160	40	40	160	80	160	80	160
B/Dakar/08/2015	3	2015-05-05	C0/MDCK2	5120	160	1280	320	80	160	320	160	320	320	320
B/Dakar/09/2015	3	2015-05-12	C0/MDCK2	2560	80	160	160	80	80	160	80	160	160	160
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-17. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-08-11

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre Post-infection ferret antisera B/Phuket[1,3] 3073/13 SH614	B/Fl[1] 4/06 F1/10	B/Bris[1] 3/07 F38/14	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F06/15	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F42/14	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[4] 3417/14 Egg St Judes F715/14
Genetic Group					1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	1280	320	640	160	160	40	640	160	160	20	160
B/Brisbane/3/2007	2	2007-09-03	E2/E3	1280	320	320	80	160	40	640	80	160	10	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	2560	160	160	320	160	20	320	40	160	40	160
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	160	80	160	10	160	40	80	20	160
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	1280	160	160	160	40	160	160	320	160	40	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E3	1280	320	640	80	160	40	1280	160	160	10	160
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	640	320	160	160	160	640	320	160	40	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	2560	320	320	320	160	20	320	40	320	40	160
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	5120	160	160	320	160	320	320	320	640	640	320
B/Hong Kong/3417/2014	3	2014-06-04	E4/E1	1280	80	80	80	20	10	80	40	80	10	160
TEST VIRUSES														
B/Estonia/91207/2015		2015-01-21	SIAT2/MDCK1	2560	160	160	320	80	80	320	160	160	160	160
B/South Africa/R0342/2015	3	2015-01-31	MDCK2/MDCK1	5120	320	320	320	160	160	320	160	640	640	320
B/Estonia/92315/2015		2015-02-20	SIAT2/MDCK1	5120	160	160	320	160	160	320	320	320	640	320
B/Poland/1062/2015	3	2015-02-26	MDCK1/MDCK1	5120	160	320	320	160	160	640	320	640	640	320
B/Estonia/92621/2015	3	2015-02-27	SIAT1/MDCK1	2560	80	160	320	80	160	320	160	320	320	160
B/Estonia/92596/2015	3	2015-03-05	SIAT1/MDCK1	2560	80	160	160	40	40	160	80	160	160	160
B/Poland/2477/2015	3	2015-03-16	MDCK1/MDCK1	5120	160	320	320	160	320	320	320	640	640	320
B/Estonia/93061/2015	3	2015-03-27	SIAT1/MDCK1	5120	160	320	320	160	160	320	160	320	320	320
B/Poland/1054/2015	3	2015-03-27	MDCK1/MDCK1	2560	160	160	160	80	40	160	80	160	160	160
B/Estonia/93249/2015	3	2015-04-07	SIAT2/MDCK1	5120	160	320	320	160	320	640	320	640	640	320
B/Estonia/93274/2015	3	2015-04-09	SIAT2/MDCK1	2560	160	160	160	80	80	160	80	160	160	320
B/Estonia/93275/2015	3	2015-04-09	SIAT2/SIAT2	5120	640	640	640	320	640	640	640	1280	1280	320
B/South Africa/R3192/2015	3	2015-06-08	MDCK1/MDCK1	5120	160	160	160	80	80	160	80	320	160	320
B/South Africa/R3183/2015	3	2015-06-10	MDCK1/MDCK1	5120	320	160	320	80	80	160	80	320	160	320
B/South Africa/R3762/2015	3	2015-06-29	MDCK1/MDCK1	2560	80	160	160	80	80	320	80	320	160	160
B/South Africa/R3769/2015		2015-06-29	MDCK1/MDCK1	5120	160	320	320	80	80	320	80	320	160	320
										Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE serum pre-absorbed with TRBC

Sequences in phylogenetic tree

Table 13-18. Summary of HI reactivity of cell-propagated influenza B/Yamagata lineage Test viruses*

HI Table	No of test viruses	Fold reduction compared to homologous titre	Antiserum to Virus	Number of test viruses reacting within 2-fold and 4-fold of the antiserum homologous titre									
				B/Fl 4/06	B/Bris 3/07	B/Wis 1/10	B/Stock 12/11	B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Phuket 3073/13	B/Phuket 3073/13	B/HK 3417/14
			Host	Egg	Egg	Egg	Egg	Cell	Egg	Cell	Egg	Cell	Egg
			Genetic group	1	2	3	3	2	2	2	3	3	3
X-4	1	2-fold		0	0	0	1	0	0	0	1	0	1
		4-fold		0	0	1	1	0	0	0	1	0	1
X-5	1	2-fold		0	0	1	1	0	0	1	1	1	1
		4-fold		0	1	1	1	0	0	1	1	1	1
X-6	25	2-fold		1	4	25	13	0	1	4	22	4	25
		4-fold		7	16	25	20	4	4	18	25	15	25
X-7	26	2-fold		1	8	26	3	11	2	6	22	14	26
		4-fold		10	13	26	16	22	6	12	26	23	26
X-8	13	2-fold		0	2	13	10	13	0	3	13	9	13
		4-fold		1	13	13	13	13	3	10	13	13	13
X-9	30	2-fold		2	11	26	30	13	0	9	28	8	30
		4-fold		16	22	30	30	21	7	16	30	14	30
X-10	23	2-fold		4	9	19	20	17	2	13	20	13	23
		4-fold		17	15	22	23	19	13	17	22	17	23
X-11	34	2-fold		10	1	29	26	24	7	12	33	12	34
		4-fold		26	11	34	33	33	16	19	34	19	34
X-12	19	2-fold		0	0	14	17	10	2	0	19	0	19
		4-fold		6	14	19	19	19	12	3	19	5	19
X-13	2	2-fold		0	0	0	0	2	0	0	2	0	2
		4-fold		1	0	2	0	2	0	1	2	1	2
X-14	21	2-fold		13	14	21	13	13	11	12	21	12	21
		4-fold		21	21	21	21	21	14	17	21	15	21
X-15	11	2-fold		2	3	11	7	4	3	4	11	4	11
		4-fold		10	10	11	7	7	7	11	11	10	11
X-16	15	2-fold		0	1	14	9	2	0	2	15	2	15
		4-fold		5	6	15	15	4	3	8	15	8	15
X-17	14	2-fold		11	14	14	13	12	3	7	14	7	14
		4-fold		14	14	14	14	14	9	14	14	14	14
Total No	**235**												
	Number reacting within 2-fold			44	67	213	163	121	31	73	222	86	235
	Number reacting within 4-fold			134	156	234	213	179	94	147	234	155	235
	Percentage reacting within 2-fold			18.72	28.51	90.64	69.36	51.49	13.19	31.06	94.47	36.60	100.00
	Percentage reacting within 4-fold			57.02	66.38	99.57	90.64	76.17	40.00	62.55	99.57	65.96	100.00
									Vaccine NH2014-15		Vaccine SH2015 NH2015-16		

* Only viruses with collection dates after 2015-01-31 included

Figure 17. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

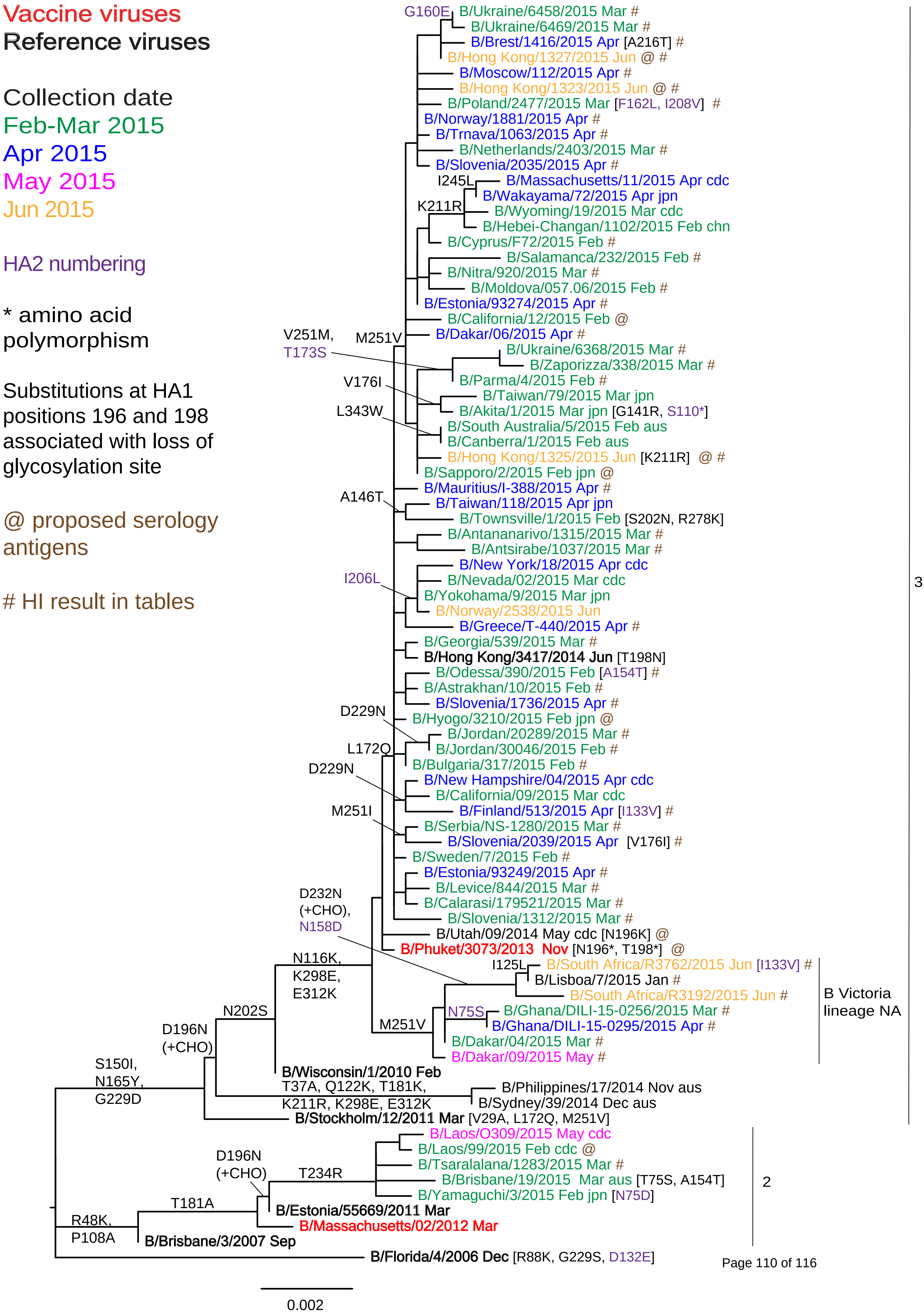

Figure 18. HA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

3V: these viruses are B/Yamagata HA clade 3 with B/Victoria NA.

```
                                  10        20        30        40        50        60        70        80        90       100
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006          DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSYFANLKGTRTRGKLCPDCLNCTDLDVALGRPMCVGTTPSAKASILHEVKPVTSGCFPIMHD
B/Brisbane/3/2007         ...............................................K.......................................R............ 2
B/Massachusetts/02/2012   ...............................................K.......................................R............ 2
B/Estonia/55669/2011      ...............................................K.......................................R............ 2
B/Yamaguchi/3/2015        ...............................................K.......................................R............ 2
B/Brisbane/19/2015        ...............................................K..........................S............R............ 2
B/Tsaralalana/1283/2015   ...............................................K.......................................R............ 2
B/Laos/0309/2015          ...............................................K.......................................R............ 2
B/Stockholm/12/2011       ............................A..........................................................R............ 3
B/Sydney/39/2014          ....................................A..................................................R............ 3
B/Wisconsin/1/2010        .......................................................................................R............ 3
B/Dakar/09/2015           .......................................................................................R............ 3V
B/Ghana/DILI-15-0295/2015 .......................................................................................R............ 3V
B/South Africa/R3762/2015 .......................................................................................R............ 3V
B/Phuket/3073/2013        .......................................................................................R............ 3
B/Estonia/93249/2015      .......................................................................................R............ 3
B/Slovenia/2039/2015      .......................................................................................R............ 3
B/Finland/513/2015        .......................................................................................R............ 3
B/Jordan/20289/2015       .......................................................................................R............ 3
B/Slovenia/1736/2015      .......................................................................................R............ 3
B/Hong Kong/3417/2014     .......................................................................................R............ 3
B/Norway/2538/2015        .......................................................................................R............ 3
B/Taiwan/118/2015         .......................................................................................R............ 3
B/Canberra/1/2015         .......................................................................................R............ 3
B/Taiwan/79/2015          .......................................................................................R............ 3
B/Zaporizza/338/2015      .......................................................................................R............ 3
B/Hebei-Changan/1102/2015 .......................................................................................R............ 3
B/Massachusetts/11/2015   .......................................................................................R............ 3
B/Hong Kong/1323/2015     .......................................................................................R............ 3
B/Ukraine/6469/2015       .......................................................................................R............ 3

                                 110       120       130       140       150       160       170       180       190       200
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006          RTKIRQLPNLLRGYENIRLSTQNVIDAEKAPGGPYRLGTSGSCPNATSKSGFFATMAWAVPKDNNKNATNPLTVEVPYICTEGEDQITVWGFHSDDKTQM
B/Brisbane/3/2007         .......A............................................................................................
B/Massachusetts/02/2012   .......A........................................................................A...................
B/Estonia/55669/2011      .......A........................................................................A..............N....
B/Yamaguchi/3/2015        .......A........................................................................A..............N....
B/Brisbane/19/2015        .......A.............................................T..........................A..............N....
B/Tsaralalana/1283/2015   .......A........................................................................A..............N....
B/Laos/0309/2015          .......A........................................................................A..............N....
B/Stockholm/12/2011       .................................................I..............Y......Q............................
B/Sydney/39/2014          .....................K...........................I..............Y...............K..............N....
B/Wisconsin/1/2010        .................................................I..............Y..............................N....
B/Dakar/09/2015           ...............K.................................I..............Y..............................N....
B/Ghana/DILI-15-0295/2015 ...............K.................................I..............Y..............................N....
B/South Africa/R3762/2015 ...............K........L........................I..............Y..............................N....
B/Phuket/3073/2013        ...............K.................................I..............Y..............................X.X..
B/Estonia/93249/2015      ...............K.................................I..............Y......Q.......................N....
B/Slovenia/2039/2015      ...............K.................................I..............Y......Q...I...................N....
B/Finland/513/2015        ...............K.................................I..............Y......Q.......................N....
B/Jordan/20289/2015       ...............K.................................I..............Y......Q.......................N....
B/Slovenia/1736/2015      ...............K.................................I..............Y......Q.......................N....
B/Hong Kong/3417/2014     ...............K.................................I..............Y......Q.......................N.N..
B/Norway/2538/2015        ...............K.................................I..............Y......Q.......................N....
B/Taiwan/118/2015         ...............K.............................T...I..............Y......Q.......................N....
B/Canberra/1/2015         ...............K.................................I..............Y......Q.......................N....
B/Taiwan/79/2015          ...............K.................................I..............Y......Q...I...................N....
B/Zaporizza/338/2015      ...............K.................................I..............Y......Q.......................N....
B/Hebei-Changan/1102/2015 ...............K.................................I..............Y......Q.......................N....
B/Massachusetts/11/2015   ...............K.................................I..............Y......Q.......................N....
B/Hong Kong/1323/2015     ...............K.................................I..............Y......Q.......................N....
B/Ukraine/6469/2015       ...............K.................................I..............Y......Q.......................N....

                                 210       220       230       240       250       260       270       280       290       300
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006          KNLYGDSNPQKFTSSANGVTTHYVSQIGSFPDQTEDGGLPQSGRIVVDYMMQKPGKTGTIVYQRGVLLPQKVWCASGRSKVIKGSLPLIGEADCLHEKYG
B/Brisbane/3/2007         ............................G.......................................................................
B/Massachusetts/02/2012   ............................G.......................................................................
B/Estonia/55669/2011      ............................G.......................................................................
B/Yamaguchi/3/2015        ............................G....R..................................................................
B/Brisbane/19/2015        ............................G....R..................................................................
B/Tsaralalana/1283/2015   ............................G....R..................................................................
B/Laos/0309/2015          ............................G....R..................................................................
B/Stockholm/12/2011       ............................D.....................V.................................................
B/Sydney/39/2014          ..........R.................D.....................................................................E..
B/Wisconsin/1/2010        .S..........................D.....................................................................E..
B/Dakar/09/2015           .S..........................D.....................V..............................................E..
B/Ghana/DILI-15-0295/2015 .S..........................D.....................V..............................................E..
B/South Africa/R3762/2015 .S..........................D..N..................V..............................................E..
B/Phuket/3073/2013        .S..........................D.....................................................................E..
B/Estonia/93249/2015      .S..........................D.....................................................................E..
B/Slovenia/2039/2015      .S..........................D.....................I..............................................E..
B/Finland/513/2015        .S..........................N.....................................................................E..
B/Jordan/20289/2015       .S..........................N.....................................................................E..
B/Slovenia/1736/2015      .S..........................D.....................................................................E..
B/Hong Kong/3417/2014     .S..........................D.....................................................................E..
B/Norway/2538/2015        .S..........................D.....................................................................E..
B/Taiwan/118/2015         .S..........................D.....................................................................E..
B/Canberra/1/2015         .S..........................D.....................V..............................................E..
B/Taiwan/79/2015          .S..........................D.....................V..............................................E..
B/Zaporizza/338/2015      .S..........................D.....................................................................E..
B/Hebei-Changan/1102/2015 .S........R.................D.....................V..............................................E..
B/Massachusetts/11/2015   .S........R.................D...............L.....V..............................................E..
B/Hong Kong/1323/2015     .S..........................D.....................V..............................................E..
B/Ukraine/6469/2015       .S..........................D.....................V..............................................E..
```

```
                                  310       320       330       340       350       360       370       380       390       400
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006         GLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLNS
B/Brisbane/3/2007        ....................................................................................................
B/Massachusetts/02/2012  ....................................................................................................
B/Estonia/55669/2011     ....................................................................................................
B/Yamaguchi/3/2015       ....................................................................................................
B/Brisbane/19/2015       ....................................................................................................
B/Tsaralalana/1283/2015  ....................................................................................................
B/Laos/0309/2015         ....................................................................................................
B/Stockholm/12/2011      ....................................................................................................
B/Sydney/39/2014         ...........K........................................................................................
B/Wisconsin/1/2010       ....................................................................................................
B/Dakar/09/2015          ...........K........................................................................................
B/Ghana/DILI-15-0295/2015 ...........K........................................................................................
B/South Africa/R3762/2015 ...........K........................................................................................
B/Phuket/3073/2013       ...........K........................................................................................
B/Estonia/93249/2015     ...........K........................................................................................
B/Slovenia/2039/2015     ...........K........................................................................................
B/Finland/513/2015       ...........K........................................................................................
B/Jordan/20289/2015      ...........K........................................................................................
B/Slovenia/1736/2015     ...........K........................................................................................
B/Hong Kong/3417/2014    ...........K........................................................................................
B/Norway/2538/2015       ...........K........................................................................................
B/Taiwan/118/2015        ...........K........................................................................................
B/Canberra/1/2015        ...........K..............................W.........................................................
B/Taiwan/79/2015         ...........K........................................................................................
B/Zaporizza/338/2015     ...........K........................................................................................
B/Hebei-Changan/1102/2015 ...........K........................................................................................
B/Massachusetts/11/2015  ...........K........................................................................................
B/Hong Kong/1323/2015    ...........K........................................................................................
B/Ukraine/6469/2015      ...........K........................................................................................

                                  410       420       430       440       450       460       470       480       490       500
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006         LSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIAA
B/Brisbane/3/2007        .............................................................................D......................
B/Massachusetts/02/2012  .............................................................................D......................
B/Estonia/55669/2011     .............................................................................D......................
B/Yamaguchi/3/2015       ....................D........................................................D......................
B/Brisbane/19/2015       .............................................................................D......................
B/Tsaralalana/1283/2015  .............................................................................D......................
B/Laos/0309/2015         .............................................................................D......................
B/Stockholm/12/2011      .............................................................................D......................
B/Sydney/39/2014         .............................................................................D......................
B/Wisconsin/1/2010       .............................................................................D......................
B/Dakar/09/2015          .............................................................................D......................
B/Ghana/DILI-15-0295/2015 ....................S........................................................D......................
B/South Africa/R3762/2015 .............................................................................D......................
B/Phuket/3073/2013       .............................................................................D......................
B/Estonia/93249/2015     .............................................................................D......................
B/Slovenia/2039/2015     .............................................................................D......................
B/Finland/513/2015       .............................................................................DV.....................
B/Jordan/20289/2015      .............................................................................D......................
B/Slovenia/1736/2015     .............................................................................D......................
B/Hong Kong/3417/2014    .............................................................................D......................
B/Norway/2538/2015       .............................................................................D......................
B/Taiwan/118/2015        .............................................................................D......................
B/Canberra/1/2015        .............................................................................D......................
B/Taiwan/79/2015         .............................................................................D......................
B/Zaporizza/338/2015     .............................................................................D......................
B/Hebei-Changan/1102/2015 .............................................................................D......................
B/Massachusetts/11/2015  .............................................................................D......................
B/Hong Kong/1323/2015    .............................................................................D......................
B/Ukraine/6469/2015      .............................................................................D......................

                                  510       520       530       540       550       560
                         ....|....|....|....|....|....|....|....|....|....|....|....|....|....
B/Florida/4/2006         GTFNAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMLAIFIVYMVSRDNVSCSICL
B/Brisbane/3/2007        .....................................................................
B/Massachusetts/02/2012  .....................................................................
B/Estonia/55669/2011     .....................................................................
B/Yamaguchi/3/2015       .....................................................................
B/Brisbane/19/2015       .....................................................................
B/Tsaralalana/1283/2015  .....................................................................
B/Laos/0309/2015         .....................................................................
B/Stockholm/12/2011      .....................................................................
B/Sydney/39/2014         .....................................................................
B/Wisconsin/1/2010       .....................................................................
B/Dakar/09/2015          .....................................................................
B/Ghana/DILI-15-0295/2015 .....................................................................
B/South Africa/R3762/2015 ...D.........................K.......................................
B/Phuket/3073/2013       .....................................................................
B/Estonia/93249/2015     .....................................................................
B/Slovenia/2039/2015     .....................................................................
B/Finland/513/2015       .....................................................................
B/Jordan/20289/2015      .....................................................................
B/Slovenia/1736/2015     .....................................................................
B/Hong Kong/3417/2014    .....................................................................
B/Norway/2538/2015       ...................................................L.................
B/Taiwan/118/2015        .....................................................................
B/Canberra/1/2015        .....................................................................
B/Taiwan/79/2015         .....................................................................
B/Zaporizza/338/2015     ..................S..................................................
B/Hebei-Changan/1102/2015 .....................................................................
B/Massachusetts/11/2015  .....................................................................
B/Hong Kong/1323/2015    .....................................................................
B/Ukraine/6469/2015      .....E...............................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

3V: these viruses are B/Yamagata HA clade 3 with B/Victoria NA.

Figure 19. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

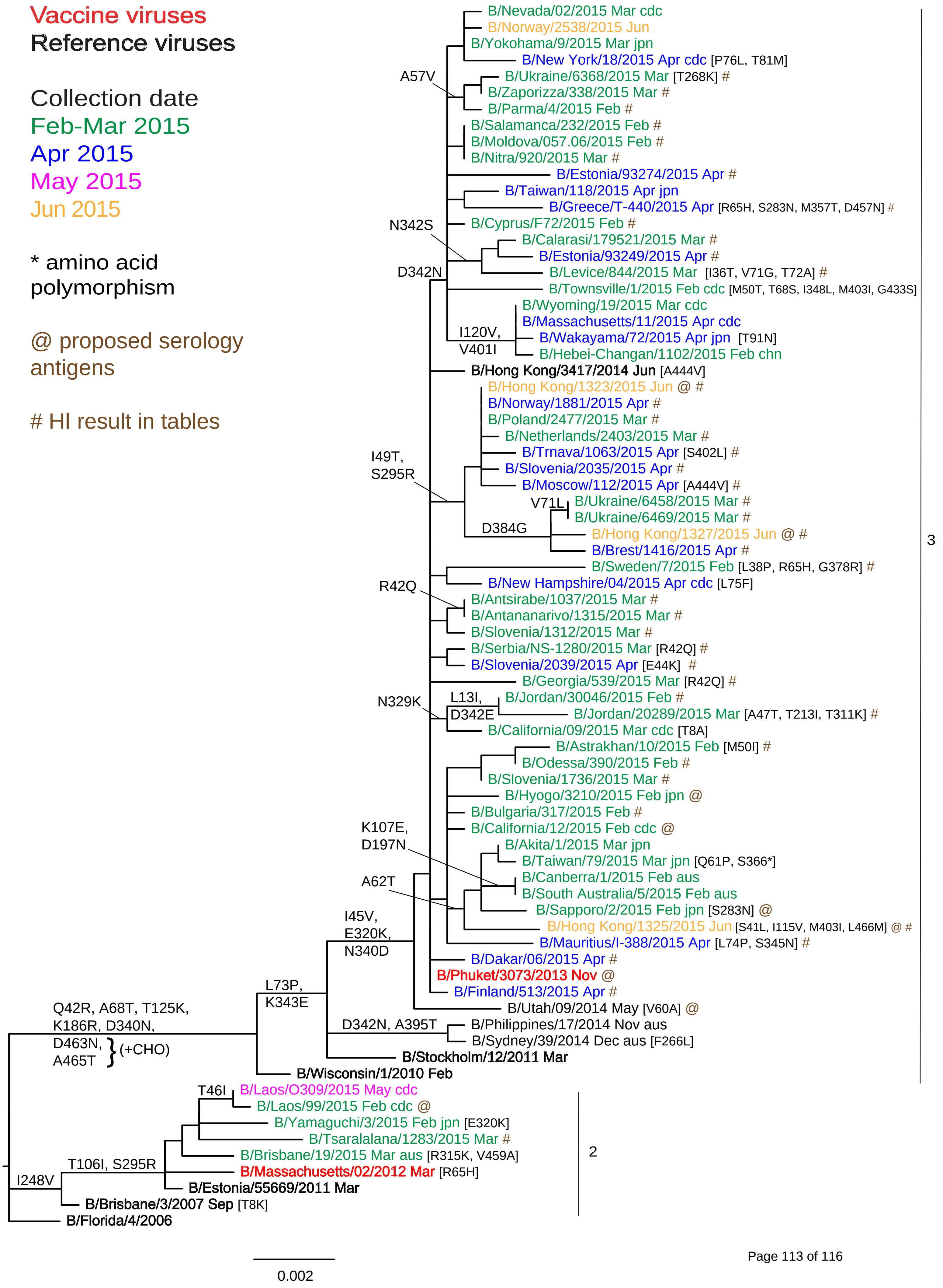

Figure 20. NA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site
3V: these viruses are B/Yamagata HA clade 3 with B/Victoria NA.

```
                                   10        20        30        40        50        60        70        80        90       100
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006           MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFSQTEITAPIMPLDCANASNVQAVNRSAAKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Brisbane/3/2007          .......K............................................................................................ 2
B/Massachusetts/02/2012    ................................................................H................................... 2
B/Estonia/55669/2011       .................................................................................................... 2
B/Yamaguchi/3/2015         .................................................................................................... 2
B/Brisbane/19/2015         .................................................................................................... 2
B/Tsaralalana/1283/2015    .................................................................................................... 2
B/Laos/0309/2015           .............................................I...................................................... 2
B/Stockholm/12/2011        .........................................R.........................T....P........................... 3
B/Sydney/39/2014           .........................................R.........................T....P........................... 3
B/Wisconsin/1/2010         .........................................R.........................T................................ 3
B/Dakar/09/2015            .........................................P..T...TV.................T..L............................. 3V
B/Ghana/DILI-15-0295/2015  .........................................P..T...TV.................T..L............................. 3V
B/South_Africa/R3762/2015  .........................................P..T...TV.................T..L............................. 3V
B/Phuket/3073/2013         .........................................R..V......................T....P........................... 3
B/Estonia/93249/2015       .........................................R..V......................T....P........................... 3
B/Slovenia/2039/2015       .........................................R.KV......................T....P........................... 3
B/Finland/513/2015         .........................................R..V......................T....P........................... 3
B/Jordan/20289/2015        ............I............................R..V.T....................T....P........................... 3
B/Slovenia/1736/2015       .........................................R..V......................T....P........................... 3
B/Hong Kong/3417/2014      .........................................R..V......................T....P........................... 3
B/Norway/2538/2015         .........................................R..V......................T....P........................... 3
B/Taiwan/118/2015          .........................................R..V......................T....P........................... 3
B/South_Australia/5/2015   .........................................R..V................T.....T....P........................... 3
B/Taiwan/79/2015           .........................................R..V...............PT.....T....P........................... 3
B/Zaporizza/338/2015       .........................................R..V...........V..........T....P........................... 3
B/Hebei-Changan/1102/2015  .........................................R..V......................T....P........................... 3
B/Massachusetts/11/2015    .........................................R..V...T..................T....P........................... 3
B/Hong Kong/1323/2015      .........................................R..V...T..................T....P........................... 3
B/Ukraine/6469/2015        .........................................R..V...T..................T..L.P........................... 3

                                  110       120       130       140       150       160       170       180       190       200
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006           HRFGETKGNSAPLIIREPFIACGPTECKHFALTHYAAQPGGYYNGTREDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDSNA
B/Brisbane/3/2007          ....................................................................................................
B/Massachusetts/02/2012    .....I..............................................................................................
B/Estonia/55669/2011       .....I..............................................................................................
B/Yamaguchi/3/2015         .....I..............................................................................................
B/Brisbane/19/2015         .....I..............................................................................................
B/Tsaralalana/1283/2015    .....I..............................................................................................
B/Laos/0309/2015           .....I..............................................................................................
B/Stockholm/12/2011        ........................K............................................................R..............
B/Sydney/39/2014           ........................K............................................................R..............
B/Wisconsin/1/2010         ........................K............................................................R..............
B/Dakar/09/2015            ........................N......................G.................................................N..
B/Ghana/DILI-15-0295/2015  ........................N......................G.................................................N..
B/South_Africa/R3762/2015  ........................N......................G....M............................................N..
B/Phuket/3073/2013         ........................K............................................................R..............
B/Estonia/93249/2015       ........................K............................................................R..............
B/Slovenia/2039/2015       ........................K............................................................R..............
B/Finland/513/2015         ........................K............................................................R..............
B/Jordan/20289/2015        ........................K............................................................R..............
B/Slovenia/1736/2015       ........................K............................................................R..............
B/Hong Kong/3417/2014      ........................K............................................................R..............
B/Norway/2538/2015         ........................K............................................................R..............
B/Taiwan/118/2015          ........................K............................................................R..............
B/South_Australia/5/2015   ......E.................K............................................................R..........N...
B/Taiwan/79/2015           ........................K............................................................R..............
B/Zaporizza/338/2015       ........................K............................................................R..............
B/Hebei-Changan/1102/2015  ...................V....K............................................................R..............
B/Massachusetts/11/2015    ...................V....K............................................................R..............
B/Hong Kong/1323/2015      ........................K............................................................R..............
B/Ukraine/6469/2015        ........................K............................................................R..............

                                  210       220       230       240       250       260       270       280       290       300
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006           LLKIKYGEAYTDTYHSYAKNILRTQESACNCIGGDCYLMITDGPASGISECRFLKIREGRIIKEIFPTGRVKHTEECTCGFASNKTIECACRDNSYTAKR
B/Brisbane/3/2007          ...............................................V....................................................
B/Massachusetts/02/2012    ...............................................V..............................................R.....
B/Estonia/55669/2011       ...............................................V..............................................R.....
B/Yamaguchi/3/2015         ...............................................V..............................................R.....
B/Brisbane/19/2015         ...............................................V..............................................R.....
B/Tsaralalana/1283/2015    ...............................................V..............................................R.....
B/Laos/0309/2015           ...............................................V..............................................R.....
B/Stockholm/12/2011        ....................................................................................................
B/Sydney/39/2014           .................................................................L..................................
B/Wisconsin/1/2010         ....................................................................................................
B/Dakar/09/2015            ...V..............NK..............N........S...V..............................................R.....
B/Ghana/DILI-15-0295/2015  ...V..............NK..............N........S...V..............................................R.....
B/South_Africa/R3762/2015  ...V..............NK..............N........S...V..............................................R.....
B/Phuket/3073/2013         ....................................................................................................
B/Estonia/93249/2015       ....................................................................................................
B/Slovenia/2039/2015       ....................................................................................................
B/Finland/513/2015         ....................................................................................................
B/Jordan/20289/2015        ............I.......................................................................................
B/Slovenia/1736/2015       ....................................................................................................
B/Hong Kong/3417/2014      ....................................................................................................
B/Norway/2538/2015         ....................................................................................................
B/Taiwan/118/2015          ....................................................................................................
B/South_Australia/5/2015   ....................................................................................................
B/Taiwan/79/2015           ....................................................................................................
B/Zaporizza/338/2015       ....................................................................................................
B/Hebei-Changan/1102/2015  ....................................................................................................
B/Massachusetts/11/2015    ....................................................................................................
B/Hong Kong/1323/2015      ..............................................................................................R.....
B/Ukraine/6469/2015        ..............................................................................................R.....
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                                   310       320       330       340       350       360       370       380       390       400
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006           PFVKLNVETDTAEIRLMCTETYLDTPRPNDGSITGPCESDGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWTDSEALALSGVM
B/Brisbane/3/2007          ....................................................................................................
B/Massachusetts/02/2012    ....................................................................................................
B/Estonia/55669/2011       ....................................................................................................
B/Yamaguchi/3/2015         ...................K................................................................................
B/Brisbane/19/2015         ..............K.....................................................................................
B/Tsaralalana/1283/2015    ....................................................................................................
B/Laos/0309/2015           ....................................................................................................
B/Stockholm/12/2011        .......................................N..E.........................................................
B/Sydney/39/2014           .......................................N.NE...................................................T.....
B/Wisconsin/1/2010         .......................................N............................................................
B/Dakar/09/2015            ...................D......................E..............K..............E...............A..D...F....
B/Ghana/DILI-15-0295/2015  ...................D......................E..............K..............E...............A..D...F....
B/South_Africa/R3762/2015  ...................D......................E..............K..............E...............A..D...F....
B/Phuket/3073/2013         ...................K......................E.........................................................
B/Estonia/93249/2015       ...................K.....................SE.........................................................
B/Slovenia/2039/2015       ...................K......................E.........................................................
B/Finland/513/2015         ...................K......................E.........................................................
B/Jordan/20289/2015        ..........K........K........K............EE.........................................................
B/Slovenia/1736/2015       ...................K......................E.........................................................
B/Hong Kong/3417/2014      ...................K......................E.........................................................
B/Norway/2538/2015         ...................K.....................NE.........................................................
B/Taiwan/118/2015          ...................K.....................NE.........................................................
B/South_Australia/5/2015   ...................K......................E.........................................................
B/Taiwan/79/2015           ...................K......................E......................X..................................
B/Zaporizza/338/2015       ...................K.....................NE.........................................................
B/Hebei-Changan/1102/2015  ...................K.....................NE.........................................................
B/Massachusetts/11/2015    ...................K.....................NE.........................................................
B/Hong Kong/1323/2015      ...................K......................E.........................................................
B/Ukraine/6469/2015        ...................K......................E........................................G................

                                   410       420       430       440       450       460
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Florida/4/2006           VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKTTWHSAATAIYCLMGSGQLLWDTVTGVDMAL
B/Brisbane/3/2007          ..................................................................
B/Massachusetts/02/2012    ..................................................................
B/Estonia/55669/2011       ..................................................................
B/Yamaguchi/3/2015         ..................................................................
B/Brisbane/19/2015         ..........................................................A.......
B/Tsaralalana/1283/2015    ..................................................................
B/Laos/0309/2015           ..................................................................
B/Stockholm/12/2011        ..............................................................N.T.
B/Sydney/39/2014           ..............................................................N.T.
B/Wisconsin/1/2010         ..............................................................N.T.
B/Dakar/09/2015            ...K...............................E..............................
B/Ghana/DILI-15-0295/2015  ...K...............................E..............................
B/South_Africa/R3762/2015  ...K...............................E..............................
B/Phuket/3073/2013         ..............................................................N.T.
B/Estonia/93249/2015       ..............................................................N.T.
B/Slovenia/2039/2015       ..............................................................N.T.
B/Finland/513/2015         ..............................................................N.T.
B/Jordan/20289/2015        ..............................................................N.T.
B/Slovenia/1736/2015       ..............................................................N.T.
B/Hong Kong/3417/2014      ...........................................V..................N.T.
B/Norway/2538/2015         ..............................................................N.T.
B/Taiwan/118/2015          ..............................................................N.T.
B/South_Australia/5/2015   ..............................................................N.T.
B/Taiwan/79/2015           ..............................................................N.T.
B/Zaporizza/338/2015       ..............................................................N.T.
B/Hebei-Changan/1102/2015  I.............................................................N.T.
B/Massachusetts/11/2015    I.............................................................N.T.
B/Hong Kong/1323/2015      ..............................................................N.T.
B/Ukraine/6469/2015        ..............................................................N.T.
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Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

3V: these viruses are B/Yamagata HA clade 3 with B/Victoria NA.

Table 14. Antiviral (NAI) testing of isolates with collection dates in 2014 and 2015

Month of Collection	Number of viruses tested						
	A(H1N1)pdm09		A(H3N2)		Flu B		Total
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2014							
January	132 (1)	1[a] + 1*	186 (3)	1[e]	15		336 (4)
February	122 (2)	1[b]	147 (4)	5[f,g,h,i,j]	20		295 (6)
March	72		112 (1)	1[k]	30 (1)		215 (2)
April	26		60		9		95
May	10		21		19 (2)		50 (2)
June	12 (3)	1[c]	40 (2)		18 (2)		71 (7)
July	11 (3)		49 (9)		25 (10)		85 (22)
August	0		25 (10)		16 (16)		41 (26)
Totals	385 (9)	4	640 (29)	7	152 (31)		1188 (69)
September	6 (2)		43 (37)		13 (10)		62 (49)
October	11 (3)		56 (29)		33 (16)		100 (48)
November	14 (9)		51 (24)		35 (14)		100 (47)
December	70 (18)		209 (133)		89 (23)		368 (174)
2015							
January	80 (69)		199 (184)		77 (70)	1[l]	357 (323)
February	68		147		126	1[m]	342
March	78		53		92		223
April	33		15		36		84
May	11	1[d]	10		5		27
June	11		33		12		56
July			7				7
Totals	382 (101)	1	823 (407)		518 (133)	2	1726 (641)
Overall Totals	767 (110)*	5	1463 (436)	7	670 (164)	2	2914 (710)

Viruses were subjected to phenotypic testing against oseltamivir and zanamivir

a A(H1N1)pdm09 virus (A/Belgium/14G0006/2014) carries I223R	Fold difference = os 40 zan 4	
b A(H1N1)pdm09 virus (A/Cyprus/F79/2014) carries S247N	Fold difference = os 7 zan 4	Outlier
c A(H1N1)pdm09 virus (A/Paris/1823/2014) carries H275Y	Fold difference = os 446 zan 3	
d A(H1N1)pdm09 virus (A/Norway/2227/2015) carries H275Y	Not tested	
* A(H1N1)pdm09 virus (A/Hungary/366/2014) carries H275Y	Not tested	
e A(H3N2) virus (A/Serbia/NS-613/2014) carries Q391H	Fold difference = os 8 zan 9	Outlier
f A(H3N2) virus (A/Milano/84/2014) carries V215I and S331R (-CHO)	Fold difference = os 10 zan 8	
g A(H3N2) virus (A/Ukraine/86/2014) carries S331R (-CHO)	Fold difference = os 10 zan 8	
h A(H3N2) virus (A/Stockholm/14/2014) carries I194V, S331R (-CHO) and E381G	Fold difference = os 26 zan 9	
i A(H3N2) virua (A/Israel/Z-1213/2014) carries E74K, I312T and N329K (-CHO)	Fold difference = os 13 zan 3	
j A(H3N2) virus (A/Ukraine/125/2014) carries I312T and S331R (-CHO)	Fold difference = os 10 zan 1	
k A(H3N2) virus (A/Calarasi/162804/2014) carries S331R (-CHO) and D339N	Fold difference = os 17 zan 1	
l B/Yam virus (B/Ukraine/6295/2015) carries D197N	Fold difference = os 6 zan 4	
m B/Yam virus (B/Extremadura/728/2015) carries D197G	Fold difference = os 4 zan 2	Outlier

* Numbers in parentheses indicate the number of viruses assayed since the February 2015 VCM